Skip Connections Unlock d^2_{model} Parameters per Layer

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Abstract

It was recently shown that skip connections change the function class of a layer. We formally prove that they unlock at least d^2_{model} functional dimensions per layer (resp. $\approx d^2_{\text{model}}/2$ under pre-RMSNorm). More precisely, in a standard residual transformer of any depth consisting of modern analytic layers such as attention, GELU or SwiGLU, the only dimensional redundancy consists of the continuous layerwise symmetries, together with a single global $\mathrm{GL}(d_{\text{model}})$ (resp. $\mathrm{O}(d_{\text{model}})$) one. This answers a conjecture of Henry (2026) and corrects a statement of Wang and Wang (2025). From skipless networks, by contrast, we remove d^2_{model} (resp. $\approx d^2_{\text{model}}/2$) parameters per layer without changing the network function. This is a lossless reduction of about 8.5 percent of the parameters of the skipless ViT-B of Ji et al. (2026), which indicates that it outperforms its residual counterpart with fewer parameters.

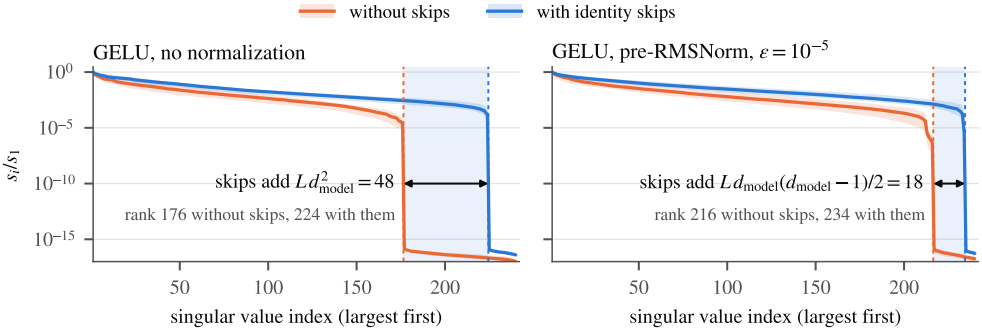


Figure 1: Singular values of the Jacobian of the parameter-to-output map on 4000 Gaussian inputs, with and without skips at the same weights, for $L = 3$ GELU blocks of $d_{\text{hidden}} = 8$ hidden units, with $d_{\text{model}} = 4$; median and range over 20 draws of weights and inputs. Skips raise the rank, the functional dimension.

1 Introduction

Skip connections are a building block of almost every deep network trained today (He et al., 2016). The usual reason is training: skips make deep networks easier to optimize (Balduzzi et al., 2017; He et al., 2016; Orhan and Pitkow, 2018). Mijoski and Karbevski (2026) show that skips also change which functions a network can represent: for most kinds of block, almost every network with skips computes a function that no network of such blocks without skips can compute. Their result establishes that there is a difference in representation, but does not quantify it. In the same direction, we show that skip connections add at least d^2_{model} degrees of freedom per block, and about half as

*Author contributions. Marko Karbevski (MK) conceived and led the study, proved the results at depth (Theorems 4.9, 4.12 and 4.13), and worked on the symmetry breaking across layers and its counting, the parameter counts at model scale, the experiments and the applications. Antonij Mijoski (AM) worked on the writing. All of MK's work on the results, the proofs and the experiments was done with AI assistants.

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many under pre-RMSNorm (Figure 1), where d_{model} is the model dimension. We motivate this study by realizing residual connections as a reduction of invertible (resp. orthogonal, under pre-RMSNorm) learned linear skips (Section 3.2).

Skipless networks have recently been trained to varying degrees of success. He et al. (2023) train deep transformers without skips by modifying the attention, and He and Hofmann (2024) remove the skips of transformer blocks without losing training speed. Ji et al. (2025) find that, with standard training, the skips around the MLP blocks of a vision transformer can be removed with little loss of accuracy. Most notably (and recently), Ji et al. (2026) remove the skips around the attention blocks as well, with a suitable initialization, and their ViT-B with no skips at all outperforms the residual one on ImageNet-1k, at the same parameter budget. Their explanation is that a skip keeps bringing the information of early layers back into later ones, which disrupts the hierarchy of features that depth should build.

Where the degrees of freedom come from. Without skips, the basis in which a hidden state is written is free: if the weights that write the output of one block are multiplied by an invertible matrix, and the weights through which the next block reads it by the inverse, the function does not change. Such a change of basis can set a $d_{\text{model}} \times d_{\text{model}}$ submatrix of the weights to the identity, so every hidden state of a skipless network carries d_{model}^2 redundant parameters. A skip adds each hidden state to the output of the next block, and the sum is the next hidden state, so the function is preserved only when two consecutive hidden states have the same change of basis. With skips, one change of basis is therefore left for the whole network, whereas a skipless network of L blocks has one at each of its $L+1$ hidden states, from the one after the embedding to the one before the readout. The difference is L changes of basis: d_{model}^2 parameters per block. Under pre-RMSNorm only orthogonal changes of basis preserve the function, so d_{model}^2 becomes $d_{\text{model}}(d_{\text{model}}-1)/2$, per hidden state and per block. In the skipless ViT-B of Ji et al. (2026), these redundant parameters make up 8.5 percent of the model (Section 3.4).

Functional dimension. By the degrees of freedom of a network we mean its *functional dimension*, the number of independent directions in which the parameters move the function (Grigsby et al., 2025). We use the two terms interchangeably. At generic parameters, the functional dimension is the local dimension of the set of functions that the network can represent. The changes of basis are symmetries of the network: they move the parameters and not the function. Blocks have symmetries of their own, such as the rescaling of a hidden unit of a ReLU block (Section 4.2). So the functional dimension is at most the number of parameters minus the number that the symmetries make redundant. This is only an upper bound: a network can lose more dimensions than its symmetries account for. Grigsby et al. (2023) show it for ReLU networks on sets of parameters of positive measure. It happens with skips as well, for ReLU networks of two or more blocks on an open set of weights (Appendix A.8).

1.1 Contributions

Main result. Residual networks attain the upper bound from the symmetries at any depth when their blocks are attention, $\tanh$, GELU, SwiGLU or GeGLU blocks (Theorem 4.9). These five kinds of block are real-analytic, and we call them *analytic* blocks. The theorem holds at generic parameters, with or without pre-RMSNorm, within the scope of Section 2: in particular, for networks without biases and for continuous inputs, which excludes one-hot tokens. So, for each block, the parameters that a change of basis makes redundant in the skipless network move the function in the residual network. In this sense the skips unlock them (Figure 1). The skips raise the functional dimension by at least the number of these parameters, and by exactly that number when the skipless network attains its bound too. We prove that it does for networks of one block without normalization, and, at any depth and with or without pre-RMSNorm, for networks of $\tanh$, SwiGLU or GeGLU blocks. For networks of GELU blocks we prove it at any depth under pre-RMSNorm, and without normalization when every block has at least $2d_{\text{model}}$ hidden units (Section 4.3). For attention-only transformers with skip connections, the main result answers Conjecture 2.4 of Henry (2026): the redundancy is that of the blocks, as conjectured, together with the one change of basis of the whole network

(Corollary 4.11). The same holds with MLP blocks and under pre-RMSNorm.

1. *Changes of basis, in one form for all blocks* (Section 3). We write every block in one form, input weights followed by a fixed map and by output weights. In this form the change of basis is one symmetry for all blocks: it holds for MLP blocks and for attention blocks, with and without pre-RMSNorm, and it extends to blocks whose skip is a trainable matrix (item 3).
2. *The one change of basis that the skips leave* (Sections 3.3 and 3.4). We prove that changes of basis, one per hidden state, preserve the function of every residual network if and only if they are all equal and, under pre-RMSNorm, orthogonal (Theorems 3.6 and 3.7). The one change of basis that remains also changes the embedding, so it is not a product of the symmetries of the blocks that Wang and Wang (2025) characterize (Appendix C).
3. *Learned skips* (Section 3.2). A learned skip multiplies the hidden state by a trainable $d_{\text{model}} \times d_{\text{model}}$ matrix A before adding it, and the usual skip, with $A = I$, is an *identity skip*. A change of basis acts on a learned skip as well. Without normalization, when every A is invertible, changes of basis that telescope along the network turn every learned skip into the identity, so the network computes the same function as a residual network with Ld_{model}^2 fewer parameters (Theorem 3.4). Under pre-RMSNorm, a network with orthogonal learned skips computes the same function as a residual network, and invertible learned skips can be made upper triangular (Section 3.4).
4. *Measured ranks* (Section 4.5). We compute the rank of the Jacobian of the parameter-to-output map for MLPs and for small transformers of several blocks. It attains the bound in all cases but two, both without skips: pre-RMSNorm ReGLU networks, and one of eight ReLU networks.
5. *Function classes and limits* (Section 4.4). Mijoski and Karbevski (2026) show that, at generic parameters, a residual network of analytic blocks computes a function that no skipless network of such blocks computes, with one exception: a residual network of GELU blocks without normalization computes the same function as a skipless network with d_{model} more hidden units per block. They also show that, without normalization, every skipless network is a limit of residual networks of the same widths. In the other direction, we show that almost no residual network is a limit of skipless networks of the same widths, for analytic blocks, with or without pre-RMSNorm (Theorem 4.13(i)). The reason is the functional dimension: these skipless networks, together with their limits, form a family of functions of lower dimension than the residual networks. For GELU blocks without normalization, the same comparison shows that a skipless network of L blocks needs at least $Ld_{\text{model}}/2$ more hidden units in total to compute the function of a residual network of L blocks at generic parameters (Theorem 4.13(ii)). The construction of Mijoski and Karbevski (2026) uses Ld_{model} more.
6. *Local identifiability*. For analytic blocks, we show that a network attains its bound exactly when it is *locally identifiable* at generic parameters: nearby parameters with the same function differ only by changes of basis and by symmetries of the blocks (Proposition 4.7). So the residual networks of the main result are locally identifiable (Corollary 4.10).

1.2 Related work

The change of basis as a symmetry. Godfrey et al. (2022) and, independently, Zhao et al. (2023) formalize the change of basis as a symmetry of MLPs (their Propositions 3.4 and C.1). To our knowledge, Graef (2024) is the first to use it to remove parameters, independently of both: in skipless transformers without normalization, the reduction sets the output projection of every attention block and one of its query, key and value matrices to the identity. When this matrix is the value matrix, the reduced block has identity value and projection matrices, as the skipless attention blocks of He and Hofmann (2024) have by design. We use the symmetry for every kind of block and at every hidden state, with and without pre-RMSNorm (Section 3).

One change of basis for a residual network. Earlier work uses one change of basis for all the hidden states of a residual network. The residual stream has no privileged basis (Elhage et al., 2021). SliceGPT proves that an RMSNorm transformer computes the same function after one rotation of every matrix that touches the stream (Ashkboos et al., 2024a, Theorem 1), QuaRot and SpinQuant use this invariance for quantization (Ashkboos et al., 2024b; Liu et al., 2025), and model merging aligns streams with one global permutation (Verma and Elbayad, 2024, Section 3.3) or orthogonal matrix

Proven	Value	Blocks	Result
Parameters explicitly removed without changing the function of the network, post-training			
without skips	$(L+1)d_{\text{model}}^2$	any	Prop. 3.2
with pre-RMSNorm	$(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$	any	Thm. 3.7(i)
learned skips	$(L+1)d_{\text{model}}^2$	any	Rem. 3.5
with pre-RMSNorm	$(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$	any	Rem. A.12
identity skips	d_{model}^2	any	Thm. 3.6(iii)
with pre-RMSNorm	$d_{\text{model}}(d_{\text{model}}-1)/2$	any	Thm. 3.7(iii)
Dimensions of symmetry removed by the skips			
any L	Ld_{model}^2	any	Thm. 3.6(ii)
with pre-RMSNorm	$Ld_{\text{model}}(d_{\text{model}}-1)/2$	any	Thm. 3.7(ii)
Functional dimension added by the skips			
$L=1$	exactly d_{model}^2	any	Thm. 4.8(i)
with pre-RMSNorm	at least $d_{\text{model}}(d_{\text{model}}-1)/2$	any	Thm. 4.8(ii)
$L\geq 2$	at least Ld_{model}^2	analytic	Thm. 4.9
with pre-RMSNorm	at least $Ld_{\text{model}}(d_{\text{model}}-1)/2$	analytic	Thm. 4.9
with $\varepsilon=0$	at least $Ld_{\text{model}}(d_{\text{model}}-1)/2+L$	analytic	Thm. A.26
$L\geq 2$	exactly Ld_{model}^2	$\tanh$	Thm. 4.12(i)
with pre-RMSNorm	exactly $Ld_{\text{model}}(d_{\text{model}}-1)/2$	$\tanh$	Thm. 4.12(i)
with $\varepsilon=0$	exactly $Ld_{\text{model}}(d_{\text{model}}-1)/2+L$	$\tanh$	Thm. A.34
$L\geq 2$	exactly Ld_{model}^2	GELU, $d_{\text{hidden}}\geq 2d_{\text{model}}$	Thm. 4.12(ii)
with pre-RMSNorm	exactly $Ld_{\text{model}}(d_{\text{model}}-1)/2$	GELU	Thm. 4.12(ii)
with $\varepsilon=0$	exactly $Ld_{\text{model}}(d_{\text{model}}-1)/2+L$	GELU	Thm. A.34
$L\geq 2$	exactly Ld_{model}^2	SwiGLU, GeGLU	Thm. 4.12(iii)
with pre-RMSNorm	exactly $Ld_{\text{model}}(d_{\text{model}}-1)/2$	SwiGLU, GeGLU	Thm. 4.12(iii)
Residual networks that equal, or are limits of, skipless networks			
same widths, any L	measure zero	analytic	Thm. 4.13(i)
with pre-RMSNorm	measure zero	analytic	Thm. 4.13(i)
wider by $< Ld_{\text{model}}/2$	measure zero	GELU	Thm. 4.13(ii)

Table 1: Summary of the results, for networks of L blocks at generic parameters (scope in Section 2). "Analytic" blocks are attention, $\tanh$, GELU, SwiGLU and GeGLU blocks. "Any" block is an analytic block or a ReLU, ReGLU or ReLU^2 (the square of ReLU) block.

(Theus et al., 2025). van Nierop (2024) counts the symmetries of LayerNorm transformers, with one rotation of the stream for the whole model. In residual ReLU networks, skips are known to tie the permutations and rescalings of the hidden states (Godfrey et al., 2022, Remark 3.6). In the language of gauge theory, skips turn a symmetry that is local in depth into a global one (Hashimoto et al., 2024). Where a block outweighs its skip, the local changes of basis remain approximate symmetries (Gong et al., 2026, Section 6.4). Zhao et al. (2023) remark that their proposition can be adapted to networks with skip connections. Theorems 3.6 and 3.7 work this case out for changes of basis, in both directions.

Learned and emulated skips. LAuReL trains skips that are the identity plus a low-rank matrix (Menghani et al., 2025), RMNet emulates the skips of convolutional networks with ReLU hidden units (Meng et al., 2021), and SliceGPT inserts rotations into identity skips and deletes rows and columns, approximating the model (Ashkboos et al., 2024a, Sections 3.3 and 3.4). Item 3 concerns trainable linear skips, and its reduction is exact.

Functional dimension and the Jacobian. Grigsby et al. (2025) bound the functional dimension of ReLU networks by their rescaling symmetries, and it is observed to fall during training (Bona-Pellissier et al., 2024). The functional dimension bounds a local pseudodimension from below (Grigsby and Lindsey, 2024), and it accounts for the memorization capacity of attention layers better than the parameter count does (Henry, 2026). For polynomial networks it is the dimension of the

functional variety, a measure of expressive power that Kileel et al. (2019) compute, as we do, as the rank of a Jacobian over a finite field. MacDonald et al. (2023) show that identity skips keep the Jacobian of the parameter-to-output map of full row rank on linearly independent inputs. We study the kernel of the same Jacobian. Marcotte et al. (2025) show that a block has the same conservation laws of gradient flow with and without its skip. This is consistent with our result, which needs an embedding and a readout around the block: without them, the skip does not change the derivative of the block with respect to its parameters.

Identifiability and the symmetries of attention. For skipless networks of attention blocks with one head and without softmax, Henry et al. (2025) prove that, at generic parameters, the function determines the weights up to scalings of the blocks and changes of basis inside and between them. With softmax, they prove for one block that the function determines the attention matrix, the product of the query and key matrices, and the value matrix, and they conjecture that at any depth it determines the weights up to the changes of basis inside and between the blocks. The remark after Theorem 4.8 uses their theorem for one block with one head, and for residual networks item 6 proves the corresponding local statement at any depth, with several heads. Wang and Wang (2025) prove that an attention block with several heads of total width d_{model} has, at generic parameters, no symmetries beyond the changes of basis inside its heads and the permutations of its heads. The symmetries of a block are the same with and without skips, so they cancel when a residual and a skipless network are compared (Section 4.2). For one attention block on its own, without an embedding, a readout or a skip, Henry (2026, Theorem 2.1) proves that, at generic parameters, these symmetries relate all parameters with the same function. This gives the functional dimension of the block, and the remark after Theorem 4.8 uses it for several heads. The same paper conjectures that an attention-only transformer with skip connections has, at any depth, the redundancy of its blocks and no more (Conjecture 2.4 there). Corollary 4.11 answers it: this holds for the blocks alone, and with an embedding and a readout the network has one more change of basis. Dahlgren et al. (2026)[1] prove that residual networks with polynomial activations, whose blocks have no output weights, are generically identifiable up to finitely many sign flips. The changes of basis that the skips remove do not arise there.

2 Setup

This section fixes the networks that we compare. We write every block, MLP or attention, in one form. We then define the networks without skips, with identity skips and with learned skips, add normalization, and state the scope of the results: wide MLP blocks ($d_{\text{hidden}} \geq d_{\text{model}}$), attention blocks whose heads have total width d_{model}, no biases, and continuous inputs.

Blocks. A hidden state is a matrix $h \in \mathbb{R}^{T \times d_{\text{model}}}$ whose rows are the T tokens of a sequence. A block reads it with a trainable matrix W_{read}, the *read*, applies a fixed map f, and writes the result with a trainable matrix W_{write}, the *write*:

$$B(h) = f(hW_{\text{read}})\, W_{\text{write}}. \tag{1}$$

In an MLP block with d_{hidden} hidden units, its width, the read is a $d_{\text{model}} \times d_{\text{hidden}}$ matrix, the write is a $d_{\text{hidden}} \times d_{\text{model}}$ matrix, and f applies an activation to every hidden unit of every token: $\tanh$, GELU, ReLU, or its square, ReLU^2. In a gated block the read has $2d_{\text{hidden}}$ columns, which give each token a gate g_i and a value v_i for each hidden unit, and f maps them to $\sigma(g_i)\, v_i$, with σ the ReLU in ReGLU, SiLU in SwiGLU and GELU in GeGLU (Shazeer, 2020). In an attention block with H heads of width d_h, the read holds the query, key and value matrices W_Q, W_K and W_V of all heads, f is the attention map, and the write is the output projection W_O; the width of the block is Hd_h.

Networks. A network has a linear embedding $E \in \mathbb{R}^{d_{\text{in}} \times d_{\text{model}}}$, then L blocks $B_1, \ldots, B_L$, each with a skip A_ℓ, then a linear readout, or unembedding, $D \in \mathbb{R}^{d_{\text{model}} \times d_{\text{out}}}$, and an output function C.

[1] Concurrently with and independently of our first version (7 October 2026, https://github.com/MarkoKarbevski/skips_fix_gauges/commit/8128617).

It maps an input $x \in \mathbb{R}^{T \times d_{\text{in}}}$ to

$$\mathcal{M}(x) = C \circ D \circ (A_L + B_L) \circ \cdots \circ (A_1 + B_1) \circ E\,(x), \tag{2}$$

where E, D and A_ℓ stand for the maps $x \mapsto xE$, $h \mapsto hD$ and $h \mapsto hA_\ell$: $A_\ell = 0$ in a *skipless* network, $A_\ell = I$ in a *residual* network, and A_ℓ is a trainable $d_{\text{model}} \times d_{\text{model}}$ matrix in a network with *learned skips*. So $h_0 = xE$, $h_\ell = h_{\ell-1}A_\ell + B_\ell(h_{\ell-1})$ and $y = h_L D$. The output function C is set by the task, for example the identity for regression and the softmax for multiclass classification (Bishop, 2006, Sections 5.1 and 5.2). It has no weights, and we ignore it: functions, equality and functional dimension below are those of the map $x \mapsto y$. The skipless and the residual network have the same P parameters. A transformer layer has two blocks, its attention and MLP sub-layers: a transformer of 12 layers has $L = 24$, and *per block* means per sub-layer.

Normalization. RMSNorm with the constant $\varepsilon > 0$ is the map N_ε that acts on every row $v \in \mathbb{R}^{d_{\text{model}}}$ of a hidden state h as

$$N_\varepsilon(v) \;=\; \frac{v}{\sqrt{\dfrac{\|v\|^2}{d_{\text{model}}} + \varepsilon}},$$

so that $N_\varepsilon(h)$ is the matrix of the normalized rows. The constant ε is fixed and is not a parameter. In a *pre-RMSNorm* network the blocks and the readout read $N_\varepsilon(h)$ in place of h. Skips read h itself, so a residual network computes $h_\ell = h_{\ell-1} + B_\ell\big(N_\varepsilon(h_{\ell-1})\big)$ and $y = N_\varepsilon(h_L)D$. A learned gain after N_ε is folded into the next read. Below, *under RMSNorm* means in a pre-RMSNorm network.

Scope. Networks have no biases, and their inputs are continuous: they range over all of $\mathbb{R}^{T \times d_{\text{in}}}$, which excludes one-hot tokens. Every MLP block has $d_{\text{hidden}} \geq d_{\text{model}}$, and all MLP blocks are of one kind. All attention blocks have the same number H of heads, of total width $Hd_h = d_{\text{model}}$, and act on sequences of at least two tokens. The input and the output satisfy $d_{\text{in}}, d_{\text{out}} \geq d_{\text{model}} \geq 2$, which excludes a classifier with fewer classes than d_{model}. The reductions of Section 3 hold for any inputs, one-hot tokens included, and with biases. For one-hot tokens the counts of Section 4 are measured (Section 4.5), and with biases the skips are measured to add the same Ld_{model}^2 (Table 6). The appendix states each result in greater generality. A statement holds at *generic* parameters if it holds outside a set of Lebesgue measure zero, so with probability one for weights drawn from a density. Attention, $\tanh$, GELU, SwiGLU and GeGLU blocks are real-analytic, *analytic* for short, and for them any open set of inputs gives the same results.

3 Changes of basis that preserve the function

The results of this section rest on one symmetry, which the first two blocks of a skipless network without normalization already show. Let block ℓ have the read W_{read}^ℓ, the fixed map f_ℓ and the write W_{write}^ℓ. Block 1 writes the hidden state $h_1 = f_1(h_0 W_{\text{read}}^1)\, W_{\text{write}}^1$, and block 2 reads it as

$$h_1 W_{\text{read}}^2 \;=\; f_1(h_0 W_{\text{read}}^1)\; W_{\text{write}}^1 W_{\text{read}}^2. \tag{3}$$

The write of block 1 and the read of block 2 enter the function only through their product, and for any invertible $d_{\text{model}} \times d_{\text{model}}$ matrix G we have the invariance

$$\big(W_{\text{write}}^1 G\big)\big(G^{-1} W_{\text{read}}^2\big) \;=\; W_{\text{write}}^1 W_{\text{read}}^2. \tag{4}$$

So replacing W_{write}^1 by $W_{\text{write}}^1 G$ and W_{read}^2 by $G^{-1} W_{\text{read}}^2$ does not change the function of the network. It changes the hidden state h_1 into $h_1 G$: the same tokens, written in another basis.

Definition 3.1 (Change of basis)**.** In a skipless and in a residual network alike, the writes into the hidden state h_ℓ are W_{write}^ℓ, or E if $\ell = 0$, and the reads of h_ℓ are $W_{\text{read}}^{\ell+1}$, or D if $\ell = L$. Let G be an invertible $d_{\text{model}} \times d_{\text{model}}$ matrix. The *change of basis* G at h_ℓ maps

$$W \mapsto WG \;\text{ for the writes into } h_\ell, \qquad W \mapsto G^{-1} W \;\text{ for the reads of } h_\ell. \tag{5}$$

An identity skip has no weights, and a change of basis leaves it in place.

What is known. This section builds on results that are known, in scattered and sometimes partial form. For MLPs, Godfrey et al. (2022) show that the weights of two consecutive layers can be changed in this way by every matrix A for which the activation σ between them satisfies $\sigma \circ A = B \circ \sigma$ for some matrix B (their Proposition 3.4). With the identity as activation this is every invertible matrix, and the symmetry is the invariance (4). Zhao et al. (2023) show the same independently (their Proposition C.1). They include activations that rescale a vector by a function of its norm, $\sigma(z) = f(\|z\|)\, z$, for which A is orthogonal. RMSNorm is such an activation, applied to every token, with $f(r) = (r^2/d_{\text{model}} + \varepsilon)^{-1/2}$ (Section 3.4). They name model compression as a prospect of such symmetries, and stop short of using them to remove parameters. Such a removal is moreover exact, a free lunch: the network keeps its function. To our knowledge, Graef (2024) is the first to remove parameters in this way, independently and without explicitly mentioning the symmetry framework. Moreover, the work is restricted to skipless transformers without normalization, and applies the reduction only to the attention blocks (Section 3.1). For skipless networks of attention blocks with one head, Henry et al. (2025) state the symmetry between consecutive blocks and count the dimensions that it makes redundant. For residual networks, earlier work uses one change of basis for all hidden states (Section 1.2).

What this section builds. We use the symmetry to remove parameters from every block of Section 2, MLP blocks and attention blocks alike: a change of basis turns a $d_{\text{model}} \times d_{\text{model}}$ submatrix of the weights into the identity (Section 3.1). The computation (3) uses only the form (1) of a block. For an attention block, $W_{\text{read}} = [\, W_Q \mid W_K \mid W_V \,]$ and $W_{\text{write}} = W_O$. A change of basis also acts on a learned skip, and changes of basis that telescope along the network turn invertible learned skips into identity skips (Section 3.2). With identity skips, changes of basis preserve the function only when they are the same at all hidden states (Section 3.3). Under pre-RMSNorm only orthogonal changes of basis preserve the function, and the submatrix becomes triangular in place of the identity (Section 3.4).

3.1 Skipless networks: one change of basis per hidden state

We start with networks that have neither skips nor normalization, where the symmetry is largest. We show on an example how one change of basis removes d_{model}^2 parameters, and then do the same at every hidden state (Proposition 3.2).

In a skipless network without normalization, the invariance (4) holds at every hidden state, so every change of basis preserves the function (Figure 2a). A change of basis that turns a $d_{\text{model}} \times d_{\text{model}}$ submatrix of the weights into the identity removes d_{model}^2 parameters: the submatrix no longer requires storage or computation.

Example (Choosing the submatrix)**.** Take $1 \le \ell < L$ and write $W_{\text{read}}^{\ell+1} = [\, W_1 \mid W_2 \,]$, where W_1 holds the first d_{model} columns. If W_1 is invertible, the change of basis $G = W_1$ at h_ℓ maps

$$W_{\text{read}}^{\ell+1} \;\mapsto\; W_1^{-1} \,[\, W_1 \mid W_2 \,] = [\, I \mid W_1^{-1} W_2 \,], \qquad W_{\text{write}}^{\ell} \;\mapsto\; W_{\text{write}}^{\ell}\, W_1.$$

The identity in the new read holds d_{model}^2 fixed entries. We take the first d_{model} columns for convenience: any d_{model} columns that form an invertible matrix serve as W_1.

Proposition 3.2 (Exact reduction without skips)**.** *Without normalization, in a skipless network, let d_{model} columns of the read of h_ℓ ($W_{\text{read}}^{\ell+1}$, or D if $\ell = L$) form an invertible matrix W_1. The change of basis $G = W_1$ at h_ℓ preserves the function and turns W_1 into $G^{-1}W_1 = I$. At generic parameters every hidden state has such columns, and these changes of basis at all $L+1$ hidden states remove $(L+1)d_{\text{model}}^2$ parameters.*

A special case of Proposition A.9, proved in Appendix A.3.

Rows of the matrix W_{write}^{ℓ} work too, or rows of E if $\ell = 0$: if d_{model} of them form an invertible matrix W_1, the change of basis $G = W_1^{-1}$ at h_ℓ turns them into $W_1 G = I$.

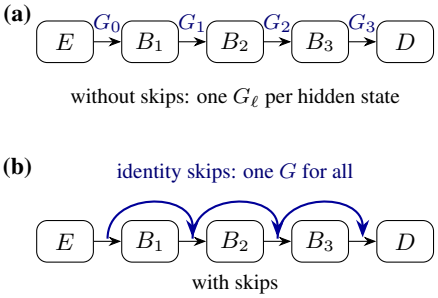


Figure 2: (a) Without skips, every hidden state has its own change of basis G_ℓ. (b) Identity skips carry each hidden state into the next, so one change of basis is left.

Example 3.3 (Choosing rows)**.** Let $d_{\text{model}} = 2$ and $d_{\text{hidden}} = 3$, and let W_1 consist of rows 1 and 3 of W^ℓ_{write}, outlined in red. The change of basis $G = W_1^{-1}$ at h_ℓ maps

$$W^\ell_{\text{write}} = \begin{bmatrix} \boxed{\begin{matrix} 2 & 1 \end{matrix}} \\ \begin{matrix} 0 & 3 \end{matrix} \\ \boxed{\begin{matrix} 1 & 1 \end{matrix}} \end{bmatrix} \quad \longmapsto \quad W^\ell_{\text{write}} G = \begin{bmatrix} \boxed{\begin{matrix} 1 & 0 \end{matrix}} \\ \begin{matrix} -3 & 6 \end{matrix} \\ \boxed{\begin{matrix} 0 & 1 \end{matrix}} \end{bmatrix}, \qquad G = W_1^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix},$$

and the next read $W^{\ell+1}_{\text{read}}$ to $G^{-1}W^{\ell+1}_{\text{read}} = W_1 W^{\ell+1}_{\text{read}}$. The two outlined rows now hold $d^2_{\text{model}} = 4$ fixed entries, and the other row has changed. Permuting the hidden units of the block moves the chosen rows next to each other. At generic parameters any d_{model} rows form an invertible matrix, so in practice one takes a $d_{\text{model}} \times d_{\text{model}}$ block of the matrix, for example its first d_{model} rows.

Take an attention block B_ℓ of a skipless transformer with $Hd_h = d_{\text{model}}$, so that W_Q and W_O are $d_{\text{model}} \times d_{\text{model}}$ matrices, invertible at generic parameters. The change of basis $G = W_Q$ at $h_{\ell-1}$ turns W_Q into $W_Q^{-1}W_Q = I$, and the change of basis $G = W_O^{-1}$ at h_ℓ turns W_O into $W_O W_O^{-1} = I$. Together they remove $2d^2_{\text{model}}$ parameters per layer. This is the reduction of Graef (2024), who sets W_O and one of W_Q, W_K and W_V to the identity. By the count given there, it would remove 15 percent of the weights of Mistral-7B if the model had no skip connections and no normalization. Proposition 3.2 removes d^2_{model} more, at the hidden state before the readout, and holds for every kind of block.

3.2 Telescoping (generic) learned skips to identity skips

In Section 3.1 the change of basis at each hidden state was chosen on its own. We now turn to networks with learned skips, $h_\ell = h_{\ell-1}A_\ell + B_\ell(h_{\ell-1})$. A change of basis acts on the skips as well. We first write this action out, and then choose the changes of basis so that every invertible skip becomes the identity (Theorem 3.4). This motivates our study: identity skips, the usual residual connections, are a reduction of generic learned linear skips. The choice is no longer local: the change of basis at one hidden state is a product of all the later skips. Networks have no normalization here, and Section 3.4 treats pre-RMSNorm.

The skip A_ℓ multiplies $h_{\ell-1}$, and the result is added to h_ℓ, so A_ℓ is at once a read of $h_{\ell-1}$ and a write into h_ℓ. With changes of basis $G_{\ell-1}$ at $h_{\ell-1}$ and G_ℓ at h_ℓ,

$$h_\ell G_\ell \;=\; \big(h_{\ell-1}G_{\ell-1}\big)\,\big(G_{\ell-1}^{-1}A_\ell G_\ell\big) \;+\; f_\ell\Big(\big(h_{\ell-1}G_{\ell-1}\big)\,G_{\ell-1}^{-1}W^\ell_{\text{read}}\Big)\,W^\ell_{\text{write}}G_\ell. \tag{6}$$

So the maps (5) of Definition 3.1, together with $A_\ell \mapsto G_{\ell-1}^{-1}A_\ell G_\ell$ for every skip, preserve the function of the network: every hidden state h_ℓ becomes $h_\ell G_\ell$.

Let every A_ℓ be invertible, as it is at generic parameters. The new skip $G_{\ell-1}^{-1}A_\ell G_\ell$ is the identity when $G_{\ell-1} = A_\ell G_\ell$. Take $G_L = I$, which leaves the readout unchanged, and go from the last block down to the first:

$$G_\ell \;=\; A_{\ell+1}A_{\ell+2}\cdots A_L \;=:\; \Lambda_\ell, \qquad \text{so that} \qquad A_\ell\Lambda_\ell = \Lambda_{\ell-1}. \tag{7}$$

The products telescope: with $G_\ell = \Lambda_\ell$ at every hidden state, the skip of block ℓ becomes $\Lambda_{\ell-1}^{-1} A_\ell \Lambda_\ell = I$. The embedding becomes $E\Lambda_0 = E\, A_1 \cdots A_L$.

Theorem 3.4 (Learned skips reduce to identity skips)**.** *Without normalization, let every learned skip A_ℓ be invertible, and let $\Lambda_\ell = A_{\ell+1} \cdots A_L$, with $\Lambda_L = I$. The network with learned skips computes the same function as the residual network with the embedding $E\Lambda_0$, the reads $\Lambda_{\ell-1}^{-1} W_{\text{read}}^\ell$, the writes $W_{\text{write}}^\ell \Lambda_\ell$ and the readout D.*

A special case of Theorem A.5, proved in Appendix A.2.

So the networks with invertible learned skips have Ld_{model}^2 more parameters than the residual networks and compute exactly the same functions, with the same functional dimension (Theorem A.5(ii)). The theorem does not cover the skipless networks, which are the networks with learned skips $A_\ell = 0$ for every ℓ.

Remark 3.5 (The reduction is not unique). In a network with learned skips, the changes of basis can also be chosen as in Section 3.1, each on its own: at every hidden state h_ℓ, a submatrix of $W_{\text{read}}^{\ell+1}$ or of W_{write}^ℓ becomes the identity, or one of D or of E at the two ends. The skips then stay learned, as $G_{\ell-1}^{-1} A_\ell G_\ell$. This removes $(L+1)d_{\text{model}}^2$ parameters, as many as Theorem 3.4 removes together with the one change of basis that a residual network keeps (Section 3.3). Remark A.10 writes it out.

3.3 Identity skips: one change of basis for the whole network

Section 3.2 turned invertible learned skips into identity skips. We now ask which changes of basis remain once the skips are identities. We compute what two different changes of basis do to one residual block, and read off the answer (Theorem 3.6).

A residual network keeps one change of basis, the same at all its hidden states. With a change of basis $G_{\ell-1}$ at $h_{\ell-1}$ and G_ℓ at h_ℓ, a residual block computes

$$h_{\ell-1} G_{\ell-1} + B_\ell(h_{\ell-1})\, G_\ell \;=\; h_\ell G_\ell + h_{\ell-1}(G_{\ell-1} - G_\ell). \tag{8}$$

The skip carries $h_{\ell-1}$ in the basis $G_{\ell-1}$, and the block writes in the basis G_ℓ. Unless the two agree, the defect $h_{\ell-1}(G_{\ell-1} - G_\ell)$ reaches every later block.

Theorem 3.6 (One change of basis for the whole network)**.** *Without normalization, let $G_0, \ldots, G_L$ be invertible.*
(i) *The map $E \mapsto EG_0$, $W_{\text{read}}^\ell \mapsto G_{\ell-1}^{-1} W_{\text{read}}^\ell$, $W_{\text{write}}^\ell \mapsto W_{\text{write}}^\ell G_\ell$, $D \mapsto G_L^{-1} D$ preserves the function of every residual network if and only if $G_0 = G_1 = \cdots = G_L$.*
(ii) *The changes of basis that preserve the function at all parameters form a group of dimension $(L+1)d_{\text{model}}^2$ in a skipless network and d_{model}^2 in a residual one.*
(iii) *In a residual network, let d_{model} columns of one of the reads $W_{\text{read}}^1, \ldots, W_{\text{read}}^L, D$ form an invertible matrix W_1. The same change of basis $G = W_1$ at all hidden states preserves the function and turns W_1 into the identity.*

Proof in Appendix A.2.

So the L skips remove Ld_{model}^2 dimensions from this group, and Section 4 shows when they become functional dimension.

3.4 RMSNorm and LayerNorm: orthogonal changes of basis

Up to here the networks had no normalization. We now add pre-RMSNorm and go through the same steps: the block form, the changes of basis that preserve the function, and the parameters that they remove. One thing changes: the changes of basis must be orthogonal (Theorem 3.7). LayerNorm reduces to this case.

With pre-RMSNorm, the network with learned skips computes

$$h_\ell = h_{\ell-1} A_\ell + B_\ell\big(N_\varepsilon(h_{\ell-1})\big), \qquad y = N_\varepsilon(h_L)\, D. \tag{9}$$

The skip reads $h_{\ell-1}$ itself and the block reads $N_\varepsilon(h_{\ell-1})$. A change of basis G at $h_{\ell-1}$ turns these into $h_{\ell-1}G$ and $N_\varepsilon(h_{\ell-1}G)$. For an orthogonal Q,

$$N_\varepsilon(hQ) \;=\; N_\varepsilon(h)\,Q,$$

since Q preserves the norm of every row, so Q is undone by $A_\ell \mapsto Q^\top A_\ell$ and $W^\ell_{\text{read}} \mapsto Q^\top W^\ell_{\text{read}}$. For an invertible G that is not orthogonal, $N_\varepsilon(hG) = N_\varepsilon(h)M$ fails for every matrix M at some h, since $\varepsilon > 0$ (Lemma A.2). So W^ℓ_{read} can compensate only orthogonal changes of basis. Every hidden state of a skipless network, or of one with learned skips, therefore keeps an orthogonal change of basis, $d_{\text{model}}(d_{\text{model}}-1)/2$ dimensions. It cannot turn a submatrix into the identity, but a QR factorization makes it triangular.

Theorem 3.7 (Orthogonal changes of basis under RMSNorm)**.** *Consider pre-RMSNorm networks.*
(i) *In a skipless network, let d_{model} columns of the read of h_ℓ ($W^{\ell+1}_{\text{read}}$, or D if $\ell = L$) form a matrix $W_1 = QR$, with Q orthogonal and R upper triangular. The change of basis Q at h_ℓ preserves the function and turns W_1 into R; at all $L+1$ hidden states, they remove $(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$ parameters.*
(ii) *The changes of basis that preserve the function at all parameters are orthogonal, and in a residual network they agree at all hidden states. They form a group of dimension $(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$ in a skipless network and $d_{\text{model}}(d_{\text{model}}-1)/2$ in a residual one.*
(iii) *In a residual network, let d_{model} columns of one of the reads $W^1_{\text{read}}, \dots, W^L_{\text{read}}, D$ form a matrix $W_1 = QR$. The same change of basis Q at all hidden states preserves the function and turns W_1 into R.*
(iv) Triangular skips. *If every learned skip A_ℓ is invertible, there are orthogonal $Q_1, \dots, Q_L$, with $Q_0 = I$, such that every $Q^\top_{\ell-1}A_\ell Q_\ell$ is upper triangular, and the network with these skips, the reads $Q^\top_{\ell-1}W^\ell_{\text{read}}$, the writes $W^\ell_{\text{write}}Q_\ell$ and the readout $Q^\top_L D$ computes the same function.*

Proof in Appendix A.4.

Learned skips under RMSNorm. In (9) the skip A_ℓ reads $h_{\ell-1}$ itself, so orthogonal changes of basis $Q_{\ell-1}$ at $h_{\ell-1}$ and Q_ℓ at h_ℓ map it as in (6), to $Q^\top_{\ell-1}A_\ell Q_\ell$. If every A_ℓ is orthogonal, so is every Λ_ℓ of (7), and the choice $Q_\ell = \Lambda_\ell$ turns every skip into the identity: the network computes the same function as a residual network (Theorem A.11(ii)). When the skips are invertible, Theorem 3.7(iv) chooses the Q_ℓ so that every $Q^\top_{\ell-1}A_\ell Q_\ell = R_\ell$ is upper triangular, which removes the $d_{\text{model}}(d_{\text{model}}-1)/2$ entries below its diagonal. For $L = 1$, the $d_{\text{model}}(d_{\text{model}}+1)/2$ entries of R_1 that are left are not redundant: the functional dimension with a learned skip exceeds the functional dimension with an identity skip by exactly $d_{\text{model}}(d_{\text{model}}+1)/2$ (Theorem 4.8(ii)).

LayerNorm. LayerNorm is RMSNorm after a projection that removes the mean of each row, so every RMSNorm count above holds with $d_{\text{model}} - 1$ in place of d_{model} (Proposition A.13).

Trained models. The reduction applies to trained weights. In the skipless ViT-B (Dosovitskiy et al., 2021) of Ji et al. (2026), with standard pre-LayerNorm blocks, the changes of basis at its 25 hidden states remove 7.3M parameters, 8.5 percent of the model, without changing its function. A residual model keeps one change of basis, and Appendix D gives the counts for the residual ViT-B and for Llama-3.1 (Grattafiori et al., 2024).

4 Functional dimension with and without skips

The reductions of Section 3 remove parameters without changing the function. Removing the skips is different: Mijoski and Karbevski (2026) show that, for most kinds of block, it changes the family of functions. Their result compares the residual and the skipless family through the functions that they share, and does not tell the extent of the change. We compare the degrees of freedom of the two families: their functional dimension. The same measure tells whether redundancy remains even after the stated reductions, a question that Section 3 leaves open.

4.1 Definition and a linear example

We first define the functional dimension and the redundancy, and compute both for a linear network, where they can be read off by hand.

Definition 4.1 (Functional dimension (Grigsby et al., 2025))**.** Let F_θ be the function computed at parameters $\theta \in \mathbb{R}^P$. For a finite batch $X = (x_1, \ldots, x_n)$ of inputs, let $F_X(\theta) = (F_\theta(x_1), \ldots, F_\theta(x_n))$ be the outputs on X, and $\mathrm{D}F_X(\theta)$ the Jacobian of $\theta \mapsto F_X(\theta)$. The *functional dimension* and the *redundancy* at θ are

$$\mathrm{fd}(\theta) \;=\; \max_X \operatorname{rank} \mathrm{D}F_X(\theta), \qquad\qquad \mathrm{red}(\theta) \;=\; P - \mathrm{fd}(\theta),$$

where the maximum is over all finite batches X.

A continuous symmetry adds to the redundancy: if a continuous family of maps of the parameters preserves the function, the directions in which it moves θ lie in the kernel of every such Jacobian.

Example 4.2 (Not all reductions are maximal)**.** The linear network $x \mapsto xM_1M_2M_3$, with three $d_{\mathrm{model}} \times d_{\mathrm{model}}$ matrices, has $3d_{\mathrm{model}}^2$ parameters. Its functions are the linear maps $x \mapsto xM$ with $M = M_1M_2M_3$, so its functional dimension is d_{model}^2 at generic parameters and its redundancy is $2d_{\mathrm{model}}^2$. Merging M_2 and M_3 into $M_2' = M_2M_3$ gives $x \mapsto xM_1M_2'$, with $2d_{\mathrm{model}}^2$ parameters, the same functions and a redundancy of d_{model}^2. Merging once more, into $M = M_1M_2'$, leaves d_{model}^2 parameters and no redundancy.

Example 4.3 (Fewer degrees of freedom than stored entries)**.** The orthogonal $d_{\mathrm{model}} \times d_{\mathrm{model}}$ matrices form a manifold of dimension $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2$. Yet no entry of an orthogonal matrix is fixed, so storing one in memory without any computation at read time takes d_{model}^2 entries.

4.2 Upper bounds from the symmetries

We now bound the functional dimension from above. A direction in which a symmetry moves the parameters does not change the function, so each dimension of symmetry is one dimension of redundancy. We count the symmetries, state the bounds with and without skips (Theorem 4.4), and give a name to the case in which a bound is attained. We then relate this case to identifiability.

Two kinds of symmetry bound the functional dimension. The changes of basis of one hidden state form a group $\mathcal{G}$: $\mathrm{GL}(d_{\mathrm{model}})$ without normalization, of dimension d_{model}^2, and $\mathrm{O}(d_{\mathrm{model}})$ under RMSNorm, of dimension $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2$. The *symmetries inside the blocks* change the read and the write of one block and leave the map of that block unchanged. Permutations of hidden units or heads and sign flips of $\tanh$ hidden units are finite and remove no parameters (Fefferman, 1994; Sussmann, 1992). Each hidden unit of a ReLU or ReLU^2 block can be rescaled, and so can the value of each hidden unit of a gated block and, in ReGLU, its gate (Godfrey et al., 2022; Rolnick and Kording, 2020). Each attention head has a change of basis between its queries and keys and one between its values and outputs, of dimension d_h^2 each (Elhage et al., 2021; Theus et al., 2025; Wang and Wang, 2025). A ReLU block with $d_{\mathrm{hidden}} = 4d_{\mathrm{model}}$ has $4d_{\mathrm{model}}$ rescalings, against d_{model}^2 dimensions for a change of basis. Let s be the total dimension of the symmetries inside the blocks, summed over the blocks. For $\tanh$ and GELU blocks, $s = 0$. These symmetries leave the map of every block unchanged, so they preserve the function with skips and without them, and s is the same number for both networks.

Theorem 4.4 (Upper bounds)**.** *Let $\mathcal{G}$ be the group of changes of basis of one hidden state, and s the dimension of the symmetries inside the blocks, the same with and without skips. With or without pre-RMSNorm, at generic parameters, the functional dimension is at most $P - (L+1)\dim\mathcal{G} - s$ without skips and at most $P - \dim\mathcal{G} - s$ with identity skips.*

A special case of Theorem A.14, proved in Appendix A.6.

Definition 4.5 (Equality condition)**.** A network satisfies the *equality condition* if its bound in Theorem 4.4 is attained at generic parameters, that is, if its symmetries are its only redundancy.

When the residual network satisfies it, each skip unlocks $\dim \mathcal{G}$ parameters, d_{model}^2 without normalization and $d_{\text{model}}(d_{\text{model}} - 1)/2$ under RMSNorm: they are redundant in the skipless network and count in the residual one.

Definition 4.6 (Identifiability up to symmetries)**.** A network is *identifiable up to its symmetries* at θ if every θ' with $F_{\theta'} = F_\theta$ is the image of θ under a symmetry: a change of basis that preserves the function, a symmetry inside a block, or a composition of these. It is *locally identifiable* at θ if this holds for every θ' in some neighborhood of θ.

Vlačić and Bölcskei (2020) define identifiability up to an equivalence relation on networks. For ReLU networks, Stock and Gribonval (2021) define identifiability up to permutations and rescalings of the hidden units and local identifiability up to rescalings, and Bona-Pellissier et al. (2022) define local identifiability in the same way. In Definition 4.6 the equivalence is given by the symmetries of Theorem 4.4.

Functional dimension and identifiability. The equality condition can be stated as local identifiability.

Proposition 4.7 (The equality condition and local identifiability)**.** *Let the blocks be analytic, with or without pre-RMSNorm, within the scope of Section 2. A skipless or a residual network satisfies the equality condition if and only if, at generic parameters, it is locally identifiable up to its symmetries. Every symmetry that is needed is then close to the identity.*

Proof in Appendix A.6.

The proposition compares only parameters that are close to each other. A network that is identifiable at generic parameters is locally identifiable there, so it satisfies the equality condition. In the other direction, the equality condition does not exclude a distant parameter with the same function that no symmetry reaches.

The gain in a transformer. Under the equality condition, a residual network with P parameters has the functional dimension of a skipless network with $L \dim \mathcal{G}$ more parameters. Under normalization, this is about a twelfth more for transformer layers with an MLP block of $d_{\text{hidden}} = 4d_{\text{model}}$ (Appendix D). What functional dimension predicts is only partly known (Section 1.2), and its link to accuracy is open. The skipless ViT-B of Ji et al. (2026) outperforms the residual one. If it satisfies the equality condition, it does so with less functional dimension, by the changes of basis at 24 of its 25 hidden states.

4.3 When the bounds are attained

The equality condition is a hypothesis, and it can fail (Section 4.5). We prove it in three cases.

Theorem 4.8 (Equality for one block)**.** *Let $L = 1$, with a* tanh*, GELU, ReLU, ReLU^2, ReGLU, SwiGLU, GeGLU or attention block.*
(i) *Without normalization, the skipless and the residual network satisfy the equality condition, and the skip raises the functional dimension by exactly d_{model}^2.*
(ii) *Under RMSNorm, the residual network satisfies the equality condition, and the skip raises the functional dimension by at least $d_{\text{model}}(d_{\text{model}} - 1)/2$. A learned skip raises it by exactly $d_{\text{model}}(d_{\text{model}} + 1)/2$ more than an identity skip.*

A special case of Theorem B.1, proved in Appendix B.

Remark (One attention block without a skip). For an attention block without a skip, the equality condition of Theorem 4.8(i) also follows from earlier work: from Theorem 3.9 of Henry et al. (2025) for one head, and from Theorem 2.1 of Henry (2026) for several heads and sequences of at least three tokens. Let head i have the query, key and value matrices $W_{Q,i}, W_{K,i}, W_{V,i} \in \mathbb{R}^{d_{\text{model}} \times d_h}$ and the rows $W_{O,i} \in \mathbb{R}^{d_h \times d_{\text{model}}}$ of W_O. Let x_t be row t of the input x, and let

$$\Xi_i \;=\; E W_{Q,i} W_{K,i}^\top E^\top / \sqrt{d_h} \;\in\; \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}, \qquad \Psi_i \;=\; E W_{V,i} W_{O,i} D \;\in\; \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}.$$

The skipless network gives token t the output $y_t = \sum_{i=1}^{H} \sum_{s=1}^{T} a_{ts}^{(i)}\, x_s \Psi_i$, where $(a_{t1}^{(i)}, \dots, a_{tT}^{(i)})$ is the softmax of the scores $(x_t \Xi_i x_1^\top, \dots, x_t \Xi_i x_T^\top)$. The two theorems say that the map $x \mapsto y$ determines the pairs (Ξ_i, Ψ_i), up to the order of the heads: the first for one head and all matrices Ξ_1 and $\Psi_1 \neq 0$ of these sizes, the second at generic parameters. The second is stated for $d_{\text{in}} = d_{\text{out}}$, and padding Ξ_i and Ψ_i with zeros reduces the other cases to it.

At generic parameters, Ξ_i and Ψ_i are products of two factors of rank d_h each: $\sqrt{d_h}\,\Xi_i = (EW_{Q,i})(EW_{K,i})^\top$ and $\Psi_i = (EW_{V,i})(W_{O,i}D)$. A factorization of a matrix of rank r into two factors of rank r is unique up to an invertible matrix between the factors. The heads have total width d_{model} (Section 2), so the factors $EW_{Q,i}$ of all heads, side by side, form a matrix of rank d_{model} with the factors E and W_Q, with the same E for the keys and the values. The same holds for the factors W_O and D. So the weights with the same pairs (Ξ_i, Ψ_i) are exactly the following, with G_0 and G_1 any invertible $d_{\text{model}} \times d_{\text{model}}$ matrices and C_i and C_i' any invertible $d_h \times d_h$ matrices:

$$\begin{aligned} &EG_0, \qquad G_0^{-1} W_{Q,i} C_i, \qquad G_0^{-1} W_{K,i} C_i^{-\top}, \\ &G_0^{-1} W_{V,i} C_i', \qquad C_i'^{-1} W_{O,i} G_1, \qquad G_1^{-1} D. \end{aligned}$$

Here G_0 and G_1 change the basis of h_0 and of h_1, and C_i and C_i' act inside head i. These are symmetries of the network, and so is a change in the order of the heads (Section 4.2). So the network is identifiable up to its symmetries at generic parameters, which implies the equality condition (Proposition 4.7). The proof of Theorem 4.8 does not use these theorems. It also covers sequences of two tokens and the residual network, the latter under RMSNorm as well.

For a $\tanh$ block and for a ReLU block without a skip, the equality condition of Theorem 4.8(i) also follows from earlier work on networks with one hidden layer: from the theorem of Sussmann (1992), and from the results of Stock and Gribonval (2021) (Remark B.4).

Theorem 4.9 (Equality for residual networks at any depth)**.** *Let the blocks be analytic, with or without pre-RMSNorm. The residual network satisfies the equality condition: at generic parameters its functional dimension is $P - \dim \mathcal{G} - s$. So the skips raise the functional dimension by at least $L \dim \mathcal{G}$, and by exactly $L \dim \mathcal{G}$ when the skipless network satisfies the equality condition too.*

Proof in Appendix A.7.

Corollary 4.10 (Residual networks are locally identifiable)**.** *Let the blocks be analytic, with or without pre-RMSNorm, and let θ be generic parameters of the residual network. Every θ' near θ with the same function is obtained from θ by symmetries inside the blocks and by one matrix $G \in \mathcal{G}$, all close to the identity. The matrix G is the same at all hidden states: for every block ℓ,*

$$E \mapsto EG, \qquad W_{\text{read}}^{\ell} \mapsto G^{-1} W_{\text{read}}^{\ell}, \qquad W_{\text{write}}^{\ell} \mapsto W_{\text{write}}^{\ell} G, \qquad D \mapsto G^{-1} D.$$

This is Theorem 4.9 together with Proposition 4.7. Whether these networks are identifiable, and not only locally, is open.

Corollary 4.11 (Attention-only transformers: Conjecture 2.4 of Henry, 2026)**.** *Without normalization, consider a residual network of L attention blocks at generic parameters, and let $s = 2Ld_{\text{model}}^2/H$.*
(i) *With an embedding and a readout, its functional dimension is $P - d_{\text{model}}^2 - s$.*
(ii) *Without an embedding and a readout, so that $h_0 = x \in \mathbb{R}^{T \times d_{\text{model}}}$ and $y = h_L$, its functional dimension is $P - s = L\left(4d_{\text{model}}^2 - 2d_{\text{model}}^2/H\right)$.*

Proof in Appendix A.7.

Henry (2026) proves that one attention block without a skip has $2d_{\text{model}}^2/H$ redundant dimensions, the changes of basis inside its H heads, and conjectures that an attention-only transformer of L blocks with skip connections has L times as many (Corollary 2.3 and Conjecture 2.4 there). The corollary answers the conjecture. For the blocks alone the count is the conjectured one: s is L times the count for one block, and $P - s$ is the functional dimension stated there (ii). A transformer also has an embedding and a readout, and then the one change of basis of the whole network adds d_{model}^2 redundant dimensions (i). This is the symmetry that the product of the groups of the blocks leaves out

(Appendix C). Theorem 4.9 gives the same count with MLP blocks, and under pre-RMSNorm with $d_{\text{model}}(d_{\text{model}}-1)/2$ in place of d_{model}^2. The corollary gives the dimension of the set of parameters with the same function near generic parameters. Whether the symmetries reach all parameters with the same function, as Theorem 2.1 there shows for one block, is open.

Theorem 4.12 (Exact gain for $\tanh$, GELU, SwiGLU and GeGLU)**.** *With or without pre-RMSNorm, in the three cases below the skipless network satisfies the equality condition too, so the skips raise the functional dimension by exactly $L \dim \mathcal{G}$: by Ld_{model}^2 without normalization, and by $Ld_{\text{model}}(d_{\text{model}}-1)/2$ under RMSNorm.*
(i) *The blocks are* $\tanh$ *blocks.*
(ii) *The blocks are GELU blocks, with $d_{\text{hidden}} \geq 2d_{\text{model}}$ if there is no normalization.*
(iii) *The blocks are SwiGLU blocks, or GeGLU blocks.*

Proof in Appendices A.7 and A.9.

By Proposition 4.7, the skipless networks of Theorem 4.12 are locally identifiable at generic parameters θ. For $\tanh$ and GELU blocks $s=0$, so every θ' near θ with the same function is obtained from θ by matrices $G_0, \ldots, G_L \in \mathcal{G}$ close to the identity, one per hidden state:

$$E \mapsto EG_0, \qquad W_{\text{read}}^{\ell} \mapsto G_{\ell-1}^{-1} W_{\text{read}}^{\ell}, \qquad W_{\text{write}}^{\ell} \mapsto W_{\text{write}}^{\ell} G_{\ell}, \qquad D \mapsto G_L^{-1} D.$$

In the residual network of Corollary 4.10, these matrices are all equal. For SwiGLU and GeGLU blocks, the rescalings of the value of each hidden unit come in addition (Section 4.2).

Proof sketches. With one block, the parameters enter the function only through the products ED, $W_{\text{write}}D$ and EW_{read}. A factorization of a matrix of rank d_{model} into two factors is unique up to a change of basis between them, so the skipless network keeps one change of basis at h_0 and one at h_1, and the term xED of the skip ties them into one. At any depth, replace every W_{write}^{ℓ} by $\tau W_{\text{write}}^{\ell}$ with a small τ. To first order in τ, every block reads $h_0 = xE$, and the residual network computes $xED + \tau \sum_{\ell} B_{\ell}(xE)\, D$: the linear map xED plus one wide block that holds all its hidden units and heads. This brings the one-block argument to every depth, and for analytic blocks a rank that is reached at one choice of weights is reached at almost all. For skipless $\tanh$, SwiGLU and GeGLU networks, and for skipless GELU networks under pre-RMSNorm, take a direction of the parameters along which the function does not change. It gives an identity in the input of the first block, which extends to every row by analyticity, and we follow this input to infinity along a ray, as Sussmann (1992) does for $\tanh$ networks with one hidden layer. There the hidden units of a $\tanh$ block saturate at different exponential rates, and the hidden units of a GELU block and the gates of a SwiGLU or GeGLU block differ from linear or vanishing ones by small terms of different rates. This separates the hidden units of the block: the direction acts on the weights of the block as a symmetry does, and what is left is the same statement for the network after this block (Appendix A.9). Without normalization, a SwiGLU or GeGLU block is quadratic along the ray, so the inputs of the later blocks grow faster than its own and their small terms are negligible. A GELU block is linear along the ray, the small terms of all blocks are of the same order, and this argument does not apply to it. For skipless $\tanh$ networks without normalization, the equality condition also follows from the theorem of Fefferman (1994), by which the function determines the product of each write with the next read, up to permutations and sign flips. For skipless GELU networks without normalization, the proof starts from a residual network and absorbs each skip with d_{model} more hidden units per block, so it reaches only blocks with $d_{\text{hidden}} \geq 2d_{\text{model}}$.

RMSNorm with $\varepsilon = 0$. The networks of Section 2 have $\varepsilon > 0$. For $\varepsilon = 0$, RMSNorm keeps only the direction of a token, $N_{\varepsilon}(v) = \sqrt{d_{\text{model}}}\, v/\|v\|$ for a row $v \neq 0$, so $N_{\varepsilon}(c\,hQ) = N_{\varepsilon}(h)\, Q$ for every number $c > 0$ and every orthogonal Q. Besides the symmetries inside its blocks, a residual network then has the symmetries

$$E \mapsto c\, EQ, \qquad W_{\text{read}}^{\ell} \mapsto Q^{\top} W_{\text{read}}^{\ell}, \qquad W_{\text{write}}^{\ell} \mapsto c\, W_{\text{write}}^{\ell} Q, \qquad D \mapsto Q^{\top} D,$$

with the same c and Q for every block ℓ: one orthogonal change of basis and one scale for all hidden states. The matrices cQ form the group $\mathrm{CO}(d_{\text{model}})$, of dimension $d_{\text{model}}(d_{\text{model}}-1)/2+1$.

Theorem 4.9 extends to this case with $\mathcal{G} = \mathrm{CO}(d_{\mathrm{model}})$: for analytic blocks, at generic parameters, the residual network has functional dimension $P - d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 - 1 - s$, and the skips raise the functional dimension by at least $L\,(d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1)$ (Theorem A.26). For $\tanh$ blocks and for GELU blocks of any width, the skipless network satisfies the equality condition as well, with a number c_ℓ and a matrix Q_ℓ of its own at every hidden state h_ℓ, and the skips raise the functional dimension by exactly this number (Theorem A.34).

4.4 Residual and skipless networks compute different functions

The gain in functional dimension says something about the functions themselves: it separates what skipless networks can approximate from what residual networks compute. We first say what a limit is, then recall what is known, and then state the result (Theorem 4.13).

Definition (Pointwise limit)**.** Let $\mathcal{X} = \mathbb{R}^{T \times d_{\mathrm{in}}}$ be the set of inputs, and let $\mathcal{F}$ be a set of networks with inputs in $\mathcal{X}$. A network F is a *pointwise limit* of $\mathcal{F}$ if

$$\forall n \geq 1,\ \ \forall (x_1, \ldots, x_n) \in \mathcal{X}^n,\ \ \forall \delta > 0,\ \ \exists F' \in \mathcal{F},\ \ \forall i \in \{1, \ldots, n\} :\ \ \|F'(x_i) - F(x_i)\| < \delta.$$

The network F' may depend on n, on $(x_1, \ldots, x_n)$ and on δ.

If a sequence of networks of $\mathcal{F}$ converges to F at every input, then F is a pointwise limit of $\mathcal{F}$. This is the case when the sequence converges to F uniformly on compact sets of inputs. Mijoski and Karbevski (2026) show that, without normalization, every skipless network is a limit of residual networks of the same widths, uniformly on bounded sets of inputs. They also show that a residual network of analytic blocks at generic parameters equals no skipless network of blocks of the same kind, whatever its depth and widths, except for GELU blocks without normalization. We show that, for all analytic blocks, with or without pre-RMSNorm, a residual network at generic parameters is not even a pointwise limit of skipless networks of the same widths.

By Definition 4.1 and Theorem 4.9, the Jacobian of the residual network has rank $P - \dim \mathcal{G} - s$ on some batch $X = (x_1, \ldots, x_n)$ of inputs, at some parameters. As the blocks are analytic, it has this rank on X at generic parameters. Let $\mathcal{S}_X$ and $\mathcal{R}_X$ be the images of the map $\theta \mapsto (F_\theta(x_1), \ldots, F_\theta(x_n))$ of Definition 4.1 for the skipless and for the residual networks of the same widths: the sets of their outputs on X. The maps $\tanh$, GELU, SiLU, the softmax and RMSNorm are formulas in the exponential and the error function, with quotients and square roots. This makes both sets *definable in an o-minimal structure* (Speissegger, 1997): they belong to a collection of sets that is closed under finite unions, complements and projections, and whose members on the real line are finite unions of points and intervals (Definition A.17 and Lemma A.18). Every definable set has a dimension (Definition A.19), and for $\mathcal{S}_X$ and $\mathcal{R}_X$ it is the rank of the Jacobian of the map at generic θ (Lemma A.21(i)). So Theorem 4.4 and the choice of X give

$$\dim \mathcal{S}_X \ \leq\ P - (L+1)\dim \mathcal{G} - s \ <\ P - \dim \mathcal{G} - s \ =\ \dim \mathcal{R}_X. \tag{10}$$

A pointwise limit of these skipless networks has its outputs on X in the closure $\overline{\mathcal{S}_X}$ of $\mathcal{S}_X$. A definable set and its closure have the same dimension (Lemma A.20, from van den Dries, 1998, Chapter 4, Theorem (1.8)), so $\dim \overline{\mathcal{S}_X} = \dim \mathcal{S}_X < \dim \mathcal{R}_X$ by (10). A definable set of lower dimension than $\mathcal{R}_X$ contains the outputs of the residual network on X only for parameters in a set of measure zero (Lemma A.21(ii)). The closure $\overline{\mathcal{S}_X}$ is such a set, so at generic parameters the residual network is not a pointwise limit of these skipless networks.

Theorem 4.13 (Different function classes)**. (i)** *For analytic blocks, with or without pre-RMSNorm, a residual network at generic parameters equals no skipless network of the same widths, and is not a pointwise limit of such networks. In particular, it cannot be locally uniformly approximated by such networks.*

(ii) Too few extra hidden units. *For networks of GELU blocks without normalization, a residual network of L blocks at generic parameters equals no skipless network of L blocks with, in total, fewer than $Ld_{\mathrm{model}}/2$ more hidden units, and is not a pointwise limit of such networks.*

A special case of Theorem A.25, proved in Appendix A.7.

For GELU blocks without normalization, Mijoski and Karbevski (2026) show that almost every residual network equals a skipless network with d_{model} more hidden units per block. Part (ii) bounds the number of extra hidden units from below. A skipless network with k more hidden units than the residual network has $2d_{\text{model}}k$ more parameters, so its bound in Theorem 4.4 is $P + 2d_{\text{model}}k - (L+1)d_{\text{model}}^2$, with $\dim \mathcal{G} = d_{\text{model}}^2$ and $s = 0$. The argument for (10) applies as long as this bound stays below $\dim \mathcal{R}_X$:

$$P + 2d_{\text{model}}k - (L+1)d_{\text{model}}^2 \;<\; P - d_{\text{model}}^2, \qquad \text{that is,} \quad k < Ld_{\text{model}}/2. \tag{11}$$

So the smallest number of extra hidden units with which a skipless network of L blocks equals a residual network at generic parameters lies between $Ld_{\text{model}}/2$ and Ld_{model}.

4.5 Numerical ranks at depth, and a conjecture

For skipless networks at depth, Theorem 4.12 gives the equality condition for $\tanh$, GELU, SwiGLU and GeGLU blocks, with one exception: GELU blocks with $d_{\text{hidden}} < 2d_{\text{model}}$ without normalization. For this case and for the other kinds of block we turn to numerical ranks, and we state what they suggest as a conjecture.

At depth we compute the rank of the Jacobian on a large batch of inputs (Appendix E), which is a lower bound on the functional dimension. Where this rank meets the upper bound of Theorem 4.4, the functional dimension is known exactly. The two meet in every row of Table 2, and also for SwiGLU and GeGLU networks and for nanoGPT (Karpathy, 2023) at other depths and without normalization (Tables 6 and 7). After training on four datasets, 398 of 436 small MLPs keep the rank that they have at random weights (Figure 5), and 36 of the other 38 are the skipless ReGLU networks under RMSNorm treated below.

In two cases the measured rank stays below the upper bound. In skipless ReGLU networks under RMSNorm with $\varepsilon = 10^{-5}$, some singular values fall below double precision. In one of eight skipless ReLU networks, five hidden units active on the same inputs act as one linear map, as can happen on sets of positive measure (Grigsby et al., 2023).

Conjecture 1 (Equality without skips)**.** *Without normalization, skipless networks of GELU blocks with $d_{\text{model}} \le d_{\text{hidden}} < 2d_{\text{model}}$ satisfy the equality condition. By Theorem 4.9, the skips then raise the functional dimension by exactly Ld_{model}^2.*

The analogue of the conjecture is known for polynomial activations. Shahverdi et al. (2026, Theorem 4.1) prove that an MLP whose activation is a generic polynomial of large enough degree has as many dimensions of functions as weights. Such an MLP is a skipless network with $d_{\text{hidden}} = d_{\text{model}}$ in which each product $W_{\text{write}}^{\ell} W_{\text{read}}^{\ell+1}$ is one weight matrix, so this is the equality condition for it, without normalization. For ReLU^2 and ReGLU blocks we expect the equality condition to hold only for residual networks: without skips, a hidden unit can be inactive at every input on an open set of weights (Proposition A.33). For ReLU blocks the condition fails on an open set of weights with skips as well (Appendix A.8).

					Rank		
Measured	d_{model}	L	**Normalization**	P	**no skips**	**skips**	**Added**
$\tanh$, GELU	4	3	none	240	176	224	48
$\tanh$	4	3	RMSNorm	240	216	234	18
$\tanh$	16	6	RMSNorm	3712	2872	3592	720
ReLU^2, exact rank	4	8	none	560	352	480	128
ReGLU, exact rank	4	8	none	816	544	672	128
nanoGPT: attention and GELU	8	8	LayerNorm	3336	2532	2700	168

Table 2: Rank of the Jacobian at random weights, without biases (Appendix E); exact rank: computed modulo a prime. In every row the skips add $L \dim \mathcal{G}$, where $\dim \mathcal{G}$ is d_{model}^2 without normalization, $d_{\text{model}}(d_{\text{model}}-1)/2$ under RMSNorm and $(d_{\text{model}}-1)(d_{\text{model}}-2)/2$ under LayerNorm.

One-hot tokens. With one-hot tokens under RMSNorm, the rows of the embedding table can be moved one by one to undo any change of basis near the identity, so the change of basis at the embedding of a skipless network need not be orthogonal, which adds $d_{\text{model}}(d_{\text{model}}+1)/2$ dimensions of symmetry (Table 6). The nanoGPT models take one-hot tokens too, but add a position embedding to each token embedding, and show no such addition (Table 7).

5 Conclusion

Skip connections change how a network uses its parameters. Without skips, the basis of every hidden state is free, and d_{model}^2 parameters per hidden state are redundant. With skips, all hidden states share one basis. Where the symmetries are the only redundancy, the parameters that are redundant without skips become functional dimension. We prove this gain in the cases of Table 1, exactly or as a lower bound, and measure it in further networks. Mijoski and Karbevski (2026) establish that networks with and without skips differ in what they can represent, and this gain gives the difference a size. Residual networks with analytic blocks are also locally identifiable at generic parameters: nearby parameters with the same function differ only by the symmetries (Corollary 4.10). The changes of basis also apply to trained models. Without changing the function, they remove the redundant parameters of a skipless network, and without normalization they turn every invertible learned skip into an identity skip.

Open questions. The equality condition at depth is open for the skipless networks of Conjecture 1, and for residual ReLU^2 and ReGLU networks. Whether residual networks are identifiable, and not only locally, is open. Whether the reduced forms, such as triangular learned skips, train better is untested. Input-dependent skips, as in mHC (Xie et al., 2025), are nonlinear in the hidden state, and we do not treat them.

Use of AI

The conception of the study is the authors' own. One author obtained their results, proofs and experiments working with AI assistants, through continuous conversation; the assistants were used to find, verify and simplify proofs and to write the code of the experiments, which the authors validated.

Code availability

The notebook that reproduces the numbers, tables and figures of this paper (Appendix E) will be released in a public Git repository, coming soon.

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A Proofs

Throughout, the tokens of a hidden state are its rows; where every block acts on each token separately, as MLP blocks do, it suffices to take one token, $T = 1$, so that $h \in \mathbb{R}^{1\times d_{\text{model}}}$ is a row vector. Biases, when present, enter as $B_\ell(h) = f_\ell(hW^\ell_{\text{read}} + b^\ell_{\text{read}})W^\ell_{\text{write}} + b^\ell_{\text{write}}$, with row vectors b added to every token. Each subsection recalls the results of the main text named in its title, explains what the proof does, and then proves them; auxiliary results are stated where they are first used. Mixed blocks, which hold a learned skip as d_{model} more hidden units, are defined in Appendix A.1. Further, $\mathcal{G}$ is $\mathrm{GL}(d_{\text{model}})$ without normalization, $\mathrm{O}(d_{\text{model}})$ under RMSNorm with $\varepsilon > 0$, and $\mathrm{CO}(d_{\text{model}})$, the positive multiples of orthogonal matrices, under RMSNorm with $\varepsilon = 0$. The one-block theorem, Theorem B.1, is proved in Appendix B.

A.1 Changes of basis at junctions

Mixed blocks. A learned skip adds hA to the output of a block, in place of h. It acts as d_{model} more hidden units with the identity as activation, the read A and the write I. We call them *identity units* (Lin and Jegelka, 2018, Section 4). A *mixed block* is a block with d_{model} identity units whose read U and write V are both trainable $d_{\text{model}} \times d_{\text{model}}$ matrices (Figure 3). Let the read of the block have p columns and its write q rows, so that f maps $\mathbb{R}^{T\times p}$ to $\mathbb{R}^{T\times q}$, and define the fixed map

$$\tilde{f} : \mathbb{R}^{T\times d_{\text{model}}} \times \mathbb{R}^{T\times p} \to \mathbb{R}^{T\times(d_{\text{model}}+q)}, \qquad \tilde{f}\big([\,u \mid z\,]\big) = \big[\,u \mid f(z)\,\big],$$

which is the identity on the first d_{model} columns and f on the others. The mixed block computes

$$hUV + f(hW_{\text{read}})\,W_{\text{write}} \;=\; \tilde{f}\big(h\,[\,U \mid W_{\text{read}}\,]\big)\begin{bmatrix} V \\ W_{\text{write}} \end{bmatrix}. \tag{12}$$

So it has the form (1), with the read $[\,U \mid W_{\text{read}}\,]$ and the write that stacks V on W_{write}, and it is a block with the learned skip $A = UV$. A network with learned skips A_ℓ is the network of mixed

blocks with $U_\ell = A_\ell$ and $V_\ell = I$. Under pre-RMSNorm the skip reads h and the block reads $N_\varepsilon(h)$, and the mixed block computes

$$hUV + f\big(N_\varepsilon(h)\,W_{\text{read}}\big)\,W_{\text{write}} \;=\; \tilde{f}\left(\left[\,h \mid N_\varepsilon(h)\,\right]\begin{bmatrix} U & 0 \\ 0 & W_{\text{read}} \end{bmatrix}\right)\begin{bmatrix} V \\ W_{\text{write}} \end{bmatrix}. \tag{13}$$

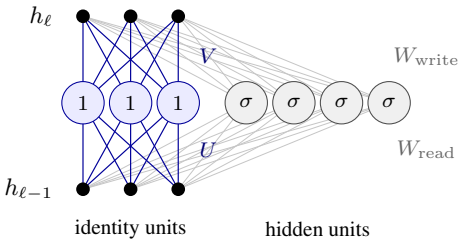


Figure 3: A mixed block: d_{model} identity units with the read U and the write V, beside the hidden units of the block. A learned skip A is the case $U = A$, $V = I$.

The main text uses changes of basis at hidden states (Sections 3.1 and 3.2); the proofs also use them at the identity units of a mixed block. A *junction* is a place where one linear map feeds the next: a hidden state, or the identity units of a mixed block. The following proposition states both kinds of change of basis, for skipless networks and networks of mixed blocks.

Proposition A.1 (Junctions)**. (i) Only UV matters.** *In a mixed block, $(U, V) \mapsto (UK, K^{-1}V)$ preserves the function for every $K \in \mathrm{GL}(d_{\text{model}})$.*
(ii) Hidden states are junctions. *In a skipless network, and in a network of mixed blocks, the map $W \mapsto WG$ on the matrices that write a hidden state and $W \mapsto G^{-1}W$ on those that read it preserves the function for every $G \in \mathcal{G}$. For $\varepsilon = 0$ and $G = cQ$, a read through RMSNorm maps to $Q^\top W$ instead.*
(iii) One group per junction. *The maps of (i) and (ii) commute when they act at different junctions. A skipless network therefore carries one copy of $\mathcal{G}$ per hidden state, and a network of mixed blocks one more copy of $\mathrm{GL}(d_{\text{model}})$ per block.*

Overview. The only real work is under normalization, where a read sees $N_\varepsilon(hG)$ rather than hG: Lemma A.2 determines which changes of basis can be moved through N_ε.

Lemma A.2 (RMSNorm equivariance)**.** *Let $G \in \mathrm{GL}(d_{\text{model}})$.*
(i) *There is a matrix M with $N_\varepsilon(hG) = N_\varepsilon(h)M$ for all row vectors $h \neq 0$ if and only if $G \in \mathrm{CO}(d_{\text{model}})$ when $\varepsilon = 0$, and $G \in \mathrm{O}(d_{\text{model}})$ when $\varepsilon > 0$. Then $M = Q$, where $G = cQ$ with Q orthogonal.*
(ii) *If $\varepsilon > 0$ and $G = cQ$, then $\|N(hG) - N_\varepsilon(h)Q\|/\|N(h)\| = \frac{\varepsilon}{2r^2}|1 - c^{-2}| + O(\varepsilon^2)$, where $r^2 = \|h\|^2/d_{\text{model}}$.*

Proof. The case $\varepsilon = 0$. Here $N_\varepsilon(h) = \sqrt{d}_{\text{model}}\, h/\|h\|$. If $G = cQ$, the factor c cancels in $hG/\|hG\|$, so $N_\varepsilon(hG) = N_\varepsilon(h)Q$. Conversely, if $hG/\|hG\| = hM/\|h\|$ for all $h \neq 0$, taking norms gives $\|hM\| = \|h\|$, so M is orthogonal, and $hGM^\top = (\|hG\|/\|h\|)\,h$: every nonzero row vector is a left eigenvector of $GM^\top$, so $GM^\top = cI$ with $c = \|hG\|/\|h\| > 0$, and $G = cM$.

The case $\varepsilon > 0$. For $G = cQ$, $N_\varepsilon(h\,cQ) = c\,hQ/\sqrt{c^2\|h\|^2/d_{\text{model}} + \varepsilon}$ and $N_\varepsilon(h)Q = hQ/\sqrt{\|h\|^2/d_{\text{model}} + \varepsilon}$; these agree for all h if and only if $c = 1$. If $N_\varepsilon(hG) = N_\varepsilon(h)M$ for all h with $\varepsilon > 0$, letting $\|h\| \to \infty$ along rays reduces to the case $\varepsilon = 0$, so $G = cM$ with M orthogonal, and the computation just made forces $c = 1$. The expansion of the deviation is the Taylor expansion of $c\sqrt{r^2 + \varepsilon}/\sqrt{c^2r^2 + \varepsilon} - 1$ in ε. □

For positively homogeneous blocks and $\varepsilon = 0$, more changes of basis preserve the function: every invertible one before the last hidden state. The next lemma shows this and measures how $\varepsilon > 0$ breaks it; Theorem A.14, Theorem B.1(iv) and the checks of Appendix E use it.

Lemma A.3 (Homogeneous blocks under RMSNorm)**.** *Let a skipless network with pre-RMSNorm, $h_\ell = B_\ell(N_\varepsilon(h_{\ell-1}))$ for $\ell = 1, \dots, L$ and $y = N_\varepsilon(h_L)D$, have blocks that are positively homogeneous of degree k, and let $r_i^2 = \|h_i\|^2/d_{\text{model}}$. Apply the change of basis $G \in \mathrm{GL}(d_{\text{model}})$ at h_ℓ with $\ell < L$, or the scaling $G = cI$ at h_L with its read D unchanged. At every input whose hidden states are nonzero, the output becomes Λy with a scalar $\Lambda > 0$, and*
(i) *$\Lambda = 1$ if $\varepsilon = 0$ or G is orthogonal;*
(ii) *for $G = e^{tS}$ with S symmetric and $\ell < L$,*

$$\frac{d}{dt}\log\Lambda\Big|_{t=0} = -\,k^{L-\ell}\,\frac{h_\ell S h_\ell^\top}{d_{\text{model}}\,(r_\ell^2+\varepsilon)}\prod_{i=\ell+1}^{L}\frac{\varepsilon}{r_i^2+\varepsilon},$$

and for the scaling $c = e^t$ at h_L it is $\varepsilon/(r_L^2+\varepsilon)$.

Proof. Since $N_\varepsilon(h)$ is a positive multiple of h,

$$N_\varepsilon(h_\ell G)\,G^{-1} = \gamma N_\varepsilon(h_\ell), \qquad \gamma = \sqrt{\frac{r_\ell^2+\varepsilon}{\|h_\ell G\|^2/d_{\text{model}}+\varepsilon}}.$$

Block $\ell+1$ thus reads $\gamma N_\varepsilon(h_\ell)$ and, being homogeneous, writes $\mu_{\ell+1}h_{\ell+1}$ with $\mu_{\ell+1} = \gamma^k$; for the scaling at h_L, set $\mu_L = c$. If a hidden state becomes $\mu_i h_i$ with $\mu_i > 0$, then $N_\varepsilon(\mu_i h_i) = \beta_i N_\varepsilon(h_i)$ with

$$\beta_i = \mu_i\sqrt{\frac{r_i^2+\varepsilon}{\mu_i^2 r_i^2+\varepsilon}},$$

so the next block writes $\mu_{i+1}h_{i+1}$ with $\mu_{i+1} = \beta_i^k$, and the readout gives $\Lambda = \beta_L$.

(i) If $\varepsilon = 0$, every β_i is 1. If G is orthogonal, $\gamma = 1$, so every μ_i and β_i is 1.

(ii) At $\mu_i = 1$, $\partial\log\beta_i/\partial\log\mu_i = \varepsilon/(r_i^2+\varepsilon)$, and $\frac{d}{dt}\log\gamma\big|_{t=0} = -h_\ell S h_\ell^\top/(d_{\text{model}}(r_\ell^2+\varepsilon))$, since $\|h_\ell e^{tS}\|^2 = \|h_\ell\|^2 + 2t\,h_\ell S h_\ell^\top + O(t^2)$. The chain rule along $\log\mu_{\ell+1} = k\log\gamma$, $\log\mu_{i+1} = k\log\beta_i$ and $\log\Lambda = \log\beta_L$ gives the product. □

Proof of Proposition A.1. (i) $(hUK)K^{-1}V = hUV$, and no other part of the network involves U or V. (ii) Fix a hidden state h_ℓ and apply the change of basis G there (Section 3.1). The writes into h_ℓ produce $h_\ell G$ in place of h_ℓ. A read that sees h_ℓ itself, such as the read of the identity units, computes $(h_\ell G)G^{-1}W = h_\ell W$. With pre-RMSNorm reads, a read sees $N_\varepsilon(h_\ell G)$, which by Lemma A.2 equals $N_\varepsilon(h_\ell)Q$ exactly when $G = cQ$ with Q orthogonal and $c = 1$ if $\varepsilon > 0$, and is then absorbed by $W \mapsto Q^\top W$. So nothing downstream changes, for every G in $\mathrm{GL}(d_{\text{model}})$ without normalization, in $\mathrm{CO}(d_{\text{model}})$ for $\varepsilon = 0$ and in $\mathrm{O}(d_{\text{model}})$ for $\varepsilon > 0$. With biases and without normalization, the affine change $h_\ell \mapsto h_\ell G + g$ is absorbed by $b_{\text{write}} \mapsto b_{\text{write}}G + g$ on the writes and by $(W, b) \mapsto (G^{-1}W,\ b - gG^{-1}W)$ on the reads, since $h_\ell W + b = (h_\ell G + g)G^{-1}W + b - gG^{-1}W$; this gives $\mathrm{Aff}(d_{\text{model}})$. (iii) The change of basis at the identity units of block ℓ multiplies U_ℓ on the right and V_ℓ on the left. The change at $h_{\ell-1}$ multiplies the reads of block ℓ, U_ℓ included, on the left, and the change at h_ℓ multiplies the writes of block ℓ, V_ℓ included, on the right. Multiplications on the left and on the right commute, and every matrix is multiplied on each side by the change at one junction only, so all these changes commute, and a matrix of the skipless network is touched by one hidden state only. The symmetries inside blocks (Section 4.2) multiply W_{read} on the right and W_{write} on the left, and commute with all of the above. □

Under RMSNorm with $\varepsilon > 0$, the orthogonal changes of basis are the only ones, also for skipless networks.

Lemma A.4 (Only orthogonal changes of basis pass)**.** *Consider a skipless pre-RMSNorm network with $\varepsilon > 0$ whose blocks are those of Section 2, and let $G_0, \dots, G_L$ be invertible. Multiply the writes into every hidden state h_ℓ by G_ℓ on the right and its reads by G_ℓ^{-1} on the left. If this preserves the function at all parameters, every G_ℓ is orthogonal.*

Proof. For a row u and an invertible G, let $N_\varepsilon^G(u) = N_\varepsilon(uG)\,G^{-1} = u/\sqrt{\|uG\|^2/d_{\text{model}} + \varepsilon}$. The changed network is the original one with $N_\varepsilon^{G_\ell}$ in place of N_ε at the hidden state h_ℓ, for every ℓ. Take inputs whose tokens are all equal. Then the tokens of every hidden state are equal, an MLP block acts on each of them, and an attention block maps equal rows a to aS, with S the sum of its value–output products. Let u_ℓ and $\tilde{u}_\ell$ be a row of h_ℓ in the original network and, before the factor G_ℓ, in the changed one. We use three facts.

(a) Scaling. If $N_\varepsilon^G(c\tilde{u}) = N_\varepsilon(cu)$ for all $c > 0$, then $\tilde{u} = u$ and $\|uG\| = \|u\|$. Indeed, both sides are positive multiples of $\tilde{u}$ and of u, so $\tilde{u} = \lambda u$ with $\lambda > 0$, or both vanish. Then $\lambda^2(c^2\|u\|^2/d_{\text{model}} + \varepsilon) = c^2\lambda^2\|uG\|^2/d_{\text{model}} + \varepsilon$ for all $c > 0$. The constant terms give $\lambda = 1$, as $\varepsilon > 0$, and the terms in c^2 give $\|uG\| = \|u\|$.

(b) Blocks separate their inputs. If rows a and a' satisfy $B(a) = B(a')$ for all parameters of a block B, then $a = a'$. For an attention block on equal tokens, $aS = a'S$ for every matrix S. For a vanilla block, $\sigma(aw) = \sigma(a'w)$ for every column w. If $a \neq a'$, choose w with $\alpha = aw \neq a'w = \beta$, and replace w by tw: $\sigma(t\alpha) = \sigma(t\beta)$ for all t. If $\beta = -\alpha$, σ is even. Otherwise $\sigma(z) = \sigma(qz)$ for all z with some $|q| < 1$, so $\sigma(z) = \sigma(q^k z) \to \sigma(0)$, and σ is constant. None of $\tanh$, GELU, ReLU and ReLU^2 is even or constant. For a gated block, $\sigma(ag)\,(av) = \sigma(a'g)\,(a'v)$ for all columns g and v, so $\sigma(ag)\,a = \sigma(a'g)\,a'$ for all g. If $a = 0$ then $a' = 0$. Otherwise $a' = \kappa a$ for a number $\kappa \neq 0$, and $\sigma(z) = \kappa\,\sigma(\kappa z)$ for all z. For the ReLU gate this gives $\kappa = 1$. The SiLU and GELU gates have $\sigma'(0) = 1/2$, so $\kappa^2 = 1$, and they are not odd, so $\kappa = 1$.

(c) Hidden states reach every row. As the input and the parameters vary, u_ℓ takes every row of $\mathbb{R}^{1\times d_{\text{model}}}$. For $u_0 = xE$ this is clear. If $u_{\ell-1} \neq 0$, some read of block ℓ gives hidden units that do not all vanish, and its write then sends them to any row.

Induction from the last hidden state. The outputs agree for all D, so $N_\varepsilon^{G_L}(\tilde{u}_L) = N_\varepsilon(u_L)$. Multiplying the write of block L by $c > 0$ multiplies $\tilde{u}_L$ and u_L by c, so (a) gives $\tilde{u}_L = u_L$ and $\|u_L G_L\| = \|u_L\|$. By (c), u_L is any row, so G_L is orthogonal. Now suppose that $\tilde{u}_\ell = u_\ell$ at all parameters, for some $\ell \geq 1$. As $u_\ell = B_\ell\big(N_\varepsilon(u_{\ell-1})\big)$ and $\tilde{u}_\ell = B_\ell\big(N_\varepsilon^{G_{\ell-1}}(\tilde{u}_{\ell-1})\big)$, where the inputs of B_ℓ do not depend on its parameters, (b) gives $N_\varepsilon^{G_{\ell-1}}(\tilde{u}_{\ell-1}) = N_\varepsilon(u_{\ell-1})$. Multiplying the write of block $\ell-1$, or the embedding if $\ell = 1$, by $c > 0$, (a) gives $\tilde{u}_{\ell-1} = u_{\ell-1}$ and $\|u_{\ell-1}G_{\ell-1}\| = \|u_{\ell-1}\|$. By (c), $u_{\ell-1}$ is any row, so $G_{\ell-1}$ is orthogonal. □

A.2 The gauge fix (Theorems 3.4 and 3.6)

A symmetry of the parameters that preserves the function is also called a *gauge*, and using it to fix parameters, as the sweep of Section 3.2 does, a *gauge fix*. Theorems 3.4 and 3.6 collect what the main text uses of the following theorem. We write the parameters of a network of mixed blocks as $\theta = \big(E,\ (U_\ell, V_\ell, W^\ell_{\text{read}}, W^\ell_{\text{write}})_{\ell=1}^L,\ D\big)$, and those of a network with learned skips as $\big(E,\ (A_\ell, W^\ell_{\text{read}}, W^\ell_{\text{write}})_{\ell=1}^L,\ D\big)$. For invertible $G_0, \dots, G_L$ and $K_1, \dots, K_L$, let $T_{G,K}$ be the map

$$\begin{aligned} E \mapsto EG_0, \qquad W^\ell_{\text{read}} \mapsto G_{\ell-1}^{-1} W^\ell_{\text{read}}, \qquad W^\ell_{\text{write}} \mapsto W^\ell_{\text{write}} G_\ell, \qquad D \mapsto G_L^{-1} D, \\ U_\ell \mapsto G_{\ell-1}^{-1} U_\ell K_\ell, \qquad V_\ell \mapsto K_\ell^{-1} V_\ell G_\ell, \end{aligned} \tag{14}$$

which applies the change of basis G_ℓ at h_ℓ and K_ℓ at the identity units of block ℓ. On a network with learned skips, T_G acts by $A_\ell \mapsto G_{\ell-1}^{-1} A_\ell G_\ell$ in place of the second line. When every skip $A_\ell = U_\ell V_\ell$ is invertible, let $\Lambda_\ell = A_{\ell+1} \cdots A_L$, with $\Lambda_L = I$, and let

$$\Phi(\theta) = \big(E\Lambda_0,\ (\Lambda_{\ell-1}^{-1} W^\ell_{\text{read}},\ W^\ell_{\text{write}} \Lambda_\ell)_{\ell=1}^L,\ D\big), \tag{15}$$

a parameter of the residual network, with the same formula for learned skips. With biases, Φ also sends the embedding bias b_E to $b_E\Lambda_0$ and each write bias b^ℓ_{write} to $b^\ell_{\text{write}}\Lambda_\ell$, and keeps each read bias. We write F_θ for the function of the network of mixed blocks at θ, and $F^{\text{res}}_{\Phi(\theta)}$ for that of the residual network at $\Phi(\theta)$.

Theorem A.5 (The gauge fix)**.** *Consider networks without normalization in (i) and (ii), and parameters θ whose skips A_ℓ are invertible.*

(i) Same function. *The maps $T_{G,K}$ preserve the function. With $G_\ell = \Lambda_\ell$ and $K_\ell = U_\ell^{-1}\Lambda_{\ell-1}$, $T_{G,K}$ sets every U_ℓ and V_ℓ to I and the other parameters to $\Phi(\theta)$, so $F_\theta = F^{\mathrm{res}}_{\Phi(\theta)}$. For learned skips, T_G with $G_\ell = \Lambda_\ell$ sets every A_ℓ to I in the same way.*
(ii) What the gauge fix identifies. *Every parameter of the residual network is $\Phi(\theta)$ for some θ, and $\Phi(\tilde\theta) = \Phi(\theta)$ exactly when $\tilde\theta = T_{G,K}(\theta)$ for some G and K with $G_L = I$. The derivative of Φ is onto at every θ. So the networks of mixed blocks with invertible skips and the residual networks compute the same set of functions, with the same functional dimension at θ and at $\Phi(\theta)$. The same holds for learned skips, without the K_ℓ, and for networks with biases.*
(iii) One change of basis is left. *Let $d_{\mathrm{model}} \geq 2$, and let every block be nonzero for some value of its parameters. Without normalization or with pre-RMSNorm and $\varepsilon > 0$, let $G_0, \dots, G_L$ be invertible. The map T_G of the embedding, the reads, the writes and the readout preserves the function of every residual network if and only if $G_0 = G_1 = \cdots = G_L \in \mathcal{G}$. Under RMSNorm with $\varepsilon = 0$, the same holds for $G_\ell = c_\ell Q_\ell \in \mathrm{CO}(d_{\mathrm{model}})$, with $c_\ell > 0$ and Q_ℓ orthogonal, when T_G multiplies the reads and the readout by $Q_{\ell-1}^\top$ and $Q_L^\top$ in place of $G_{\ell-1}^{-1}$ and G_L^{-1}.*

Overview. Part (i) is an induction along the network: in the coordinates $h'_\ell = h_\ell\Lambda_\ell$ the identity units become identity skips, and we check that the residual recursion with transformed weights produces exactly these coordinates. Part (ii) lifts every direction on the residual side to one with the identity units held fixed, which makes the derivative of Φ onto, and solves for the maps $T_{G,K}$ that relate two parameters with the same image. Part (iii) switches off all branches but one, so that the output compares the bases on the two sides of that block, and a branch that is not identically zero forces them to agree.

Proof of Theorem A.5. Part (i), with biases and a skip bias. Merge the identity units, $A_\ell = U_\ell V_\ell$, and write the network as $h_\ell = h_{\ell-1}A_\ell + k_\ell + f_\ell(h_{\ell-1}W^\ell_{\mathrm{read}} + b^\ell_{\mathrm{read}})W^\ell_{\mathrm{write}} + b^\ell_{\mathrm{write}}$ with $h_0 = xE + b_E$, where $k_\ell \in \mathbb{R}^{1\times d_{\mathrm{model}}}$ is a skip bias ($k_\ell = 0$ is allowed). Every A_ℓ is invertible, hence every Λ_ℓ, and $\Lambda_{\ell-1} = A_\ell\Lambda_\ell$. Define new variables and new weights

$$h'_\ell = h_\ell\Lambda_\ell, \qquad E' = E\Lambda_0, \quad b'_E = b_E\Lambda_0, \qquad W'^\ell_{\mathrm{read}} = \Lambda_{\ell-1}^{-1}W^\ell_{\mathrm{read}}, \quad b'^\ell_{\mathrm{read}} = b^\ell_{\mathrm{read}},$$
$$W'^\ell_{\mathrm{write}} = W^\ell_{\mathrm{write}}\Lambda_\ell, \quad b'^\ell_{\mathrm{write}} = (b^\ell_{\mathrm{write}} + k_\ell)\Lambda_\ell, \qquad D' = D.$$

We show by induction that the residual network with primed weights computes h'_ℓ. For $\ell = 0$, $h'_0 = (xE + b_E)\Lambda_0 = xE' + b'_E$. Assume $h'_{\ell-1} = h_{\ell-1}\Lambda_{\ell-1}$. The residual recursion with primed weights gives

$$\begin{aligned} &h'_{\ell-1} + f_\ell(h'_{\ell-1}W'^\ell_{\mathrm{read}} + b'^\ell_{\mathrm{read}})W'^\ell_{\mathrm{write}} + b'^\ell_{\mathrm{write}} \\ &\quad = h_{\ell-1}\Lambda_{\ell-1} + f_\ell(h_{\ell-1}W^\ell_{\mathrm{read}} + b^\ell_{\mathrm{read}})W^\ell_{\mathrm{write}}\Lambda_\ell + (b^\ell_{\mathrm{write}} + k_\ell)\Lambda_\ell, \end{aligned}$$

and since $\Lambda_{\ell-1} = A_\ell\Lambda_\ell$ the right-hand side equals $h_\ell\Lambda_\ell$. Finally $h'_L D' = h_L D$ because $\Lambda_L = I$. This is the map $T_{G,K}$ of (14) with $G_\ell = \Lambda_\ell$ and $K_\ell = U_\ell^{-1}\Lambda_{\ell-1}$: it sends U_ℓ to $G_{\ell-1}^{-1}U_\ell K_\ell = I$ and V_ℓ to $K_\ell^{-1}V_\ell G_\ell = \Lambda_{\ell-1}^{-1}A_\ell\Lambda_\ell = I$, and the other matrices as above.

Part (ii). The identity units carry no biases of their own, since these would only add to the write bias. For parameters θ of the mixed network with invertible A_ℓ, the map Φ of (15) is the change of variables of part (i) with $k_\ell = 0$, so $F_\theta = F^{\mathrm{res}}_{\Phi(\theta)}$. Every parameter of the residual network is reached: the θ with the same entries and $U_\ell = V_\ell = I$ has every $\Lambda_\ell = I$, so Φ returns these entries. The Fréchet derivative $\mathrm{D}_\theta\Phi(\theta)$ is onto at every θ. A direction η' of the residual parameters, with the entries η'_E, $\eta'^\ell_{\mathrm{read}}$, $\eta'^\ell_{\mathrm{write}}$ and η'_D at E', W'^ℓ_{read}, W'^ℓ_{write} and D', is $\mathrm{D}_\theta\Phi(\theta)\,\eta$ for the direction η of θ that is zero at every U_ℓ and V_ℓ and has the entries $\eta'_E\Lambda_0^{-1}$ at E, $\Lambda_{\ell-1}\eta'^\ell_{\mathrm{read}}$ at W^ℓ_{read}, $\eta'^\ell_{\mathrm{write}}\Lambda_\ell^{-1}$ at W^ℓ_{write} and η'_D at D, since along it every Λ_ℓ stays fixed; the biases are treated in the same way. The maps $T_{G,K}$ with $G_L = I$ send A_ℓ to $G_{\ell-1}^{-1}A_\ell G_\ell$ and hence Λ_ℓ to $G_\ell^{-1}\Lambda_\ell$, so they leave Φ unchanged. Conversely, let $\Phi(\tilde\theta) = \Phi(\theta)$ for a second parameter $\tilde\theta$ of the mixed network, whose entries and products $\tilde\Lambda_\ell$ carry a tilde. Set $G_\ell = \Lambda_\ell\tilde\Lambda_\ell^{-1}$, so that $G_L = I$, and $K_\ell = U_\ell^{-1}G_{\ell-1}\tilde U_\ell$. Then $\tilde E = EG_0$, $\widetilde{W}^\ell_{\mathrm{read}} = G_{\ell-1}^{-1}W^\ell_{\mathrm{read}}$, $\widetilde{W}^\ell_{\mathrm{write}} = W^\ell_{\mathrm{write}}G_\ell$ and $\tilde D = D$; $\tilde U_\ell = G_{\ell-1}^{-1}U_\ell K_\ell$ by the choice of K_ℓ; and $\tilde A_\ell = \tilde\Lambda_{\ell-1}\tilde\Lambda_\ell^{-1} = G_{\ell-1}^{-1}A_\ell G_\ell$ gives $\tilde V_\ell = \tilde U_\ell^{-1}\tilde A_\ell = K_\ell^{-1}V_\ell G_\ell$. So $\tilde\theta = T_{G,K}(\theta)$.

Since $F_\theta = F^{\mathrm{res}}_{\Phi(\theta)}$ and every residual parameter is reached, the two networks compute the same functions. For every batch of inputs, the Jacobian of the mixed network at θ is the Jacobian of the residual network at $\Phi(\theta)$ times the derivative of Φ, by the chain rule; a matrix times a matrix that is onto has the rank of the first, so the two Jacobians have the same rank, and the two networks the same functional dimension. For learned skips the same argument applies without the K_ℓ.

Part (iii). A single global G is a symmetry: the skip carries $h_{\ell-1}G$ to $h_{\ell-1}G$, and the branch becomes $B_\ell(\,\cdot\, G^{-1})\,G$. Conversely, let $G_0, \dots, G_L$ preserve the function for all parameter values; the change of basis conjugates each branch, $B_\ell \mapsto B_\ell(\,\cdot\, G_{\ell-1}^{-1})\,G_\ell$. First set every write W^k_{write}, and its bias if there is one, to zero. The network computes $x \mapsto xED$ and the transformed network $x \mapsto xEG_0G_L^{-1}D$, and these agree for all D and E only if $G_L = G_0$. Now suppose $G_0 = \cdots = G_{\ell-1} =: G$ and set every write except that of block ℓ to zero. Then $h_L = h_{\ell-1} + B_\ell(h_{\ell-1})$ with $h_{\ell-1} = xE$, block ℓ reads $h_{\ell-1}GG_{\ell-1}^{-1} = h_{\ell-1}$, and the transformed network outputs $\big(h_{\ell-1}G + B_\ell(h_{\ell-1})\,G_\ell\big)G^{-1}D$. Equality with $\big(h_{\ell-1} + B_\ell(h_{\ell-1})\big)D$ for all D gives $B_\ell(h)\,(G_\ell G^{-1} - I) = 0$ for all h and all parameters of block ℓ. As the write of block ℓ ranges over all matrices, the outputs of a block whose computation does not vanish identically span $\mathbb{R}^{d_{\mathrm{model}}}$, so $G_\ell = G$, and by induction all G_ℓ are equal.

Part (iii) under RMSNorm with $\varepsilon > 0$. A global orthogonal G is a symmetry, since $N_\varepsilon(hG) = N_\varepsilon(h)G$ (Proposition A.1(ii)). Conversely, with all writes zero the network computes $x \mapsto N_\varepsilon(xE)D$ and the transformed network $x \mapsto N_\varepsilon(xEG_0)G_L^{-1}D$. These agree for all E and D only if $N_\varepsilon(hG_0) = N_\varepsilon(h)G_L$ for all rows h, so G_0 is orthogonal and $G_L = G_0$, by Lemma A.2. Now suppose $G_0 = \cdots = G_{\ell-1} = G_L =: G$, and set every write except that of block ℓ to zero. With $h = xE$ and $b = B_\ell(N_\varepsilon(h))$, the network computes $N_\varepsilon(h + b)D$. In the transformed network block ℓ reads $N_\varepsilon(hG)G^{-1}W^\ell_{\mathrm{read}} = N_\varepsilon(h)W^\ell_{\mathrm{read}}$ and writes bG_ℓ, so the output is $N_\varepsilon(hG + bG_\ell)G^{-1}D = N_\varepsilon(h + bG_\ell G^{-1})D$. For $\varepsilon > 0$ the map N_ε is injective, so equality for all D gives $b\,(G_\ell G^{-1} - I) = 0$, and $G_\ell = G$ as above.

Part (iii) under RMSNorm with $\varepsilon = 0$. Now $N_\varepsilon(chQ) = N_\varepsilon(h)Q$ for $c > 0$ and orthogonal Q. With all writes zero the transformed network computes $N_\varepsilon(xE)Q_0Q_L^\top D$, so $Q_L = Q_0$. Suppose $G_0 = \cdots = G_{\ell-1} = cQ$ and $Q_L = Q$, and set every write except that of block ℓ to zero. With h and b as above, the transformed network computes $N_\varepsilon(chQ + c_\ell bQ_\ell)Q^\top D = N_\varepsilon(ch + c_\ell bR)D$ with $R = Q_\ell Q^\top$. Equality with $N_\varepsilon(h + b)D$ for all D makes $ch + c_\ell bR$ a positive multiple of $h + b$. Fix a token, and an input and reads at which the row of $f(N_\varepsilon(h)W^\ell_{\mathrm{read}})$ of this token is nonzero. As the write varies, the row b of this token takes every value. So, with $u = h + b$, the row $c_\ell\, uR + (ch - c_\ell\, hR)$ is a positive multiple of u for every $u \neq 0$. Large u give $R = I$. Then $(c - c_\ell)\, h$ is parallel to every u, so $c_\ell = c$. So $G_\ell = cQ$, and by induction all G_ℓ are equal. □

Remark A.6 (The forward sweep). The sweep of part (i) can also run forward, with $G_\ell = \Pi_\ell^{-1}$ at h_ℓ, with $\Pi_\ell = A_1 \cdots A_\ell$ and $\Pi_0 = I$. By the same induction it gives the residual network with embedding E, reads $\Pi_{\ell-1}W^\ell_{\mathrm{read}}$, writes $W^\ell_{\mathrm{write}}\Pi_\ell^{-1}$, write biases $(b^\ell_{\mathrm{write}} + k_\ell)\Pi_\ell^{-1}$ and readout $\Pi_L D$. Any linear change of variables $h'_\ell = h_\ell G'_\ell$ that yields identity skips must satisfy $G'^{-1}_{\ell-1}A_\ell G'_\ell = I$, so $G'_\ell = \Lambda_\ell G'_L$: two such changes, the forward and the backward sweep among them, differ by a global change of basis and, when biases are present, by a translation at each hidden state (Lemma A.8).

Remark A.7 (The changes of basis of Zhao et al.). For vanilla MLP blocks, the maps $T_{G,K}$ are the changes of basis of Proposition C.1 of Zhao et al. (2023), transposed to row vectors. Read the network of mixed blocks as a feedforward network whose hidden layers alternate between the hidden states h_ℓ, with the identity as activation, and the hidden layers of the blocks, with the mixed activation $\tilde f\big(\big[\, u \;\big|\; z \,\big]\big) = \big[\, u \;\big|\; f(z) \,\big]$ of (12). The identity commutes with every matrix in $\mathrm{GL}(d_{\mathrm{model}})$ (their Example C.3), and $\tilde f$ with $\mathrm{diag}(K, I)$ for every $K \in \mathrm{GL}(d_{\mathrm{model}})$, so their proposition gives these changes of basis G_ℓ and K_ℓ; for other blocks the same computation does. Propagating a change of basis through a skip is thus their change of basis at the identity units, chosen together with the changes of basis at the two hidden states that the skip joins.

We recall Theorem 3.4 from Section 3.2, where $\Lambda_\ell = A_{\ell+1} \cdots A_L$, and Theorem 3.6 from Section 3.3.

Theorem 3.4 (Learned skips reduce to identity skips)**.** *Without normalization, let every learned skip A_ℓ be invertible, and let $\Lambda_\ell = A_{\ell+1} \cdots A_L$, with $\Lambda_L = I$. The network with learned skips computes the same function as the residual network with the embedding $E\Lambda_0$, the reads $\Lambda_{\ell-1}^{-1} W_{\text{read}}^\ell$, the writes $W_{\text{write}}^\ell \Lambda_\ell$ and the readout D.*

Proof of Theorem 3.4. This is the statement for learned skips in Theorem A.5(i), with $\Phi(\theta)$ written out by (15). □

Theorem 3.6 (One change of basis for the whole network)**.** *Without normalization, let $G_0, \ldots, G_L$ be invertible.*
(i) *The map $E \mapsto EG_0$, $W_{\text{read}}^\ell \mapsto G_{\ell-1}^{-1} W_{\text{read}}^\ell$, $W_{\text{write}}^\ell \mapsto W_{\text{write}}^\ell G_\ell$, $D \mapsto G_L^{-1} D$ preserves the function of every residual network if and only if $G_0 = G_1 = \cdots = G_L$.*
(ii) *The changes of basis that preserve the function at all parameters form a group of dimension $(L+1)d_{\text{model}}^2$ in a skipless network and d_{model}^2 in a residual one.*
(iii) *In a residual network, let d_{model} columns of one of the reads $W_{\text{read}}^1, \ldots, W_{\text{read}}^L, D$ form an invertible matrix W_1. The same change of basis $G = W_1$ at all hidden states preserves the function and turns W_1 into the identity.*

Proof of Theorem 3.6. Part (i) is Theorem A.5(iii) with $\mathcal{G} = \mathrm{GL}(d_{\text{model}})$, part (ii) is Theorem A.14(i) with $\dim \mathcal{G} = d_{\text{model}}^2$, and part (iii) is Proposition A.9(ii). □

With biases every hidden state also carries a translation, which the skips do not tie.

Lemma A.8 (Translations are not tied)**.** *In a residual network with biases and without normalization, adding a row g to the hidden states $h_\ell, \ldots, h_L$ preserves the function once the biases are adjusted, for every $\ell = 0, \ldots, L$. The symmetry group generated by changes of basis, translations and the symmetries inside blocks therefore has dimension $d_{\text{model}}^2 + (L+1)d_{\text{model}} + s$, against $(L+1)(d_{\text{model}}^2 + d_{\text{model}}) + s$ without skips, and the skips remove Ld_{model}^2 dimensions, as without biases.*

Proof. Add g to the write bias of block ℓ, or to the embedding bias when $\ell = 0$; subtract gW_{read}^k from the read bias of every block $k > \ell$; subtract gD from the readout bias. The skips carry g forward unchanged, so every later hidden state is shifted by g, and every later read removes it. So the function is unchanged. These $L+1$ translations move the biases in independent directions, as many as in a skipless network, where each hidden state has its own translation (Proposition A.1(ii)). A skip ties only the linear part of $\mathrm{Aff}(d_{\text{model}})$, which gives the counts. □

Table 6 confirms these counts for $\tanh$ networks with biases.

A.3 Exact reduction without skips (Proposition 3.2)

The following proposition is the general form of Proposition 3.2. Let W_1 be an invertible $d_{\text{model}} \times d_{\text{model}}$ submatrix formed by d_{model} columns of a matrix that reads h_ℓ, or by d_{model} rows of a matrix that writes it.

Proposition A.9 (Exact reduction without skips)**.** **(i)** *In a skipless network or a network of mixed blocks, the map $W \mapsto WG$ on the matrices that write h_ℓ and $W \mapsto G^{-1}W$ on those that read it turns W_1 into the identity for $G = W_1$ if W_1 is part of a read, and for $G = W_1^{-1}$ if it is part of a write. Under RMSNorm with $\varepsilon > 0$, $G = Q$ from a QR factorization $W_1 = QT$, or $W_1^\top = QT$ for a write, turns W_1 into a triangular matrix. Done at every hidden state, this removes d_{model}^2, respectively $d_{\text{model}}(d_{\text{model}} - 1)/2$, parameters per hidden state without changing the function.*
(ii) *In a residual network, the same map with one G at all hidden states preserves the function and turns one such submatrix, in any read or write of the network, into the identity or a triangular matrix.*

Proof of Proposition A.9. By Proposition A.1, a change of basis $G \in \mathcal{G}$ at h_ℓ preserves the function and replaces every read W of h_ℓ by $G^{-1}W$ and every write W into h_ℓ by WG. So the columns of a read that form W_1 become $G^{-1}W_1$, and the rows of a write that form W_1 become $W_1 G$. Without normalization $\mathcal{G} = \mathrm{GL}(d_{\text{model}})$, and $G = W_1$ for a read, or $G = W_1^{-1}$ for a write, turns W_1 into I.

Under RMSNorm with $\varepsilon > 0$, $\mathcal{G} = \mathrm{O}(d_{\mathrm{model}})$ (Lemma A.2). For a read, let $W_1 = QT$ be a QR factorization, with T upper triangular; $G = Q$ gives $Q^\top W_1 = T$. For a write, let $W_1^\top = QT$ be a QR factorization; then $W_1 = T^\top Q^\top$, and $G = Q$ gives $W_1 Q = T^\top$, which is lower triangular. The identity fixes all d_{model}^2 entries of W_1, and the triangular form fixes the $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2$ entries on one side of its diagonal. The change of basis at h_ℓ touches only the writes into h_ℓ and the reads of h_ℓ, and a matrix is touched on each side by one hidden state only (Proposition A.1(iii)), so the reductions at all hidden states can be made together, one submatrix per hidden state. In a residual network, one change of basis $G \in \mathcal{G}$ applied at every hidden state preserves the function (Theorem A.5(iii)); it replaces every read W by $G^{-1}W$ and every write W by WG, and leaves the identity skips unchanged, since $G^{-1}IG = I$. The same choice of G then turns one submatrix W_1, of any one read or write, into the identity or into a triangular matrix. □

Remark A.10 (Learned skips: the local reduction). Theorem 3.4 is not the only reduction of a network with learned skips. Write the skip of block ℓ as a product of two matrices,

$$A_\ell \;=\; A_\ell^{\mathrm{read}} A_\ell^{\mathrm{write}}, \qquad \text{at first with} \quad A_\ell^{\mathrm{read}} = A_\ell, \quad A_\ell^{\mathrm{write}} = I.$$

These are the matrices U_ℓ and V_ℓ of a mixed block (12), which has the read $[\, A_\ell^{\mathrm{read}} \mid W_{\mathrm{read}}^\ell \,]$ and the write that stacks A_ℓ^{write} on W_{write}^ℓ. Changes of basis $G_0, \dots, G_L$ at the hidden states preserve the function (Proposition A.1(ii)) and map, for every block ℓ,

$$\big[\, A_\ell^{\mathrm{read}} \mid W_{\mathrm{read}}^\ell \,\big] \;\mapsto\; G_{\ell-1}^{-1}\big[\, A_\ell^{\mathrm{read}} \mid W_{\mathrm{read}}^\ell \,\big], \qquad \begin{bmatrix} A_\ell^{\mathrm{write}} \\ W_{\mathrm{write}}^\ell \end{bmatrix} \;\mapsto\; \begin{bmatrix} A_\ell^{\mathrm{write}} \\ W_{\mathrm{write}}^\ell \end{bmatrix} G_\ell.$$

Choose each G_ℓ on its own, as in Proposition A.9(i), so that d_{model} columns of $W_{\mathrm{read}}^{\ell+1}$ (of D if $\ell = L$) or d_{model} rows of W_{write}^ℓ (of E if $\ell = 0$) become the identity. The two matrices of the identity units have no activation between them, and they merge into one:

$$\big(G_{\ell-1}^{-1} A_\ell^{\mathrm{read}}\big)\big(A_\ell^{\mathrm{write}} G_\ell\big) \;=\; G_{\ell-1}^{-1} A_\ell \, G_\ell.$$

The result is a network with the learned skips $G_{\ell-1}^{-1} A_\ell G_\ell$ in which a $d_{\mathrm{model}} \times d_{\mathrm{model}}$ submatrix of W_{read} or W_{write} is the identity at every hidden state. It has $(L+1)d_{\mathrm{model}}^2$ fewer parameters, as many as Theorem 3.4 and Theorem 3.6(iii) remove together.

We recall Proposition 3.2 from Section 3.1.

Proposition 3.2 (Exact reduction without skips)**.** *Without normalization, in a skipless network, let d_{model} columns of the read of h_ℓ ($W_{\mathrm{read}}^{\ell+1}$, or D if $\ell = L$) form an invertible matrix W_1. The change of basis $G = W_1$ at h_ℓ preserves the function and turns W_1 into $G^{-1}W_1 = I$. At generic parameters every hidden state has such columns, and these changes of basis at all $L+1$ hidden states remove $(L+1)d_{\mathrm{model}}^2$ parameters.*

Proof of Proposition 3.2. This is Proposition A.9(i) without normalization, for a submatrix of a read. □

A.4 RMSNorm (Theorem 3.7)

Theorem 3.7(iv) is a case of the following theorem. Let $\Pi_\ell = A_1 \cdots A_\ell$, with $\Pi_0 = I$.

Theorem A.11 (RMSNorm with learned skips)**.** *Consider pre-RMSNorm networks with $\varepsilon \geq 0$, and the network with learned skips of (9), whose skips A_ℓ are invertible in (i) and (ii).*
(i) Triangular or symmetric skips. *Let $Q_0 = I$, and for $\ell = 1, \dots, L$ let $Q_{\ell-1}^\top A_\ell = R_\ell Q_\ell^\top$ be an RQ factorization, with R_ℓ upper triangular with positive diagonal and Q_ℓ orthogonal. The network with the skips R_ℓ, the reads $Q_{\ell-1}^\top W_{\mathrm{read}}^\ell$, the writes $W_{\mathrm{write}}^\ell Q_\ell$ and the readout $Q_L^\top D$ computes the same function. So does the network built in the same way from the polar decompositions $Q_{\ell-1}^\top A_\ell = S_\ell Q_\ell^\top$, with S_ℓ symmetric positive definite in place of R_ℓ.*
(ii) Orthogonal skips can be absorbed. *If every Π_ℓ lies in $\mathcal{G}$, write $\Pi_\ell = c_\ell P_\ell$ with $c_\ell > 0$ and P_ℓ*

orthogonal, where $c_\ell = 1$ if $\varepsilon > 0$. Then the network computes the same function as the residual network with the embedding E, the reads $P_{\ell-1}W^\ell_{\text{read}}$, the writes $W^\ell_{\text{write}}\Pi_\ell^{-1}$ and the readout $P_L D$. Conversely, let the reads and the readout have rank d_{model}, and let invertible $G_0, \dots, G_L$ turn every skip into the identity, $G_{\ell-1}^{-1} A_\ell G_\ell = I$. If the network, written in the variables $h_\ell G_\ell$, is a pre-RMSNorm residual network, then every Π_ℓ lies in $\mathcal{G}$.

(iii) What each skip adds. *Under the equality condition for the networks with learned skips, with identity skips and without skips, the functional dimension of the network with learned skips exceeds that of the residual network by $L\,(d^2_{\text{model}} - \dim \mathcal{G})$, and that of the residual network exceeds that of the skipless network by $L \dim \mathcal{G}$. These are $L\, d_{\text{model}}(d_{\text{model}}+1)/2$ and $L\, d_{\text{model}}(d_{\text{model}}-1)/2$ for $\varepsilon > 0$, and $L\,(d_{\text{model}}(d_{\text{model}}+1)/2 - 1)$ and $L\,(d_{\text{model}}(d_{\text{model}}-1)/2 + 1)$ for $\varepsilon = 0$. For $\varepsilon = 0$ and positively homogeneous blocks, the skipless network does not satisfy the equality condition (Lemma A.3).*

Overview. Part (i) runs the sweep in the proof of Theorem A.5(i) forward, with orthogonal matrices only, which remove the orthogonal factor of each skip and leave its triangular or symmetric factor. Part (ii) reruns the change of variables of that proof and locates the only step that can fail, the read through N_ε; Lemma A.2 says exactly when a matrix can be moved through N_ε. Part (iii) counts the symmetries with Proposition A.1 and Theorem A.14.

Proof of Theorem A.11. Part (i). For every $\varepsilon \geq 0$, multiplying the writes into h_ℓ, the skip A_ℓ included, on the right by an orthogonal Q_ℓ, and the reads of h_ℓ on the left by $Q_\ell^\top$, preserves the function (Proposition A.1(ii)): the raw read $A_{\ell+1}$ directly, and the reads through N_ε because $N_\varepsilon(hQ_\ell) = N_\varepsilon(h)Q_\ell$ (Lemma A.2). With $Q_0 = I$ and the Q_ℓ of the statement, the skip A_ℓ becomes $Q_{\ell-1}^\top A_\ell Q_\ell = R_\ell$; the RQ factorization of the invertible matrix $Q_{\ell-1}^\top A_\ell$ exists, and is unique. The same computation with the polar decompositions gives $Q_{\ell-1}^\top A_\ell Q_\ell = S_\ell$: every invertible matrix X is $X = SQ^\top$ with $S = (XX^\top)^{1/2}$ symmetric positive definite and $Q^\top = S^{-1}X$ orthogonal, uniquely. The readout absorbs the last factor, $D \mapsto Q_L^\top D$.

Part (ii). Let the network be $h_\ell = h_{\ell-1}A_\ell + f_\ell(N_\varepsilon(h_{\ell-1})W^\ell_{\text{read}})W^\ell_{\text{write}}$ and set $h'_\ell = h_\ell \Pi_\ell^{-1}$, the forward sweep of Remark A.6, which turns every skip into the identity. The read of block ℓ in the new variables is $N_\varepsilon(h'_{\ell-1}\Pi_{\ell-1})W^\ell_{\text{read}}$. If every Π_ℓ, $\ell = 0, \dots, L$, lies in $\mathcal{G}$, Lemma A.2 gives $N_\varepsilon(h'\Pi_{\ell-1}) = N_\varepsilon(h')P_{\ell-1}$, so the reads $P_{\ell-1}W^\ell_{\text{read}}$, the writes $W^\ell_{\text{write}}\Pi_\ell^{-1}$ and the readout $P_L D$, since the readout reads $N_\varepsilon(h_L)$, reproduce the induction of that proof, and the result is a pre-RMSNorm residual network. Conversely, $G_{\ell-1}^{-1}A_\ell G_\ell = I$ for every ℓ gives $G_\ell = \Pi_\ell^{-1}G_0$, so in the variables $h'_\ell = h_\ell G_\ell$ block ℓ reads $N_\varepsilon(h'_{\ell-1}G_0^{-1}\Pi_{\ell-1})W^\ell_{\text{read}}$, and the readout reads $N_\varepsilon(h'_L G_0^{-1}\Pi_L)$. In a pre-RMSNorm residual network every read has the form $N_\varepsilon(h')W$; as the reads and the readout have rank d_{model}, this gives matrices M_ℓ with $N_\varepsilon(h'G_0^{-1}\Pi_\ell) = N_\varepsilon(h')M_\ell$ for all h', and Lemma A.2 puts every $G_0^{-1}\Pi_\ell$ in $\mathcal{G}$. For $\ell = 0$ this puts G_0 in $\mathcal{G}$, and so every Π_ℓ lies in $\mathcal{G}$.

Part (iii). Apply Proposition A.1 to the pre-RMSNorm network with learned skips: the skip A_ℓ absorbs any change of basis, and the normalized reads restrict it to $\mathcal{G}$, which is $\mathrm{CO}(d_{\text{model}})$ when $\varepsilon = 0$ and $\mathrm{O}(d_{\text{model}})$ when $\varepsilon > 0$ (Lemma A.2), so each of the $L+1$ hidden states carries its own copy of $\mathcal{G}$. The network with learned skips has Ld^2_{model} more parameters than the residual network and L more copies of $\mathcal{G}$ in its symmetry group. If both satisfy the equality condition, the difference of their functional dimensions is therefore $L\,(d^2_{\text{model}} - \dim \mathcal{G})$, which is $L\, d_{\text{model}}(d_{\text{model}}+1)/2$ when $\varepsilon > 0$ and $L\,(d_{\text{model}}(d_{\text{model}}+1)/2 - 1)$ when $\varepsilon = 0$. The difference between the residual and the skipless network is $L \dim \mathcal{G}$ by Theorem A.14(iii), which is $L\, d_{\text{model}}(d_{\text{model}}-1)/2$ when $\varepsilon > 0$ and $L\,(d_{\text{model}}(d_{\text{model}}-1)/2 + 1)$ when $\varepsilon = 0$. □

Remark A.12 (Other starting points and tied readouts). The recursion of part (i) can also start at any h_k with any orthogonal Q_k, running forward as above for $\ell > k$ and backward for $\ell \leq k$ with the QR factorization $A_\ell Q_\ell = Q_{\ell-1}R_\ell$ or the polar decomposition $A_\ell Q_\ell = Q_{\ell-1}S_\ell$; the embedding absorbs Q_0. With Q_k the rotation of Proposition A.9(i) for one submatrix of a read or a write of h_k, that submatrix becomes triangular too, and $(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$ parameters are removed in all. If Q_0 and Q_L are fixed, or tied to each other as by a tied readout $D = E^\top$, which forces $Q_L = Q_0$, the recursions from h_0 forward and from h_L backward meet at any one skip, which keeps its full

matrix; without normalization the same holds in $\mathrm{GL}(d_{\mathrm{model}})$, with identities in place of the R_ℓ and with $G_L = G_0^{-\top}$ for a tied readout.

We recall Theorem 3.7 from Section 3.4.

Theorem 3.7 (Orthogonal changes of basis under RMSNorm)**.** *Consider pre-RMSNorm networks.*
(i) *In a skipless network, let d_{model} columns of the read of h_ℓ ($W_{\mathrm{read}}^{\ell+1}$, or D if $\ell = L$) form a matrix $W_1 = QR$, with Q orthogonal and R upper triangular. The change of basis Q at h_ℓ preserves the function and turns W_1 into R; at all $L+1$ hidden states, they remove $(L+1)d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2$ parameters.*
(ii) *The changes of basis that preserve the function at all parameters are orthogonal, and in a residual network they agree at all hidden states. They form a group of dimension $(L+1)d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2$ in a skipless network and $d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2$ in a residual one.*
(iii) *In a residual network, let d_{model} columns of one of the reads $W_{\mathrm{read}}^1, \dots, W_{\mathrm{read}}^L, D$ form a matrix $W_1 = QR$. The same change of basis Q at all hidden states preserves the function and turns W_1 into R.*
(iv) Triangular skips. *If every learned skip A_ℓ is invertible, there are orthogonal $Q_1, \dots, Q_L$, with $Q_0 = I$, such that every $Q_{\ell-1}^\top A_\ell Q_\ell$ is upper triangular, and the network with these skips, the reads $Q_{\ell-1}^\top W_{\mathrm{read}}^\ell$, the writes $W_{\mathrm{write}}^\ell Q_\ell$ and the readout $Q_L^\top D$ computes the same function.*

Proof of Theorem 3.7. For $\varepsilon > 0$, $\mathcal{G} = \mathrm{O}(d_{\mathrm{model}})$ (Lemma A.2). Part (i) is Proposition A.9(i). In part (ii), the changes of basis are orthogonal by Lemma A.4 for the skipless network and by Theorem A.5(iii) for the residual one, where they also agree at all hidden states, and the dimensions are Theorem A.14(i) with $\dim \mathcal{G} = d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2$. Part (iii) is Proposition A.9(ii), and part (iv) is Theorem A.11(i). □

A.5 LayerNorm reduces to RMSNorm (Proposition A.13)

The reduction of LayerNorm to RMSNorm is standard: the centering folds into the adjacent weights (Elhage et al., 2021), and centering every write gives an RMSNorm network whose hidden states are mean-free (Ashkboos et al., 2024a; Theus et al., 2025). We state it with the constants that the counts need. It is used in Section 3.4 and for ViT-B in Section 3.4 and Appendix D.

Proposition A.13 (LayerNorm reduces to RMSNorm)**.** *Let $Z = I - \frac{1}{d_{\mathrm{model}}}\mathbf{1}\mathbf{1}^\top$ and $\mathcal{V} = \mathbf{1}^\perp$, and consider a pre-LayerNorm network, skipless or residual, whose reads and readout see $\mathrm{LN}(h) = \gamma \odot hZ/\sqrt{\|hZ\|^2/d_{\mathrm{model}} + \varepsilon} + \beta$, with h, γ and β row vectors.*
(i) *Replacing each read W by $\mathrm{diag}(\gamma)W$ with the added bias βW, and each write W, write bias b and embedding E by WZ, bZ and EZ, gives a network that computes the same function, whose hidden states lie in $\mathcal{V}$, and whose reads see $N_\varepsilon(h) = h/\sqrt{\|h\|^2/d_{\mathrm{model}} + \varepsilon}$.*
(ii) *In an orthonormal basis of $\mathcal{V}$, this network is a pre-RMSNorm network of dimension $d_{\mathrm{model}} - 1$, with every read multiplied by $\sqrt{d_{\mathrm{model}}/(d_{\mathrm{model}}-1)}$ and ε replaced by $\varepsilon d_{\mathrm{model}}/(d_{\mathrm{model}}-1)$.*
(iii) *If the gains γ have nonzero entries, the difference in redundancy between the skipless and the residual version of the LayerNorm network equals the difference for the reduced networks of (ii).*

Overview. LayerNorm sees only the mean-free part of the stream, so the mean can be removed from every write without changing the function. On the mean-free subspace LayerNorm is RMSNorm up to constants. The map from the parameters of the LayerNorm network to those of the reduced network has a derivative that is onto, and it is the same for the skipless and the residual version, so it preserves the difference in redundancy.

Proof. (i) Since $\mathrm{LN}(h) = \gamma \odot N_\varepsilon(hZ) + \beta$, a read satisfies $\mathrm{LN}(h)\,W = N_\varepsilon(hZ)\,\mathrm{diag}(\gamma)W + \beta W$. Every hidden state is a sum of writes, directly in a skipless network and through the skips in a residual one, so replacing every write, write bias and the embedding by its image under Z replaces every hidden state h by hZ. Since $Z^2 = Z$, every read then sees the same $N_\varepsilon(hZ)$ as before, and the function is unchanged; the hidden states lie in $\mathcal{V}$. (ii) Let the columns of $\bar{U} \in \mathbb{R}^{d_{\mathrm{model}} \times (d_{\mathrm{model}}-1)}$ be an orthonormal basis of $\mathcal{V}$ and $h = y\bar{U}^\top$ with a row vector y. Then $\|h\|^2/d_{\mathrm{model}} + \varepsilon = \frac{d_{\mathrm{model}}-1}{d_{\mathrm{model}}}\big(\|y\|^2/(d_{\mathrm{model}} -$

$1) + \varepsilon d_{\text{model}}/(d_{\text{model}} - 1)\big)$, so $N_\varepsilon(y\bar{U}^\top) = \sqrt{d_{\text{model}}/(d_{\text{model}} - 1)}\, N'(y)\bar{U}^\top$, where N' is RMSNorm of dimension $d_{\text{model}} - 1$ with constant $\varepsilon d_{\text{model}}/(d_{\text{model}} - 1)$. A read $N_\varepsilon(y\bar{U}^\top)W$ is the read $\sqrt{d_{\text{model}}/(d_{\text{model}} - 1)}\,\bar{U}^\top W$ of $N'(y)$, and a write WZ is the write $W\bar{U}$ in the coordinates y. (iii) The reduced parameters are $\sqrt{d_{\text{model}}/(d_{\text{model}} - 1)}\,\bar{U}^\top \text{diag}(\gamma)W$ and $b + \beta W$ for each read, $W\bar{U}$ and $b\bar{U}$ for each write, $E\bar{U}$ for the embedding, and the analogous expressions for the readout. For γ with nonzero entries each is linear and onto in W, b or E, so the derivative of the map from the parameters of the LayerNorm network to those of the reduced network is onto. The LayerNorm network computes the reduced network's function composed with this map, so by the chain rule every batch Jacobian of the LayerNorm network is that of the reduced network times this derivative, and at corresponding parameters both have the same functional dimension, and the redundancy of the LayerNorm network is that of the reduced network plus the difference of the parameter counts. That difference is the same for the skipless and the residual versions, which have the same shapes. □

Section 6 of the notebook (Appendix E) checks each step on a residual pre-LayerNorm network with $d_{\text{model}} = 6$: folding the gains changes the function by a relative 2×10^{-16}, centering the writes by 2×10^{-16}, and an orthogonal change of basis on $\mathcal{V}$ by 8×10^{-16}, while a generic orthogonal change that moves $\mathbf{1}$ changes it by 0.75.

A.6 Symmetry counts and upper bounds (Theorem 4.4)

Theorem 4.4 and Theorem 3.6(ii) are parts of the following theorem. A symmetry inside a block (Section 4.2) changes the read and the write of that block and leaves its map unchanged. So it preserves the function of a skipless and of a residual network alike, with or without normalization, and the total dimension s of these symmetries is the same number for both. The changes of basis and the symmetries inside blocks act *locally freely* at θ if no continuous family of them, other than the identity, fixes θ.

Theorem A.14 (Symmetry counts and upper bounds)**. (i)** *Without normalization or with pre-RMSNorm and $\varepsilon > 0$, the changes of basis that preserve the function at all parameters form a group of dimension $(L + 1)\dim \mathcal{G}$ in a skipless network and $\dim \mathcal{G}$ in a residual one, so the L skips remove $L \dim \mathcal{G}$ dimensions of symmetry. For $\varepsilon = 0$, the changes of basis in $\mathcal{G}$ at the hidden states preserve the function of a skipless network, and they preserve that of every residual network only if they agree at all hidden states.*
(ii) *If in both networks the symmetries act locally freely at generic parameters, the functional dimension at these parameters is at most $P - (L + 1)\dim \mathcal{G} - s$ without skips and $P - \dim \mathcal{G} - s$ with them.*
(iii) *If in addition both networks satisfy the equality condition, the skips raise the functional dimension by exactly $L \dim \mathcal{G}$, which is Ld_{model}^2 without normalization.*

For $\varepsilon = 0$ and positively homogeneous blocks, a skipless network has more symmetries than the changes of basis in $\mathcal{G}$: all of $\text{GL}(d_{\text{model}})$ at $h_0, \ldots, h_{L-1}$ (Lemma A.3). Its functional dimension then lies below the bound of (ii), so it does not satisfy the equality condition; for one block the skip adds at least d_{model}^2 (Theorem B.1(iv)).

Overview. Part (i) counts copies of $\mathcal{G}$: one per hidden state without skips, by Proposition A.1 and Lemma A.4, and one with them, by Theorem A.5(iii). Part (ii) bounds the rank of every batch Jacobian by the number of parameters minus the number of independent directions in which the symmetries move the parameters, which is the dimension of the symmetries when they act locally freely; Lemma A.15 proves that they do when the embedding has rank d_{model} and every block has width at least d_{model}. Part (iii) turns both bounds into equalities under the equality condition and subtracts. In the last row of Table 3, the change of basis $I + \tau X$ moves the fixed submatrix $W_1 = I$ by $-\tau X$, so its directions fill the fixed entries: fixing them removes d_{model}^2 parameters and no rank.

Lemma A.15 (Locally free action)**.** *Consider an MLP network, skipless or residual, without normalization or with pre-RMSNorm, with vanilla MLP blocks with $\tanh$, GELU, ReLU or ReLU^2 activations, or with gated blocks with any gate, and the symmetries inside blocks of Section 4.2. If the*

Table 3: Parameters, dimension of the symmetries and, under the equality condition, functional dimension of the networks of Section 2, without normalization or biases; P counts the parameters of the skipless network. Each step of the gauge fix removes as many parameters as dimensions of symmetry, while deleting the identity units removes $2Ld_{\text{model}}^2$ parameters but only Ld_{model}^2 dimensions of symmetry. Given the value for the residual network, those of the mixed network, of learned skips and of the last row hold at corresponding parameters without further assumption (Theorem A.5(ii), Proposition A.9(ii)).

Network	Parameters	Symmetry	Functional dimension
skipless	P	$(L+1)d_{\text{model}}^2 + s$	$P - (L+1)d_{\text{model}}^2 - s$
mixed, with trainable identity units	$P + 2Ld_{\text{model}}^2$	$(2L+1)d_{\text{model}}^2 + s$	$P - d_{\text{model}}^2 - s$
learned skips	$P + Ld_{\text{model}}^2$	$(L+1)d_{\text{model}}^2 + s$	$P - d_{\text{model}}^2 - s$
residual	P	$d_{\text{model}}^2 + s$	$P - d_{\text{model}}^2 - s$
residual, one submatrix fixed	$P - d_{\text{model}}^2$	s	$P - d_{\text{model}}^2 - s$

embedding has rank d_{model} and every block has width at least d_{model}, then at generic parameters the changes of basis and the symmetries inside blocks act locally freely. The same holds if the readout has rank d_{model} and every block has width at least d_{model}. With an embedding of rank d_{model}, it also holds for networks with attention blocks whose heads have width $d_h \leq d_{\text{model}}$ and total width $Hd_h \geq d_{\text{model}}$.

Proof. Permutations and sign flips are isolated, so it suffices to consider changes of basis and rescalings close to the identity. Let changes of basis $G_0, \dots, G_L$ and positive rescalings c_ℓ of hidden units (for ReLU and ReLU^2), of gates and values (for ReGLU) or of values (for the other gated blocks) fix a generic θ; for $\tanh$ and GELU there are no c_ℓ. The embedding transforms as $E \mapsto EG_0$, and E has rank d_{model}, so $G_0 = I$. Suppose $G_{\ell-1} = I$. The read of block ℓ then transforms as $W_{\text{read}}^\ell \mapsto W_{\text{read}}^\ell \operatorname{diag}(c_\ell)$, with entries 1 on the columns that are not rescaled, and no column of a generic W_{read}^ℓ vanishes, so $c_\ell = \mathbf{1}$. The write then transforms as $W_{\text{write}}^\ell \mapsto W_{\text{write}}^\ell G_\ell$, and W_{write}^ℓ has rank d_{model} because the block has width at least d_{model}, so $G_\ell = I$. By induction only the identity fixes θ. In a residual network the G_ℓ are equal (Theorem A.5(iii)) and the same argument applies. The second statement follows by the mirror induction from the readout, $D \mapsto G_L^{-1} D$: rescalings act freely on the rows of a generic write, except that for ReGLU a row fixes only the product cc' of the gate and value scales of its unit; each read, of rank d_{model} and with generic columns, then forces $G_{\ell-1} = I$ and the remaining scales to be 1. With RMSNorm the changes of basis lie in $\mathrm{CO}(d_{\text{model}})$ or $\mathrm{O}(d_{\text{model}})$, and the argument is unchanged. For an attention block, once $G_{\ell-1} = I$ the changes of basis inside its heads multiply $W_{Q,i}$ and $W_{V,i}$ on the right. At generic parameters these have full column rank, as $d_h \leq d_{\text{model}}$, so the changes of basis inside the heads are the identity. The write, which stacks the $W_{O,i}$, then transforms as $W_O \mapsto W_O G_\ell$ and has rank d_{model}, as $Hd_h \geq d_{\text{model}}$, so $G_\ell = I$. □

Remark A.16 (The width condition; attention blocks). The width condition is needed: at a hidden state of a skipless network that is written and read at rank $m < d_{\text{model}}$, the changes of basis that fix the parameters form a family of dimension $(d_{\text{model}} - m)^2$, so they move the parameters in only $2d_{\text{model}}m - m^2$ independent directions, as Zhao et al. (2023, Proposition C.3) compute for one hidden layer. The full group of the symmetries inside attention heads is characterized by Wang and Wang (2025). A numerical check on small skipless transformers confirms a locally free action, also for heads of total width $Hd_h < d_{\text{model}}$ followed by an MLP block of width at least d_{model}, which the induction does not reach (Section 6 of the notebook, Appendix E).

Proof of Theorem A.14. Parts (i) and (ii). If a family of transformations of the parameters leaves the function unchanged, then the function does not change to first order along any direction in which the family moves θ, so these directions lie in the kernel of every batch Jacobian at θ, and the functional dimension is at most P minus their number. For a skipless network, Proposition A.1 gives $L+1$ independent copies of $\mathcal{G}$, one per hidden state, and the directions in which they move θ are linearly independent of one another and of those of the symmetries inside blocks at generic parameters: the change of basis at h_ℓ multiplies W_{write}^ℓ on the right and $W_{\text{read}}^{\ell+1}$ on the left, the symmetries inside block ℓ multiply W_{read}^ℓ on the right and W_{write}^ℓ on the left, and by hypothesis

they act locally freely at generic parameters (Lemma A.15). The number of independent directions is therefore $(L+1)\dim\mathcal{G}+s$. For a residual network, Theorem A.5(iii) leaves one copy, and the same argument gives $\dim\mathcal{G}+s$. For part (i), no other changes of basis preserve the function at all parameters: without normalization $\mathcal{G}$ is all of $\mathrm{GL}(d_{\text{model}})$, and for $\varepsilon>0$ this is Lemma A.4 for the skipless network and Theorem A.5(iii) for the residual one. *Part (iii).* Under the hypothesis, the inequalities of part (ii) are equalities, so the generic functional dimensions are $P-(L+1)\dim\mathcal{G}-s$ and $P-\dim\mathcal{G}-s$, and the difference is $L\dim\mathcal{G}$, which is d_{model}^2 per skip without normalization, and under RMSNorm $d_{\text{model}}(d_{\text{model}}-1)/2$ when $\varepsilon>0$ and $d_{\text{model}}(d_{\text{model}}-1)/2+1$ when $\varepsilon=0$ (Lemma A.2). □

We recall Theorem 4.4 from Section 4.2.

Theorem 4.4 (Upper bounds)**.** *Let $\mathcal{G}$ be the group of changes of basis of one hidden state, and s the dimension of the symmetries inside the blocks, the same with and without skips. With or without pre-RMSNorm, at generic parameters, the functional dimension is at most $P-(L+1)\dim\mathcal{G}-s$ without skips and at most $P-\dim\mathcal{G}-s$ with identity skips.*

Proof of Theorem 4.4. Within the scope of Section 2, the symmetries act locally freely at generic parameters (Lemma A.15). So the theorem is Theorem A.14(ii), and the gain under the equality condition is Theorem A.14(iii). □

The next proposition relates the equality condition to identifiability (Definition 4.6).

Proposition 4.7 (The equality condition and local identifiability)**.** *Let the blocks be analytic, with or without pre-RMSNorm, within the scope of Section 2. A skipless or a residual network satisfies the equality condition if and only if, at generic parameters, it is locally identifiable up to its symmetries. Every symmetry that is needed is then close to the identity.*

Proof. Let k be the dimension of the symmetries of the network, $(L+1)\dim\mathcal{G}+s$ without skips and $\dim\mathcal{G}+s$ with them. For a finite batch X, write $F_X(\theta)$ for the outputs on X, and let r be the largest rank that the Jacobian of F_X attains on any batch at any parameters. Fix a batch X that attains it. The outputs are analytic in θ, so the rank on X is r outside the zero set of the sum of the squares of the minors of size r, a set of measure zero. Hence r is the functional dimension at generic parameters, $r\leq P-k$ by Theorem 4.4, and the equality condition is $r=P-k$.

Take θ at which the rank on X is r and the symmetries act locally freely, as they do at generic parameters (Lemma A.15). By lower semicontinuity the rank on X is r on a neighborhood U of θ. By the constant rank theorem, after shrinking U, the set $Z=\{\theta'\in U : F_X(\theta')=F_X(\theta)\}$ is a connected manifold of dimension $P-r$, whose tangent space at each point is the kernel of the Jacobian. For any further input x, the Jacobian on $X\cup\{x\}$ has rank r on U as well, so it has the same kernel, and $\theta'\mapsto F_{\theta'}(x)$ is constant on Z. So $Z=\{\theta'\in U : F_{\theta'}=F_\theta\}$. The changes of basis that preserve the function, the rescalings and the changes of basis inside the heads form a Lie group of dimension k, which acts smoothly. It acts locally freely at θ, so $g\mapsto g\cdot\theta$ is an immersion near the identity, and it maps a neighborhood of the identity onto a manifold $O\subset Z$ of dimension k.

If $r=P-k$, then O is open in Z, since the two have the same dimension. So every θ' close to θ with $F_{\theta'}=F_\theta$ is the image of θ under a symmetry close to the identity. Conversely, let every θ' in a neighborhood of θ in Z be the image of θ under a symmetry. With the permutations and the sign flips, the symmetries form a Lie group of dimension k with finitely many components, and the images of θ under it form the image of a manifold of dimension k under a smooth map. By Sard's theorem this image contains no open subset of a manifold of larger dimension, so $P-r\leq k$, which with $r\leq P-k$ is the equality condition. □

A.7 Depth (Theorems 4.9, 4.12 and 4.13)

This subsection uses Lemma B.3, Lemma B.5 and the proof of Theorem B.1 from Appendix B, which does not depend on it. Theorem A.25 below and Theorem 4.9 rest on two lemmas. The first turns a gap in functional dimension into disjointness, and for definable networks into a positive distance.

Where we apply it, F_θ is a residual network, G_η is a skipless network, and the group $\mathcal{H}$ consists of symmetries of the skipless network: changes of basis $G \in \mathcal{G}$ at some of its hidden states, together with the symmetries inside its blocks. The second shows that with small writes the Jacobian of a residual network reaches the largest rank that its symmetries allow, by the argument of Theorem B.1 for one block.

Definable sets. For limits, Lemma A.23 rests on a theorem of o-minimal geometry about the closure of a set (Lemma A.20). Lemma A.21 states the same step in terms of dimensions, which is the form of the argument in Section 4.4. We recall the definitions and the statements that we use from the book of van den Dries (1998), with its numbering. The lecture notes of Coste (1999) are an introduction that is available online; there the definition is Definition 1.4, and the results on cells, on the dimension and on the closure are Theorem 2.10, Proposition 3.17 and Theorem 3.22.

Definition A.17 (Definable sets and maps (van den Dries, 1998, Chapter 1, (2.1) and (3.2)))**.** An *o-minimal structure on the real field* is a collection of subsets of $\mathbb{R}^k$ for every $k \geq 0$, the *definable* sets, with the following three properties.

(i) *Set operations.* Finite unions, finite intersections and complements of definable subsets of $\mathbb{R}^k$ are definable. If $A \subseteq \mathbb{R}^k$ is definable, so are $\mathbb{R} \times A$ and $A \times \mathbb{R}$. The projection of a definable subset of $\mathbb{R}^{k+1}$ on its first k coordinates is definable.

(ii) *Basic sets.* The sets $\{x \in \mathbb{R}^k : x_1 = x_k\}$ and $\{(x, y) \in \mathbb{R}^2 : x < y\}$ are definable, and so are the graphs of addition and of multiplication.

(iii) *Tameness.* The definable subsets of $\mathbb{R}$ are exactly the finite unions of points and intervals.

A map is *definable* if its graph is a definable set.

The three properties are conditions on the collection as a whole, and *definable* always refers to one collection that has been fixed. Lemma A.18 fixes ours. Properties (i) and (ii) only add sets to the collection. In a collection with the three properties, a set is definable as soon as it is described by a formula built from real variables and constants, definable functions, the relations $=$ and $<$, the words *and*, *or* and *not*, and the quantifiers *there exists* and *for all* over real variables. Here *or*, *and* and *not* stand for a union, an intersection and a complement, *there exists* stands for a projection, and *for all* is *not*, *there exists*, *not*. For example, the image of a definable map H is the set of all y for which there exists a β with $y = H(\beta)$. In this way sums, products and compositions of definable maps are definable, and so are the image of a definable set under a definable map (van den Dries, 1998, Chapter 1, Lemma (2.3)) and the closure of a definable set (van den Dries, 1998, Chapter 1, Lemma (3.4)). Property (iii) excludes sets, among them the graph of the sine: if this graph were definable, then by an intersection and a projection so would be the zero set $\{j\pi : j \in \mathbb{Z}\}$ of the sine, which is not a finite union of points and intervals.

Lemma A.18 (The outputs of the networks are definable)**.** *There is an o-minimal structure on the real field in which the exponential and the error function are definable. In this structure, for every network of attention,* $\tanh$*, GELU, SwiGLU or GeGLU blocks, with or without pre-RMSNorm with* $\varepsilon > 0$*, skipless or residual, and for every batch* X *of inputs, the map from the weights to the outputs on* X *is definable, and so is the set of these outputs.*

Proof. Speissegger (1997) constructs an o-minimal structure on the real field, the Pfaffian closure of the real field, with the following two properties, which are the special case and the corollary of the theorem in the introduction of that paper. If a function $u : \mathbb{R} \to \mathbb{R}$ of class C^1 satisfies $u'(t) = \Phi(t, u(t))$ for all t, with Φ definable and of class C^1, then u is definable. If $\varphi : \mathbb{R} \to \mathbb{R}$ is definable and continuous, then $t \mapsto \int_0^t \varphi(\xi)\, d\xi$ is definable. The first property, with $\Phi(t, y) = y$, makes the exponential definable. The second then makes the error function definable:

$$\operatorname{erf}(t) = \frac{2}{\sqrt{\pi}} \int_0^t e^{-\xi^2}\, d\xi.$$

The maps of the networks are formulas in these two functions,

$$\tanh z = \frac{e^{2z} - 1}{e^{2z} + 1}, \qquad \operatorname{SiLU}(z) = \frac{z}{1 + e^{-z}}, \qquad \operatorname{GELU}(z) = \frac{z}{2}\Big(1 + \operatorname{erf}\Big(\frac{z}{\sqrt{2}}\Big)\Big),$$

$$\mathrm{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \qquad N_\varepsilon(v) = \frac{v}{\sqrt{\dfrac{\|v\|^2}{d_{\mathrm{model}}} + \varepsilon}},$$

and the rest of a network consists of sums and products of matrices. A quotient and a square root are definable where they are defined: $y = \frac{a}{b}$ is the formula $yb = a$, and $y = \sqrt{a}$ is the formula $y^2 = a$ and $y \geq 0$. So the map from the weights to the outputs on X is a composition of definable maps, and the set of outputs is its image. □

Definition A.19 (Cells and dimension (van den Dries, 1998, Chapter 3, (2.3), and Chapter 4, (1.1)))**.** The *cells* of $\mathbb{R}$ are the points and the open intervals. A cell of $\mathbb{R}^{k+1}$ lies over a cell of $\mathbb{R}^k$: it is the graph of a continuous definable function on that cell, or the band between the graphs of two such functions, where the lower one may be $-\infty$ and the upper one $+\infty$. A cell is a *d-cell* if d intervals and bands occur in its construction, from $\mathbb{R}$ up to $\mathbb{R}^k$. The *dimension* of a definable set is the largest d for which it contains a d-cell, and $-\infty$ if the set is empty.

In the plane, a point is a 0-cell, the graph of a function over an interval and a vertical segment over a point are 1-cells, and a band over an interval is a 2-cell. The k-cells of $\mathbb{R}^k$ are open sets, every open box is a k-cell, and the other cells of $\mathbb{R}^k$ are not open. So a definable subset of $\mathbb{R}^k$ has dimension k exactly when it has nonempty interior. A suitable coordinate projection maps a d-cell homeomorphically onto an open cell of $\mathbb{R}^d$ (van den Dries, 1998, Chapter 3, (2.7)). Every definable set is a union of finitely many cells (van den Dries, 1998, Chapter 3, Theorem (2.11)). The dimension of a union of two definable sets is the larger of their dimensions, and a definable bijection preserves the dimension (van den Dries, 1998, Chapter 4, Proposition (1.3)).

Lemma A.20 (The closure of a definable set)**.** *Let $S \subseteq \mathbb{R}^k$ be a nonempty definable set, with closure $\bar{S}$. Then $\dim(\bar{S} \setminus S) < \dim S$ (van den Dries, 1998, Chapter 4, Theorem (1.8)), and therefore $\dim \bar{S} = \dim S$. If S has Lebesgue measure zero, then so has $\bar{S}$.*

Proof. The inequality is the cited theorem. The closure is the union of S and $\bar{S} \setminus S$, which are both definable, so its dimension is the larger of their dimensions, which is $\dim S$. Now let S have measure zero. Then S has empty interior, so $\dim S < k$, and hence $\dim \bar{S} < k$. So $\bar{S}$ is a union of finitely many cells of $\mathbb{R}^k$, none of them open. A cell of $\mathbb{R}^k$ that is not open has measure zero, by induction on k. In $\mathbb{R}$ it is a point. In $\mathbb{R}^k$ it is the graph of a function over a cell of $\mathbb{R}^{k-1}$, or a band over a cell of $\mathbb{R}^{k-1}$ that is not open. Both have measure zero by Fubini's theorem, the band by the induction hypothesis. □

Lemma A.21 (Images of definable analytic maps)**.** *Let $H : \mathbb{R}^p \to \mathbb{R}^n$ be definable and real-analytic, and let ρ be the largest rank of its Jacobian.*
(i) *The Jacobian has rank ρ outside a set of measure zero, and the image $H(\mathbb{R}^p)$ has dimension ρ.*
(ii) *If $C \subseteq \mathbb{R}^n$ is a definable set of dimension less than ρ, then $H(\theta) \notin C$ for almost every $\theta \in \mathbb{R}^p$.*

Proof. (i) The rank is ρ outside the zero set of the sum of the squares of the minors of size ρ of the Jacobian. This sum is analytic and not identically zero, so its zero set has measure zero. The image $H(\mathbb{R}^p)$ is definable. Suppose that it contained a d-cell with $d > \rho$. A suitable coordinate projection would map this cell onto an open subset of $\mathbb{R}^d$. The composition of H with this projection has a Jacobian of rank at most $\rho < d$ at every point, so its image has measure zero by Sard's theorem, and contains no open set. So the dimension of $H(\mathbb{R}^p)$ is at most ρ. Conversely, the rank is ρ near a point where it is ρ. By the constant rank theorem, H maps a small ball B around such a point onto a piece $H(B)$ of a manifold of dimension ρ, which a suitable coordinate projection q maps diffeomorphically onto an open subset of $\mathbb{R}^\rho$. The set $H(B)$ is a definable subset of $H(\mathbb{R}^p)$. A definable bijection preserves the dimension, and a nonempty open subset of $\mathbb{R}^\rho$ has dimension ρ. So the dimension of $H(\mathbb{R}^p)$ is at least ρ.

(ii) Take B and q as in (i). The set $H(B) \cap C$ is definable and has dimension less than ρ, and so has its image under q, since q is a definable bijection on $H(B)$. A definable subset of $\mathbb{R}^\rho$ of dimension less than ρ is a union of finitely many cells that are not open, so it has measure zero, as in

the proof of Lemma A.20. As q is a diffeomorphism from $H(B)$ onto an open set, $H(B) \cap C$ has measure zero in the manifold $H(B)$. So has its preimage in B, because H has constant rank ρ on B. Countably many such balls cover the set of points where the rank is ρ, and by (i) the other points form a set of measure zero. □

Remark A.22 (Without definability the closure can gain dimension). The curve $\beta \mapsto (\sin \beta, \sin \sqrt{2}\,\beta)$ in $\mathbb{R}^2$ is the image of a map with one parameter. As $\sqrt{2}$ is irrational, the curve comes arbitrarily close to every point of the square $[-1, 1]^2$, so its closure is the whole square. The curve is the set of outputs, on the two inputs $(1, 0)$ and $(0, 1)$, of the family $x \mapsto \sin(\beta)\, x_1 + \sin(\sqrt{2}\,\beta)\, x_2$ with the single parameter β. Every map $x \mapsto a_1 x_1 + a_2 x_2$ with $|a_1| \le 1$ and $|a_2| \le 1$ is a pointwise limit of this family, in the sense of Section 4.4. So a family of functional dimension 1 has among its limits a family of functional dimension 2. With $\tanh$ in place of the sine, the curve tends to $(1, 1)$ and $(-1, -1)$ as $\beta \to \pm\infty$, and its closure adds only these two points.

Lemma A.23 (A rank criterion). *Let $\{F_\theta\}_{\theta \in \mathbb{R}^P}$ and $\{G_\eta\}_{\eta \in \mathbb{R}^{P'}}$ be two parametrized families of maps $\mathbb{R}^{T \times d_{\text{in}}} \to \mathbb{R}^{T \times d_{\text{out}}}$ such that, for every input x, $F_\theta(x)$ is real-analytic in θ and $G_\eta(x)$ is real-analytic in η. Let a Lie group $\mathcal{H}$ act smoothly on $\mathbb{R}^{P'}$, written $\eta \mapsto a \cdot \eta$ for $a \in \mathcal{H}$, such that $G_{a \cdot \eta} = G_\eta$ for all $a \in \mathcal{H}$ and all $\eta \in \mathbb{R}^{P'}$, and such that the orbit $\mathcal{H} \cdot \eta = \{a \cdot \eta : a \in \mathcal{H}\}$ has dimension at least k at generic η. Suppose that at some θ_0 the Jacobian of $\theta \mapsto (F_\theta(x))_{x \in X}$ has rank greater than $P' - k$ on some finite batch X.*
(i) *For almost every θ, no η satisfies $F_\theta = G_\eta$.*
(ii) *If the outputs $G_\eta(x)$, $x \in X$, are definable functions of η, then for almost every θ there is a $\delta > 0$ such that every η has $\|F_\theta(x) - G_\eta(x)\| \ge \delta$ at some $x \in X$.*

Proof. Write $F_X(\theta), G_X(\eta) \in \mathbb{R}^n$ for the outputs on X. At generic η the map G_X is constant along the orbit $\mathcal{H} \cdot \eta$, since $G_{a \cdot \eta} = G_\eta$, and this orbit has dimension at least k. So the Jacobian of G_X has rank at most $r_0 = P' - k$; the rank is lower semicontinuous, so the bound holds at every η. The Jacobian of F_X reaches its largest rank r^*, which exceeds r_0 by hypothesis, outside the zero set of the sum of the squares of its minors of size r^*. This function is analytic and not identically zero, so its zero set has measure zero. Take θ outside it. By the constant rank theorem, F_X maps a neighborhood U of θ onto an embedded manifold Y of dimension r^*, and after shrinking U a linear projection $\pi : \mathbb{R}^n \to \mathbb{R}^{r^*}$ maps Y diffeomorphically onto an open set. The map $\pi \circ G_X$ has rank at most $r_0 < r^*$, so all its values are critical, and by Sard's theorem its image has measure zero. So $G_X(\mathbb{R}^{P'}) \cap Y$ has measure zero in Y, and so does its preimage under F_X in U, which contains every θ' in U with $F_{\theta'} = G_\eta$ for some η. Countably many such U cover the complement of the zero set, which gives (i).

For (ii), let $S = G_X(\mathbb{R}^{P'})$, with closure $\bar{S}$. The image $\pi(S)$ of $\pi \circ G_X$ is definable and has measure zero, so its closure has measure zero (Lemma A.20). As π is continuous, this closure contains $\pi(\bar{S} \cap Y)$. So $\bar{S} \cap Y$ has measure zero in Y, and so does its preimage under F_X in U. For θ' in U outside this preimage, $F_X(\theta')$ does not lie in $\bar{S}$, so some $\delta > 0$ has $\|F_X(\theta') - G_X(\eta)\| \ge \delta\sqrt{|X|}$ for every η, and then $\|F_{\theta'}(x) - G_\eta(x)\| \ge \delta$ at some $x \in X$. Countably many such U cover the complement of the zero set again. □

Lemma A.24 (Small writes). *Let $d_{\text{model}} \ge 2$, let E and D have rank d_{model}, and let each block, in any order and of any width, be an attention block with heads of width at most d_{model}, causal or not, on sequences of at least two tokens, or an MLP block, with all MLP blocks* $\tanh$*, all GELU, all SwiGLU or all GeGLU blocks. For generic parameters*

$$\theta \;=\; \Big(E,\; \big\{W^\ell_{\text{read}},\; W^\ell_{\text{write}}\big\}_{\ell=1}^{L},\; D\Big)$$

and a number τ, let

$$\theta_\tau \;=\; \Big(E,\; \big\{W^\ell_{\text{read}},\; \tau\, W^\ell_{\text{write}}\big\}_{\ell=1}^{L},\; D\Big).$$

With biases, θ also holds b_E, b^ℓ_{read}, b^ℓ_{write} and b_D, and θ_τ holds $\tau\, b^\ell_{\text{write}}$ in place of b^ℓ_{write}. For all small $\tau \neq 0$, the residual network has

$$\begin{aligned}
&\mathrm{fd}(\theta_\tau) = P - d_{\text{model}}^2 - s && \textit{no normalization, no biases},\\
&\mathrm{fd}(\theta_\tau) = P - d_{\text{model}}(d_{\text{model}}-1)/2 - s && \textit{pre-RMSNorm with } \varepsilon > 0,\ \textit{no biases},\\
&\mathrm{fd}(\theta_\tau) = P - d_{\text{model}}(d_{\text{model}}-1)/2 - 1 - s && \textit{pre-RMSNorm with } \varepsilon = 0,\ \textit{no biases},\\
&\mathrm{fd}(\theta_\tau) = P - d_{\text{model}}^2 - (L+1)d_{\text{model}} - s && \textit{MLP blocks with biases, no normalization},\\
&\mathrm{fd}(\theta_\tau) = P - s && \textit{no embedding, readout, biases or normalization}.
\end{aligned}$$

Proof. Let $F_X(\theta) = (F_\theta(x))_{x\in X}$ be the outputs on a finite batch X, and let $J_\tau = \mathrm{D}_\theta F_X(\theta_\tau)$ be its Jacobian at θ_τ. Let S_τ be the diagonal matrix on $\mathbb{R}^P$ with the entry $1/\tau$ at the coordinates of the reads W^ℓ_{read} (for an attention block, of W_Q, W_K and W_V) and of the read biases b^ℓ_{read}, and the entry 1 at all other coordinates. Then

$$\operatorname{rank} J_\tau \;=\; \operatorname{rank}(J_\tau S_\tau) \quad \text{for } \tau \neq 0, \qquad \text{and we let} \qquad J_0 \;=\; \lim_{\tau\to 0} J_\tau S_\tau,$$

a limit that the expansions below show to exist. For a direction $\eta \in \mathbb{R}^P$, write $(J_0\eta)(x)$ for the part of $J_0\,\eta$ at the input x, which does not depend on the rest of the batch. As in Appendix B, η is *flat* if $(J_0\eta)(x) = 0$ for every input x.

Let $W_{\text{read}} = [\,W^1_{\text{read}} \mid \cdots \mid W^L_{\text{read}}\,]$ place the reads of the MLP blocks side by side, and let W_{write} stack their writes $W^1_{\text{write}}, \dots, W^L_{\text{write}}$, all at θ, so that $n \mapsto f(nW_{\text{read}})W_{\text{write}}$ is one wide block that holds all hidden units of the network. Likewise, let the heads of all attention blocks form one attention block.

MLP blocks without normalization. At θ_τ every hidden state is $h_\ell = xE + b_E + O(\tau)$, so the output is

$$y = xED + b_E D + b_D + \tau \sum_{\ell=1}^{L} \Big(f\big((xE + b_E)W^\ell_{\text{read}} + b^\ell_{\text{read}}\big)W^\ell_{\text{write}} + b^\ell_{\text{write}} \Big) D + O(\tau^2).$$

Let c_i be the columns of EW_{read}, β_i the entries of $b_E W_{\text{read}} + b_{\text{read}}$ and a_i the rows of $W_{\text{write}}D$, so that unit i has the pre-activation $z_i = xc_i + \beta_i$ and adds $\tau\,\sigma(z_i)\,a_i$ to y. Differentiating along η and letting $\tau \to 0$,

$$(J_0\eta)(x) \;=\; x\psi_M + \psi_\gamma + \sum_i \big[\sigma(z_i)\,\psi_{a_i} + \sigma'(z_i)\,(x\psi_{c_i} + \psi_{\beta_i})\,a_i\big],$$

where ψ_{c_i}, ψ_{β_i} and ψ_{a_i} are the columns of ψ_C, the entries of ψ_β and the rows of ψ_A, and

$$\begin{gathered}
\psi_M = \eta_E D + E\eta_D, \qquad \psi_\gamma = \eta_{b_E} D + b_E \eta_D + \eta_{b_D} + \textstyle\sum_\ell \eta_{b^\ell_{\text{write}}} D,\\
\psi_C = E\eta_{\text{read}}, \qquad \psi_\beta = b_E\eta_{\text{read}} + \eta_{b_{\text{read}}}, \qquad \psi_A = \eta_{\text{write}} D.
\end{gathered}$$

The terms $x\psi_M + \psi_\gamma$ come from the part of y without τ and from the write biases $\tau\, b^\ell_{\text{write}}$. The terms $\sigma(z_i)\,\psi_{a_i}$ come from the writes $\tau\, W^\ell_{\text{write}}$. The terms with $\sigma'(z_i)$ come from the reads and the read biases: they are of order τ, and S_τ divides them by τ. For the gated blocks, the read of a unit has a gate and a value: c_i and β_i belong to the gate, and e_i, β'_i, ψ_{e_i} and $\psi_{\beta'_i}$ are the same quantities for the value. By (57),

$$(J_0\eta)(x) \;=\; \mathrm{D}_\phi G(\phi, x)\,\psi \qquad \text{at} \quad M = ED, \quad \gamma = b_E D + b_D,$$

for the one-block function G of Lemma B.3. At generic parameters the hypotheses of Lemma B.3 hold for the units of all blocks together, as for one block (Appendix B). So for a flat η the lemma gives $\psi_M = 0$, $\psi_\gamma = 0$ and a rescaling at every unit. As E and D have rank d_{model}, $\psi_M = 0$ gives $\eta_E = EX$ and $\eta_D = -XD$ for some $X \in \mathbb{R}^{d_{\text{model}}\times d_{\text{model}}}$. The rescalings determine η_{read} and η_{write} through ψ_C and ψ_A, and then $\eta_{b_{\text{read}}}$ through ψ_β. Finally $\psi_\gamma = 0$ determines η_{b_D} once η_{b_E} and

the $\eta_{b^\ell_{\text{write}}}$ are chosen. So the flat directions form a space of dimension $d^2_{\text{model}} + s$ without biases, and $d^2_{\text{model}} + (L+1)d_{\text{model}} + s$ with biases. Without an embedding and a readout, $M = I$ is fixed, so $\psi_M = 0$ for every direction, and the flat directions are the rescalings alone, of dimension s.

Attention blocks without normalization. Without biases, $y = xED + \tau \sum_\ell g_\ell(xE)\, D + O(\tau^2)$, where g_ℓ is block ℓ at θ. So $(J_0\eta)(x)$ is the derivative of the residual network with one block that holds all heads and all hidden units, in the direction with the entries ψ_M, ψ_C and ψ_A above and, for each head i,

$$\psi_{B_i} = E\big(\eta_{Q,i}W_{K,i}^\top + W_{Q,i}\eta_{K,i}^\top\big)E^\top/\sqrt{d_h}, \qquad \psi_{C_i} = E\big(\eta_{V,i}W_{O,i} + W_{V,i}\eta_{O,i}\big)D,$$

at $B_i = EW_{Q,i}W_{K,i}^\top E^\top/\sqrt{d_h}$ and $C_i = EW_{V,i}W_{O,i}D$ as in Appendix B, where $W_{O,i}$ holds the rows of W^ℓ_{write} that belong to head i of block ℓ.

First take an input whose tokens all equal x. Head i then gives every token the output xC_i, whatever its scores are, so ψ_{B_i} does not enter, and at every token

$$(J_0\eta)(x,\dots,x) \;=\; x\Big(\psi_M + \textstyle\sum_i \psi_{C_i}\Big) + \sum_j \big[\sigma(z_j)\,\psi_{a_j} + \sigma'(z_j)\,(x\psi_{c_j})\,a_j\big],$$

where i runs over the heads and j over the hidden units. This is $\mathrm{D}_\phi G(\phi, x)\,\psi$ with $M + \sum_i C_i$ in place of M, so for a flat η Lemma B.3 gives a rescaling at every unit. The terms of the units then vanish at every input: they act on each token separately, and they are the derivative along a rescaling, which does not change the function.

Next take an input whose tokens other than the second all equal the first. Head i gives the second token the output of (58), with $k = 1$ if its block is causal and $k = T - 1$ otherwise. The constants $-\frac{1}{2}\log k$ act as biases of the $\tanh$ units, which Lemma B.3 allows. So Step 2 for attention blocks in Appendix B gives $\psi_M = 0$, $\psi_{B_i} = 0$ and $\psi_{C_i} = 0$ for every head. As E has full column rank and D has full row rank, the last two are

$$\eta_{Q,i}W_{K,i}^\top + W_{Q,i}\eta_{K,i}^\top = 0, \qquad \eta_{V,i}W_{O,i} + W_{V,i}\eta_{O,i} = 0,$$

and, as in Step 1 there, their solutions are

$$\eta_{Q,i} = W_{Q,i}\Gamma_i, \quad \eta_{K,i} = -W_{K,i}\Gamma_i^\top, \qquad \eta_{V,i} = W_{V,i}\Gamma_i', \quad \eta_{O,i} = -\Gamma_i' W_{O,i},$$

with $d_h \times d_h$ matrices Γ_i and Γ_i': the changes of basis inside head i. So the flat directions form a space of dimension $d^2_{\text{model}} + s$, and of dimension s without an embedding and a readout.

Under pre-RMSNorm with $\varepsilon > 0$. At θ_τ, $h_\ell = xE + O(\tau)$ and every block reads $N_\varepsilon(xE) + O(\tau)$, with N_ε applied to every token. Let $g = \sum_\ell g_\ell$ be the wide block at θ. The output is

$$y = N_\varepsilon(xE)\, D + \tau\, \mathrm{D}N_\varepsilon(xE)\,\Big(g\big(N_\varepsilon(xE)\big)\Big)\, D + O(\tau^2).$$

So, with $h = xE$ and $n = N_\varepsilon(h)$,

$$(J_0\eta)(x) \;=\; \mathrm{D}N_\varepsilon(h)\,\big(x\eta_E + \mathrm{D}_\theta g(n)\,\eta\big)\, D + N_\varepsilon(h)\,\eta_D,$$

where $\mathrm{D}_\theta g(n)\,\eta$ is the derivative of the wide block at the input n along the entries of η at its reads and writes. This is the equation of the proof of Theorem B.1(ii) with $h_1 = xE$ and $\mathrm{D}_\theta h_1\,\eta = x\eta_E + \mathrm{D}_\theta g(n)\,\eta$, and its Steps 1 to 5 go through with three changes. In Step 2, $\mathrm{D}_\theta g(n)\,\eta$ is bounded, since $\|n_t\| < \sqrt{d_{\text{model}}}$ for every token t. In Step 3, the direction η^Y has the entries EY at E and $-YD$ at D and no others, since $\mathrm{D}N_\varepsilon(h)\,(hY) = N_\varepsilon(h)\,Y$ for antisymmetric Y. In Step 4, the hidden units are handled first, on inputs with equal tokens, as above. So the flat directions are the η^Y, the rescalings of the units and the changes of basis inside the heads, a space of dimension $d_{\text{model}}(d_{\text{model}} - 1)/2 + s$.

Under pre-RMSNorm with $\varepsilon = 0$. Now $N_\varepsilon(h) = \sqrt{d_{\text{model}}}\,h/\|h\|$ on every nonzero row, and the inputs are those whose tokens x_t all have $x_tE \neq 0$. On a finite batch of such inputs, every hidden

state at θ_τ has nonzero rows for small τ, and the expansion of y and the formula for $(J_0\eta)(x)$ hold as for $\varepsilon > 0$, with

$$\mathrm{D}N_\varepsilon(h)\,k \;=\; \frac{\sqrt{d_{\text{model}}}}{\|h\|}\Big(k - \frac{\langle h,k\rangle}{\|h\|^2}\,h\Big)$$

for rows $h \neq 0$ and k. This derivative vanishes exactly when k is a multiple of h. Let η be flat. As in Step 1 of the proof of Theorem B.1(ii), split $\eta_D = -YD + Z$; then $Z = 0$, because the rows $N_\varepsilon(h)$ span $\mathbb{R}^{d_{\text{model}}}$, and at every token

$$\mathrm{D}N_\varepsilon(h)\,\big(x\eta_E + \mathrm{D}_\theta g(n)\,\eta\big) \;=\; N_\varepsilon(h)\,Y.$$

The left side is orthogonal to h, so $hYh^\top = 0$ for all rows h, and Y is antisymmetric. Then $\mathrm{D}N_\varepsilon(h)\,(hY) = N_\varepsilon(h)\,Y$, so at every token t

$$x_t\eta_E + \big(\mathrm{D}_\theta g(n)\,\eta\big)_t - x_tEY \;=\; \mu_t(x)\,x_tE \qquad \text{for a number } \mu_t(x).$$

Replace x by λx with $\lambda > 0$. Then n and $\mathrm{D}_\theta g(n)\,\eta$ do not change, and

$$\lambda\,x_t\,(\eta_E - EY) + \big(\mathrm{D}_\theta g(n)\,\eta\big)_t \;=\; \lambda\,\mu_t(\lambda x)\,x_tE \qquad \text{for all } \lambda > 0.$$

With $\lambda = 1$ and $\lambda = 2$, both rows on the left are multiples of x_tE:

$$x_t\,(\eta_E - EY) \in \mathbb{R}\,x_tE, \qquad \big(\mathrm{D}_\theta g(n)\,\eta\big)_t \in \mathbb{R}\,n_t.$$

The first relation holds for every row x_t with $x_tE \neq 0$. If $x_0E = 0$, then $x_t\,(\eta_E - EY)$ and $(x_t + x_0)\,(\eta_E - EY)$ are both multiples of x_tE, so $x_0\,(\eta_E - EY)$ is a multiple of every nonzero row of $\mathbb{R}^{d_{\text{model}}}$, hence zero. So $\eta_E - EY = E\Lambda$ for a matrix $\Lambda \in \mathbb{R}^{d_{\text{model}}\times d_{\text{model}}}$, and every nonzero row v of $\mathbb{R}^{d_{\text{model}}}$ has $v\Lambda \in \mathbb{R}\,v$. Hence $\Lambda = cI$ for a number c, and $\eta_E = E\,(Y + cI)$. The second relation holds for every sequence $n = (n_1, \ldots, n_T)$ of rows of the sphere $\mathbb{S}$ of Lemma B.5, since E has rank d_{model}. By that lemma, applied to the wide block g, the entries of η at the reads and the writes are a direction of the symmetries inside the blocks. Conversely, a direction with the entries $E\,(Y + cI)$ at E and $-YD$ at D, with Y antisymmetric, is flat, since $\mathrm{D}N_\varepsilon(h)\,\big(h\,(Y + cI)\big) = N_\varepsilon(h)\,Y$, and so are the directions of the symmetries inside the blocks. So the flat directions form a space of dimension $d_{\text{model}}(d_{\text{model}} - 1)/2 + 1 + s$.

The rank. In each case let m be the dimension of the space of flat directions. The kernel of J_0 decreases as the batch X grows, and its intersection over all batches is the space of flat directions, so on some finite batch $\operatorname{rank} J_0 = P - m$. The rank is lower semicontinuous and $J_\tau S_\tau \to J_0$, so on this batch, for all small $\tau \neq 0$,

$$\operatorname{rank} J_\tau \;=\; \operatorname{rank}(J_\tau S_\tau) \;\geq\; \operatorname{rank} J_0 \;=\; P - m.$$

Conversely $\operatorname{rank} J_\tau \leq P - m$ on every batch, since the symmetries of the residual network move θ_τ in m independent directions when E has rank d_{model} and the queries and values of every head have full column rank. These symmetries are the global change of basis (Theorem A.5(iii)), the translations (Lemma A.8), the rescalings of the units and the changes of basis inside the heads. So $\mathrm{fd}(\theta_\tau) = P - m$. □

Remark (Open sets of inputs). For analytic blocks $(J_0\eta)(x)$ is real-analytic in the input x, so a direction with $(J_0\eta)(x) = 0$ on an open set of inputs is flat. The proof above uses rays and sequences of equal tokens, which need not lie in that set.

Theorem 4.13 is a case of the following theorem.

Theorem A.25 (GELU and analytic networks at depth)**. (i)** *For GELU blocks and almost every choice of weights, a composition of $L \geq 2$ residual blocks equals no composition of L blocks without skips of the same widths.*
(ii) *With an embedding and a readout, the same holds for networks of analytic blocks of any depth $L \geq 1$, also under pre-RMSNorm with $\varepsilon > 0$.*

(iii) Too few extra hidden units. *For GELU blocks with an embedding and a readout, without normalization, and almost every choice of weights, a residual network of L blocks equals no skipless network of L blocks with, in total, fewer than $Ld_{\text{model}}/2$ more hidden units.*
(iv) No limits. *In (ii), and in (iii) for skipless networks whose blocks all have at least d_{model} hidden units, almost every choice of weights of the residual network F has a finite set X of inputs and a $\delta > 0$ such that every such skipless network G has $\|G(x) - F(x)\| \geq \delta$ at some $x \in X$. So F is not a pointwise limit of these skipless networks, in the sense of Section 4.4.*

Proof of Theorem A.25(i) and (ii). By Lemma A.24, a residual network without biases has functional dimension $P - d_{\text{model}}^2 - s$ at some parameters, where s counts the changes of basis inside the heads and the rescalings of the values of gated units, and a network of GELU blocks without an embedding and a readout has functional dimension P. In a skipless network of the same widths, with embedding E' and readout D', the changes of basis at h_0 and h_L, those inside the heads and the rescalings have orbits of dimension $2d_{\text{model}}^2 + s$ at generic parameters, where E' and D' have rank d_{model} and the queries and values of every head have full column rank. As $P - d_{\text{model}}^2 - s > P - 2d_{\text{model}}^2 - s$, Lemma A.23 gives the claim. Without an embedding, a readout and attention blocks, for $L \geq 2$ the changes of basis at h_1 have orbits of positive dimension, since every block has at least one hidden unit, so Lemma A.23 applies again. Under pre-RMSNorm with $\varepsilon > 0$ the networks stay analytic, Lemma A.24 gives the residual network functional dimension $P - d_{\text{model}}(d_{\text{model}} - 1)/2 - s$, and the orthogonal changes of basis at h_0 and h_L, which commute with the normalization, have orbits of dimension $d_{\text{model}}(d_{\text{model}} - 1) + s$ in the skipless network; as $d_{\text{model}} \geq 2$, Lemma A.23 applies once more. □

Proof of Theorem A.25(iii). Let the residual network have blocks with $d_{\text{hidden},1}, \ldots, d_{\text{hidden},L}$ hidden units and P parameters, and let a skipless network have blocks with $d'_{\text{hidden},1}, \ldots, d'_{\text{hidden},L}$ hidden units. If some $d'_{\text{hidden},\ell} < d_{\text{model}}$, the hidden state h_ℓ of the skipless network lies in a subspace of dimension $d'_{\text{hidden},\ell}$, so the Jacobian of its output with respect to the input has rank below d_{model} at every input. That of the residual network is $ED + O(\tau)$ at the parameters θ_τ of Lemma A.24, of rank d_{model} for small τ, so by analyticity it has rank d_{model} at some input for almost every choice of weights, and the two networks differ. So let every $d'_{\text{hidden},\ell} \geq d_{\text{model}}$. The skipless network then has $P' = P + 2d_{\text{model}} \sum_\ell (d'_{\text{hidden},\ell} - d_{\text{hidden},\ell})$ parameters, and its changes of basis at $h_0, \ldots, h_L$ have orbits of dimension $(L+1)d_{\text{model}}^2$ at generic parameters (Lemma A.15). By Lemma A.24 the residual network has functional dimension $P - d_{\text{model}}^2$ at some parameters, which exceeds $P' - (L+1)d_{\text{model}}^2$ exactly when $\sum_\ell (d'_{\text{hidden},\ell} - d_{\text{hidden},\ell}) < Ld_{\text{model}}/2$. For each of the finitely many such choices of widths, Lemma A.23 gives a set of weights of measure zero, and so does their union. □

Proof of Theorem A.25(iv). For the networks of (ii) and (iii), the outputs on a batch are definable functions of the weights (Lemma A.18). So where the proofs above apply Lemma A.23(i) to them, Lemma A.23(ii) applies as well, on a batch X on which the Jacobian of the residual network reaches the rank used there. In (iii) this covers the finitely many choices of widths with every $d'_{\text{hidden},\ell} \geq d_{\text{model}}$, on the same batch, and the smallest of their δ serves for all. □

We recall Theorem 4.13 from Section 4.4.

Theorem 4.13 (Different function classes)**. (i)** *For analytic blocks, with or without pre-RMSNorm, a residual network at generic parameters equals no skipless network of the same widths, and is not a pointwise limit of such networks. In particular, it cannot be locally uniformly approximated by such networks.*
(ii) Too few extra hidden units. *For networks of GELU blocks without normalization, a residual network of L blocks at generic parameters equals no skipless network of L blocks with, in total, fewer than $Ld_{\text{model}}/2$ more hidden units, and is not a pointwise limit of such networks.*

Proof of Theorem 4.13. In the main text every MLP block has at least d_{model} hidden units (Section 2). Part (i) is Theorem A.25(ii) with (iv), and part (ii) is Theorem A.25(iii) with (iv). □

We recall Theorems 4.9 and 4.12 from Section 4.3.

Theorem 4.9 (Equality for residual networks at any depth)**.** *Let the blocks be analytic, with or without pre-RMSNorm. The residual network satisfies the equality condition: at generic parameters its functional dimension is $P - \dim \mathcal{G} - s$. So the skips raise the functional dimension by at least $L \dim \mathcal{G}$, and by exactly $L \dim \mathcal{G}$ when the skipless network satisfies the equality condition too.*

Proof of Theorem 4.9. The equality condition. Let r be P minus the dimension of the symmetries of the residual network, the value in Lemma A.24. At generic parameters the symmetries move the parameters in independent directions, so the rank of the Jacobian is at most r on every batch. Lemma A.24 gives parameters and a finite batch on which it equals r. On this batch the rank is therefore at least r outside the zero set of the sum of the squares of the minors of size r, an analytic function that is not identically zero, so this zero set has measure zero. At generic parameters the functional dimension is thus r, which is the equality condition.

Attention blocks (Corollary 4.11). Part (i) is the equality condition for attention blocks without normalization, where $\dim \mathcal{G} = d_{\text{model}}^2$. For part (ii), without an embedding and a readout, the last value of Lemma A.24 is $r = P - s$, and at generic parameters the changes of basis inside the heads move the parameters in s independent directions. The argument above then gives the functional dimension $P - s$ at generic parameters. With heads of total width $Hd_h = d_{\text{model}}$, a block has $4d_{\text{model}}^2$ parameters and H heads, each with two changes of basis of dimension $d_h^2 = d_{\text{model}}^2/H^2$. So $P = 4Ld_{\text{model}}^2$ and $s = 2Ld_{\text{model}}^2/H$.

The gain. When every block has width at least d_{model}, the functional dimension of the skipless network is at most $P - (L+1)\dim \mathcal{G} - s$ (Theorem A.14(ii), Lemma A.15). This gives the gain of at least $L \dim \mathcal{G}$, with equality when the skipless network satisfies the equality condition. □

The case $\varepsilon = 0$ of Lemma A.24 gives the equality condition for residual networks under RMSNorm with $\varepsilon = 0$, where $\mathcal{G} = \mathrm{CO}(d_{\text{model}})$.

Theorem A.26 (Residual networks under RMSNorm with $\varepsilon = 0$)**.** *Let $d_{\text{in}}, d_{\text{out}} \geq d_{\text{model}} \geq 2$, and consider networks under pre-RMSNorm with $\varepsilon = 0$, without biases, whose blocks are those of Lemma A.24. At generic parameters the residual network has functional dimension*

$$P - \dim \mathrm{CO}(d_{\text{model}}) - s \;=\; P - d_{\text{model}}(d_{\text{model}} - 1)/2 - 1 - s,$$

which is the equality condition. If every block has width at least d_{model}, the skips raise the functional dimension by at least $L\,(d_{\text{model}}(d_{\text{model}} - 1)/2 + 1)$, and by exactly this number when the skipless network satisfies the equality condition too.

Proof. Let $r = P - \dim \mathrm{CO}(d_{\text{model}}) - s$. By Lemma A.24, at generic θ and for small $\tau \neq 0$, the Jacobian at θ_τ has rank r on a finite batch X whose tokens x_t all have $x_t E \neq 0$. With $\varepsilon = 0$ the outputs F_X on this batch are defined, and real-analytic in the parameters, where no row of a hidden state vanishes. We need the following lemma.

> **Lemma A.27** (The domain)**.** *Let X be a finite batch of inputs with nonzero tokens, and let $\mathcal{O}_X \subset \mathbb{R}^P$ be the set of parameters at which every row of every hidden state $h_0, \ldots, h_L$ is nonzero, for every $x \in X$. Then $\mathcal{O}_X$ is open and connected, and its complement has measure zero.*
>
> *Proof.* Let $\mathcal{O}^0$ be the set of parameters with $x_t E \neq 0$ for all tokens of all $x \in X$, and for $\ell \geq 1$ let $\mathcal{O}^\ell$ be the set of parameters in $\mathcal{O}^{\ell-1}$ at which every row of h_ℓ is nonzero for every $x \in X$. Then $\mathcal{O}_X = \mathcal{O}^L$, and h_ℓ is real-analytic on $\mathcal{O}^{\ell-1}$. The complement of $\mathcal{O}^0$ is a finite union of linear subspaces of codimension $d_{\text{model}} \geq 2$, so $\mathcal{O}^0$ is open and connected, and its complement has measure zero. Let $\ell \geq 1$, fix $x \in X$ and a token t, and let $f_{\ell,t}$ be row t of $f_\ell\big(N_\varepsilon(h_{\ell-1})W_{\text{read}}^\ell\big)$, so that row t of h_ℓ is
>
> $$h_{\ell,t} \;=\; h_{\ell-1,t} + f_{\ell,t}\, W_{\text{write}}^\ell.$$
>
> Neither $h_{\ell-1,t}$ nor $f_{\ell,t}$ depends on W_{write}^ℓ. At a point of $\mathcal{O}^{\ell-1}$ with $h_{\ell,t} = 0$ we have $f_{\ell,t}\, W_{\text{write}}^\ell = -h_{\ell-1,t} \neq 0$, so $f_{\ell,t} \neq 0$, and the derivative of $h_{\ell,t}$ with respect to W_{write}^ℓ, the map $\eta_{\text{write}} \mapsto f_{\ell,t}\, \eta_{\text{write}}$, is onto $\mathbb{R}^{d_{\text{model}}}$. So the parameters in $\mathcal{O}^{\ell-1}$ with $h_{\ell,t} = 0$ form

a closed submanifold of $\mathcal{O}^{\ell-1}$ of codimension $d_{\text{model}} \geq 2$. Removing finitely many closed submanifolds of codimension at least two from a connected open set leaves a connected open set, and removes a set of measure zero. So every $\mathcal{O}^{\ell}$ is open and connected, and its complement has measure zero. □

For small τ, every hidden state at θ_τ is $xE + O(\tau)$, so $\theta_\tau \in \mathcal{O}_X$. The sum of the squares of the minors of size r of the Jacobian of F_X is real-analytic on the connected open set $\mathcal{O}_X$ and does not vanish at θ_τ, so its zero set has measure zero. Hence the rank on X is at least r at almost every parameter. At generic parameters it is at most r on every batch: the changes of basis in $\mathrm{CO}(d_{\text{model}})$ (Theorem A.5(iii)) and the symmetries inside the blocks move the parameters in $\dim \mathrm{CO}(d_{\text{model}}) + s$ independent directions, as in the proof of Lemma A.24. So the functional dimension is r at generic parameters. When every block has width at least d_{model}, the functional dimension of the skipless network is at most $P - (L+1)\dim \mathrm{CO}(d_{\text{model}}) - s$ (Theorem A.14(ii), Lemma A.15), which gives the gain. □

For $\tanh$ blocks and for GELU blocks, the skipless network satisfies the equality condition too, so the gain is exactly $L\,(d_{\text{model}}(d_{\text{model}}-1)/2 + 1)$ (Theorem A.34).

Theorem 4.12 (Exact gain for $\tanh$, GELU, SwiGLU and GeGLU)**.** *With or without pre-RMSNorm, in the three cases below the skipless network satisfies the equality condition too, so the skips raise the functional dimension by exactly $L \dim \mathcal{G}$: by Ld_{model}^2 without normalization, and by $Ld_{\text{model}}(d_{\text{model}}-1)/2$ under RMSNorm.*
(i) *The blocks are* $\tanh$ *blocks.*
(ii) *The blocks are GELU blocks, with $d_{\text{hidden}} \geq 2d_{\text{model}}$ if there is no normalization.*
(iii) *The blocks are SwiGLU blocks, or GeGLU blocks.*

Proof. By Theorem 4.9 the residual network satisfies the equality condition, so it remains to show it for the skipless network; the gain is then $L \dim \mathcal{G}$. For $\tanh$, SwiGLU and GeGLU blocks, with or without pre-RMSNorm, and for GELU blocks under pre-RMSNorm, this is Theorem A.34 (Appendix A.9). What is left is part (ii) without normalization, where $\dim \mathcal{G} = d_{\text{model}}^2$.

Let block ℓ have $d_{\text{hidden},\ell} \geq 2d_{\text{model}}$ hidden units, and take a residual network with blocks of $d_{\text{hidden},\ell} - d_{\text{model}}$ hidden units at the parameters θ_τ of Lemma A.24, in which every W^{ℓ}_{write} is replaced by $\tau\, W^{\ell}_{\text{write}}$. Since $\sigma(z) - \sigma(-z) = z$, each of its blocks equals a block without skip with d_{model} more hidden units: for d_{model} hidden units i whose reads w_i form an invertible matrix, add the row γ_i of its inverse to the write of unit i, and add a hidden unit with the read $-w_i$ and the write $-\gamma_i$. This construction is smooth in the residual parameters near these, where the chosen reads stay invertible, and gives a skipless network that computes the residual one. By the chain rule, the columns of the skipless Jacobian therefore span those of the residual Jacobian, which, with the read columns divided by τ, tend to the limit J_0 of Lemma A.24, of rank $P_R - d_{\text{model}}^2$ on some finite batch, where $P_R = P - 2Ld_{\text{model}}^2$ counts the parameters of the residual network. As $\sigma(z) - \sigma(-z) = z$ gives $\sigma'(-z) = 1 - \sigma'(z)$, and every block is the identity at $\tau = 0$, the reads of the added units give the further columns

$$\big(\sigma'(xEw_i) - 1\big)(xE\delta_i)\,\gamma_i D, \qquad \delta_i \in \mathbb{R}^{d_{\text{model}}},$$

in the limit. Suppose that $J_0\,\eta$ plus a sum of these vanishes for all x. Each added term has the Gaussian part of the term of ψ_{c_i} in Lemma B.3, with $E\delta_i$ in place of ψ_{c_i} and the row $\gamma_i D$ in place of a_i. In the proof of that lemma for GELU, on the rays $x = tn$, the coefficient of t^2 for the slowest unit j becomes $u_j\big((n\psi_{c_j})\,a_j + (nE\delta_j)\,\gamma_j D\big)$. At generic parameters $\gamma_j D$ is not parallel to a_j, so $nE\delta_j = 0$ for all n in an open set, and $\delta_j = 0$ as E has rank d_{model}; the proof then goes on as before. So, on some finite batch, the added reads raise the rank in the limit by Ld_{model}^2, to $P_R - d_{\text{model}}^2 + Ld_{\text{model}}^2 = P - (L+1)d_{\text{model}}^2$. By lower semicontinuity the skipless network has this rank for small τ, which is the most its symmetries allow, and an analytic network keeps its largest rank at generic parameters. □

Remark (Skipless $\tanh$ networks without normalization, from the theorem of Fefferman). Without normalization, the equality condition for skipless $\tanh$ networks also follows from Theorem I.B.9 of

Fefferman (1994), with or without biases. With read biases b_ℓ and write biases c_ℓ, which may vanish, the skipless network computes

$$x^0 = x, \qquad x^\ell = \tanh\left(x^{\ell-1}A_\ell + \beta_\ell\right) \ (1 \le \ell \le L), \qquad y = x^L A_{L+1} + \beta_{L+1},$$

with $A_1 = EW_1$, $A_\ell = V_{\ell-1}W_\ell$ and $A_{L+1} = V_L D$, where W_ℓ and V_ℓ are the read and the write of block ℓ, and $\beta_1 = b_E W_1 + b_1$, $\beta_\ell = c_{\ell-1}W_\ell + b_\ell$ and $\beta_{L+1} = c_L D + b_D$. Two such networks that agree on an open set of inputs agree on all inputs, by analyticity. As $\sigma(z) = \tanh(z/2)$ is injective, two such networks have the same output exactly when the nets of that paper with the hidden layers $x^\ell = \sigma(2x^{\ell-1}A_\ell + 2\beta_\ell)$ and the output layer $\sigma(x^L A_{L+1} + \beta_{L+1})$ do, and then so do, for every input coordinate and every output, the nets with one input and one output that keep only these. At generic parameters every entry of every A_ℓ is nonzero, and no ratio of the weights that one unit sends to two units of a layer of width d_{hidden} is a fraction p/q with $1 \le q \le 100\, d_{\text{hidden}}^2$. These conditions are open, and they are those of Theorem I.B.9 (pages 517 and 518 there), which is stated for nets with one input and one output and puts no condition on the thresholds. So near generic parameters, two networks with the same output give, for every input coordinate and every output, nets with one input and one output that agree up to permutations of the units of each hidden layer and sign flips. These are the identity: at the first layer where a unit is moved or flipped, the weights it receives would move by a fixed amount, since none vanishes and the weights that one unit sends to two units differ in absolute value. So the two networks have the same A_ℓ and β_ℓ. When E, D and every read and write have rank d_{model}, the parameters with the same A_ℓ and β_ℓ are those reached by the changes of basis at the $L+1$ hidden states, and with biases by the translations there.

By analyticity, the functional dimension at generic parameters is the largest rank r that the Jacobian attains on any batch at any parameters. Take generic parameters and a batch on which the rank is r. Near these parameters the rank on this batch is at least r, by lower semicontinuity, and so equal to r. Its kernel is then the space of directions that are flat for every input, and the parameters with the same outputs on the batch form a manifold of dimension $P - r$ along which the function does not change. So $P - r$ is at most the dimension of the symmetries, and the symmetries give the reverse inequality (Theorem A.14(ii), Lemma A.8). The proof of Theorem A.34 does not use this theorem.

Lemma A.28 (ReLU-type blocks with small writes)**.** *Let the blocks be ReLU, ReLU^2 or ReGLU blocks, without normalization or biases, let E and D have rank d_{model}, and let inputs range over all of $\mathbb{R}^{1\times d_{\text{in}}}$, or over an open set that contains 0. For generic parameters θ, let θ_τ be the parameters θ with every W^ℓ_{write} replaced by $\tau\, W^\ell_{\text{write}}$, as in Lemma A.24. For all small $\tau \neq 0$, the Jacobian of the residual network at θ_τ has rank $P - d^2_{\text{model}} - s$ on some finite batch on which no pre-activation vanishes. So the residual network has functional dimension $P - d^2_{\text{model}} - s$ on an open set of weights that contains these, and at almost every point of this set the skips add at least Ld^2_{model} when every block has width at least d_{model}.*

Proof. We follow the proof of Lemma A.24 for MLP blocks without normalization. The limit J_0 is again the derivative of the one-block function G of Lemma B.3, whose units are those of all blocks, with the pre-activations xEW^ℓ_{read}. On each region of inputs on which these pre-activations keep their signs, $J_0\,\eta$ is a polynomial in x, and the regions are cones with apex 0. So a direction that is flat on a neighborhood of 0 is flat at every input at which no pre-activation vanishes, and by Lemma B.3, which covers these units, the flat directions form a space of dimension $d^2_{\text{model}} + s$, as in Lemma A.24. Fix a finite batch of such inputs, inside the set of inputs, on which the kernel of J_0 is this space; it exists, since the kernels decrease as the batch grows.

With the writes τV_ℓ, the pre-activations of block ℓ are $h_{\ell-1}W^\ell_{\text{read}}$ with $h_{\ell-1} = xE + O(\tau)$. So for $|\tau|$ below some $\tau_0 > 0$, which depends on the parameters and on the batch, every pre-activation on the batch has the sign of its limit. The outputs on the batch are then polynomial in the parameters and in τ, the Jacobian on the batch, with its read columns divided by τ, tends to J_0, and by lower semicontinuity it has rank at least $P - d^2_{\text{model}} - s$ for all small $\tau \neq 0$.

Near such parameters the signs on the batch do not change, E has rank d_{model} and no read of a unit vanishes. There the change of basis of the hidden states and the rescalings of the units move the

parameters in $d_{\text{model}}^2 + s$ independent directions, which lie in the kernel of the Jacobian on every batch on which no pre-activation vanishes. So the rank on the batch, which is at least $P - d_{\text{model}}^2 - s$ by lower semicontinuity, is exactly $P - d_{\text{model}}^2 - s$, and it is the functional dimension. The gain follows from Theorem A.14(ii), which bounds the functional dimension of the skipless network by $P - (L+1)d_{\text{model}}^2 - s$ at generic parameters. □

For ReLU blocks the count fails on another open set of weights, where a block folds the hidden state and units of later blocks are never active (Appendix A.8).

A.8 ReLU blocks: folding and dead units

This section explains why the count of Lemma A.28 holds on one open set of weights and fails on another, so that it does not hold at almost every parameter. The blocks are ReLU blocks, without normalization. An MLP block acts on each token separately, so we take inputs with one token, $x \in \mathbb{R}^{1\times d_{\text{in}}}$, and hidden states $h \in \mathbb{R}^{1\times d_{\text{model}}}$. In this section inputs range over all of $\mathbb{R}^{1\times d_{\text{in}}}$, and E has rank d_{model}.

Active and dead units. Unit i of block ℓ has the read r_i, column i of W_{read}^{ℓ}, and the write $\hat{r}_i$, row i of W_{write}^{ℓ}. Its *pre-activation* at the input x is the number $z_i(x) = h_{\ell-1}(x)\, r_i$, and the unit adds $\mathrm{ReLU}(z_i(x))\, \hat{r}_i$ to the output of the block. The unit is *active* at x if $z_i(x) > 0$. Where $z_i(x) < 0$, the unit adds nothing, and the derivatives of the output with respect to r_i and $\hat{r}_i$ vanish. A unit is *dead* if it is active at no input. A dead unit can be deleted, or its write changed, without changing the function.

Definition A.29 (Folding)**.** Let $\Phi_\ell(h) = h + B_\ell(h)$ be the map of $\mathbb{R}^{1\times d_{\text{model}}}$ computed by residual block ℓ. The block *folds* if Φ_ℓ is not injective: two different hidden states give the same next hidden state.

For example, take $d_{\text{hidden}} = d_{\text{model}}$, $W_{\text{read}} = -I$ and $W_{\text{write}} = 2I$. Then $B(h) = 2\,\mathrm{ReLU}(-h)$, and coordinate j of $h + B(h)$ is h_j if $h_j \geq 0$ and $-h_j$ if $h_j < 0$. So the block maps h to $(|h_1|, \ldots, |h_{d_{\text{model}}}|)$. It folds every coordinate axis at 0, like a sheet folded along a line, and its image is the nonnegative orthant. Wherever no coordinate vanishes, its Jacobian is $\mathrm{diag}(\pm 1)$, so a block can fold although its Jacobian is invertible wherever it exists.

Proposition A.30 (Blocks that do not fold keep units alive)**.** *Consider a residual network of ReLU blocks without normalization.*
(i) *If block ℓ does not fold, Φ_ℓ is a homeomorphism of $\mathbb{R}^{1\times d_{\text{model}}}$.*
(ii) *If no block folds, every unit with a nonzero read is active on a nonempty open set of inputs and inactive on another. In particular, no unit with a nonzero read is dead.*
(iii) *A block with $\|W_{\text{read}}\|\,\|W_{\text{write}}\| < 1$, in spectral norm, does not fold.*

Proof. (i) The map Φ_ℓ is continuous, and $\Phi_\ell(ch) = c\,\Phi_\ell(h)$ for $c > 0$, since there are no biases. As $\Phi_\ell(0) = 0$ and Φ_ℓ is injective, $\Phi_\ell(h) \neq 0$ for $h \neq 0$. So $c_0 = \min_{\|h\|=1} \|\Phi_\ell(h)\|$ is positive, and $\|\Phi_\ell(h)\| \geq c_0 \|h\|$. Hence the preimage of a bounded set is bounded, and the image of Φ_ℓ is closed. By invariance of domain, an injective continuous map of $\mathbb{R}^{d_{\text{model}}}$ to itself is open. So the image is also open, hence all of $\mathbb{R}^{1\times d_{\text{model}}}$, and the inverse is continuous. (ii) The map $x \mapsto xE$ is onto $\mathbb{R}^{1\times d_{\text{model}}}$, and by (i) so is the continuous map $x \mapsto h_{\ell-1}(x)$. The sets $\{h : hr_i > 0\}$ and $\{h : hr_i < 0\}$ are open and not empty, and so are their preimages. (iii) As ReLU is 1-Lipschitz, $\|B_\ell(h) - B_\ell(h')\| \leq \|W_{\text{read}}\|\,\|W_{\text{write}}\|\,\|h - h'\|$. If $\Phi_\ell(h) = \Phi_\ell(h')$, then $\|h - h'\| = \|B_\ell(h') - B_\ell(h)\|$, so $h = h'$. □

By (iii), small enough writes never fold, which is the regime of Lemma A.28. The condition of (iii) is the one under which Behrmann et al. (2019) invert residual networks (their Theorem 1), and it is only sufficient. With $W_{\text{read}} = I$ and $W_{\text{write}} = 2I$, the block maps each coordinate h_j to h_j if $h_j \leq 0$ and to $3h_j$ if $h_j > 0$, so it does not fold, although $\|W_{\text{read}}\|\,\|W_{\text{write}}\| = 2$.

Proposition A.31 (Folding blocks can kill units and lower the functional dimension)**.** *Consider residual networks of $L \geq 2$ ReLU blocks without normalization.*

(i) *There is a nonempty open set of weights on which block* 1 *folds and block* 2 *has a dead unit.*
(ii) *If* ν *units are dead, then on every batch on which no pre-activation vanishes, the Jacobian has rank at most* $P - d_{\text{model}}^2 - s - (2d_{\text{model}} - 1)\nu$.

Proof. (i) Let block 1 hold the d_{model} units of the example above, with $W_{\text{read}} = -I$ and $W_{\text{write}} = 2I$. Give its other units zero writes. Let a unit of block 2 have a read r whose entries are all negative. Then $\Phi_1(h)\, r < 0$ for every $h \neq 0$, since $\Phi_1(h)$ is a nonzero vector with no negative entry. By compactness of the unit sphere, $\max_{\|h\|=1} \Phi_1(h)\, r < 0$ persists under small changes of all the weights. Since the perturbed Φ_1 is still positively homogeneous, $\Phi_1(h)\, r < 0$ then holds for every $h \neq 0$. As $h_1(x) = \Phi_1(xE)$ and $\Phi_1(0) = 0$, the unit is active at no input, so it is dead. And Φ_1 misses the half-space $\{h : hr > 0\}$, so it is not onto, and block 1 folds by Proposition A.30(i). (ii) The columns of the Jacobian that belong to the read and the write of a dead unit vanish, $2d_{\text{model}}\nu$ columns in all. The other columns are those of the network with the dead units deleted. That network has $P - 2d_{\text{model}}\nu$ parameters and $s - \nu$ unit rescalings, since s counts one rescaling per ReLU unit. Its function does not change under the change of basis of the hidden states and these rescalings. At generic parameters they act with orbits of dimension $d_{\text{model}}^2 + s - \nu$, since E has rank d_{model}. On the open set of parameters at which no pre-activation vanishes on the batch, the outputs on the batch are polynomial in the parameters. So, as in the proof of Lemma A.23, the rank is at most $P - 2d_{\text{model}}\nu - d_{\text{model}}^2 - (s - \nu)$ at generic points of this set, and by lower semicontinuity at all of them. □

So on the open set of Proposition A.31(i) the residual network has functional dimension below $P - d_{\text{model}}^2 - s$. For ReLU networks, the bound of Theorem 4.4 is therefore attained on one open set of weights (Lemma A.28) and missed on another.

Remark A.32 (Units that are always active). A unit that is active at every input with $xE \neq 0$ lowers the bound in the same way, on the open set of weights on which it stays so; a unit of block 2 whose read has positive entries is one, after block 1 of Proposition A.31(i). It adds a linear map of rank one to the skip of its block, which for generic weights is then an invertible learned skip, and by Theorem 3.4 the network equals a residual network without this unit, which has $2d_{\text{model}}$ fewer parameters and one rescaling less. On this open set the Jacobian is that of the smaller network times the derivative of a smooth map of the weights, so its rank is at most $(P - 2d_{\text{model}}) - d_{\text{model}}^2 - (s - 1)$.

Without skips, units also die in ReLU^2 and ReGLU blocks, and at every width.

Proposition A.33 (Dead units without skips)**.** *Consider skipless networks of* $L \geq 2$ *ReLU, ReLU*^{2} *or ReGLU blocks without normalization. There is a nonempty open set of weights on which a unit of block* 2 *is dead. On it the functional dimension is at most* $P - (L+1)d_{\text{model}}^2 - s - (2d_{\text{model}} - 1)$*, and at most* $P - (L+1)d_{\text{model}}^2 - s - (3d_{\text{model}} - 2)$ *for ReGLU, so the equality condition fails there.*

Proof. Let unit k of block 1 have the read g_k and the write $\hat{r}_k$, and let a unit of block 2 have the read r. For ReGLU, g_k and r are gate reads, and v_k is the value read of unit k.

ReLU and ReLU^2. At $h_0 = u$ the pre-activation of the unit of block 2 is $\sum_k \sigma(ug_k)\, \hat{r}_k r$, with $\sigma \geq 0$. It is positive at no input when $\hat{r}_k r < 0$ for all k, which holds on a nonempty open set of weights.

ReGLU. The pre-activation of the gate is $z(u) = \sum_k \mathrm{ReLU}(ug_k)\,(uv_k)\,\hat{r}_k r$, and the sign of a term depends on the value uv_k. We choose weights at which every gate opens where its value is positive. Let d_{hidden} be the width of block 1 and $0 < \delta < 1/(d_{\text{hidden}} d_{\text{model}})$. Let $g_k = e_{i(k)}$, a standard basis column, and $v_k = e_{i(k)} - \delta \sum_{j \neq i(k)} e_j$, with every index $i \leq d_{\text{model}}$ used by some unit, and let $\hat{r}_k r = -1$ for all k. Then

$$z(u) = -\sum_k \mathrm{ReLU}(u_{i(k)})\Big(u_{i(k)} - \delta \sum_{j \neq i(k)} u_j\Big) \;\leq\; -(1 - \delta\, d_{\text{hidden}} d_{\text{model}}) \sum_{i\,:\,u_i > 0} u_i^2,$$

so $z(u) < 0$ if u has a positive entry, and $z(u) = 0$ if it has none. Suppose that, at weights that converge to these, the unit is active at unit rows u_n, and let u be a limit of the u_n. Then $z(u) \geq 0$, so u has no positive entry. For large n, a unit k with $u_{i(k)} < 0$ has a closed gate at u_n. A unit k

with $u_{i(k)} = 0$ has a positive value at u_n, since its value at u is $-\delta \sum_{j \neq i(k)} u_j > 0$, so its term is not positive. Hence the pre-activation at u_n is not positive, a contradiction. The pre-activation is positively homogeneous in u, so the unit is dead at all weights close to the chosen ones.

The rank. As in the proof of Proposition A.31(ii), the columns of the Jacobian that belong to the dead unit vanish, $2d_{\text{model}}$ of them, or $3d_{\text{model}}$ for ReGLU. The other columns are those of the network without this unit, whose symmetries are the changes of basis of its $L+1$ hidden states and $s-1$ rescalings, or $s-2$ for ReGLU. □

With skips, no unit of a ReLU^2 or ReGLU block with a nonzero read is dead. Such a block is of second order at 0: its Lipschitz constant on a ball of radius ρ is $O(\rho)$. So $h \mapsto h + B(h)$ maps a small ball onto a neighborhood of 0, and every hidden state of a residual network takes all values near 0.

Whether almost every residual ReLU network whose blocks all satisfy $\|W_{\text{read}}\| \, \|W_{\text{write}}\| < 1$ has functional dimension $P - d_{\text{model}}^2 - s$ is open.

A.9 Skipless networks that attain their bound (Theorem 4.12)

This subsection proves that skipless networks of $\tanh$, SwiGLU or GeGLU blocks, and of GELU blocks under pre-RMSNorm, satisfy the equality condition (Theorem A.34). This gives Theorem 4.12(i), Theorem 4.12(ii) under pre-RMSNorm, Theorem 4.12(iii), and the exact gain for $\varepsilon = 0$, which Theorem A.26 gives only as a lower bound. We show that the function changes along every direction of the parameters that is not a direction of the symmetries, by removing the blocks one at a time (Claim (D_k) below). A direction along which the function does not change gives an identity in the input of a block, and we follow this input to infinity. On the way, the hidden units of a $\tanh$ block saturate at different exponential rates, which separates them (Lemma A.40), and the hidden units of a GELU block and the gates of a SwiGLU or GeGLU block differ from linear or vanishing ones by small terms of different rates. Without normalization and for $\varepsilon > 0$, the inputs of a block fill an open set of rows, so the identity holds at every real row, and the way to infinity is a ray (Lemmas A.41 and A.46). For $\varepsilon = 0$ the inputs of a block lie on a sphere, and we follow a great circle of the sphere to complex angles, as in the proof of Lemma B.5 (Lemmas A.52 and A.53). The blocks are MLP blocks without biases, with the activation σ or, in a SwiGLU or GeGLU block, the gate σ. An MLP block acts on each token separately, so we take inputs with one token, $x \in \mathbb{R}^{1 \times d_{\text{in}}}$.

Theorem A.34 (Skipless networks that attain their bound)**.** *Let $d_{\text{in}}, d_{\text{out}} \geq d_{\text{model}} \geq 2$, and consider skipless networks without biases whose blocks are all of one kind, $\tanh$, GELU, SwiGLU or GeGLU, of width at least d_{model}. Let s be the dimension of the symmetries inside the blocks (Section 4.2): $s = 0$ for $\tanh$ and GELU blocks, and s is the number of hidden units of the network for SwiGLU and GeGLU blocks. In the following cases the functional dimension at generic parameters is $P - (L+1)\dim \mathcal{G} - s$, which is the equality condition, so the skips raise the functional dimension by exactly $L \dim \mathcal{G}$.*
(i) $\tanh$ *blocks: without normalization, where $\mathcal{G} = \mathrm{GL}(d_{\text{model}})$; under pre-RMSNorm with $\varepsilon > 0$, where $\mathcal{G} = \mathrm{O}(d_{\text{model}})$; and under pre-RMSNorm with $\varepsilon = 0$, where $\mathcal{G} = \mathrm{CO}(d_{\text{model}})$ and $\dim \mathcal{G} = d_{\text{model}}(d_{\text{model}} - 1)/2 + 1$.*
(ii) *GELU blocks: under pre-RMSNorm with $\varepsilon > 0$, and under pre-RMSNorm with $\varepsilon = 0$.*
(iii) *SwiGLU blocks, and GeGLU blocks: without normalization, and under pre-RMSNorm with $\varepsilon > 0$.*

For GELU blocks the theorem needs no condition $d_{\text{hidden}} \geq 2d_{\text{model}}$. It does not cover GELU blocks without normalization, which Appendix A.7 treats by another argument, for $d_{\text{hidden}} \geq 2d_{\text{model}}$. On a ray, a GELU hidden unit and the gate of a SwiGLU or GeGLU hidden unit differ from linear or vanishing ones by small terms, and the step from one block to the next is an induction over all products of these terms (Lemma A.46). Under RMSNorm the later blocks read a normalized row, which stays bounded along the ray, so the small terms come from one block at a time. Without normalization the inputs of the later blocks grow along the ray, and their hidden units add small terms of their own. A SwiGLU or GeGLU block is quadratic along the ray up to its small terms, so the

inputs of the later blocks grow like the square of its own input. Their small terms are then of a smaller order, and the network after the block can be replaced by a polynomial map (Lemma A.45). A GELU block is linear along the ray, the small terms of all blocks are of the same order, and the block cannot be removed on its own. For GELU, SwiGLU and GeGLU blocks we prove the claim at one explicit choice of parameters (Lemma A.47), and the theorem follows by analyticity.

The three cases. We write (n) for the case without normalization, (+) for pre-RMSNorm with $\varepsilon > 0$, and (0) for pre-RMSNorm with $\varepsilon = 0$. Let N be the map that a block applies to a hidden state before it reads it: the identity in the case (n), and N_ε in the cases (+) and (0). The table gives N, the set of its values, which are the inputs of the blocks, the group $\mathcal{G}$ of the changes of basis, and its Lie algebra $\mathfrak{g}$: the set of the matrices Z with $\exp(\tau Z) \in \mathcal{G}$ for all numbers τ.

Case	$N(h)$	Values of N	$\mathcal{G}$	$\mathfrak{g}$
(n)	h	all rows $n \in \mathbb{R}^{1\times d_{\text{model}}}$	$\mathrm{GL}(d_{\text{model}})$	all matrices Z
(+)	$h\left(\|h\|^2/d_{\text{model}} + \varepsilon\right)^{-1/2}$	the ball $\|n\|^2 < d_{\text{model}}$	$\mathrm{O}(d_{\text{model}})$	antisymmetric Z
(0)	$\sqrt{d_{\text{model}}}\, h/\|h\|$, for $h \neq 0$	the sphere $\mathbb{S}$: $\|n\|^2 = d_{\text{model}}$	$\mathrm{CO}(d_{\text{model}})$	$Z = Y + \beta I$

In the last row Y is antisymmetric and β is a number, and $\mathbb{S}$ is the set of the rows $n \in \mathbb{R}^{1\times d_{\text{model}}}$ with $\|n\|^2 = d_{\text{model}}$.

Blocks. For two rows of the same length, $\odot$ is their entrywise product. Let block ℓ have q_ℓ hidden units, and let $a_1, \dots, a_{q_\ell}$ be the rows of its write $W^\ell_{\text{write}} \in \mathbb{R}^{q_\ell \times d_{\text{model}}}$. A $\tanh$ or GELU block has the activation σ, and its read $W^\ell_{\text{read}} \in \mathbb{R}^{d_{\text{model}}\times q_\ell}$ has the columns $c_1, \dots, c_{q_\ell}$. A SwiGLU or GeGLU block, a *gated* block, has the read $W^\ell_{\text{read}} = [\, W^\ell_g \mid W^\ell_v \,]$, where $W^\ell_g \in \mathbb{R}^{d_{\text{model}}\times q_\ell}$ holds the gate columns $c_1, \dots, c_{q_\ell}$ and $W^\ell_v \in \mathbb{R}^{d_{\text{model}}\times q_\ell}$ the value columns $e_1, \dots, e_{q_\ell}$. Its gate σ is GELU in a GeGLU block and SiLU, $\sigma(z) = z/(1+e^{-z})$, in a SwiGLU block. Block ℓ maps its input n to

$$g_\ell(n) \;=\; \sigma(n\, W^\ell_{\text{read}})\, W^\ell_{\text{write}} \;=\; \sum_{j=1}^{q_\ell} \sigma(nc_j)\, a_j \qquad \text{for a } \tanh \text{ or GELU block,}$$

$$g_\ell(n) \;=\; \left(\sigma(n\, W^\ell_g) \odot n\, W^\ell_v\right) W^\ell_{\text{write}} \;=\; \sum_{j=1}^{q_\ell} \sigma(nc_j)\,(ne_j)\, a_j \qquad \text{for a gated block.}$$

We treat gated blocks in the cases (n) and (+) only.

Symmetries. The skipless network computes $h_0 = xE$, $h_\ell = g_\ell\big(N(h_{\ell-1})\big)$ and $F_\theta(x) = N(h_L)\, D$; in the case (0), at the inputs x for which no h_ℓ vanishes. For $G \in \mathcal{G}$ let $\widehat{G} = G$ in the cases (n) and (+), and $\widehat{G} = Q$ for $G = \lambda Q$ in the case (0), with a number $\lambda > 0$ and an orthogonal Q. Then $N(hG) = N(h)\,\widehat{G}$ in all three cases. So for $G_0, \dots, G_L \in \mathcal{G}$ the parameters

$$\Big(EG_0,\;\; \big\{\widehat{G}^{-1}_{\ell-1} W^\ell_{\text{read}},\;\; W^\ell_{\text{write}} G_\ell\big\}^L_{\ell=1},\;\; \widehat{G}^{-1}_L D\Big)$$

give the same function, with the hidden states $h_\ell G_\ell$: one matrix of $\mathcal{G}$ per hidden state. For $Z \in \mathfrak{g}$ let $\widehat{Z} = Z$ in the cases (n) and (+), and $\widehat{Z} = Y$ for $Z = Y + \beta I$ in the case (0). For $Z_0, \dots, Z_L \in \mathfrak{g}$ let

$$\eta^Z \;=\; \Big(EZ_0,\;\; \big\{-\widehat{Z}_{\ell-1} W^\ell_{\text{read}},\;\; W^\ell_{\text{write}} Z_\ell\big\}^L_{\ell=1},\;\; -\widehat{Z}_L D\Big) \in \mathbb{R}^P, \tag{16}$$

the derivative of these parameters at $\tau = 0$ for $G_\ell = \exp(\tau Z_\ell)$. So $\mathrm{D}_\theta F(\theta, x)\, \eta^Z = 0$ at every input at which F_θ is defined. When E and every W^ℓ_{write} have rank d_{model}, the map $(Z_0, \dots, Z_L) \mapsto \eta^Z$ is injective.

A gated block has, in addition, one rescaling per hidden unit. For diagonal matrices $\Delta_1, \dots, \Delta_L$, with Δ_ℓ of size q_ℓ, the parameters

$$\Big(EG_0,\;\; \big\{G^{-1}_{\ell-1} W^\ell_g,\;\; G^{-1}_{\ell-1} W^\ell_v\, e^{-\Delta_\ell},\;\; e^{\Delta_\ell}\, W^\ell_{\text{write}} G_\ell\big\}^L_{\ell=1},\;\; G^{-1}_L D\Big)$$

give the same function as well: the diagonal matrix e^{Δ_ℓ} multiplies the write row of every hidden unit of block ℓ by a positive number and divides its value column by the same number. For $Z_0, \dots, Z_L \in \mathfrak{g}$ let

$$\eta^{Z,\Delta} \;=\; \Big(EZ_0,\; \big\{ -Z_{\ell-1}W_g^\ell,\; -Z_{\ell-1}W_v^\ell - W_v^\ell\Delta_\ell,\; \Delta_\ell W_{\mathrm{write}}^\ell + W_{\mathrm{write}}^\ell Z_\ell \big\}_{\ell=1}^{L},\; -Z_L D \Big) \in \mathbb{R}^P, \tag{17}$$

the derivative of these parameters at $\tau = 0$ for $G_\ell = \exp(\tau Z_\ell)$ and with $\tau\Delta_\ell$ in the place of Δ_ℓ. So $\mathrm{D}_\theta F(\theta, x)\, \eta^{Z,\Delta} = 0$ at every input x. When E and every W_{write}^ℓ have rank d_{model} and no value column vanishes, $\eta^{Z,\Delta} = 0$ only if every Z_ℓ and every Δ_ℓ vanishes: $EZ_0 = 0$ gives $Z_0 = 0$, then $W_v^1\Delta_1 = 0$ gives $\Delta_1 = 0$, then $W_{\mathrm{write}}^1 Z_1 = 0$ gives $Z_1 = 0$, and so on.

Let $s = 0$ for $\tanh$ and GELU blocks and $s = q_1 + \cdots + q_L$ for gated blocks. Under these conditions the directions of the symmetries form a space of dimension $(L+1)\dim\mathcal{G} + s$, so on every batch the Jacobian has rank at most

$$P - (L+1)\dim\mathcal{G} - s. \tag{18}$$

Notation. For rows u, v, real or complex, $\langle u, v\rangle = uv^\top$, without complex conjugation. For $k = 0, \dots, L$ and a row h, with $h \neq 0$ in the case (0), let $\mathcal{B}_{>k}(h)$ be the output of the network when its hidden state h_k is set to h:

$$\mathcal{B}_{>L}(h) = N(h)\, D, \qquad \mathcal{B}_{>k-1}(h) = \mathcal{B}_{>k}\big(g_k(N(h))\big), \qquad \text{so that} \quad F_\theta(x) = \mathcal{B}_{>0}(xE).$$

$\mathcal{B}_{>k}$ depends on the weights of the blocks $k+1, \dots, L$ and on D. We write $\mathrm{D}\mathcal{B}_{>k}(h)\, v$ for its derivative at h along a row v, and $\mathrm{D}_\theta\mathcal{B}_{>k}(h)\, \eta$ for its derivative at fixed h along a direction η of these weights. In the cases (+) and (0), for a $d_{\mathrm{model}} \times d_{\mathrm{model}}$ matrix Z and $n = N(h)$,

$$\mathrm{D}N(h)\,(hZ) \;=\; nZ - \frac{\langle nZ, n\rangle}{d_{\mathrm{model}}}\, n. \tag{19}$$

In the case (0), $\mathcal{B}_{>k}(\lambda h) = \mathcal{B}_{>k}(h)$ for $\lambda > 0$, so

$$\mathrm{D}\mathcal{B}_{>k}(h)\, h = 0. \tag{20}$$

We call θ *regular* if

(R1) E, D and every W_{write}^ℓ have rank d_{model};
(R2) in the case (0), for every ℓ and every $n \in \mathbb{S}$: $g_\ell(n) \neq 0$;
(R3) for every ℓ: in the cases (n) and (+), $\mathrm{D}g_\ell(n)$ is invertible at some row n; in the case (0), there is an $n \in \mathbb{S}$ such that no row $v \neq 0$ with $\langle v, n\rangle = 0$ has $\mathrm{D}g_\ell(n)\, v \in \mathbb{R}\, g_\ell(n)$.

In the cases (n) and (+), every $\mathcal{B}_{>k}$ is real-analytic on all rows h. In the case (0), by (R2), it is defined and real-analytic on all rows $h \neq 0$.

Lemma A.35 (The derivative of $\mathcal{B}_{>k}$)**.** *Let θ be regular. For every k there is an open dense set $\mathcal{U}_k$ of rows such that, for $h \in \mathcal{U}_k$, $\mathrm{D}\mathcal{B}_{>k}(h)\, v = 0$ only if $v = 0$, in the cases (n) and (+), and only if $v \in \mathbb{R}\, h$, in the case (0). In the case (0), $\mathcal{U}_k$ consists of rows $h \neq 0$ and is closed under multiplication by positive numbers.*

Proof. The cases (n) and (+). N is the identity, or a diffeomorphism of $\mathbb{R}^{1\times d_{\mathrm{model}}}$ onto the ball of the rows n with $\|n\|^2 < d_{\mathrm{model}}$, with the inverse $n \mapsto \sqrt{\varepsilon}\; n \left(1 - \|n\|^2/d_{\mathrm{model}}\right)^{-1/2}$. So $\mathrm{D}N(h)$ is invertible at every h. As D is injective on rows, $\mathrm{D}\mathcal{B}_{>L}(h)\, v = \big(\mathrm{D}N(h)\, v\big) D$ vanishes only for $v = 0$: $\mathcal{U}_L$ is the set of all rows. Let $k < L$. The determinant of $\mathrm{D}g_{k+1}(n)$ is real-analytic in n and does not vanish identically, by (R3). So g_{k+1} is a local diffeomorphism on an open dense set of rows n, and $g_{k+1} \circ N$ on an open dense set of rows h. Let $\mathcal{U}_k$ be the set of the h of this set with $g_{k+1}(N(h)) \in \mathcal{U}_{k+1}$. It is open and dense, and for $h \in \mathcal{U}_k$,

$$\mathrm{D}\mathcal{B}_{>k}(h)\, v \;=\; \mathrm{D}\mathcal{B}_{>k+1}\big(g_{k+1}(N(h))\big)\, \Big(\mathrm{D}\big(g_{k+1}\circ N\big)(h)\, v\Big)$$

vanishes only for $v = 0$.

The case (0). For a row $h \neq 0$, $\mathrm{D}N_\varepsilon(h)\,v = \frac{\sqrt{d_{\text{model}}}}{\|h\|}\big(v - \frac{\langle v,h\rangle}{\|h\|^2}\,h\big)$, which vanishes exactly for $v \in \mathbb{R}\,h$. For $k = L$, $\mathrm{D}\mathcal{B}_{>L}(h)\,v = \big(\mathrm{D}N_\varepsilon(h)\,v\big)D$ and D is injective on rows, so $\mathcal{U}_L$ is the set of all rows $h \neq 0$. Let $k < L$ and $g = g_{k+1}$. For $n \in \mathbb{S}$ and a row v,

$$\mathrm{D}\mathcal{B}_{>k}(n)\,v \;=\; \mathrm{D}\mathcal{B}_{>k+1}\big(g(n)\big)\,\big(\mathrm{D}g(n)\,v^\perp\big), \qquad v^\perp = v - \frac{\langle v,n\rangle}{d_{\text{model}}}\,n,$$

where $v^\perp$ is orthogonal to n and vanishes exactly for $v \in \mathbb{R}\,n$. For $n \in \mathbb{S}$ consider the linear map $v \mapsto \mathrm{D}g(n)\,v^\perp + \frac{\langle v,n\rangle}{d_{\text{model}}}\,g(n)$ of $\mathbb{R}^{1\times d_{\text{model}}}$. By (R2) it is injective exactly when n is as in (R3), and its determinant is real-analytic in n. By (R3) the determinant does not vanish identically on the connected set $\mathbb{S}$, so the n as in (R3) form an open dense subset of $\mathbb{S}$. On this subset the map $n \mapsto N_\varepsilon(g(n))$ from $\mathbb{S}$ to $\mathbb{S}$ has an injective derivative, so it is a local diffeomorphism, and the n of the subset with $g(n) \in \mathcal{U}_{k+1}$ form an open dense subset of $\mathbb{S}$ again. For such n, $\mathrm{D}\mathcal{B}_{>k}(n)\,v = 0$ gives $\mathrm{D}g(n)\,v^\perp \in \mathbb{R}\,g(n)$, so $v^\perp = 0$ and $v \in \mathbb{R}\,n$. Let $\mathcal{U}_k$ be the set of the positive multiples of these n. □

Lemma A.36 (A direction at a write)**.** *Let θ be regular, $0 \leq k \leq L$, and let A be a matrix with d_{model} columns and rank d_{model}. Let η_A be a matrix of the size of A and η a direction of the weights of the blocks $k+1,\dots,L$ and of D. If*

$$\mathrm{D}\mathcal{B}_{>k}(wA)\,(w\,\eta_A) + \mathrm{D}_\theta\mathcal{B}_{>k}(wA)\,\eta \;=\; 0 \qquad \textit{for all rows } w, \textit{ with } wA \neq 0 \textit{ in the case (0)},$$

then $\eta_A = AZ$ for a $d_{\text{model}} \times d_{\text{model}}$ matrix Z, and

$$\mathrm{D}\mathcal{B}_{>k}(h)\,(hZ) + \mathrm{D}_\theta\mathcal{B}_{>k}(h)\,\eta \;=\; 0 \qquad \textit{for all rows } h, \textit{ with } h \neq 0 \textit{ in the case (0)}. \tag{21}$$

Proof. Let $w_1 A = 0$. Subtracting the hypothesis at w and at $w + w_1$ gives $\mathrm{D}\mathcal{B}_{>k}(wA)\,(w_1\eta_A) = 0$. For $wA \in \mathcal{U}_k$, Lemma A.35 gives $w_1\eta_A = 0$ in the cases (n) and (+), and $w_1\eta_A \in \mathbb{R}\,wA$ in the case (0). In the case (0), as $\mathcal{U}_k$ is open and $d_{\text{model}} \geq 2$, wA takes two linearly independent values in $\mathcal{U}_k$, so $w_1\eta_A = 0$ as well. Let A^+ be a matrix with $A^+A = I$. For every row w, $(w - wAA^+)\,A = 0$, so $w\,\eta_A = wAA^+\eta_A$: $\eta_A = AZ$ with $Z = A^+\eta_A$. The hypothesis is now (21) at $h = wA$, which takes every value. □

Claim (D_k), for $k = 0,\dots,L$. Let Z be a $d_{\text{model}} \times d_{\text{model}}$ matrix and η a direction of the weights of the blocks $k+1,\dots,L$ and of D. If (21) holds, then $Z \in \mathfrak{g}$, and with $Z_k = Z$ there are $Z_{k+1},\dots,Z_L \in \mathfrak{g}$ such that, for $\tanh$ and GELU blocks,

$$\eta \;=\; \Big(\big\{-\widehat{Z}_{\ell-1}W^\ell_{\text{read}},\;\; W^\ell_{\text{write}}Z_\ell\big\}_{\ell=k+1}^{L},\;\; -\widehat{Z}_L D\Big). \tag{22}$$

In the case (0), with $Z_\ell = Y_\ell + \beta_\ell I$ for antisymmetric matrices $Y_k,\dots,Y_L$ and numbers $\beta_k,\dots,\beta_L$, this reads

$$Z = Y_k + \beta_k I, \qquad \eta = \Big(\big\{-Y_{\ell-1}W^\ell_{\text{read}},\;\; W^\ell_{\text{write}}(Y_\ell + \beta_\ell I)\big\}_{\ell=k+1}^{L},\;\; -Y_L D\Big). \tag{23}$$

For gated blocks there are, in addition, diagonal matrices $\Delta_{k+1},\dots,\Delta_L$, with Δ_ℓ of size q_ℓ, such that

$$\eta \;=\; \Big(\big\{-Z_{\ell-1}W^\ell_g,\;\; -Z_{\ell-1}W^\ell_v - W^\ell_v\Delta_\ell,\;\; \Delta_\ell W^\ell_{\text{write}} + W^\ell_{\text{write}}Z_\ell\big\}_{\ell=k+1}^{L},\;\; -Z_L D\Big). \tag{24}$$

Conversely, every $Z = Z_k \in \mathfrak{g}$ and η of the form (22), or (24) for gated blocks, satisfy (21): it is the derivative at $\tau = 0$ of the identity

$$\mathcal{B}_{>k}\big(h\exp(\tau Z_k)\big) \;=\; \mathcal{B}_{>k}(h),$$

in which the left side is taken at the weights moved by the symmetries with $G_\ell = \exp(\tau Z_\ell)$ for $\ell \geq k$, and for gated blocks with $\tau\Delta_\ell$ in the place of Δ_ℓ.

Proposition A.37 (The claim gives the rank)**.** *Let θ be regular, with no vanishing value column if the blocks are gated, and let (D_0) hold at θ. Then the directions $\eta \in \mathbb{R}^P$ with $\mathrm{D}_\theta F(\theta, x)\,\eta = 0$ for all x, with $xE \neq 0$ in the case (0), are the directions η^Z of (16), or $\eta^{Z,\Delta}$ of (17) for gated blocks, and the Jacobian of the network has the rank (18) on some finite batch of such inputs.*

Proof. Let $\eta = (\eta_E, \eta')$ be such a direction, with η_E its entry at E. As $F_\theta(x) = \mathcal{B}_{>0}(xE)$,

$$\mathrm{D}_\theta F(\theta, x)\,\eta \;=\; \mathrm{D}\mathcal{B}_{>0}(xE)\,(x\,\eta_E) + \mathrm{D}_\theta \mathcal{B}_{>0}(xE)\,\eta'.$$

By Lemma A.36 with $A = E$, $\eta_E = EZ$ and (21) holds for $k = 0$. By (D_0), $\eta = \eta^Z$ for some $Z_0, \dots, Z_L \in \mathfrak{g}$ with $Z_0 = Z$, or $\eta = \eta^{Z,\Delta}$ for gated blocks. The kernels of the Jacobian on finite batches are subspaces of $\mathbb{R}^P$ that can only shrink when the batch grows, so on some finite batch the kernel is the space of the directions of the symmetries, which has dimension $(L+1)\dim \mathcal{G} + s$ by the hypotheses on θ. □

Lemma A.38 (The readout)**.** *If D has rank d_{model}, then (D_L) holds.*

Proof. The case (n). Here (21) is $hZD + h\,\eta_D = 0$ for all rows h, so $\eta_D = -ZD$, and $\mathfrak{g}$ contains every Z.

The case (+). By (19), (21) is $\big(nZ - \frac{\langle nZ, n\rangle}{d_{\mathrm{model}}}\,n\big)D + n\,\eta_D = 0$ for all n in the ball, hence as a polynomial in n. Its part of degree 1 gives $\eta_D = -ZD$. Its part of degree 3 gives $\langle nZ, n\rangle\, nD = 0$, and $nD \neq 0$ for $n \neq 0$, so $\langle nZ, n\rangle = 0$ for all n: Z is antisymmetric.

The case (0). Here (21) is $\big(\mathrm{D}N_\varepsilon(h)\,(hZ)\big)D + N_\varepsilon(h)\,\eta_D = 0$. Multiplying by $\|h\|^3/\sqrt{d_{\mathrm{model}}}$ gives

$$\|h\|^2\; h\,(ZD + \eta_D) \;=\; \langle hZ, h\rangle\; hD \qquad \text{for all rows } h.$$

So $h\,(ZD + \eta_D) = \mu(h)\,hD$ with a number $\mu(h)$, for every $h \neq 0$. As $h \mapsto hD$ is injective and $d_{\mathrm{model}} \geq 2$, comparing h, h' and $h + h'$ for linearly independent h, h' shows that μ is a constant β, so $ZD + \eta_D = \beta D$. Then $\langle hZ, h\rangle = \beta\|h\|^2$ for all h, so the symmetric part of Z is βI: $Z = Y_L + \beta I$ with Y_L antisymmetric, and $\eta_D = \beta D - ZD = -Y_L D$. □

What remains is the step from (D_{k+1}) to (D_k). For $\tanh$ blocks in the cases (n) and (+) we prove it at almost all parameters (Lemma A.41). For GELU blocks in the case (+) and for gated blocks we prove it at explicit parameters (Lemmas A.46 and A.47), and in the case (0) at suitable parameters (Lemmas A.52 and A.53).

The identity for one block. Fix $k < L$, let block $k+1$ be a $\tanh$ or GELU block, and write $W = W^{k+1}_{\mathrm{read}}$, with the columns $c_1, \dots, c_q$, $V = W^{k+1}_{\mathrm{write}}$, with the rows $a_1, \dots, a_q$, and $\mathcal{B} = \mathcal{B}_{>k+1}$. Let Z and $\eta = (\eta_W, \eta_V, \eta')$ be as in (D_k), where η_W and η_V are the entries of η at W and V and η' is the rest. For rows n, $\tilde{w}$ and $\tilde{w}'$ let

$$\dot{z}(n) = \begin{cases} nZ\,W + n\,\eta_W & \text{in the case (n),} \\ \Big(nZ - \dfrac{\langle nZ, n\rangle}{d_{\mathrm{model}}}\,n\Big)\,W + n\,\eta_W & \text{in the cases (+) and (0),} \end{cases} \tag{25}$$

$$\mathcal{L}(\tilde{w}, \tilde{w}'; n) = \mathrm{D}\mathcal{B}(\tilde{w}V)\,\Big(\big(\tilde{w}' \odot \dot{z}(n)\big)\,V + \tilde{w}\,\eta_V\Big) + \mathrm{D}_\theta \mathcal{B}(\tilde{w}V)\,\eta'. \tag{26}$$

By the chain rule and (19), the left side of (21) at h is $\mathcal{L}\big(\sigma(nW), \sigma'(nW); n\big)$ at $n = N(h)$: $\tilde{w} = \sigma(nW)$ is the row of the outputs of the hidden units of block $k+1$, $\tilde{w}' = \sigma'(nW)$ the row of their derivatives, and $\dot{z}(n)$ the derivative of the row nW that they read. So (21) for k is equivalent to

$$\mathcal{L}\big(\sigma(nW),\ \sigma'(nW);\ n\big) \;=\; 0 \qquad \text{for all values } n \text{ of } N, \tag{27}$$

that is, for all rows n in the case (n), for the rows n with $\|n\|^2 < d_{\mathrm{model}}$ in the case (+), and for $n \in \mathbb{S}$ in the case (0).

Lemma A.39 (The identity at every real row)**.** *In the cases (n) and (+), (21) for k holds if and only if*

$$\mathcal{L}\big(\sigma(nW),\ \sigma'(nW);\ n\big) \;=\; 0 \qquad \textit{for every row } n \in \mathbb{R}^{1\times d_{\mathrm{model}}}. \tag{28}$$

Proof. In the case (n) this is (27). In the case (+), the left side of (28) is real-analytic in n on all of $\mathbb{R}^{1\times d_{\text{model}}}$, because $\mathcal{B}$ is real-analytic on all rows. So it vanishes on the ball if and only if it vanishes everywhere. □

In the case (0) the identity holds only on the sphere $\mathbb{S}$, which contains no ray. This is where the cases part: we follow rays in the cases (n) and (+), and a great circle at complex angles in the case (0).

Rays: no normalization, and $\varepsilon > 0$. In the cases (n) and (+) we take $n = t\,\xi$ in (28), with a row ξ, and let $t \to \infty$. The next lemma separates the hidden units of a $\tanh$ block. Its proof follows the argument by which Sussmann (1992, Lemma 3.1) proves that $\tanh$ hidden units with different arguments are linearly independent.

Lemma A.40 (Separation on a ray)**.** *Let $u_1, \dots, u_q$ be real numbers that are linearly independent over $\mathbb{Q}$, and let $\mathcal{T}_0, \dots, \mathcal{T}_K$ be real-analytic maps from $\mathbb{R}^{1\times q}$ to $\mathbb{R}^{1\times d_{\text{out}}}$. If*

$$\sum_{i=0}^{K} t^i\, \mathcal{T}_i\big(\tanh(tu_1), \dots, \tanh(tu_q)\big) \;=\; 0 \qquad \textit{for all } t > 0,$$

then $\mathcal{T}_i = 0$ on $\mathbb{R}^{1\times q}$ for every i.

Proof. No u_j is zero. Let s_j be the sign of u_j and $\zeta_j(t) = e^{-2|u_j|t} \in (0,1)$, so that $\tanh(tu_j) = s_j\,(1-\zeta_j(t))/(1+\zeta_j(t))$. The maps $\zeta \mapsto \mathcal{T}_i\big(s_1\frac{1-\zeta_1}{1+\zeta_1}, \dots, s_q\frac{1-\zeta_q}{1+\zeta_q}\big)$ are real-analytic near $\zeta = 0$, so for some $\rho \in (0,1)$ they are sums of power series $\sum_\alpha b_{i,\alpha}\,\zeta^\alpha$, over the multi-indices $\alpha = (\alpha_1, \dots, \alpha_q)$ of nonnegative integers, that converge absolutely for $|\zeta_j| \le \rho$. Choose $t_0 \ge 1$ with $\zeta_j(t_0) \le \rho$ for all j. For $t \ge t_0$ the hypothesis reads

$$\sum_{\alpha} p_\alpha(t)\, e^{-\mu_\alpha t} = 0, \qquad p_\alpha(t) = \sum_{i=0}^{K} t^i\, b_{i,\alpha}, \qquad \mu_\alpha = 2\sum_{j=1}^{q} \alpha_j\, |u_j|.$$

The numbers μ_α are distinct for distinct α, because the u_j are linearly independent over $\mathbb{Q}$, and only finitely many of them lie below any bound, because every $|u_j| > 0$. Suppose that some p_α is not zero, and among these let α^* have the smallest μ_α, say μ^*. Let $\gamma > 0$ be the distance from μ^* to the next larger μ_α. For $t \ge t_0$,

$$|p_{\alpha^*}(t)| \;\le\; \sum_{\mu_\alpha > \mu^*} |p_\alpha(t)|\, e^{-(\mu_\alpha - \mu^*)t} \;\le\; t^K\, e^{-\gamma(t-t_0)}\, e^{\mu^* t_0} \sum_{i=0}^{K}\sum_{\alpha} |b_{i,\alpha}|\, \zeta(t_0)^\alpha,$$

which tends to 0 as $t \to \infty$. A polynomial that tends to 0 is zero, a contradiction. So every $b_{i,\alpha} = 0$, and every $\mathcal{T}_i$ vanishes on the image of a neighborhood of $\zeta = 0$, which is a neighborhood of the row $(s_1, \dots, s_q)$. A real-analytic map on $\mathbb{R}^{1\times q}$ that vanishes on an open set vanishes everywhere. □

Consider, for a block ℓ with $W = W^\ell_{\text{read}}$ and $V = W^\ell_{\text{write}}$:

(T1) no row of V vanishes;

(T2) the columns of W are linearly independent over $\mathbb{Q}$.

Lemma A.41 (The step for $\tanh$ on rays)**.** *Let $\sigma = \tanh$, in the case (n) or (+), let θ be regular, and let block $k+1$ satisfy (T1) and (T2). Then (D_{k+1}) implies (D_k).*

Proof. Let Z and η satisfy (21) for k, so that (28) holds, with the notation above. Take $n = t\,\xi$ with a row ξ and $t > 0$. Then $\tilde{w} = \tanh(t\,\xi W)$ and $\tilde{w}' = 1 - \tilde{w}^2$, the row with the entries $1 - \tilde{w}_j^2$. By (25),

$$\dot{z}(t\,\xi) \;=\; t\,\xi\,(ZW + \eta_W) \;-\; t^3\,\frac{\langle \xi Z, \xi\rangle}{d_{\text{model}}}\,\xi W$$

in the case (+), and without the term in t^3 in the case (n). So (28) at $n = t\,\xi$ reads

$$t^3\,\mathcal{T}_3(\tilde{w}) + t\,\mathcal{T}_1(\tilde{w}) + \mathcal{T}_0(\tilde{w}) = 0 \qquad \text{at } \tilde{w} = \tanh(t\,\xi W),$$

with the maps

$$\mathcal{T}_3(\tilde{w}) = -\frac{\langle \xi Z, \xi \rangle}{d_{\text{model}}}\, \mathrm{D}\mathcal{B}(\tilde{w}V)\Big(\big((1-\tilde{w}^2)\odot \xi W\big)\, V\Big),$$
$$\mathcal{T}_1(\tilde{w}) = \mathrm{D}\mathcal{B}(\tilde{w}V)\Big(\big((1-\tilde{w}^2)\odot \xi\,(ZW+\eta_W)\big)\, V\Big),$$
$$\mathcal{T}_0(\tilde{w}) = \mathrm{D}\mathcal{B}(\tilde{w}V)\,(\tilde{w}\,\eta_V) + \mathrm{D}_\theta\mathcal{B}(\tilde{w}V)\,\eta',$$

where $\mathcal{T}_3 = 0$ in the case (n). They are defined and real-analytic for every row $\tilde{w} \in \mathbb{R}^{1\times q}$. By (T2), for integers $\kappa_1, \dots, \kappa_q$ that are not all zero, $\sum_j \kappa_j\, \xi c_j = \xi \sum_j \kappa_j c_j$ vanishes only for ξ in a hyperplane. So for ξ outside countably many hyperplanes the numbers $\xi c_1, \dots, \xi c_q$ are linearly independent over $\mathbb{Q}$, and Lemma A.40 gives $\mathcal{T}_3 = \mathcal{T}_1 = \mathcal{T}_0 = 0$ on $\mathbb{R}^{1\times q}$.

We use the following fact: if a row $b \in \mathbb{R}^{1\times q}$ has $\mathrm{D}\mathcal{B}(\tilde{w}V)\big(((1-\tilde{w}^2)\odot b)\, V\big) = 0$ for all rows $\tilde{w}$, then $b = 0$. Indeed, by Lemma A.35 the row $\sum_j b_j\,(1-\tilde{w}_j^2)\, a_j$ vanishes for all $\tilde{w}$ with $\tilde{w}V \in \mathcal{U}_{k+1}$. These $\tilde{w}$ form an open dense set, because V has rank d_{model}, so the row vanishes as a polynomial in $\tilde{w}$. Its coefficient of $\tilde{w}_j^2$ is $-b_j a_j$, and $a_j \neq 0$ by (T1).

(a) $\mathcal{T}_3 = 0$ gives Z. In the case (n) every matrix Z lies in $\mathfrak{g}$. In the case (+), $\xi W \neq 0$, so the fact gives $\langle \xi Z, \xi\rangle = 0$ for a dense set of rows ξ, hence for all: Z is antisymmetric. In both cases $Z \in \mathfrak{g}$ and $\widehat{Z} = Z$.

(b) $\mathcal{T}_1 = 0$ gives η_W. The fact gives $\xi\,(ZW+\eta_W) = 0$ for a dense set of rows ξ, so $\eta_W = -ZW$.

(c) $\mathcal{T}_0 = 0$ gives the rest. $\mathrm{D}\mathcal{B}(\tilde{w}V)\,(\tilde{w}\,\eta_V) + \mathrm{D}_\theta\mathcal{B}(\tilde{w}V)\,\eta' = 0$ for all rows $\tilde{w}$. By Lemma A.36 with $A = V$, $\eta_V = VZ'$ and (21) holds for $k+1$ with Z' and η'. By (D_{k+1}), $Z' \in \mathfrak{g}$ and η' is of the form (22) with $Z_{k+1} = Z'$. With (a) and (b) this is (22) for k. □

GELU and the Mills ratio. GELU is $\sigma(z) = \frac{z}{2}\big(1+\operatorname{erf}(z/\sqrt{2})\big)$, which is entire. For $z \in \mathbb{C}$ let

$$\varphi(z) = \frac{e^{-z^2/2}}{\sqrt{2\pi}}, \qquad \operatorname{mr}(z) \;=\; \sqrt{\frac{\pi}{2}}\, e^{z^2/2}\Big(1 - \operatorname{erf}\frac{z}{\sqrt{2}}\Big),$$

the normal density and the Mills ratio.

Lemma A.42 (The Mills ratio)**. (i)** *For all $z \in \mathbb{C}$: $\sigma(z) = z - \varphi(z)\, z\, \operatorname{mr}(z)$ and $\sigma'(z) = 1 - \varphi(z)\,\big(\operatorname{mr}(z) - z\big)$.*
(ii) *For $\operatorname{Re} z > 0$ and every integer $m \geq 0$, with $(-1)!! = 1$,*

$$\Big|\, \operatorname{mr}(z) - \sum_{l=0}^{m-1} \frac{(-1)^l\,(2l-1)!!}{z^{2l+1}} \,\Big| \;\leq\; \frac{(2m-1)!!}{(\operatorname{Re} z)^{2m+1}}.$$

Proof. (i) follows from $\varphi(z)\,\operatorname{mr}(z) = \frac{1}{2}\big(1-\operatorname{erf}(z/\sqrt{2})\big)$ and $\sigma'(z) = \frac{1}{2}\big(1+\operatorname{erf}(z/\sqrt{2})\big) + z\,\varphi(z)$. For (ii): for real z, $\operatorname{mr}(z) = e^{z^2/2}\int_z^\infty e^{-y^2/2}\,dy$, and the substitution $y = z+u$ gives $\operatorname{mr}(z) = \int_0^\infty e^{-zu-u^2/2}\,du$. Both sides are entire, so this holds for all z. For a number $y \geq 0$, $\big|e^{-y} - \sum_{l<m}\frac{(-y)^l}{l!}\big| \leq \frac{y^m}{m!}$. With $y = u^2/2$, and with $\int_0^\infty e^{-zu}u^{2l}\,du = \frac{(2l)!}{z^{2l+1}}$ and $\frac{(2l)!}{2^l\,l!} = (2l-1)!!$, this gives (ii). □

We say that a map Θ on an interval (t_0, ∞) is *meromorphic at infinity* if $\Theta(t) = t^l\,\Theta_0(1/t)$ for an integer l and a map Θ_0 that is holomorphic near 0. Such a map vanishes identically if it tends to 0 faster than every power of $1/t$.

Lemma A.43 (The Mills ratio at infinity)**.** *Let $z(t) = \omega\, t + \omega'/t$ with $\omega, \omega' \in \mathbb{C}$ and $\operatorname{Re}\omega > 0$, and let Θ_1 and Θ_2 be meromorphic at infinity, with values in $\mathbb{C}^{1\times d_{\text{out}}}$. If $\Theta_1(t)\,\operatorname{mr}(z(t)) + \Theta_2(t)$ tends to 0 faster than every power of $1/t$ as $t \to +\infty$, then $\Theta_1 = 0$ and $\Theta_2 = 0$.*

Proof. Suppose that a component of Θ_1 is not zero, and let Θ_1, Θ_2 stand for this component of both. Then $\operatorname{mr}(z(t)) = -\Theta_2(t)/\Theta_1(t) + O(t^{-m})$ for every m. For large t, $t = \frac{z}{2\omega}\big(1+\sqrt{1-4\omega\omega'/z^2}\big)$ with $z = z(t)$, so $1/t$ is a convergent power series in $1/z$ without constant term. Hence $-\Theta_2/\Theta_1$

is, for large t, a convergent series in powers of $1/z$ with finitely many positive powers of z, and this series is an asymptotic expansion of $\mathrm{mr}(z)$ along the curve $z = z(t)$. On this curve $\mathrm{Re}\, z$ is at least a positive constant times $|z|$ for large t, so by Lemma A.42(ii) the series $\sum_l (-1)^l (2l-1)!!\, z^{-2l-1}$ is one as well. Asymptotic expansions are unique, so the two series have the same coefficients. But the numbers $(2l-1)!!$ grow faster than every geometric sequence, so the second series converges for no z, a contradiction. Hence $\Theta_1 = 0$, then Θ_2 tends to 0 faster than every power, and $\Theta_2 = 0$. □

A gate on a ray. GELU and SiLU both satisfy $\sigma(z) - \sigma(-z) = z$, and so $\sigma'(z) + \sigma'(-z) = 1$. Let $\delta(y) = y - \sigma(y)$, with the derivative $\delta'(y) = 1 - \sigma'(y)$. For every real number $y > 0$,

$$\sigma(y) = y - \delta(y), \qquad \sigma(-y) = -\delta(y), \qquad \sigma'(y) = 1 - \delta'(y), \qquad \sigma'(-y) = \delta'(y). \tag{29}$$

By Lemma A.42(i), for GELU $\delta(y) = \varphi(y)\, y\ \mathrm{mr}(y)$ and $\delta'(y) = \varphi(y)\,\big(\mathrm{mr}(y) - y\big)$, and for SiLU

$$\delta(y) = \frac{y}{1+e^y}, \qquad \delta'(y) = \frac{1 + e^y - y\, e^y}{(1+e^y)^2}.$$

Let $p = 2$ for GELU and $p = 1$ for SiLU. For every real number $y \geq 2$,

$$\tfrac{1}{6}\, e^{-y^p/p} \;\leq\; \delta(y) \;\leq\; y\, e^{-y^p/p}, \qquad |\delta'(y)| \;\leq\; 4\, y\, \delta(y). \tag{30}$$

For GELU, Lemma A.42(ii) gives $\mathrm{mr}(y) \leq 1/y$ and $|\mathrm{mr}(y) - 1/y| \leq 1/y^3$, so $\frac{3}{4} \leq y\,\mathrm{mr}(y) \leq 1$ and $|\mathrm{mr}(y) - y| \leq 2y$. For SiLU, $\frac{1}{2}\, y\, e^{-y} \leq \delta(y) \leq y\, e^{-y}$ and $|\delta'(y)| \leq (2+y)\, e^{-y}$. So on a ray a GELU hidden unit, and the gate of a hidden unit of a gated block, differ from a linear or a vanishing one by the small term δ.

Lemma A.44 (The rest of the network on a ray)**.** *In the case (+), let $k \leq L$, let $h^\circ \neq 0$ be a row, let $v_1, \ldots, v_i$ be rows, and let η be a direction of the weights of the blocks $k+1, \ldots, L$ and of D. With D^i the derivative of order i in h, the maps*

$$t \;\mapsto\; t^{\,i}\, \mathrm{D}^i \mathcal{B}_{>k}(t\, h^\circ)\,(v_1, \ldots, v_i) \qquad \textit{and} \qquad t \;\mapsto\; t^{\,i}\, \mathrm{D}^i\big(\mathrm{D}_\theta \mathcal{B}_{>k}(\cdot)\,\eta\big)(t\, h^\circ)\,(v_1, \ldots, v_i)$$

are real-analytic functions of $1/t$ near $1/t = 0$, so they are meromorphic at infinity. Moreover, there are numbers $\varrho, t_0 > 0$ such that the maps $u \mapsto \mathcal{B}_{>k}\big(t\,(h^\circ + u)\big)$ and $u \mapsto \mathrm{D}_\theta \mathcal{B}_{>k}\big(t\,(h^\circ + u)\big)\,\eta$ extend to holomorphic maps on the rows $u \in \mathbb{C}^{1 \times d_{\mathrm{model}}}$ with $|u_j| \leq \varrho$ for all j, which are bounded uniformly in $t \geq t_0$.

Proof. $\mathcal{B}_{>k} = \mathcal{B}^\sharp \circ N_\varepsilon$, with $\mathcal{B}^\sharp(n) = nD$ if $k = L$ and $\mathcal{B}^\sharp = \mathcal{B}_{>k+1} \circ g_{k+1}$ otherwise, which is real-analytic on all rows. For $t > 0$ and a row u,

$$N_\varepsilon\big(t\,(h^\circ + u)\big) \;=\; \frac{h^\circ + u}{\sqrt{\|h^\circ + u\|^2/d_{\mathrm{model}} + \varepsilon\, t^{-2}}},$$

and the right side is a real-analytic function of $(1/t, u)$ near $(0, 0)$, also at $1/t = 0$, because $h^\circ \neq 0$. So $\mathcal{B}_{>k}\big(t\,(h^\circ + u)\big)$ is a real-analytic function of $(1/t, u)$ near $(0, 0)$. It extends to a holomorphic and bounded function on a complex neighborhood of $(0, 0)$, which gives ϱ and t_0. A derivative of order i in $v = t\, u$ at $v = 0$ is t^{-i} times the derivative of order i in u at $u = 0$. The weights enter analytically as well, which gives the same for $\mathrm{D}_\theta \mathcal{B}_{>k}(\cdot)\,\eta$. □

GELU and gated blocks in one form. The rest of the treatment of rays is for GELU, SwiGLU and GeGLU blocks. A GELU block has the form of a GeGLU block in which the value ne_j of every hidden unit is replaced by the number 1. So in a GELU block let $W_g^\ell = W_{\mathrm{read}}^\ell$, and call $c_1, \ldots, c_{q_\ell}$ its gate columns. For a row n let $\iota^\ell(n) \in \mathbb{R}^{1 \times q_\ell}$ be the row of the values of its hidden units:

$$\iota^\ell(n) = n\, W_v^\ell \;\text{ in a gated block,} \qquad \iota^\ell(n) = (1, \ldots, 1) \;\text{ in a GELU block,}$$
$$g_\ell(n) \;=\; \big(\sigma(n\, W_g^\ell) \odot \iota^\ell(n)\big)\, W_{\mathrm{write}}^\ell \;\text{ in both.}$$

Let $m = 2$ for gated blocks and $m = 1$ for GELU blocks, so that $\iota^\ell(\lambda n) = \lambda^{m-1}\,\iota^\ell(n)$ for every number $\lambda > 0$. The exponent p of (30) is 2 for GELU and GeGLU blocks and 1 for SwiGLU blocks.

Cones and good parameters. A *cone* is a set of rows that contains λh for each of its rows h and every number $\lambda > 0$. Let $\mathcal{C}_0, \dots, \mathcal{C}_L$ be open convex cones in $\mathbb{R}^{1\times d_{\text{model}}}$ that do not contain 0, one for each hidden state. We call θ *good* for these cones if (R1) holds and every block ℓ, with its gate columns c_j, its write rows a_j and the entries $\iota_j(n)$ of $\iota^\ell(n)$, satisfies:

(K1) $nc_j \neq 0$ for every $n \in \mathcal{C}_{\ell-1}$ and every j. So every gate has one sign on $\mathcal{C}_{\ell-1}$. Let $\mathcal{A}_\ell$ be the set of the hidden units j with $nc_j > 0$ on $\mathcal{C}_{\ell-1}$, the *active* ones, and let

$$g_\ell^\circ(n) \;=\; \sum_{j\in\mathcal{A}_\ell} (nc_j)\,\iota_j(n)\,a_j$$

be the block with its gates *frozen*. It is a linear map for a GELU block and a quadratic map for a gated block: $g_\ell^\circ(\lambda n) = \lambda^m\, g_\ell^\circ(n)$.

(K2) $g_\ell^\circ(n) \in \mathcal{C}_\ell$ for every $n \in \mathcal{C}_{\ell-1}$, and $\mathrm{D}g_\ell^\circ(n)$ is invertible at some row n.

(K3) No a_j vanishes, and in a gated block no value column e_j vanishes. For GELU and GeGLU blocks the matrices $c_1c_1^\top, \dots, c_{q_\ell}c_{q_\ell}^\top$ are linearly independent over $\mathbb{Q}$, and for SwiGLU blocks the columns $c_1, \dots, c_{q_\ell}$ are.

The determinant of $\mathrm{D}g_\ell^\circ(n)$ is a polynomial in n, so by (K2) g_ℓ° is a local diffeomorphism on an open dense set of rows. A good θ is regular. Indeed, take $\xi \in \mathcal{C}_{\ell-1}$ with $\mathrm{D}g_\ell^\circ(\xi)$ invertible. For a row v, by (29) and (30), in a gated block

$$\begin{aligned}\frac{1}{t}\,\mathrm{D}g_\ell(t\,\xi)\,v \;&=\; \sum_j \Big[\sigma'(t\,\xi c_j)\,(vc_j)\,(\xi e_j) + \frac{\sigma(t\,\xi c_j)}{t}\,(ve_j)\Big]\,a_j \\ &\longrightarrow \sum_{j\in\mathcal{A}_\ell} \Big[(vc_j)\,(\xi e_j) + (\xi c_j)\,(ve_j)\Big]\,a_j \;=\; \mathrm{D}g_\ell^\circ(\xi)\,v \qquad \text{as } t\to\infty,\end{aligned}$$

and in a GELU block $\mathrm{D}g_\ell(t\,\xi)\,v = \sum_j \sigma'(t\,\xi c_j)\,(vc_j)\,a_j$ tends to $\sum_{j\in\mathcal{A}_\ell}(vc_j)\,a_j = \mathrm{D}g_\ell^\circ(\xi)\,v$. So $\mathrm{D}g_\ell(t\,\xi)$ is invertible for large t, which is (R3).

The frozen network. In the case (n), for gated blocks and a good θ, let $\mathcal{B}_{>k}^\circ$ be the network after h_k with the gates of all its blocks frozen:

$$\mathcal{B}_{>L}^\circ(h) = hD, \qquad \mathcal{B}_{>k-1}^\circ(h) = \mathcal{B}_{>k}^\circ\big(g_k^\circ(h)\big).$$

It is a polynomial map of h and of the weights. Lemmas A.35 and A.36 hold for $\mathcal{B}_{>k}^\circ$ in the place of $\mathcal{B}_{>k}$, with an open dense set $\mathcal{U}_k^\circ$ of rows and with the same proofs, in which g_{k+1}° takes the place of g_{k+1}.

The identity for one GELU or gated block. Fix $k < L$ and write $W_g = W_g^{k+1}$, with the columns $c_1, \dots, c_q$, $V = W_{\text{write}}^{k+1}$, with the rows $a_1, \dots, a_q$, $\iota = \iota^{k+1}$ and $\mathcal{B} = \mathcal{B}_{>k+1}$, and in a gated block $W_v = W_v^{k+1}$, with the columns $e_1, \dots, e_q$. Let Z and $\eta = (\eta_g, \eta_v, \eta_V, \eta')$ be as in (D_k), where η_g, η_v and η_V are the entries of η at W_g, W_v and V and η' is the rest; in a GELU block there is no η_v. For a row n let $\tilde n = nZ$ in the case (n) and $\tilde n = nZ - \frac{\langle nZ, n\rangle}{d_{\text{model}}}\,n$ in the case (+). For rows n and $\tilde w, \tilde w' \in \mathbb{R}^{1\times q}$ let

$$\dot z^g(n) = \tilde n\,W_g + n\,\eta_g, \qquad \dot z^v(n) = \begin{cases} \tilde n\,W_v + n\,\eta_v & \text{in a gated block,} \\ 0 & \text{in a GELU block,}\end{cases} \tag{31}$$

$$\mathcal{L}(\tilde w, \tilde w'; n) = \mathrm{D}\mathcal{B}\big(\hat w V\big)\,\Big(\big(\tilde w' \odot \dot z^g(n) \odot \iota(n) + \tilde w \odot \dot z^v(n)\big)\,V + \hat w\,\eta_V\Big) + \mathrm{D}_\theta\mathcal{B}\big(\hat w V\big)\,\eta', \tag{32}$$

where $\hat w = \tilde w \odot \iota(n)$, and $\dot z_j^g(n)$ and $\dot z_j^v(n)$ are the entries of the rows (31). Here $\tilde w$ stands for the row $\sigma(nW_g)$ of the gates, $\tilde w'$ for the row $\sigma'(nW_g)$ of their derivatives, and $\hat w$ for the row of the outputs of the hidden units, and $\dot z^g(n)$ and $\dot z^v(n)$ are the derivatives of the rows nW_g and $\iota(n)$. In a GELU block $\hat w = \tilde w$, and these are the maps (25) and (26). By the chain rule and (19), the left side of (21)

at h is $\mathcal{L}\big(\sigma(nW_g), \sigma'(nW_g); n\big)$ at $n = N(h)$. As in Lemma A.39, (21) for k therefore holds if and only if

$$\mathcal{L}\big(\sigma(nW_g),\ \sigma'(nW_g);\ n\big) \ = \ 0 \qquad \text{for every row } n \in \mathbb{R}^{1\times d_{\text{model}}}. \tag{33}$$

We use two properties of $\mathcal{L}$. The row $\tilde{w}'$ enters only through the terms $\tilde{w}'_j\, \dot{z}^g_j(n)\, \iota_j(n)\, a_j$, so

$$\frac{\partial \mathcal{L}}{\partial \tilde{w}'_j}(\tilde{w}, \tilde{w}'; n) \ = \ \dot{z}^g_j(n)\ \iota_j(n)\ \mathrm{D}\mathcal{B}\big(\hat{w}V\big)\, a_j. \tag{34}$$

And in a gated block $\dot{z}^g$ and $\dot{z}^v$ are odd in n, while $\hat{w}$ does not change when $\tilde{w}$ and n both change their sign, so

$$\mathcal{L}(-\tilde{w}, \tilde{w}'; -n) \ = \ \mathcal{L}(\tilde{w}, \tilde{w}'; n) \qquad \text{in a gated block.} \tag{35}$$

Lemma A.45 (The network after h_k far out in its cone)**.** *In the case (n), for gated blocks, let θ be good, $k \le L$, $h^\circ \in \mathcal{C}_k$, and let η be a direction of the weights of the blocks $k+1, \dots, L$ and of D. There are numbers $\varrho, \kappa, t_0 > 0$ such that, for $t \ge t_0$, the maps $u \mapsto \mathcal{B}_{>k}\big(t\,(h^\circ + u)\big)$ and $u \mapsto \mathrm{D}_\theta \mathcal{B}_{>k}\big(t\,(h^\circ + u)\big)\,\eta$ extend to holomorphic maps on the rows $u \in \mathbb{C}^{1\times d_{\text{model}}}$ with $|u_j| \le \varrho$ for all j, and for these u, at $h = t\,(h^\circ + u)$,*

$$\|\mathcal{B}_{>k}(h) - \mathcal{B}^\circ_{>k}(h)\| \ \le \ e^{-\kappa t^p}, \qquad \|\mathrm{D}_\theta \mathcal{B}_{>k}(h)\,\eta - \mathrm{D}_\theta \mathcal{B}^\circ_{>k}(h)\,\eta\| \ \le \ e^{-\kappa t^p}.$$

Proof. One gate. Let $z \in \mathbb{C}$ with $\operatorname{Re} z \ge 2$ and $|\operatorname{Im} z| \le \frac{1}{2} \operatorname{Re} z$, so that $\operatorname{Re}(z^2) \ge \frac{3}{4}(\operatorname{Re} z)^2$ and $|z| \le 2 \operatorname{Re} z$. For GELU, by Lemma A.42, $\sigma(z) - z = -\varphi(z)\, z \operatorname{mr}(z)$ and $|z \operatorname{mr}(z)| \le |z| / \operatorname{Re} z \le 2$, so $|\sigma(z) - z| \le e^{-\frac{3}{8}(\operatorname{Re} z)^2}$. For SiLU, $\sigma(z) - z = -z/(1+e^z)$ and $|1 + e^z| \ge \frac{1}{2} e^{\operatorname{Re} z}$, so $|\sigma(z) - z| \le 4 \operatorname{Re} z\, e^{-\operatorname{Re} z}$. Both gates are holomorphic at z and at $-z$, and $\sigma(-z) = \sigma(z) - z$. So there is a number $\kappa_1 > 0$ such that, for large $\operatorname{Re} z$,

$$|\sigma(z) - z| \ \le \ e^{-\kappa_1 (\operatorname{Re} z)^p} \qquad \text{and} \qquad |\sigma(-z)| \ \le \ e^{-\kappa_1 (\operatorname{Re} z)^p}. \tag{36}$$

The first bound, by induction on $L - k$. We prove it at the weights $\theta + \tau\eta$ as well, for all complex numbers τ with $|\tau| \le \tau_0$ and a small $\tau_0 > 0$, where the hidden units whose gates are frozen are those of θ. For $k = L$, $\mathcal{B}_{>L}(h) = hD = \mathcal{B}^\circ_{>L}(h)$. Let the bound hold for some $k \ge 1$, and let $h^\circ \in \mathcal{C}_{k-1}$. By (K1), $h^\circ c_j \ne 0$ for the gate columns c_j of block k at θ. Let ϱ and τ_0 be small, and let $h = t\,(h^\circ + u)$. For every j, with c_j taken at the weights $\theta + \tau\eta$, the number $z = hc_j$ or $z = -hc_j$ with the sign of $h^\circ c_j$ satisfies $\operatorname{Re} z \ge \frac{t}{2} |h^\circ c_j|$ and $|\operatorname{Im} z| \le \frac{1}{2} \operatorname{Re} z$. By (36), g_k is holomorphic at these h, and

$$\|g_k(h) - g^\circ_k(h)\| \ \le \ e^{-\kappa_2 t^p}, \qquad g^\circ_k(h) \ = \ t^2 \left(g^\circ_k(h^\circ) + u'\right),$$

with a number $\kappa_2 > 0$, where g_k and g°_k are taken at the weights $\theta + \tau\eta$ and $g^\circ_k(h^\circ)$ at θ, and where the entries of the row u' are as small as we like when ϱ and τ_0 are small. So $g_k(h)$ is $t^2 \left(g^\circ_k(h^\circ) + u''\right)$ with a row u'' of the same kind. By (K2), $g^\circ_k(h^\circ) \in \mathcal{C}_k$. So the bound for k, at the row $g^\circ_k(h^\circ)$ and with t^2 in the place of t, gives $\|\mathcal{B}_{>k}(g_k(h)) - \mathcal{B}^\circ_{>k}(g_k(h))\| \le e^{-\kappa t^{2p}}$. The derivative of the polynomial map $\mathcal{B}^\circ_{>k}$ is at most a fixed power of t at the rows $t^2\,(g^\circ_k(h^\circ) + u')$, so $\|\mathcal{B}^\circ_{>k}(g_k(h)) - \mathcal{B}^\circ_{>k}(g^\circ_k(h))\|$ is at most a fixed power of t times $e^{-\kappa_2 t^p}$. As $\mathcal{B}_{>k-1} = \mathcal{B}_{>k} \circ g_k$ and $\mathcal{B}^\circ_{>k-1} = \mathcal{B}^\circ_{>k} \circ g^\circ_k$, this gives the bound for $k-1$, with a smaller κ.

The second bound. At $h = t\,(h^\circ + u)$ the difference of $\mathcal{B}_{>k}(h)$ and $\mathcal{B}^\circ_{>k}(h)$ at the weights $\theta + \tau\eta$ is holomorphic in τ for $|\tau| \le \tau_0$ and at most $e^{-\kappa t^p}$. By Cauchy's estimate its derivative at $\tau = 0$, which is $\mathrm{D}_\theta \mathcal{B}_{>k}(h)\,\eta - \mathrm{D}_\theta \mathcal{B}^\circ_{>k}(h)\,\eta$, is at most $e^{-\kappa t^p}/\tau_0$. □

Lemma A.46 (The step for GELU and gated blocks on rays)**.** *Let the blocks be GELU blocks in the case (+), or gated blocks in the case (n) or (+). Let θ be good for the cones $\mathcal{C}_0, \dots, \mathcal{C}_L$, and let $k < L$. Then (D_{k+1}) implies (D_k).*

Proof. Let Z and η satisfy (21) for k, so that (33) holds, with the notation above. Let $\mathcal{A} = \mathcal{A}_{k+1}$ and $g^\circ = g^\circ_{k+1}$. Let s_j be the sign of nc_j on $\mathcal{C}_k$ and $\bar{s} = (s_1, \dots, s_q)$, let $I_\mathcal{A}$ be the diagonal matrix with the entries 1 at the hidden units of $\mathcal{A}$ and 0 elsewhere, and $1_\mathcal{A}$ the row with these entries. Let

$$\mathcal{B}^\infty = \mathcal{B}_{>k+1} \ \text{ in the case (+)}, \qquad \mathcal{B}^\infty = \mathcal{B}^\circ_{>k+1} \ \text{ in the case (n)},$$

and let $\mathcal{L}^\infty$ be (32) with $\mathcal{B}^\infty$ in the place of $\mathcal{B}$. It is real-analytic in $(\tilde{w}, \tilde{w}', n)$ on all rows, a polynomial in the case (n), and it satisfies (34), and (35) in a gated block, with $\mathcal{B}^\infty$. Lemmas A.35 and A.36 hold for $\mathcal{B}^\infty$, and we write $\mathcal{U}^\infty$ for its open dense set $\mathcal{U}_{k+1}$ or $\mathcal{U}^\circ_{k+1}$.

The ray. Let $\xi \in \mathcal{C}_k$ and $n = t\,\xi$ with $t > 0$. Let

$$r_j = |\xi c_j|^p, \qquad \delta_j(t) = \delta\big(t\,|\xi c_j|\big), \qquad \delta_j'(t) = \delta'\big(t\,|\xi c_j|\big),$$

and let $\delta(t)$ and $\delta'(t)$ be the rows with these entries. By (30), with a constant κ, for large t

$$\tfrac{1}{6}\,e^{-r_j t^p/p} \;\le\; \delta_j(t) \;\le\; \kappa\, t\, e^{-r_j t^p/p}, \qquad |\delta_j'(t)| \;\le\; \kappa\, t\, \delta_j(t). \tag{37}$$

By (29), $\sigma(nW_g) = n\,W_g I_{\mathcal{A}} - \delta(t)$ and $\sigma'(nW_g) = 1_{\mathcal{A}} - \bar{s} \odot \delta'(t)$. So (33) at $n = t\,\xi$ is

$$\mathcal{L}\big(t\,\xi W_g I_{\mathcal{A}} - \delta(t),\;\; 1_{\mathcal{A}} - \bar{s} \odot \delta'(t);\;\; t\,\xi\big) \;=\; 0 \qquad \text{for all } t > 0. \tag{38}$$

Without the small terms the hidden units of the block give the row $t\,\xi W_g I_{\mathcal{A}} \odot \iota(t\,\xi)$, and the block gives $g^\circ(t\,\xi) = t^m\,h^\circ$ with $h^\circ = g^\circ(\xi)$, which lies in $\mathcal{C}_{k+1}$ by (K2), so $h^\circ \neq 0$. With a row δ in the place of $\delta(t)$ the block gives $t^m\,(h^\circ + u)$ with $u = -\frac{1}{t}\,\big(\delta \odot \iota(\xi)\big)\,V$.

The expansion. For a multi-index $\alpha = (\alpha_1, \dots, \alpha_q)$ of nonnegative integers let $\delta^\alpha = \prod_j \delta_j^{\alpha_j}$, $|\alpha| = \sum_j \alpha_j$, $\alpha! = \prod_j \alpha_j!$ and $r_\alpha = \sum_j \alpha_j r_j$, and let 1_j be the multi-index with the entry 1 at j and 0 elsewhere. Let

$$\mathcal{L}_\alpha(t) = \frac{1}{\alpha!}\,\frac{\partial^{|\alpha|}\mathcal{L}^\infty}{\partial \tilde{w}^\alpha}, \qquad \mathcal{L}'_{\alpha,i}(t) = \frac{1}{\alpha!}\,\frac{\partial^{|\alpha|}}{\partial \tilde{w}^\alpha}\,\frac{\partial \mathcal{L}^\infty}{\partial \tilde{w}_i'}, \qquad \text{both at } \big(t\,\xi W_g I_{\mathcal{A}},\; 1_{\mathcal{A}};\; t\,\xi\big).$$

We write $\mathcal{L}_j = \mathcal{L}_{1_j}$ and $\mathcal{L}'_j = \mathcal{L}'_{0,j}$. As $\mathcal{L}^\infty$ is real-analytic in $\tilde{w}$ and affine in $\tilde{w}'$,

$$\mathcal{L}^\infty\big(t\,\xi W_g I_{\mathcal{A}} - \delta,\;\; 1_{\mathcal{A}} - \bar{s} \odot \delta';\;\; t\,\xi\big) \;=\; \sum_\alpha (-1)^{|\alpha|}\,\delta^\alpha \Big[\mathcal{L}_\alpha(t) - \sum_{i=1}^q s_i\,\mathcal{L}'_{\alpha,i}(t)\,\delta_i'\Big] \tag{39}$$

for small rows δ and all rows δ'. By (37), at $\delta = \delta(t)$ and $\delta' = \delta'(t)$ the term of δ^α is of the order of $e^{-r_\alpha t^p/p}$, up to powers of t, and the term of $\delta^\alpha \delta_i'$ of the order of $e^{-(r_\alpha + r_i)t^p/p}$. We call r_α and $r_\alpha + r_i$ the *exponents* of these terms. We use three facts.

(a) Every $\mathcal{L}_\alpha$ and $\mathcal{L}'_{\alpha,i}$ is a real-analytic function of $t \in \mathbb{R}$ that is meromorphic at infinity. In the case (n) it is a polynomial in t. In the case (+) it is built from derivatives of $\mathcal{B}$ and of $\mathrm{D}_\theta \mathcal{B}(\cdot)\,\eta'$ at $t^m\,h^\circ$ and from polynomials in t, so this is Lemma A.44 with t^m in the place of t.

(b) At $\delta = \delta(t)$ and $\delta' = \delta'(t)$, the left side of (39) vanishes in the case (+), by (38). In the case (n), where the blocks are gated and $m = 2$, it is at most $e^{-\kappa' t^{2p}}$ for large t, with a number $\kappa' > 0$. Indeed, by Lemma A.45 for $k+1$, at the row h° and with t^2 in the place of t, the maps $\mathcal{B} - \mathcal{B}^\infty$ and $\mathrm{D}_\theta \mathcal{B}(\cdot)\,\eta' - \mathrm{D}_\theta \mathcal{B}^\infty(\cdot)\,\eta'$ are holomorphic and at most $e^{-2\kappa' t^{2p}}$ at the rows $t^2\,(h^\circ + u)$ with $|u_j| \le \varrho$, and by Cauchy's estimate so is the derivative of the first, for large t, where $|u_j| \le \varrho/2$. At $\delta = \delta(t)$ the block gives such a row, with $u \to 0$, and the rows to which these derivatives are applied in (32) are at most a fixed power of t. So $\mathcal{L}$ and $\mathcal{L}^\infty$ differ at the point of (38) by at most a fixed power of t times $e^{-2\kappa' t^{2p}}$, and $\mathcal{L}$ vanishes there.

(c) For large t the series (39) converges absolutely at $\delta(t)$, $\delta'(t)$, and the sum of any set of its terms is at most a fixed power of t times $e^{-r t^p/p}$, where r is the smallest exponent of a term of the set. In the case (n) the series is a finite sum with coefficients that are polynomials in t, and this follows from (37). In the case (+), by Lemma A.44 with t^m in the place of t, the maps $u \mapsto \mathcal{B}\big(t^m(h^\circ + u)\big)$ and $u \mapsto \mathrm{D}_\theta \mathcal{B}\big(t^m(h^\circ + u)\big)\,\eta'$ are holomorphic and bounded on the rows u with $|u_j| \le \varrho$, uniformly in $t \ge t_0$, and by Cauchy's estimate so is the derivative of the first where $|u_j| \le \varrho/2$. With a row δ in the place of $\delta(t)$ the block gives $t^m\,(h^\circ + u)$, and there is a number $\varrho' > 0$ such that $|u_j| \le \varrho/2$ for the rows $\delta \in \mathbb{C}^{1\times q}$ with $|\delta_j| \le \varrho'\,t$ for all j. So the left side of (39), which is affine in δ', is holomorphic in these rows δ, and there its value at $\delta' = 0$ and its coefficients of the entries of δ' are at most a fixed power of t. By Cauchy's estimates, $\mathcal{L}_\alpha(t)$ and $\mathcal{L}'_{\alpha,i}(t)$ are at most a fixed power of t times $(\varrho'\,t)^{-|\alpha|}$. With (37), a term with the exponent r' is then at most a fixed power of t times $(\kappa/\varrho')^{|\alpha|}\,e^{-r' t^p/p}$. As $r' \ge |\alpha|\,\min_j r_j$, the terms with large $|\alpha|$ are bounded by a geometric series.

A generic ray. We now take $\xi \in \mathcal{C}_k$ such that

(i) the numbers $r_1, \dots, r_q$ are linearly independent over $\mathbb{Q}$;
(ii) $\iota_j(\xi) \neq 0$ for every j;
(iii) $g^\circ(\xi) \in \mathcal{U}^\infty$.

These ξ are dense in $\mathcal{C}_k$. For (i): if integers $\kappa_1, \dots, \kappa_q$ are not all zero, $\sum_j \kappa_j r_j$ is $\xi\big(\sum_j \kappa_j c_j c_j^\top\big)\xi^\top$ for GELU and GeGLU blocks and $\xi \sum_j \kappa_j s_j c_j$ for SwiGLU blocks, and by (K3) it vanishes only for ξ on a quadric, or in a hyperplane. For (ii): in a gated block $\iota_j(\xi) = \xi e_j$, and $e_j \neq 0$ by (K3). For (iii): g° is a local diffeomorphism on an open dense set of rows, and $\mathcal{U}^\infty$ is open and dense. By (i), different multi-indices have different exponents r_α, and only finitely many exponents lie below any bound.

The induction. We show that, for every multi-index α,

$$\mathcal{L}_\alpha(t) = 0 \quad \text{and} \quad \mathcal{L}'_{\alpha - 1_i,\, i}(t) = 0 \ \text{ for every } i \text{ with } \alpha_i \geq 1, \qquad \text{for all } t \in \mathbb{R}, \tag{40}$$

by induction on the exponent r_α. These are the coefficients of the terms of (39) with the exponent r_α. Let (40) hold for all multi-indices with a smaller exponent, and let $r^+ > r_\alpha$ be the next exponent. In (39) at $\delta(t)$, $\delta'(t)$, the terms with a smaller exponent vanish, and by (c) the terms with a larger exponent add up to at most a fixed power of t times $e^{-r^+ t^p/p}$. By (b), so do the terms with the exponent r_α, which are

$$(-1)^{|\alpha|}\, \delta(t)^\alpha \Big[\mathcal{L}_\alpha(t) + \sum_{i:\, \alpha_i \geq 1} s_i\, \mathcal{L}'_{\alpha - 1_i,\, i}(t)\, \frac{\delta_i'(t)}{\delta_i(t)}\Big].$$

By (37), $\delta(t)^\alpha \geq 6^{-|\alpha|}\, e^{-r_\alpha t^p/p}$. So

$$\mathcal{L}_\alpha(t) \ + \sum_{i:\, \alpha_i \geq 1} s_i\, \mathcal{L}'_{\alpha - 1_i,\, i}(t)\, \frac{\delta_i'(t)}{\delta_i(t)} \ \longrightarrow \ 0 \qquad \text{faster than every power of } 1/t. \tag{41}$$

No small factor, $\alpha = 0$. (41) is $\mathcal{L}_0(t) \to 0$, and by (a) $\mathcal{L}_0 = 0$.

One small factor, $\alpha = 1_j$. With $y = t\,|\xi c_j|$, (41) is $\mathcal{L}_j + s_j\, \mathcal{L}'_j\, \delta'(y)/\delta(y) \to 0$.

For GELU and GeGLU blocks, $\delta'(y)/\delta(y) = \big(\operatorname{mr}(y) - y\big)/\big(y \operatorname{mr}(y)\big)$ and $y \operatorname{mr}(y) \leq 1$, so

$$\big(y\, \mathcal{L}_j + s_j\, \mathcal{L}'_j\big)\ \operatorname{mr}(y) \ - \ s_j\, y\, \mathcal{L}'_j \ \longrightarrow \ 0 \qquad \text{faster than every power of } 1/t.$$

By (a), the maps $y\, \mathcal{L}_j + s_j \mathcal{L}'_j$ and $y\, \mathcal{L}'_j$ are meromorphic at infinity. By Lemma A.43 with $z(t) = |\xi c_j|\, t$ both vanish, so $\mathcal{L}'_j = 0$ and $\mathcal{L}_j = 0$.

For SwiGLU blocks, $y\, \delta'(y)/\delta(y) = 1 - y + \delta(y)$, and $\delta(y)$ tends to 0 faster than every power of $1/t$, so

$$f(t) \ = \ |\xi c_j|\, t\, \mathcal{L}_j(t) + s_j\, \big(1 - |\xi c_j|\, t\big)\, \mathcal{L}'_j(t)$$

tends to 0 faster than every power of $1/t$ as $t \to +\infty$. By (a), f is meromorphic at infinity, so $f(t) = 0$ for large t, and then for all $t \in \mathbb{R}$, because f is real-analytic on $\mathbb{R}$. By (35), $\mathcal{L}_j$ is odd in t and $\mathcal{L}'_j$ is even. So $0 = f(t) - f(-t) = -2\, s_j\, |\xi c_j|\, t\, \mathcal{L}'_j(t)$: $\mathcal{L}'_j = 0$, and then $\mathcal{L}_j = 0$.

For all three kinds of block, by (34), $\mathcal{L}'_j(t) = \dot z_j^{\,g}(t\,\xi)\ \iota_j(t\,\xi)\ \mathrm{D}\mathcal{B}^\infty(t^m\, h^\circ)\, a_j$. By (iii), (K3) and Lemma A.35, $\mathrm{D}\mathcal{B}^\infty(h^\circ)\, a_j \neq 0$, so the last factor, which is real-analytic in t, vanishes only at isolated t. By (ii), $\iota_j(t\,\xi) = t^{m-1}\, \iota_j(\xi) \neq 0$ for $t > 0$. So

$$\dot z_j^{\,g}(t\,\xi) = 0 \qquad \text{for all } t > 0. \tag{42}$$

Two or more small factors, $|\alpha| \geq 2$. Let $\alpha_i \geq 1$. Then $r_i < r_\alpha$, so (42) holds for the hidden unit i, and by (34) the derivative of $\mathcal{L}^\infty(\tilde w, \tilde w'; t\,\xi)$ in $\tilde w_i'$ vanishes for all rows $\tilde w$. So $\mathcal{L}'_{\alpha - 1_i,\, i} = 0$. Then (41) is $\mathcal{L}_\alpha(t) \to 0$, and by (a) $\mathcal{L}_\alpha = 0$. This ends the induction.

What the induction gives. For this ξ and every $t > 0$, all coefficients of the expansion of $(\tilde w, \tilde w') \mapsto \mathcal{L}^\infty(\tilde w, \tilde w'; t\,\xi)$ at $(t\,\xi W_g I_{\mathcal{A}}, 1_{\mathcal{A}})$ vanish, so this real-analytic map vanishes. The rows $t\,\xi$ of this kind are dense in $\mathcal{C}_k$. So, with (42), for all $n \in \mathcal{C}_k$ and all rows $\tilde w \in \mathbb{R}^{1 \times q}$, with $\hat w = \tilde w \odot \iota(n)$,

$$\dot z^g(n) = 0, \qquad \mathrm{D}\mathcal{B}^\infty(\hat w V)\, \Big(\big(\tilde w \odot \dot z^v(n)\big)\, V + \hat w\, \eta_V\Big) + \mathrm{D}_\theta \mathcal{B}^\infty(\hat w V)\, \eta' = 0. \tag{43}$$

Step 1: the gates. By (31), $\dot{z}^g(n)$ is a polynomial in n that vanishes on the open set $\mathcal{C}_k$, so it vanishes for all n. In the case (n) this is $n\,(ZW_g + \eta_g) = 0$, so $\eta_g = -ZW_g$, and $\mathfrak{g}$ contains every Z. In the case (+) its part of degree 3 is $\langle nZ, n\rangle\, nW_g = 0$, so Z is antisymmetric, as $W_g \neq 0$, and its part of degree 1 gives $\eta_g = -ZW_g$. In both cases $Z \in \mathfrak{g}$, and in a gated block $\dot{z}^v(n) = n\,(ZW_v + \eta_v)$.

Step 2: the values and the write. In a gated block, fix $n \in \mathcal{C}_k$ with $ne_j \neq 0$ for every j, and let $\Delta(n)$ be the diagonal matrix with the entries $-\dot{z}^v_j(n)/(ne_j)$, so that $\tilde{w} \odot \dot{z}^v(n) = -\hat{w}\,\Delta(n)$. In a GELU block, fix $n \in \mathcal{C}_k$ and let $\Delta(n) = 0$. The row $\hat{w}$ takes every value in $\mathbb{R}^{1\times q}$, and (43) reads

$$\mathrm{D}\mathcal{B}^\infty(\hat{w}V)\left(\hat{w}\left(\eta_V - \Delta(n)\,V\right)\right) + \mathrm{D}_\theta\mathcal{B}^\infty(\hat{w}V)\,\eta' = 0 \qquad \text{for all rows } \hat{w}.$$

By Lemma A.36 for $\mathcal{B}^\infty$, with $A = V$, there is a matrix $Z'(n)$ with

$$\eta_V - \Delta(n)\,V = V\,Z'(n), \qquad \mathrm{D}\mathcal{B}^\infty(h)\left(hZ'(n)\right) + \mathrm{D}_\theta\mathcal{B}^\infty(h)\,\eta' = 0 \quad \text{for all rows } h. \tag{44}$$

For two such rows n and n_1, subtracting gives $\mathrm{D}\mathcal{B}^\infty(h)\left(h\,(Z'(n) - Z'(n_1))\right) = 0$ for all h, so $Z'(n) = Z'(n_1)$ by Lemma A.35. Let Z' be this matrix. Then $\Delta(n)\,V = \eta_V - VZ'$ does not depend on n, and no row of V vanishes, so $\Delta(n)$ is one diagonal matrix Δ_{k+1} for all these n, with $\Delta_{k+1} = 0$ in a GELU block. So $\eta_V = \Delta_{k+1}V + VZ'$. In a gated block, $\dot{z}^v(n) = -n\,W_v\Delta_{k+1}$ on an open set of rows n, that is, $n\,(ZW_v + \eta_v + W_v\Delta_{k+1}) = 0$: $\eta_v = -ZW_v - W_v\Delta_{k+1}$.

Step 3: the rest. In the case (+), $\mathcal{B}^\infty = \mathcal{B}_{>k+1}$, and the second part of (44) is (21) for $k+1$, with Z' and η'. By (D_{k+1}), $Z' \in \mathfrak{g}$ and η' is of the form (24), or (22) for GELU blocks, with $Z_{k+1} = Z'$. With Steps 1 and 2 this is (24), or (22), for k.

In the case (n), where the blocks are gated, (44) is an identity for the frozen network. The frozen network does not change along some directions along which the network does, for example along every direction of the weights of a hidden unit that is not active, so (44) does not determine η'. Let $\eta^\flat$ be the direction (24) for k with $Z_k = Z$, with Δ_{k+1} as in Step 2, with $Z_{k+1} = Z'$, and with $Z_\ell = 0$ and $\Delta_\ell = 0$ for all later ℓ; here $\mathfrak{g}$ contains every matrix. By the converse of the claim, Z and $\eta^\flat$ satisfy (21) for k, and so do Z and η. So the matrix 0 and the direction $\eta - \eta^\flat$ satisfy it. By Steps 1 and 2, $\eta - \eta^\flat$ vanishes at the weights of block $k+1$. Let η'' be its entries at the later weights. Then (21) for k reads

$$\mathrm{D}_\theta\mathcal{B}_{>k+1}\big(g_{k+1}(h)\big)\,\eta'' = 0 \qquad \text{for all rows } h.$$

By (R3) the values of g_{k+1} contain an open set, and the left side is real-analytic, so $\mathrm{D}_\theta\mathcal{B}_{>k+1}(h)\,\eta'' = 0$ for all rows h. This is (21) for $k+1$ with the matrix 0. By (D_{k+1}), η'' is of the form (24) for $k+1$ with $Z_{k+1} = 0$. So $\eta = \eta^\flat + \eta''$ is of the form (24) for k. □

The parameters $\theta^\dagger$. Let $\mathbf{1}$ be the column of d_{model} ones. Choose numbers $\mu_1, \mu_2, \cdots > 0$ such that $1, \mu_1, \mu_2, \ldots$ are linearly independent over $\mathbb{Q}$, for SwiGLU blocks, or such that $\mu_1^2, \mu_2^2, \ldots$ are, for GELU and GeGLU blocks. Almost every choice of finitely many numbers is of this kind. Let $\theta^\dagger$ be the parameters with

$$E = \begin{bmatrix} I \\ 0 \end{bmatrix} \in \mathbb{R}^{d_{\text{in}}\times d_{\text{model}}}, \qquad D = \begin{bmatrix} I & 0 \end{bmatrix} \in \mathbb{R}^{d_{\text{model}}\times d_{\text{out}}},$$

and, in every block ℓ, with $q = q_\ell$ hidden units,

$$W_g^\ell = \left[\, I \;\middle|\; -\mu_1\mathbf{1} \;\middle|\; \cdots \;\middle|\; -\mu_{q-d_{\text{model}}}\mathbf{1} \,\right], \qquad W^\ell_{\text{write}} = \begin{bmatrix} I \\ \mathbf{1}^\top \\ \vdots \\ \mathbf{1}^\top \end{bmatrix}, \qquad W_v^\ell = \left[\, \mathbf{1} \;\middle|\; \cdots \;\middle|\; \mathbf{1} \,\right],$$

where only a gated block has W_v^ℓ. So the gate column c_j is column j of I for $j \leq d_{\text{model}}$ and $c_j = -\mu_{j-d_{\text{model}}}\,\mathbf{1}$ for $j > d_{\text{model}}$, the write row a_j is row j of I for $j \leq d_{\text{model}}$ and $a_j = \mathbf{1}^\top$ for $j > d_{\text{model}}$, and in a gated block every value column is $e_j = \mathbf{1}$. Let $\mathbb{R}^{1\times d_{\text{model}}}_{>0}$ be the positive orthant, the set of the rows n with $n_i > 0$ for every i. It is an open convex cone that does not contain 0.

Lemma A.47 ($\theta^\dagger$ is good)**.** *$\theta^\dagger$ is good for the cones $\mathcal{C}_0 = \cdots = \mathcal{C}_L = \mathbb{R}^{1\times d_{\text{model}}}_{>0}$. For every block, $\mathcal{A}_\ell = \{1, \dots, d_{\text{model}}\}$, and $g^\circ_\ell(n) = n$ for a GELU block and $g^\circ_\ell(n) = (n\mathbf{1})\, n$ for a gated block, where $n\mathbf{1} = n_1 + \cdots + n_{d_{\text{model}}}$.*

Proof. (R1): E, D and every W^ℓ_{write} contain the identity matrix. (K1): for $n \in \mathbb{R}^{1\times d_{\text{model}}}_{>0}$, $nc_j = n_j > 0$ for $j \le d_{\text{model}}$ and $nc_j = -\mu_{j-d_{\text{model}}}\, n\mathbf{1} < 0$ for $j > d_{\text{model}}$. So $\mathcal{A}_\ell = \{1, \dots, d_{\text{model}}\}$, and $g^\circ_\ell(n) = \sum_{j\le d_{\text{model}}} n_j\, \iota_j(n)\, a_j$ is n for a GELU block and $(n\mathbf{1})\, n$ for a gated block, where $\iota_j(n) = n\mathbf{1}$. (K2): $g^\circ_\ell(n)$ is a positive multiple of n. Its derivative is the identity for a GELU block, and for a gated block the map $v \mapsto (n\mathbf{1})\, v + (v\mathbf{1})\, n$, with the determinant $2\,(n\mathbf{1})^{d_{\text{model}}} \neq 0$. (K3): no a_j and no e_j vanishes. For SwiGLU blocks, $\sum_j \kappa_j c_j = 0$ with integers κ_j says that $\kappa_j = \sum_{i>d_{\text{model}}} \kappa_i\, \mu_{i-d_{\text{model}}}$ for every $j \le d_{\text{model}}$, so every κ_j vanishes, by the choice of the μ_i. For GELU and GeGLU blocks, the entries of $\sum_j \kappa_j c_j c_j^\top = 0$ off the diagonal, which exist because $d_{\text{model}} \ge 2$, give $\sum_{i>d_{\text{model}}} \kappa_i\, \mu^2_{i-d_{\text{model}}} = 0$, so $\kappa_i = 0$ for $i > d_{\text{model}}$, and then the diagonal gives $\kappa_j = 0$ for $j \le d_{\text{model}}$. □

The great circle: $\varepsilon = 0$. We now treat the case (0), for $\tanh$ and GELU blocks. The identity (27) then holds only on the sphere $\mathbb{S}$. We continue it along a great circle of $\mathbb{S}$ to complex angles, on which the inputs of the hidden units go to infinity.

RMSNorm at complex rows. For $h \in \mathbb{C}^{1\times d_{\text{model}}}$ with $\operatorname{Re}\langle h, h\rangle > 0$ let

$$N_\varepsilon(h) \;=\; \frac{\sqrt{d_{\text{model}}}\, h}{\sqrt{\langle h, h\rangle}},$$

with the principal square root. It is holomorphic, $N_\varepsilon(\lambda h) = N_\varepsilon(h)$ for $\lambda > 0$, and

$$\langle n, n\rangle = d_{\text{model}} \quad\Longrightarrow\quad N_\varepsilon(n) = n, \qquad \mathrm{D}N_\varepsilon(n)\, v = v - \frac{\langle v, n\rangle}{d_{\text{model}}}\, n. \tag{45}$$

For $\varkappa \in (0, 1]$ let

$$\mathbb{C}_\varkappa \;=\; \big\{u + \mathrm{i}u' \;:\; u, u' \in \mathbb{R}^{1\times d_{\text{model}}},\ \|u'\| < \varkappa\|u\|\big\},$$

an open cone in $\mathbb{C}^{1\times d_{\text{model}}}$ that contains every real row $h \neq 0$. On $\mathbb{C}_1$, $\operatorname{Re}\langle h, h\rangle = \|u\|^2 - \|u'\|^2 > 0$.

Lemma A.48 ($\mathcal{B}_{>k}$ on a complex cone)**.** *Let θ satisfy (R1) and (R2). For every k there is a $\varkappa_k \in (0, 1]$ such that $\mathcal{B}_{>k}$, with its derivatives $\mathrm{D}\mathcal{B}_{>k}$ and $\mathrm{D}_\theta\mathcal{B}_{>k}$, extends to a holomorphic map on $\mathbb{C}_{\varkappa_k}$ with $\mathcal{B}_{>k}(\lambda h) = \mathcal{B}_{>k}(h)$ for $\lambda > 0$. The number $\varkappa_k$ depends only on the weights of the blocks $k+1, \dots, L$ and on D.*

Proof. $\mathcal{B}_{>L}(h) = N_\varepsilon(h)\, D$ is holomorphic on $\mathbb{C}_1$: $\varkappa_L = 1$. Let $k < L$. The map g_{k+1} is holomorphic near $\mathbb{S}$ and maps the compact set $\mathbb{S}$ into the real rows $h \neq 0$, by (R2), hence into the open set $\mathbb{C}_{\varkappa_{k+1}}$. So there is a $\rho > 0$ such that $n \mapsto \mathcal{B}_{>k+1}(g_{k+1}(n))$ is holomorphic on the set of the $n \in \mathbb{C}^{1\times d_{\text{model}}}$ at distance less than ρ from $\mathbb{S}$. On the compact set of the $u + \mathrm{i}u'$ with $\|u\| = 1$ and $\|u'\| \le \frac{1}{2}$, N_ε is uniformly continuous, and $N_\varepsilon(u) \in \mathbb{S}$. So there is a $\varkappa_k$ such that N_ε maps $\mathbb{C}_{\varkappa_k}$ into the points at distance less than ρ from $\mathbb{S}$, and $\mathcal{B}_{>k} = \big(\mathcal{B}_{>k+1} \circ g_{k+1}\big) \circ N_\varepsilon$ is holomorphic on $\mathbb{C}_{\varkappa_k}$. The weights enter holomorphically as well. □

The path. For orthonormal rows $\xi, \xi' \in \mathbb{R}^{1\times d_{\text{model}}}$ let $\nu = \xi + \mathrm{i}\xi'$, $\bar\nu = \xi - \mathrm{i}\xi'$ and

$$n(t) \;=\; \frac{\sqrt{d_{\text{model}}}}{2}\Big(t\,\nu + \frac{1}{t}\,\bar\nu\Big), \qquad t \in \mathbb{C}\setminus\{0\}.$$

As $\langle\nu, \nu\rangle = \langle\bar\nu, \bar\nu\rangle = 0$ and $\langle\nu, \bar\nu\rangle = 2$, we have $\langle n(t), n(t)\rangle = d_{\text{model}}$ for all t. For $|t| = 1$ the row $n(t)$ is real and runs through the great circle of ξ and ξ' in $\mathbb{S}$, with $n(1) = \sqrt{d_{\text{model}}}\,\xi$. For real $t \ge 1$,

$$n(t) \;=\; \sqrt{d_{\text{model}}}\,\Big(\frac{t + t^{-1}}{2}\,\xi + \mathrm{i}\,\frac{t - t^{-1}}{2}\,\xi'\Big). \tag{46}$$

In the notation of the proof of Lemma B.5, this is the great circle of ξ and ξ' at the angle $\vartheta = \mathrm{i}\log t$.

The identity on the path. With the notation of (25) and (26), let $\varkappa = \varkappa_{k+1}$. The same formulas define $\dot{z}(n) \in \mathbb{C}^{1\times q}$ for $n \in \mathbb{C}^{1\times d_{\text{model}}}$, and $\mathcal{L}(\tilde{w}, \tilde{w}'; n)$ for rows $\tilde{w}, \tilde{w}' \in \mathbb{C}^{1\times q}$ with $\tilde{w}V \in \mathbb{C}_{\varkappa}$. $\mathcal{L}$ is holomorphic in $\tilde{w}$, affine in $\tilde{w}'$ and polynomial in n.

Lemma A.49 (Continuation)**.** *In the case (0), let (27) hold, and let ξ, ξ' be orthonormal. Suppose that for every real $t \geq 1$ the function σ is holomorphic at every $n(t)\, c_j$ and $\sigma(n(t)W)\, V \in \mathbb{C}_{\varkappa}$. Then*

$$\mathcal{L}\big(\sigma(n(t)W),\ \sigma'(n(t)W);\ n(t)\big) \ = \ 0 \qquad \textit{for all real } t \geq 1.$$

Proof. The left side is a holomorphic function of t on an open set that contains $[1, \infty)$, hence on a connected open neighborhood of $[1, \infty)$. This neighborhood contains an arc of the unit circle around $t = 1$, on which $n(t) \in \mathbb{S}$ and the function vanishes by (27). So it vanishes on the whole neighborhood. □

The step for $\tanh$ **on the path.** Let $\sigma = \tanh$. Consider, for a block ℓ with $W = W^{\ell}_{\text{read}}$, its columns c_j, $V = W^{\ell}_{\text{write}}$ and $\varkappa_\ell$ of Lemma A.48:

(S1) any two rows of V are linearly independent;
(S2) the columns of W are linearly independent over $\mathbb{Q}$;
(S3) there are a unit row ξ_0 and a number $\rho \in (0, 1]$ such that every $\xi_0 c_j \neq 0$ and, with s_j the sign of $\xi_0 c_j$ and $\bar{s} = (s_1, \ldots, s_q)$: the rows $\tilde{w}V$ with $|\tilde{w}_j - s_j| \leq \rho$ for all j lie in $\mathbb{C}_{\varkappa_\ell}$, and $e^{-2\sqrt{d_{\text{model}}}\,|\xi_0 c_j|} < \rho/3$ for every j.

(S3) asks for large reads: on the row $\sqrt{d_{\text{model}}}\,\xi_0$ every hidden unit of the block is close to saturation. It then stays so on the whole path (46), and the outputs of the block stay in a small set, on which the rest of the network is holomorphic.

Lemma A.50 (Separation on the path)**.** *Let $u_1, \ldots, u_q > 0$ be linearly independent over $\mathbb{Q}$, let $u'_1, \ldots, u'_q$ be real, and for $t \geq 1$ let*

$$\zeta_j(t) \ = \ \exp\Big(- \big(t + t^{-1}\big)\, u_j - \mathrm{i}\,\big(t - t^{-1}\big)\, u'_j \Big).$$

Let $\rho_0 > \max_j e^{-2u_j}$, and let $\mathcal{T}_{-K}, \ldots, \mathcal{T}_K$ be holomorphic maps from the polydisc $\{\zeta \in \mathbb{C}^q : |\zeta_j| < \rho_0\}$ to $\mathbb{C}^{1\times d_{\text{out}}}$. If

$$\sum_{i=-K}^{K} t^i\, \mathcal{T}_i\big(\zeta_1(t), \ldots, \zeta_q(t)\big) \ = \ 0 \qquad \textit{for all } t \geq 1,$$

then $\mathcal{T}_i = 0$ for every i.

Proof. This is the proof of Lemma A.40, with a phase and with negative powers. On the polydisc $\mathcal{T}_i(\zeta) = \sum_\alpha b_{i,\alpha}\, \zeta^\alpha$, over the multi-indices α of nonnegative integers, and the series converges absolutely at $|\zeta_j| = e^{-2u_j}$. Let $\mu_\alpha = \sum_j \alpha_j u_j$ and $p_\alpha(t) = \sum_i t^i\, b_{i,\alpha}$. Then $|\zeta(t)^\alpha| = e^{-(t+t^{-1})\mu_\alpha}$, and the hypothesis reads $\sum_\alpha p_\alpha(t)\, \zeta(t)^\alpha = 0$. The numbers μ_α are pairwise different, and finitely many lie below any bound. Suppose that some p_α is not zero, let α^* be the one among these with the smallest μ_α, say μ^*, and let $\gamma > 0$ be the distance from μ^* to the next larger μ_α. For $t \geq 1$, as $t + t^{-1} \geq 2$,

$$|p_{\alpha^*}(t)| \ \leq \ \sum_{\mu_\alpha > \mu^*} |p_\alpha(t)|\, e^{-(t+t^{-1})(\mu_\alpha - \mu^*)} \ \leq \ t^K\, e^{-(t+t^{-1}-2)\gamma}\, e^{2\mu^*} \sum_i \sum_\alpha |b_{i,\alpha}|\, e^{-2\mu_\alpha},$$

which tends to 0 faster than every power of $1/t$. But $p_{\alpha^*}(t) = \sum_{i=-K}^{K} t^i\, b_{i,\alpha^*}$ is not zero, so $|p_{\alpha^*}(t)|$ is at least a positive constant times t^{-K} for large t, a contradiction. □

Lemma A.51 (A change of the reads moves the output)**.** *In the case (0), let θ be regular with (S1) for block $k+1$, let $V = W^{k+1}_{\text{write}}$ have the rows $a_1, \ldots, a_q$, and let $b \in \mathbb{C}^{1\times q}$. If*

$$\mathrm{D}\mathcal{B}_{>k+1}(\tilde{w}V)\, \Big(\big((1 - \tilde{w}^2) \odot b \big)\, V \Big) \ = \ 0$$

for all $\tilde{w}$ in a nonempty open subset of $\mathbb{R}^{1\times q}$, where $1-\tilde{w}^2$ is the row with the entries $1-\tilde{w}_j^2$, then $b=0$.

Proof. At real $\tilde{w}$ the derivative is a real linear map, so the real and the imaginary part of b satisfy the hypothesis: let b be real. By Lemma A.35, for $\tilde{w}$ in an open dense subset of the open set, the row $\sum_j b_j(1-\tilde{w}_j^2)\,a_j$ is a multiple of the row $\sum_l \tilde{w}_l\,a_l$. So all 2×2 minors of these two rows vanish on an open set, hence as polynomials in $\tilde{w}$. Their part of degree 1 in $\tilde{w}$ shows that bV is a multiple of every a_l, so $bV=0$ by (S1). The coefficient of $\tilde{w}_j^2\,\tilde{w}_l$ with $l\neq j$ shows that $b_j a_j$ is a multiple of a_l, so $b_j=0$ by (S1). □

Lemma A.52 (The step for $\tanh$ on the path)**.** *Let $\sigma=\tanh$, in the case (0), let θ be regular, and let block $k+1$ satisfy (S1) to (S3). Then (D_{k+1}) implies (D_k).*

Proof. Let Z and η satisfy (21) for k, so that (27) holds, with the notation above. By (S3) there is an open set $\mathcal{P}$ of unit rows around ξ_0 such that ξc_j has the sign s_j and $e^{-2\sqrt{d_{\text{model}}}\,|\xi c_j|}<\rho/3$ for every $\xi\in\mathcal{P}$ and every j. By (S2), for integers κ_j that are not all zero, $\sum_j \kappa_j\,\xi c_j=\xi\sum_j\kappa_j c_j$ vanishes only for ξ in a hyperplane. Let $\mathcal{P}'\subset\mathcal{P}$ be the dense set of the ξ for which $\xi c_1,\dots,\xi c_q$ are linearly independent over $\mathbb{Q}$.

The path. Fix $\xi\in\mathcal{P}'$ and a unit row ξ' orthogonal to ξ, and let $n(t)$ be the path (46). Let

$$\zeta_j(t)\;=\;e^{-2s_j\,n(t)\,c_j}\;=\;\exp\Big(-\big(t+t^{-1}\big)\sqrt{d_{\text{model}}}\,|\xi c_j|-\mathrm{i}\,\big(t-t^{-1}\big)\sqrt{d_{\text{model}}}\,s_j\,\xi' c_j\Big),$$

so that $|\zeta_j(t)|<\rho/3$ for $t\geq 1$, and

$$\tanh\big(n(t)\,c_j\big)=s_j\,\frac{1-\zeta_j(t)}{1+\zeta_j(t)},\qquad \tanh'\big(n(t)\,c_j\big)=1-\tanh^2\big(n(t)\,c_j\big).$$

For ζ in the polydisc $\mathbb{D}=\{|\zeta_j|<\rho/3\}$ let $\tilde{w}(\zeta)$ be the row with the entries $s_j\frac{1-\zeta_j}{1+\zeta_j}$. As $|\tilde{w}_j(\zeta)-s_j|=\frac{2|\zeta_j|}{|1+\zeta_j|}\leq 3|\zeta_j|<\rho$, (S3) gives $\tilde{w}(\zeta)\,V\in\mathbb{C}_{\varkappa}$. The function $\tanh$ is holomorphic at $n(t)\,c_j$, as $\zeta_j(t)\neq -1$. So Lemma A.49 applies:

$$\mathcal{L}\big(\tilde{w}(\zeta(t)),\;1-\tilde{w}(\zeta(t))^2;\;n(t)\big)\;=\;0\qquad\text{for all } t\geq 1. \tag{47}$$

The powers of t. By (25), $\dot{z}(n(t))=t^3\dot{z}^{(3)}+t\,\dot{z}^{(1)}+t^{-1}\dot{z}^{(-1)}+t^{-3}\dot{z}^{(-3)}$ with rows $\dot{z}^{(i)}\in\mathbb{C}^{1\times q}$ that do not depend on t, and

$$\begin{aligned}\dot{z}^{(3)}&=-\frac{\sqrt{d_{\text{model}}}}{8}\,\langle\nu Z,\nu\rangle\,\nu W,\\ \dot{z}^{(1)}&=\frac{\sqrt{d_{\text{model}}}}{2}\Big(\nu\,(ZW+\eta_W)-\frac{1}{4}\Big(\big(\langle\nu Z,\bar{\nu}\rangle+\langle\bar{\nu}Z,\nu\rangle\big)\,\nu+\langle\nu Z,\nu\rangle\,\bar{\nu}\Big)W\Big).\end{aligned} \tag{48}$$

$\mathcal{L}$ is affine in $\dot{z}$, so (47) reads $\sum_i t^i\,\mathcal{T}_i(\zeta(t))=0$ with the maps

$$\begin{aligned}\mathcal{T}_i(\zeta)&=\mathrm{D}\mathcal{B}\big(\tilde{w}(\zeta)V\big)\,\Big(\big((1-\tilde{w}(\zeta)^2)\odot\dot{z}^{(i)}\big)\,V\Big)\qquad(i=\pm1,\pm3),\\ \mathcal{T}_0(\zeta)&=\mathrm{D}\mathcal{B}\big(\tilde{w}(\zeta)V\big)\,\big(\tilde{w}(\zeta)\,\eta_V\big)+\mathrm{D}_\theta\mathcal{B}\big(\tilde{w}(\zeta)V\big)\,\eta',\end{aligned}$$

which are holomorphic on $\mathbb{D}$. By Lemma A.50, with $u_j=\sqrt{d_{\text{model}}}\,|\xi c_j|$ and $\rho_0=\rho/3$, every $\mathcal{T}_i$ vanishes on $\mathbb{D}$. For real $\zeta\in\mathbb{D}$ the rows $\tilde{w}(\zeta)$ fill a neighborhood of $\bar{s}$ in $\mathbb{R}^{1\times q}$.

(a) $\mathcal{T}_3=0$ gives Z. Here $\dot{z}^{(3)}$ is the number $-\frac{\sqrt{d_{\text{model}}}}{8}\langle\nu Z,\nu\rangle$ times the row νW, which is not zero. By Lemma A.51, $\langle\nu Z,\nu\rangle=0$. Its imaginary part is $\xi\,(Z+Z^\top)\,\xi'^\top=0$. This holds for every unit row ξ' orthogonal to ξ, so $\xi\,(Z+Z^\top)$ is a multiple of ξ, for every $\xi\in\mathcal{P}'$ and then, by continuity, for every ξ in the open set $\mathcal{P}$. A matrix for which every row of an open set is an eigenvector is a multiple of the identity. So $Z+Z^\top=2\beta_k I$, and $Z=Y_k+\beta_k I$ with Y_k antisymmetric.

(b) $\mathcal{T}_1=0$ gives η_W. Now $\langle\nu Z,\nu\rangle=0$ and $\langle\nu Z,\bar{\nu}\rangle+\langle\bar{\nu}Z,\nu\rangle=2\beta_k\langle\nu,\bar{\nu}\rangle=4\beta_k$, so by (48) $\dot{z}^{(1)}=\frac{\sqrt{d_{\text{model}}}}{2}\,\nu\,(Y_kW+\eta_W)$. By Lemma A.51, $\dot{z}^{(1)}=0$, so $\xi\,(Y_kW+\eta_W)=0$ for every $\xi\in\mathcal{P}'$, and $\eta_W=-Y_kW$.

(c) $\mathcal{T}_0 = 0$ gives the rest. $\mathrm{D}\mathcal{B}(\tilde{w}V)\,(\tilde{w}\,\eta_V) + \mathrm{D}_\theta\mathcal{B}(\tilde{w}V)\,\eta' = 0$ for real $\tilde{w}$ near $\bar{s}$. The left side is real-analytic on the rows $\tilde{w} \in \mathbb{R}^{1\times q}$ with $\tilde{w}V \neq 0$. These form a connected set, the complement of a subspace of codimension $d_{\mathrm{model}} \geq 2$, so the identity holds on all of it. By Lemma A.36 with $A = V$, $\eta_V = VZ'$ and (21) holds for $k+1$ with Z' and η'. By (D_{k+1}), $Z' = Y_{k+1} + \beta_{k+1}I$ and η' is as in (23). With (a) and (b) this is (23) for k. □

The step for GELU on the path. Consider, for a block ℓ with $W = W^\ell_{\mathrm{read}}$, its columns c_j, $V = W^\ell_{\mathrm{write}}$, its rows a_j, and $\varkappa_\ell$ of Lemma A.48:

(P1) the columns of W are pairwise different, and the matrix WV is invertible;

(P2) there is a unit row ξ_0 such that, for every unit row ξ' orthogonal to ξ_0,

$$|\xi' c_j| < \xi_0 c_j \ \text{ for every } j, \qquad \|\xi'\,WV\| < \frac{\varkappa_\ell}{2}\,\|\xi_0\,WV\|,$$
$$\sum_j e^{-d_{\mathrm{model}}(\xi_0 c_j)^2/2}\,\|a_j\| \ < \ \frac{\varkappa_\ell\sqrt{d_{\mathrm{model}}}}{8}\,\|\xi_0\,WV\|.$$

(P2) asks for reads that are large and close to a matrix of rank one: every column of W is close to a positive multiple of $\xi_0^\top$. Then every hidden unit is close to linear on the whole path, the output of the block is close to $n(t)\,WV$, and its imaginary part stays small against its real part, so that the rest of the network is holomorphic along the path (Lemma A.48).

Lemma A.53 (The step for GELU on the path)**.** *Let σ be GELU, in the case (0), let θ be regular, and let block $k+1$ satisfy (P1) and (P2). Then (D_{k+1}) implies (D_k).*

Proof. Let Z and η satisfy (21) for k, so that (27) holds, with the notation above. Both sides of every inequality of (P2), the left sides as suprema over ξ', are continuous in ξ_0. So there is an open set $\mathcal{P}$ of unit rows around ξ_0 such that (P2) holds for every $\xi \in \mathcal{P}$ in the place of ξ_0.

The path. Fix $\xi \in \mathcal{P}$ and a unit row ξ' orthogonal to ξ, let $n(t)$ be the path (46), and

$$z_j(t) = n(t)\,c_j, \qquad r_j = (\xi c_j)^2 - (\xi' c_j)^2 > 0, \qquad \delta_j(t) = z_j - \sigma(z_j), \qquad \delta_j'(t) = 1 - \sigma'(z_j).$$

Let $z(t), \delta(t), \delta'(t)$ be the rows with these entries. For $t \geq 1$, by (46) and (P2),

$$\begin{aligned} &\mathrm{Re}\,z_j \geq \sqrt{d_{\mathrm{model}}}\,\xi c_j > 0, \qquad |\mathrm{Im}\,z_j| \leq \mathrm{Re}\,z_j, \\ &\mathrm{Re}\,\big(z_j^2\big) = d_{\mathrm{model}}\,(\xi c_j)^2 + \frac{d_{\mathrm{model}}}{4}\,\big(t - t^{-1}\big)^2\,r_j. \end{aligned} \tag{49}$$

By Lemma A.42, $\delta_j = \varphi(z_j)\,z_j\,\mathrm{mr}(z_j)$ and $\delta_j' = \varphi(z_j)\,\big(\mathrm{mr}(z_j) - z_j\big)$, with $|\mathrm{mr}(z_j)| \leq 1/\mathrm{Re}\,z_j$ and $|z_j| \leq \sqrt{2}\,\mathrm{Re}\,z_j$. So, with a constant κ,

$$|\delta_j(t)| \ \leq \ e^{-d_{\mathrm{model}}(\xi c_j)^2/2}\,e^{-\frac{d_{\mathrm{model}}}{8}(t-t^{-1})^2 r_j}, \qquad |\delta_j'(t)| \ \leq \ \kappa\,t\,e^{-\frac{d_{\mathrm{model}}}{8}(t-t^{-1})^2 r_j}. \tag{50}$$

The real and the imaginary part of $n(t)\,WV$ are $\sqrt{d_{\mathrm{model}}}\,\frac{t+t^{-1}}{2}\,\xi WV$ and $\sqrt{d_{\mathrm{model}}}\,\frac{t-t^{-1}}{2}\,\xi' WV$. By (P2), the second is shorter than $\frac{\varkappa}{2}$ times the first, and by (50) $\|\delta(t)\,V\| \leq \sum_j |\delta_j|\,\|a_j\|$ is at most $\frac{\varkappa}{8}$ times the norm of the first. Let $\varrho = \varkappa\sqrt{d_{\mathrm{model}}}\,\|\xi WV\|/\big(16\sum_j \|a_j\|\big)$. For a row $\delta \in \mathbb{C}^{1\times q}$,

$$|\delta_j| \leq \varrho\,t \ \text{ for all } j, \ \text{ or } \delta = \delta(t) \qquad \Longrightarrow \qquad \big(z(t) - \delta\big)\,V = n(t)\,WV - \delta V \ \in \mathbb{C}_\varkappa, \tag{51}$$

because then $\|\delta V\|$ is at most $\frac{\varkappa}{8}$ times the norm of the real part of $n(t)\,WV$, and $\frac{\varkappa/2+\varkappa/8}{1-\varkappa/8} < \varkappa$. As $\sigma(z) = z - \delta(t)$ and $\sigma'(z) = 1 - \delta'(t)$ on the path, Lemma A.49 applies:

$$\mathcal{L}\big(z(t) - \delta(t),\ 1 - \delta'(t);\ n(t)\big) \ = \ 0 \qquad \text{for all } t \geq 1. \tag{52}$$

The expansion. For fixed t, by (51) the map $(\delta, \delta') \mapsto \mathcal{L}\big(z(t) - \delta, 1 - \delta'; n(t)\big)$ is holomorphic in the rows δ with $|\delta_j| \leq \varrho\,t$ and affine in the row δ'. We expand it in powers of δ and δ'. Its value at 0 and its first derivatives are

$$\mathcal{L}_0(t) = \mathcal{L}\big(z(t), 1; n(t)\big), \qquad \mathcal{L}_j(t) = \frac{\partial\mathcal{L}}{\partial\tilde{w}_j}\big(z(t), 1; n(t)\big), \qquad \mathcal{L}_j'(t) = \frac{\partial\mathcal{L}}{\partial\tilde{w}_j'}\big(z(t), 1; n(t)\big),$$

so that (52) is

$$\mathcal{L}_0(t) \;-\; \sum_j \Big[\mathcal{L}_j(t)\,\delta_j(t) + \mathcal{L}'_j(t)\,\delta'_j(t)\Big] \;+\; \big(\text{terms with two or more factors}\big) \;=\; 0, \qquad (53)$$

where the factors are entries of $\delta(t)$ and $\delta'(t)$. We use two facts about the coefficients. First, every coefficient is meromorphic at infinity. Indeed, it is built from derivatives of $\mathcal{B}$ at $z(t)\,V = n(t)\,WV$ and from polynomials in $n(t)$, and

$$n(t)\,WV \;=\; \sqrt{d_{\text{model}}}\;\frac{t+t^{-1}}{2}\;\Big(\xi\,WV + \mathrm{i}\,\frac{1-t^{-2}}{1+t^{-2}}\;\xi' WV\Big),$$

where the row in the bracket is holomorphic in $1/t$ near 0, with values in $\mathbb{C}_{\varkappa/2}$, and a derivative of order i of $\mathcal{B}$ is homogeneous of degree $-i$. Second, on the rows $u + \mathrm{i}u' \in \mathbb{C}_{3\varkappa/4}$ with $\|u\| = 1$, $\mathrm{D}\mathcal{B}$ and $\mathrm{D}_\theta\mathcal{B}$ are bounded. So by Cauchy's estimates on the polydisc of (51), the coefficient of a product of i factors δ_j and of at most one factor δ'_j is at most a fixed power of t times $(\varrho\, t)^{-i}$. With (50), the series (53) therefore converges absolutely for large t, and the sum of any set of its terms is at most a power of t times $e^{-\frac{d_{\text{model}}}{8}(t-t^{-1})^2 r}$, where r is the smallest sum of the numbers r_j of the factors of a term of the set.

Step 1: the term without δ gives η'. By (53), $\mathcal{L}_0(t)$ tends to 0 faster than every power of $1/t$. It is meromorphic at infinity, so $\mathcal{L}_0(t) = 0$ for large t, and then for all $t \geq 1$, because it is holomorphic near $[1,\infty)$. At $t = 1$, where $n(t)$ is the row $n = \sqrt{d_{\text{model}}}\,\xi$, this reads, by (26), (25) and (20),

$$0 \;=\; \mathcal{L}\big(nW, 1; n\big) \;=\; \mathrm{D}\mathcal{B}(n\,WV)\,\Big(n\,\big(Z\,WV + \eta_W V + W\eta_V\big)\Big) + \mathrm{D}_\theta\mathcal{B}(n\,WV)\,\eta'.$$

The right side is real-analytic and homogeneous of degree 0 in the row $n \neq 0$, and it vanishes for ξ in the open set $\mathcal{P}$, hence for all $n \neq 0$. With $h = n\,WV$, which takes every value by (P1), this is (21) for $k+1$ with

$$Z' = (WV)^{-1}\,\big(Z\,WV + \eta_W V + W\eta_V\big) \qquad (54)$$

and η'. By (D_{k+1}), $Z' = Y_{k+1} + \beta' I$ and η' is as in (23), for antisymmetric $Y_{k+1}, \dots, Y_L$ and numbers $\beta', \beta_{k+2}, \dots, \beta_L$.

Step 2: the slowest hidden unit gives Z. On an open dense set of pairs (ξ, ξ') the numbers $r_1, \dots, r_q$ are pairwise different: $r_i = r_j$ on an open set of pairs would make $c_i c_i^\top - c_j c_j^\top$ a multiple of the identity, which is impossible for $c_i \neq \pm c_j$, and $c_i \neq c_j$ by (P1), $c_i \neq -c_j$ by (P2). Take a hidden unit j and a nonempty open set of pairs, with $\xi \in \mathcal{P}$, on which $r_j < r_i$ for all $i \neq j$, and let $\gamma = \min\big(r_j, \min_{i\neq j}(r_i - r_j)\big)$. By Step 1, all terms of (53) other than the two of unit j are, together, at most a power of t times $e^{-\frac{d_{\text{model}}}{8}(t-t^{-1})^2(r_j+\gamma)}$. Dividing by $\varphi(z_j)$, whose absolute value is a constant times $e^{-\frac{d_{\text{model}}}{8}(t-t^{-1})^2 r_j}$ by (49),

$$\big(z_j\,\mathcal{L}_j + \mathcal{L}'_j\big)\;\mathrm{mr}(z_j) \;-\; z_j\,\mathcal{L}'_j \;\longrightarrow\; 0 \qquad \text{faster than every power of } 1/t.$$

Here $z_j(t) \;=\; \frac{\sqrt{d_{\text{model}}}}{2}\big(t\,\nu c_j + \frac{1}{t}\,\bar\nu c_j\big)$ with $\mathrm{Re}(\nu c_j) \;=\; \xi c_j \;>\; 0$, and $z_j\mathcal{L}_j + \mathcal{L}'_j$ and $z_j\mathcal{L}'_j$ are meromorphic at infinity. By Lemma A.43 both vanish, so

$$\mathcal{L}'_j(t) = 0 \quad \text{and} \quad \mathcal{L}_j(t) = 0 \qquad \text{for large } t, \text{ hence for all } t \geq 1. \qquad (55)$$

By (26), with $\dot z_j$ the entry j of $\dot z$,

$$\mathcal{L}'_j(t) \;=\; \dot z_j\big(n(t)\big)\;\mathrm{D}\mathcal{B}\big(n(t)\,WV\big)\,a_j.$$

At $t = 1$ this is $\dot z_j(n)\,\mathrm{D}\mathcal{B}(n\,WV)\,a_j = 0$ at $n = \sqrt{d_{\text{model}}}\,\xi$, for ξ in an open set. By Lemma A.35, $\mathrm{D}\mathcal{B}(n\,WV)\,a_j \neq 0$ on an open dense set of n. So $\dot z_j(n) = 0$ on an open subset of $\mathbb{S}$, that is, by (25),

$$\|n\|^2\;n\,\big(Zc_j + \eta_{c_j}\big) \;=\; \langle nZ, n\rangle\;nc_j$$

on an open cone of rows n, hence for all n, where η_{c_j} is column j of η_W. For n orthogonal to c_j this gives $n\,(Zc_j + \eta_{c_j}) = 0$, so $Zc_j + \eta_{c_j} = \beta_k c_j$ for a number β_k, and then $\langle nZ, n\rangle = \beta_k \|n\|^2$ for all n. So $Z = Y_k + \beta_k I$ with Y_k antisymmetric.

Step 3: subtracting a symmetry. Let $\beta_{k+1} = \beta' - \beta_k$. The pair $Y_k + \beta_k I$ and $\big(-Y_k W,\ V(Y_{k+1} + \beta_{k+1} I),\ \eta'\big)$ is of the form (23), so it satisfies (21) for k. Subtracting it from Z and η leaves the pair 0 and $(\eta^\circ_W, \eta^\circ_V, 0)$ with

$$\eta^\circ_W = \eta_W + Y_k W, \qquad \eta^\circ_V = \eta_V - V\,(Y_{k+1} + \beta_{k+1} I),$$

which satisfies (21) for k as well, and by (54) and Step 1

$$\eta^\circ_W V + W \eta^\circ_V \;=\; WV\,(Y_{k+1} + \beta' I) - \beta_k\, WV - WV\,(Y_{k+1} + \beta_{k+1} I) \;=\; 0. \tag{56}$$

For this pair, (26) is $\mathcal{L}^\circ(\tilde w, \tilde w'; n) = \mathrm{D}\mathcal{B}(\tilde w V)\,\big((\tilde w' \odot n\eta^\circ_W)\,V + \tilde w\,\eta^\circ_V\big)$. It remains to show $\eta^\circ_W = 0$ and $\eta^\circ_V = 0$.

Step 4: the hidden units in the order of their rates. Take a nonempty open set of pairs (ξ, ξ'), with $\xi \in \mathcal{P}$, on which $r_{j_1} < r_{j_2} < \cdots < r_{j_q}$ for one order $j_1, \dots, j_q$ of the hidden units. We show by induction on i that column j_i of η°_W and row j_i of η°_V vanish. Let this hold for the units $j_1, \dots, j_{i-1}$, and let $j = j_i$. If the rows δ and δ' vanish outside the entries $j_1, \dots, j_{i-1}$, then $(1 - \delta') \odot n\eta^\circ_W = n\eta^\circ_W$ and $(z - \delta)\,\eta^\circ_V = nW\eta^\circ_V$, so by (56)

$$\mathcal{L}^\circ\big(z - \delta,\ 1 - \delta';\ n\big) \;=\; \mathrm{D}\mathcal{B}\big((z-\delta)V\big)\,\Big(n\,\big(\eta^\circ_W V + W\eta^\circ_V\big)\Big) \;=\; 0.$$

So in the expansion (53) of $\mathcal{L}^\circ$ every term has a factor $\delta_l(t)$ or $\delta'_l(t)$ of a unit l other than $j_1, \dots, j_{i-1}$. Other than the two terms of unit j, such a term has a factor of a unit $j_{i'}$ with $i' > i$, or a factor of unit j and a second factor. With $\gamma = \min(r_{j_1}, r_{j_{i+1}} - r_j)$, or $\gamma = r_{j_1}$ if $i = q$, we get (55) as in Step 2: $\mathcal{L}^{\circ\prime}_j(t) = 0$ and $\mathcal{L}^\circ_j(t) = 0$ for $t \geq 1$. Here, by (56),

$$\mathcal{L}^{\circ\prime}_j(t) = \big(n\,\eta^\circ_{c_j}\big)\ \mathrm{D}\mathcal{B}(n\,WV)\,a_j, \qquad \mathcal{L}^\circ_j(t) = \mathrm{D}\mathcal{B}(n\,WV)\,\eta^\circ_{a_j} \qquad \text{at } n = n(t),$$

with $\eta^\circ_{c_j}$ column j of η°_W and $\eta^\circ_{a_j}$ row j of η°_V. At $t = 1$, for ξ in an open set and $n = \sqrt{d_{\mathrm{model}}}\,\xi$: by Lemma A.35, $n\,\eta^\circ_{c_j} = 0$, so $\eta^\circ_{c_j} = 0$, and $\eta^\circ_{a_j}$ is a multiple of $n\,WV$ for an open set of rows n, so $\eta^\circ_{a_j} = 0$.

After $i = q$: $\eta_W = -Y_k W$ and $\eta_V = V\,(Y_{k+1} + \beta_{k+1} I)$. With Step 1 for η' and Step 2 for Z this is (23) for k. □

Parameters that satisfy all conditions, in the case (0). We choose the blocks from the last to the first. Let D and the blocks $\ell + 1, \dots, L$ be chosen, regular and with (S1) to (S3) for $\tanh$, or (P1) and (P2) for GELU. This fixes $\varkappa_\ell$ (Lemma A.48). We choose $(W, V) = (W^\ell_{\mathrm{read}}, W^\ell_{\mathrm{write}})$.

Conditions that fail on a set of measure zero of pairs (W, V). (R1), (S1), (S2) and (P1) are of this kind; (S2) fails on countably many proper subspaces. (R2): the triples (n, W, V) with $n \in \mathbb{S}$, $\sigma(nW) \neq 0$ and $\sigma(nW)\,V = 0$ form a manifold of dimension one less than that of the pairs (W, V), so their projection to the pairs has measure zero; and $\sigma(nW) = 0$ only if $nW = 0$, which requires W of rank less than d_{model}. (R3): the determinant in the proof of Lemma A.35 is real-analytic in (n, W, V), so (R3) fails on a set of measure zero once it holds for one pair. It holds for $W = [\,I \mid 0\,]$ and V with the rows of I and zero rows, at any $n \in \mathbb{S}$ with positive entries n_i: there $g_\ell(n) = \sigma(n)$, and $\mathrm{D}g_\ell(n)\,v = \mu\,g_\ell(n)$ gives $v_i = \mu\,\sigma(n_i)/\sigma'(n_i)$, so $0 = \langle v, n\rangle = \mu \sum_i n_i\,\sigma(n_i)/\sigma'(n_i)$ with a positive sum, hence $\mu = 0$ and $v = 0$.

Conditions that hold on a nonempty open set of pairs. (S3): take any V and W°, with the columns c°_j, and a unit row ξ_0 with $\xi_0 c^\circ_j \neq 0$ and $\bar s V \neq 0$ for the row $\bar s$ of the signs. As $\bar s V$ is a real nonzero row, there is a ρ as in (S3). Then $W = \lambda W^\circ$ satisfies (S3) for large $\lambda > 0$, with strict inequalities and a compact set of rows $\tilde w V$ inside the open cone, so (S3) holds near (W, V) as well. (P2): take a row α with positive entries, any V with $\alpha V \neq 0$, and $W^\circ = \xi_0^\top \alpha$. For W° the left sides of the first two inequalities of (P2) vanish and $\xi_0 c^\circ_j = \alpha_j > 0$, so these two inequalities hold for all $W = \lambda\,(W^\circ + W')$ with $\lambda > 0$ and W' small. For fixed small W', the third holds for large λ: its

left side decreases like $e^{-\kappa\lambda^2}$ with a constant $\kappa > 0$, and its right side grows like λ. All three are strict, so (P2) holds on a nonempty open set of pairs.

A nonempty open set minus a set of measure zero is not empty. So there are parameters θ^* that are regular and whose blocks all satisfy (S1) to (S3) for $\tanh$, or (P1) and (P2) for GELU.

Lemma A.54 (The domain)**.** *In the case (0), let X be a finite batch of nonzero inputs, and let $\mathcal{O}_X \subset \mathbb{R}^P$ be the set of parameters at which the hidden states $h_0, \dots, h_L$ of the skipless network are nonzero for every $x \in X$. Then $\mathcal{O}_X$ is open and connected, its complement has measure zero, and the outputs F_X on the batch are real-analytic on $\mathcal{O}_X$.*

Proof. This is Lemma A.27 for skipless networks. Let $\mathcal{O}^0$ be the set of parameters with $xE \neq 0$ for all $x \in X$, and for $\ell \geq 1$ let $\mathcal{O}^\ell$ be the set of parameters in $\mathcal{O}^{\ell-1}$ with $h_\ell \neq 0$ for all $x \in X$. The complement of $\mathcal{O}^0$ is a finite union of subspaces of codimension $d_{\text{model}} \geq 2$. Let $\ell \geq 1$ and $x \in X$. On $\mathcal{O}^{\ell-1}$, $h_\ell = \tilde{w}_\ell W^\ell_{\text{write}}$ with $\tilde{w}_\ell = \sigma\big(N_\varepsilon(h_{\ell-1}) W^\ell_{\text{read}}\big)$, and $h_{\ell-1}$ does not depend on the weights of block ℓ. For $\tanh$ and GELU, $\sigma(z) = 0$ only at $z = 0$. So $\tilde{w}_\ell = 0$ exactly where $N_\varepsilon(h_{\ell-1}) W^\ell_{\text{read}} = 0$, a closed submanifold of $\mathcal{O}^{\ell-1}$ of codimension at least $d_{\text{model}} \geq 2$, because the derivative with respect to W^ℓ_{read} is onto. Where $\tilde{w}_\ell \neq 0$, the derivative of h_ℓ with respect to W^ℓ_{write} is onto $\mathbb{R}^{1 \times d_{\text{model}}}$, so there the parameters with $h_\ell = 0$ form a closed submanifold of codimension $d_{\text{model}} \geq 2$. Removing finitely many closed submanifolds of codimension at least two, one after the other, from a connected open set leaves a connected open set and removes a set of measure zero. □

Proof of Theorem A.34. $\tanh$ *blocks in the cases (n) and (+).* Let θ satisfy (R1), let every block satisfy (T1) and (T2), and let every $W^\ell_{\text{read}} W^\ell_{\text{write}}$ be invertible. These conditions fail only on a set of measure zero; (T2) fails on countably many proper linear subspaces. Such θ are regular: $\mathrm{D}g_\ell(0)$ is the map $v \mapsto v\, W^\ell_{\text{read}} W^\ell_{\text{write}}$. By Lemma A.38 and Lemma A.41, Claim (D_k) holds at θ for $k = L$, then for $k = L - 1$, and so on down to $k = 0$. By Proposition A.37, the Jacobian at θ has the rank (18) on some finite batch, and by (R1) it has at most this rank on every batch. So the functional dimension is (18) at almost every θ.

GELU blocks in the case (+), and gated blocks in the cases (n) and (+). By Lemma A.47, $\theta^\dagger$ is good, so it is regular, and no value column vanishes. (D_L) holds at $\theta^\dagger$ (Lemma A.38), and by Lemma A.46, applied for $k = L - 1$, then for $k = L - 2$, and so on down to $k = 0$, so does (D_0). By Proposition A.37, the Jacobian at $\theta^\dagger$ has the rank (18) on some finite batch X. In the cases (n) and (+) the outputs on X are real-analytic in $\theta \in \mathbb{R}^P$, and so are the minors of the Jacobian. A minor of the size (18) does not vanish at $\theta^\dagger$, so it vanishes only on a set of measure zero. At generic parameters the Jacobian on X therefore has at least the rank (18), and it has at most this rank on every batch, because E and every W^ℓ_{write} have rank d_{model} and, in gated blocks, no value column vanishes. So the functional dimension is (18).

The case (0). By Lemma A.38 and Lemma A.52 or Lemma A.53, Claim (D_k) holds at θ^* for $k = L$, then for $k = L - 1$, and so on down to $k = 0$. By Proposition A.37 there is a finite batch X^*, with $xE \neq 0$ for all its inputs, on which the Jacobian at θ^* has the rank (18). By (R2), $\theta^* \in \mathcal{O}_{X^*}$. The sum of the squares of the minors of this size of the Jacobian of F_{X^*} is real-analytic on the connected open set $\mathcal{O}_{X^*}$ and does not vanish at θ^*, so its zero set has measure zero, and so has the complement of $\mathcal{O}_{X^*}$ (Lemma A.54). Hence the rank on X^* is at least (18) at almost every parameter. Where E and every W^ℓ_{write} have rank d_{model}, it is at most (18) on every batch. So at generic parameters the functional dimension is (18).

The gain. The number (18) is the bound of Theorem 4.4 for the skipless network, with $\mathcal{G} = \mathrm{CO}(d_{\text{model}})$ in the case (0). The residual network attains its own bound, $P - \dim \mathcal{G} - s$ with the same s (Theorem 4.9, and Theorem A.26 in the case (0)). The difference is $L \dim \mathcal{G}$. □

B The one-block theorem

This appendix states and proves the general form of Theorem 4.8. With one block, the skipless and the residual network are

$$x \mapsto f(xEW_{\text{read}})W_{\text{write}}D \qquad \text{and} \qquad x \mapsto \big(xE + f(xEW_{\text{read}})W_{\text{write}}\big)D$$

without normalization, and under pre-RMSNorm

$$x \mapsto N_\varepsilon\big(f(N_\varepsilon(xE)W_{\text{read}})W_{\text{write}}\big)D \qquad \text{and} \qquad x \mapsto N_\varepsilon\big(xEA + f(N_\varepsilon(xE)W_{\text{read}})W_{\text{write}}\big)D,$$

with the identity skip $A = I$. A learned skip $A \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ is full or upper triangular.

Theorem B.1 (One block)**.** *Let $d_{\text{model}} \geq 2$, let the input and output dimensions be at least d_{model}, let inputs range over all of $\mathbb{R}^{T \times d_{\text{in}}}$, and consider networks with one block: a* $\tanh$*, GELU, ReLU, ReLU^2, ReGLU, SwiGLU or GeGLU block, or an attention block of H heads of width $d_h \leq d_{\text{model}}$, causal or not, on sequences of at least two tokens.*
(i) *Let the network have no normalization, with or without biases, and let the block have width at least d_{model}, where the width of an attention block is Hd_h. For generic parameters the skipless network has functional dimension $P - 2d_{\text{model}}^2 - s$ and the residual network $P - d_{\text{model}}^2 - s$; with biases on the embedding, the block and the readout, counted in P, both values are $2d_{\text{model}}$ lower, and $2d_{\text{model}} + 2Hd_h$ lower for attention. Both networks satisfy the equality condition, and the skip raises the functional dimension by exactly d_{model}^2.*
(ii) *Let the network be pre-RMSNorm with $\varepsilon > 0$, without biases. For generic parameters the residual network has functional dimension $P - d_{\text{model}}(d_{\text{model}} - 1)/2 - s$, which is the equality condition. If the block has width at least d_{model}, the skip raises the functional dimension by at least $d_{\text{model}}(d_{\text{model}} - 1)/2$.*
(iii) *Let the network be pre-RMSNorm without biases, with a learned skip A, full or upper triangular. For generic parameters and $\varepsilon > 0$, its functional dimension exceeds that of the residual network of part (ii) by exactly $d_{\text{model}}(d_{\text{model}} + 1)/2$, and it satisfies the equality condition. For $\varepsilon = 0$ it exceeds that of the residual network with $\varepsilon = 0$ by at most $d_{\text{model}}(d_{\text{model}} + 1)/2 - 1$, with equality for ReLU, ReLU^2 and ReGLU blocks.*
(iv) *Let the network be pre-RMSNorm with $\varepsilon = 0$, without biases, with a ReLU, ReLU^2 or ReGLU block of width at least d_{model}. For generic parameters the residual network has functional dimension $P - d_{\text{model}}(d_{\text{model}} - 1)/2 - 1 - s$, and the skipless network at most $P - d_{\text{model}}^2 - d_{\text{model}}(d_{\text{model}} - 1)/2 - 1 - s$, with equality for ReLU and ReLU^2 blocks. So the skip raises the functional dimension by at least d_{model}^2, and by exactly d_{model}^2 for ReLU and ReLU^2 blocks. The redundancy is then that of the symmetries: besides those inside the block,* $\mathrm{GL}(d_{\text{model}})$ *at h_0 and* $\mathrm{CO}(d_{\text{model}})$ *at h_1 without the skip, and one* $\mathrm{CO}(d_{\text{model}})$ *with it.*
(v) *Let the network be pre-RMSNorm with $\varepsilon = 0$, without biases, with a* $\tanh$*, GELU, SwiGLU, GeGLU or attention block. For generic parameters the residual network has functional dimension $P - d_{\text{model}}(d_{\text{model}} - 1)/2 - 1 - s$, which is the equality condition with $\mathcal{G} = \mathrm{CO}(d_{\text{model}})$, and the skip raises the functional dimension by at least $d_{\text{model}}(d_{\text{model}} - 1)/2 + 1$, the dimension of* $\mathrm{CO}(d_{\text{model}})$.

In part (ii), for the skipless network we prove here only the bound $P - d_{\text{model}}(d_{\text{model}} - 1) - s$ of Theorem A.14(ii), which our checks attain (Appendix E). For $\tanh$, GELU, SwiGLU and GeGLU blocks, Theorem 4.12 shows that it is attained. In part (v), for $\tanh$ blocks and for GELU blocks, the skip raises the functional dimension by exactly $d_{\text{model}}(d_{\text{model}} - 1)/2 + 1$ (Theorem A.34). With parts (ii) and (iii), a learned skip under RMSNorm with $\varepsilon > 0$ adds at least d_{model}^2 over no skip if the block has width at least d_{model}, as an identity skip does without normalization. Part (i) extends to a learned absolute position embedding, $h_0 = xE + E_{\text{pos}}$, for an attention block without biases, with P counting E_{pos}, and to grouped-query attention (*Extensions of part (i)* below). With rotary embeddings, the change of basis between the queries and the keys of a head shrinks to an abelian group of dimension d_h (Tran et al., 2026, Theorem 4.2), which changes s; we do not treat this case.

Notation and flat directions. We write $F(\theta, x)$ for the output of the network at the parameters $\theta \in \mathbb{R}^P$ and the input x, and $\mathrm{D}_\theta F(\theta, x)$ for the Fréchet derivative of $F(\cdot, x)$ at θ, a linear map on $\mathbb{R}^P$. The derivative is written with an upright D so that it is not confused with the readout matrix D of the network. For a direction $\eta \in \mathbb{R}^P$, $\mathrm{D}_\theta F(\theta, x)\,\eta$ is the derivative of $\tau \mapsto F(\theta + \tau\eta, x)$ at $\tau = 0$. We write η_E, η_D, η_{read}, $\eta_{\text{write}}, \ldots$ for the entries of η at the positions of E, D, W_{read}, $W_{\text{write}}, \ldots$, and $\mathrm{D}_\theta Q\,\eta$ for the derivative along η of a matrix Q that depends on the parameters; for instance $\mathrm{D}_\theta M\,\eta = \eta_E D + E\eta_D$ for $M = ED$, by the product rule. Call η *flat* if $\mathrm{D}_\theta F(\theta, x)\,\eta = 0$

for every input x, and let $\mathcal{K}(\theta)$ be the space of flat directions. For the ReLU-type activations, $F(\theta, x)$ is differentiable in θ at the inputs x at which no hidden unit sits at its kink, which form an open set of full measure, and "every input" means every such input. With $\varepsilon = 0$, it also excludes the inputs at which a state that is normalized vanishes. For $\tanh$, GELU, SwiGLU, GeGLU and attention blocks, $\mathrm{D}_\theta F(\theta, x)\,\eta$ is real-analytic in x, so a direction that is flat on an open set of inputs is flat on all inputs, and the theorem holds for inputs in any open set. For the ReLU-type blocks it does not: on a ball of inputs that meets no kink the network is a polynomial in x. For a batch of inputs $x_1, \dots, x_n$, the Jacobian of $\theta \mapsto (F(\theta, x_1), \dots, F(\theta, x_n))$ is the matrix of the linear map $\eta \mapsto (\mathrm{D}_\theta F(\theta, x_1)\,\eta, \dots, \mathrm{D}_\theta F(\theta, x_n)\,\eta)$; it has P columns, and its kernel consists of the directions with $\mathrm{D}_\theta F(\theta, x_k)\,\eta = 0$ for the inputs of the batch. Adding inputs can only shrink this kernel, and a decreasing chain of subspaces of $\mathbb{R}^P$ stops, so some finite batch has kernel exactly $\mathcal{K}(\theta)$. By rank–nullity, the functional dimension at θ is therefore

$$P - \dim \mathcal{K}(\theta).$$

The theorem thus amounts to finding the flat directions. A statement holds for *generic* parameters if it fails only where some nonzero polynomial in the parameters, or some nonzero real-analytic function of them, vanishes, a set of measure zero.

Overview. For an MLP block, the parameters enter the function only through $M = ED$, $A = W_{\mathrm{write}}D$ and $C = EW_{\mathrm{read}}$; from part (iii) on, A is the learned skip again. Step 1 finds the directions that keep these matrices fixed: they are the directions in which the changes of basis move the parameters, two of them without the skip and one with it. Step 2 shows that a flat direction changes M, A and C only along the symmetries of the hidden units: each unit is isolated, by its kink for the ReLU-type activations, and for $\tanh$, GELU, SwiGLU and GeGLU by how fast it approaches its limit on rays $x = tn$ with $t \to \infty$. Step 3 counts. For an attention block the same steps run with one factorization per head, and inputs of two tokens turn Step 2 into the case of $\tanh$. For part (ii), large inputs first remove the final normalization and then separate the skip from the block. Part (iii) runs the same steps with a learned skip, which leaves a symmetric change of basis at h_0 in the input of the block; homogeneity or rays remove it.

Part (i) for MLP blocks

Reduction. With biases b_E, b_{read}, b_{write} and b_D, the skipless network computes $F(\theta, x) = f(xC + \beta)A + \gamma_0$ and the residual network $F(\theta, x) = xM + f(xC + \beta)A + \gamma$, where

$$M = ED, \qquad C = EW_{\mathrm{read}}, \qquad A = W_{\mathrm{write}}D,$$
$$\beta = b_E W_{\mathrm{read}} + b_{\mathrm{read}}, \qquad \gamma_0 = b_{\mathrm{write}}D + b_D, \qquad \gamma = b_E D + \gamma_0.$$

Without biases $\beta = 0$ and $\gamma = \gamma_0 = 0$. For a vanilla block let c_i be the columns of C, one per hidden unit, a_i the rows of A, β_i the entries of β, and $z_i = xc_i + \beta_i$, so that $F(\theta, x) = xM + \gamma + \sum_i \sigma(z_i)\,a_i$. For generic parameters E, D, W_{read} and W_{write} have rank d_{model}, every a_i is nonzero, and the c_i are nonzero and pairwise non-collinear: they lie in the column space of E, of dimension $d_{\mathrm{model}} \geq 2$, where two generic vectors are not collinear. For the gated blocks ReGLU, SwiGLU and GeGLU, in the same way, the value weights (e_i, β_i') of each unit, introduced below, have $e_i \neq 0$ and are not proportional to its gate weights (c_i, β_i).

Step 1: changes of basis. We need the following lemma.

> **Lemma B.2** (The directions that keep M, A and C)**.** *Let E, D, W_{read} and W_{write} have rank d_{model}. A direction η keeps C and A fixed to first order, $\mathrm{D}_\theta C\,\eta = 0$ and $\mathrm{D}_\theta A\,\eta = 0$, if and only if*
>
> $$\eta_E = EX, \qquad \eta_{\mathrm{read}} = -XW_{\mathrm{read}}, \qquad \eta_{\mathrm{write}} = W_{\mathrm{write}}Y, \qquad \eta_D = -YD$$
>
> *for some $X, Y \in \mathbb{R}^{d_{\mathrm{model}} \times d_{\mathrm{model}}}$, and it keeps M fixed as well, $\mathrm{D}_\theta M\,\eta = 0$, if and only if moreover $X = Y$. With biases, it keeps β and γ_0, respectively β and γ, fixed as well if and only*

if moreover $\eta_{b_E} = b_E X + g_0$, $\eta_{b_{\text{read}}} = -g_0 W_{\text{read}}$, $\eta_{b_{\text{write}}} = b_{\text{write}} Y + g_1$ *and* $\eta_{b_D} = -g_1 D$, *respectively* $\eta_{b_D} = -(g_0 + g_1)D$, *for some row vectors* $g_0, g_1 \in \mathbb{R}^{1 \times d_{\text{model}}}$. *The matrices* X, Y *and vectors* g_0, g_1 *are determined by the direction, so these directions form a space of dimension* $2d_{\text{model}}^2$ *without the skip and* d_{model}^2 *with it, and* $2d_{\text{model}}$ *more with biases.*

Proof. Since W_{read} has rank d_{model}, it has a right inverse R, with $W_{\text{read}} R = I$, and E, of rank d_{model} with d_{model} columns, has a left inverse. From $\mathrm{D}_\theta C\,\eta = \eta_E W_{\text{read}} + E\eta_{\text{read}} = 0$, multiplying on the right by R gives $\eta_E = EX$ with $X = -\eta_{\text{read}} R$; then $E\eta_{\text{read}} = -EXW_{\text{read}}$, and cancelling E on the left gives $\eta_{\text{read}} = -XW_{\text{read}}$. In the same way, D has a right inverse and W_{write} a left inverse, and $\mathrm{D}_\theta A\,\eta = \eta_{\text{write}} D + W_{\text{write}} \eta_D = 0$ gives $\eta_{\text{write}} = W_{\text{write}} Y$ and $\eta_D = -YD$. Conversely these directions give $\mathrm{D}_\theta C\,\eta = EXW_{\text{read}} - EXW_{\text{read}} = 0$ and $\mathrm{D}_\theta A\,\eta = 0$. In the residual network $\mathrm{D}_\theta M\,\eta = \eta_E D + E\eta_D = E(X - Y)D$, which vanishes if and only if $X = Y$, by cancelling E on the left and D on the right. With biases, set $g_0 = \eta_{b_E} - b_E X$ and $g_1 = \eta_{b_{\text{write}}} - b_{\text{write}} Y$. Then $\mathrm{D}_\theta \beta\,\eta = \eta_{b_E} W_{\text{read}} + b_E \eta_{\text{read}} + \eta_{b_{\text{read}}} = g_0 W_{\text{read}} + \eta_{b_{\text{read}}}$, which vanishes exactly when $\eta_{b_{\text{read}}} = -g_0 W_{\text{read}}$, and in the same way $\mathrm{D}_\theta \gamma_0\,\eta = g_1 D + \eta_{b_D}$ and, with $Y = X$, $\mathrm{D}_\theta \gamma\,\eta = (g_0 + g_1)D + \eta_{b_D}$. Finally $\eta_E = EX$ determines X, $\eta_D = -YD$ determines Y, and then η_{b_E} and $\eta_{b_{\text{write}}}$ determine g_0 and g_1. □

These are the directions in which the changes of basis $I + \tau X$ at h_0 and $I + \tau Y$ at h_1 move the parameters, and, with biases, the translations at the two hidden states (Lemma A.8).

Step 2: one hidden layer. We need a lemma on the function

$$G(\phi, x) = xM + \gamma + \textstyle\sum_i \sigma(z_i)\, a_i, \qquad z_i = xc_i + \beta_i,$$

of the matrices $\phi = \big(M, \gamma, (c_i, \beta_i, a_i)_i\big)$ themselves. For a direction ψ of ϕ, with entries ψ_M, ψ_γ, ψ_{c_i}, ψ_{β_i} and ψ_{a_i}, the Fréchet derivative of $G(\cdot, x)$ at ϕ gives

$$\mathrm{D}_\phi G(\phi, x)\,\psi = x\psi_M + \psi_\gamma + \sum_i \big[\sigma(z_i)\,\psi_{a_i} + \sigma'(z_i)\,(x\psi_{c_i} + \psi_{\beta_i})\,a_i\big]. \tag{57}$$

A gated unit computes $\sigma(z_i)\, w_i\, a_i$ with a gate $z_i = xc_i + \beta_i$ and a value $w_i = xe_i + \beta_i'$, where $\sigma(z) = \max(z, 0)$ for ReGLU, $z/(1 + e^{-z})$ for SwiGLU and $z\Phi(z)$ for GeGLU, so that ϕ also contains the e_i and β_i', and the terms of the unit in $\mathrm{D}_\phi G(\phi, x)\,\psi$ are $\sigma(z_i)\, w_i\, \psi_{a_i} + \sigma'(z_i)\,(x\psi_{c_i} + \psi_{\beta_i})\, w_i\, a_i + \sigma(z_i)\,(x\psi_{e_i} + \psi_{\beta_i'})\, a_i$.

Lemma B.3 (A flat direction only rescales units)**.** *Let every* $a_i \neq 0$ *and let the* c_i *be nonzero and pairwise non-collinear; for the gated blocks let also every* $e_i \neq 0$, *with* (e_i, β_i') *not proportional to* (c_i, β_i). *If* $\mathrm{D}_\phi G(\phi, x)\,\psi = 0$ *for all* x, *then* $\psi_M = 0$, $\psi_\gamma = 0$, *and for every unit* i*:*
for $\tanh$ *and GELU,* $\psi_{a_i} = 0$, $\psi_{c_i} = 0$ *and* $\psi_{\beta_i} = 0$*;*
for ReLU, $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i(c_i, \beta_i)$ *and* $\psi_{a_i} = -\lambda_i a_i$ *for some number* λ_i*;*
for ReLU^2*,* $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i(c_i, \beta_i)$ *and* $\psi_{a_i} = -2\lambda_i a_i$*;*
for ReGLU, $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i(c_i, \beta_i)$, $(\psi_{e_i}, \psi_{\beta_i'}) = -\alpha_i(e_i, \beta_i')$ *and* $\psi_{a_i} = (\alpha_i - \lambda_i)a_i$ *for some numbers* λ_i *and* α_i*;*
for SwiGLU and GeGLU, $\psi_{c_i} = 0$, $\psi_{\beta_i} = 0$, $(\psi_{e_i}, \psi_{\beta_i'}) = -\alpha_i(e_i, \beta_i')$ *and* $\psi_{a_i} = \alpha_i a_i$ *for some number* α_i.

Proof. Let $H_i = \{x : z_i = 0\}$, a hyperplane since $c_i \neq 0$. Since the c_i are pairwise non-collinear, the H_i are pairwise non-parallel, so every H_i contains points that lie on no other H_k. We use one fact from linear algebra: an affine function that vanishes on an open piece of H_i is a multiple of z_i, since it vanishes on all of H_i and H_i has codimension one. We write $J(x) = \mathrm{D}_\phi G(\phi, x)\,\psi$, a function of x alone, which vanishes by assumption.

ReLU. Here $\sigma' = \mathbf{1}[z > 0]$ jumps at 0. Near a point x_0 of H_i that lies on no other H_k, every term of (57) is continuous except $\mathbf{1}[z_i > 0](x\psi_{c_i} + \psi_{\beta_i})\, a_i$, which jumps by $(x_0\psi_{c_i} + \psi_{\beta_i})\, a_i$ as x crosses H_i at x_0. As $J = 0$ is continuous, this jump vanishes, and since $a_i \neq 0$, the affine function $x\psi_{c_i} + \psi_{\beta_i}$ vanishes on an open piece of H_i, so $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i(c_i, \beta_i)$. The terms

of unit i then add up to $\max(z_i,0)\,(\psi_{a_i}+\lambda_i a_i)$, since $z\,\mathbf{1}[z>0]=\max(z,0)$. Across H_i, the gradient of J with respect to x jumps by the matrix $c_i(\psi_{a_i}+\lambda_i a_i)$, which must vanish, so $\psi_{a_i}=-\lambda_i a_i$. With every unit removed in this way, $J(x)=x\psi_M+\psi_\gamma=0$ for all x gives $\psi_M=0$ and $\psi_\gamma=0$.

ReLU^2. The same argument one derivative higher: $\sigma'(z)=2\max(z,0)$ is continuous, and across H_i the gradient of $2\max(z_i,0)(x\psi_{c_i}+\psi_{\beta_i})\,a_i$ jumps by $2c_i(x_0\psi_{c_i}+\psi_{\beta_i})\,a_i$, which forces $(\psi_{c_i},\psi_{\beta_i})=\lambda_i(c_i,\beta_i)$. The terms of unit i then add up to $\max(z_i,0)^2(\psi_{a_i}+2\lambda_i a_i)$, whose second derivative jumps across H_i, so $\psi_{a_i}=-2\lambda_i a_i$.

ReGLU. As (e_i,β_i') is nonzero and not proportional to (c_i,β_i), the affine function w_i does not vanish on all of H_i, so it is nonzero on an open piece of H_i that meets no other H_k; we take x_0 there. Across H_i the terms of unit i jump by $(x_0\psi_{c_i}+\psi_{\beta_i})\,w_i(x_0)\,a_i$, so $(\psi_{c_i},\psi_{\beta_i})=\lambda_i(c_i,\beta_i)$ as before. The terms of unit i then add up to $\max(z_i,0)\,V_i(x)$ with the affine function $V_i(x)=w_i(x)\,(\psi_{a_i}+\lambda_i a_i)+(x\psi_{e_i}+\psi_{\beta_i'})\,a_i$. The jump of their gradient across H_i is $c_iV_i(x_0)$, so V_i vanishes on an open piece of H_i, and $V_i(x)=z_i\,\nu_i$ for a row vector ν_i. The terms of unit i then add up to $\max(z_i,0)^2\,\nu_i$, whose second derivative jumps across H_i, so $\nu_i=0$ and $V_i=0$ for all x: $w_i(x)\,(\psi_{a_i}+\lambda_i a_i)=-(x\psi_{e_i}+\psi_{\beta_i'})\,a_i$. Both sides are products of an affine function of x and a fixed row vector, and $a_i\neq 0$, so $\psi_{a_i}+\lambda_i a_i=\alpha_i a_i$ and $(\psi_{e_i},\psi_{\beta_i'})=-\alpha_i(e_i,\beta_i')$ for some number α_i. Then $\psi_M=0$ and $\psi_\gamma=0$ as for ReLU.

$\tanh$. We restrict (57) to a ray $x=tn$, $t>0$, with a direction n such that the numbers $u_k=nc_k$ are all nonzero and $|u_j|<|u_k|$ for one unit j and all $k\neq j$. For each j and each sign $s=\pm1$, the directions n with $su_j>0$ and this property form a nonempty open set: start from a point of the hyperplane $nc_j=0$ where no other nc_k vanishes, and move a little to the side s. For $z>0$,

$$\tanh z=1-2e^{-2z}+r(z),\quad 0\le r(z)\le 2e^{-4z},$$
$$\tanh' z=4e^{-2z}-q(z),\quad 0\le q(z)\le 8e^{-4z},$$

from $\tanh z=(1-e^{-2z})/(1+e^{-2z})$ and $\tanh'=1-\tanh^2$, and $\tanh$ is odd. On the ray $z_k=u_kt+\beta_k$, so for large t the terms of unit k in (57) are

$$s_k\psi_{a_k}+e^{-2|u_k|t}\,e^{-2s_k\beta_k}\big[-2s_k\psi_{a_k}+4\,(t\,n\psi_{c_k}+\psi_{\beta_k})\,a_k\big]+O\big((1+t)\,e^{-4|u_k|t}\big),$$

with s_k the sign of u_k. All terms of $J(tn)$ are therefore $t\,n\psi_M$, a constant, the bracket of unit j times $e^{-2|u_j|t}$, and terms that are $O\big((1+t)e^{-\rho t}\big)$ with $\rho=\min\big(4|u_j|,\min_{k\neq j}2|u_k|\big)>2|u_j|$. As $J(tn)=0$ for all t, dividing by t and letting $t\to\infty$ gives $n\psi_M=0$; then the constant vanishes; then multiplying by $e^{2|u_j|t}$ and letting $t\to\infty$ shows that the bracket of unit j, a polynomial of degree one in t, tends to zero, so both of its coefficients vanish: $(n\psi_{c_j})\,a_j=0$ and $2\psi_{\beta_j}a_j=s\,\psi_{a_j}$. As $a_j\neq 0$, $n\psi_{c_j}=0$ for all n in an open set, so $\psi_{c_j}=0$; and the two signs $s=\pm1$ give $\psi_{a_j}=2\psi_{\beta_j}a_j=-2\psi_{\beta_j}a_j$, so $\psi_{\beta_j}=0$ and $\psi_{a_j}=0$. This holds for every unit j, and then $J(x)=x\psi_M+\psi_\gamma=0$ gives $\psi_M=0$ and $\psi_\gamma=0$.

GELU. Here $\sigma(z)=z\Phi(z)$, with Φ the standard normal distribution function and $\varphi=\Phi'$ its density. The tail $\bar\Phi=1-\Phi$ satisfies $\varphi(z)(1/z-1/z^3)\le\bar\Phi(z)\le\varphi(z)/z$ for $z>0$, as one checks by differentiating, and $\Phi(-z)=1-\Phi(z)$ carries these bounds to $z<0$. From these, as $|z|\to\infty$ on either side,

$$\sigma(z)=\max(z,0)-\varphi(z)+O\big(\varphi(z)/z^2\big),$$
$$\sigma'(z)=\mathbf{1}[z>0]+\varphi(z)\,(z-1/z)+O\big(\varphi(z)/|z|^3\big).$$

Take a ray $x=tn$ as for $\tanh$. The terms of unit k in (57) split into a polynomial of degree at most one in t, from $\max(z,0)$ and $\mathbf{1}[z>0]$, and the Gaussian part

$$\varphi(z_k)\Big[u_k\,(n\psi_{c_k})\,t^2+\big(u_k\psi_{\beta_k}+\beta_k\,n\psi_{c_k}\big)\,t+\beta_k\psi_{\beta_k}-(n\psi_{c_k})/u_k\Big]a_k \\ -\ \varphi(z_k)\,\psi_{a_k}\ +\ O\big(\varphi(z_k)/t\big).$$

Gaussian parts tend to zero faster than any power of t, so as $J(tn) = 0$ for all t, the polynomial part vanishes, and so does the sum of the Gaussian parts. For $k \neq j$, $\varphi(z_k)/\varphi(z_j) = \exp\big(-\frac{1}{2}(u_k^2 - u_j^2)\,t^2 + O(t)\big)$ tends to zero faster than any power of t, since $u_k^2 > u_j^2$. Dividing the sum of the Gaussian parts by $\varphi(z_j)$ and letting $t \to \infty$ therefore makes the coefficients of t^2, t and 1 in the bracket of unit j vanish, in this order: $u_j(n\psi_{c_j})\,a_j = 0$, so $n\psi_{c_j} = 0$; then $u_j\psi_{\beta_j}\,a_j = 0$, so $\psi_{\beta_j} = 0$; then $\psi_{a_j} = 0$. As for $\tanh$, $n\psi_{c_j} = 0$ on an open set gives $\psi_{c_j} = 0$, and finally $\psi_M = 0$ and $\psi_\gamma = 0$.

SwiGLU and GeGLU. Take a ray $x = tn$ as for $\tanh$, and write $v_k = ne_k$, so that $w_k = v_k t + \beta'_k$; as $e_j \neq 0$, also $v_j \neq 0$ on an open set of such n. For SwiGLU, $\sigma(z) = z - z/(1 + e^z)$ gives, as $|z| \to \infty$ with s the sign of z,

$$\sigma(z) = \max(z, 0) - |z|e^{-|z|} + O\big(|z|e^{-2|z|}\big),$$
$$\sigma'(z) = \mathbf{1}[z > 0] + s\,(|z| - 1)\,e^{-|z|} + O\big(|z|e^{-2|z|}\big).$$

The terms of unit k in (57) then split into a polynomial in t, from $\max(z, 0)$ and $\mathbf{1}[z > 0]$, and a rest that is $e^{-|z_k|}$ times a polynomial in t, up to $O(t^3 e^{-2|z_k|})$; for GeGLU, with the expansions of GELU, the rest is $\varphi(z_k)$ times a polynomial in t and $1/z_k$, up to $O(\varphi(z_k)/t)$. As for $\tanh$ and GELU, the polynomial parts vanish, and dividing the sum of the rests by $e^{-|z_j|}$, respectively $\varphi(z_j)$, and letting $t \to \infty$ shows that the rest of the slowest unit j vanishes. For SwiGLU this rest is $e^{-|z_j|}$ times

$$s\,(|z_j| - 1)\,(t\,n\psi_{c_j} + \psi_{\beta_j})\,w_j\,a_j - |z_j|\big((t\,n\psi_{e_j} + \psi_{\beta'_j})\,a_j + w_j\,\psi_{a_j}\big),$$

with s the sign of u_j and $|z_j| = |u_j|t + s\beta_j$, a polynomial of degree three in t, whose coefficients therefore vanish. The coefficient of t^3 is $s|u_j|\,(n\psi_{c_j})\,v_j\,a_j$, so $\psi_{c_j} = 0$ as for $\tanh$. The coefficient of t^2 is then $-|u_j|\big((n\psi_{e_j})\,a_j + v_j(\psi_{a_j} - s\psi_{\beta_j}a_j)\big)$. As it vanishes for all n in an open set, $\psi_{e_j} = -\alpha e_j$ and $\psi_{a_j} - s\psi_{\beta_j}a_j = \alpha a_j$ for a number α, the same for both signs since ψ_{e_j} is; the two signs then give $\psi_{\beta_j} = 0$. The coefficient of t is then $-|u_j|(\psi_{\beta'_j} + \alpha\beta'_j)\,a_j$, so $\psi_{\beta'_j} = -\alpha\beta'_j$. For GeGLU the rest of unit j is $\varphi(z_j)$ times the same bracket with $z_j - 1/z_j$ in place of $s(|z_j| - 1)$ and 1 in place of $|z_j|$, and its coefficients of t^3, t^2, t and 1 give, in this order, $\psi_{c_j} = 0$, $\psi_{\beta_j} = 0$, $(\psi_{e_j}, \psi_{a_j}) = (-\alpha e_j, \alpha a_j)$ and $\psi_{\beta'_j} = -\alpha\beta'_j$. In both cases the terms of unit j then vanish; this holds for every unit, and $\psi_M = 0$ and $\psi_\gamma = 0$ as for $\tanh$. □

The directions left by the lemma are exactly those of the rescalings of the units: for ReLU, scaling the input weights and the bias of unit i by $1 + \tau\lambda_i$ and its output weights by $1 - \tau\lambda_i$ changes the function only at second order in τ. For $\tanh$, the proof on rays follows that of Lemma 3.1 of Sussmann (1992). For ReLU, the proof at the hyperplanes H_i follows those of Petzka et al. (2020, Lemma 2) and Stock and Gribonval (2021, Theorem 6).

Step 3: counting. Let η be flat. The network depends on θ only through ϕ, its M, γ and the c_i, β_i and a_i formed from C, β and A, so by the chain rule $\mathrm{D}_\theta F(\theta, x)\,\eta = \mathrm{D}_\phi G(\phi, x)\,(\mathrm{D}_\theta\phi\,\eta)$, and Lemma B.3 applies to $\psi = \mathrm{D}_\theta\phi\,\eta$. The derivatives of M, A, C, β and γ along η are thus those of a rescaling of the hidden units. A rescaling of the units by numbers λ_i is also a direction in parameter space: it scales column i of W_{read} and entry i of b_{read} by λ_i and row i of W_{write} by $-\lambda_i$ for ReLU, by $-2\lambda_i$ for ReLU^2, and similarly for ReGLU and for the values of SwiGLU and GeGLU, and along it C, β and A have the same derivatives. Subtracting it from η leaves a direction along which M, A, C, β and γ have derivative zero, which Lemma B.2 describes. So every flat direction is the sum of a direction of the changes of basis and translations of Lemma B.2 and a rescaling of the units, and conversely all of these are flat, since they come from symmetries. The two kinds are independent: a rescaling that is also a direction of Lemma B.2 has $\eta_E = EX = 0$, so $X = 0$, and then $W_{\mathrm{read}}\,\mathrm{diag}(\lambda) = 0$, so every $\lambda_i = 0$, because no column of W_{read} vanishes. Hence $\dim \mathcal{K}(\theta) = 2d_{\mathrm{model}}^2 + s$ without the skip and $d_{\mathrm{model}}^2 + s$ with it, with $2d_{\mathrm{model}}$ more for each with biases, where s is the number of rescalings: d_{hidden} for ReLU, ReLU^2, SwiGLU and GeGLU, $2d_{\mathrm{hidden}}$ for ReGLU, none for $\tanh$ and GELU. The functional dimension $P - \dim \mathcal{K}(\theta)$ has the

values of the theorem. The symmetries account for all of $\mathcal{K}(\theta)$, which is the equality condition, and the difference between the two networks is d_{model}^2.

Remark B.4 (One $\tanh$ block and one ReLU block without a skip). For a $\tanh$ block and for a ReLU block without a skip and without biases, the equality condition of Theorem 4.8(i) also follows from earlier work on networks with one hidden layer. The skipless network computes $x \mapsto \sum_i \sigma(xc_i)\, a_i$, where the c_i are the columns of $C = EW_{\text{read}}$ and the a_i the rows of $A = W_{\text{write}}D$.

$\tanh$. Theorem 2.1 of Sussmann (1992) concerns the networks $x \mapsto \tanh\big(\gamma + \sum_j \alpha_j \tanh(xc_j + \beta_j)\big)$ with one output, in which no α_j and no c_j vanishes and $(c_i, \beta_i) \neq \pm(c_j, \beta_j)$ for $i \neq j$. It says that two such networks with the same function have the same number of hidden units, and differ by a permutation of the hidden units and by sign flips $(c_j, \beta_j, \alpha_j) \mapsto (-c_j, -\beta_j, -\alpha_j)$. Albertini et al. (1993) extend this theorem to other odd activations, also in networks without biases. The outer $\tanh$ is injective, so the theorem applies to each output of our network. At generic parameters its conditions hold, and they hold at all nearby parameters. So nearby parameters with the same function have the same c_i and a_i, up to a permutation and sign flips of the hidden units, which are the identity for parameters that are close enough. They then have the same C and A. A matrix of rank d_{model} has only one factorization into two factors of rank d_{model}, up to an invertible matrix between them, so these parameters are EG_0, $G_0^{-1}W_{\text{read}}$, $W_{\text{write}}G_1$ and $G_1^{-1}D$ with invertible matrices G_0 and G_1. So the network is locally identifiable up to its symmetries, which is the equality condition (Proposition 4.7).

ReLU. Stock and Gribonval (2021) treat the networks $x \mapsto \gamma + \sum_i \max(xc_i + \beta_i,\ 0)\, a_i$ with several outputs, and call two hidden units $i \neq j$ twins if (c_i, β_i) and (c_j, β_j) are collinear. Theorem 6 there says: if a network has no twins and $\sum_{i \in T} c_i a_i \neq 0$ for every nonempty set T of hidden units, then every network with the same number of hidden units and the same function is obtained from it by a permutation of the hidden units and by rescalings $(c_i, \beta_i, a_i) \mapsto (\lambda_i c_i, \lambda_i \beta_i, a_i/\lambda_i)$ with $\lambda_i > 0$. For one output, Theorem 2 of Petzka et al. (2020) is an earlier statement of this kind. From Corollaries 2 and 3 and Lemma 15 of Stock and Gribonval (2021) one obtains the statement for the derivative. If a network has no twins, and no c_i and no a_i vanishes, then a direction of its weights along which the function does not change to first order, at the inputs at which no hidden unit sits at its kink, is a direction of these rescalings. This is Lemma B.3 for ReLU, without the linear term xM. Steps 1 and 3 above then give the count.

Part (i) for attention blocks

Reduction. Let $W_{Q,i}, W_{K,i}, W_{V,i} \in \mathbb{R}^{d_{\text{model}} \times d_h}$ and $W_{O,i} \in \mathbb{R}^{d_h \times d_{\text{model}}}$ be the projections of head i, and let

$$M = ED, \qquad B_i = EW_{Q,i}W_{K,i}^{\top}E^{\top}/\sqrt{d_h}, \qquad C_i = EW_{V,i}W_{O,i}D, \qquad S = \textstyle\sum_i C_i,$$

with M only in the residual network. Head i gives token t the scores $x_t B_i x_s^{\top}$, and the output at token t is $x_t M + \sum_i \sum_s a_{ts}^{(i)}\, x_s C_i$, so the parameters enter only through these matrices. We use inputs of two kinds: with a causal mask, any sequence, whose second token sees only the first two; without a mask, sequences whose tokens other than the second all equal the first. Let k be the number of tokens other than the second that the second token sees, $k = 1$ with a mask and $k = T - 1$ without. With $u = x_2$ and $v = x_2 - x_1$, head i puts the weight $1/(1 + k\, e^{-uB_i v^{\top}})$ on the second token and the rest on the copies of the first, and $1/(1 + e^{-z}) = \frac{1}{2} + \frac{1}{2}\tanh(z/2)$ gives the output at the second token

$$y_2 = uK - \tfrac{1}{2}\, vS + \tfrac{1}{2} \sum_i \tanh\big(\tfrac{1}{2}(uB_i v^{\top} - \log k)\big)\, vC_i, \qquad K = M + S. \tag{58}$$

With biases b_E on the embedding, $b_{Q,i}$, $b_{K,i}$, $b_{V,i}$ on the queries, keys and values, b_O on the output projection and b_D on the readout, $uB_i v^{\top}$ becomes $(uB_i + \beta_i)v^{\top}$ with $\beta_i = (b_E W_{Q,i} + b_{Q,i})W_{K,i}^{\top}E^{\top}/\sqrt{d_h}$, and the constant $\gamma = \sum_i (b_E W_{V,i} + b_{V,i})W_{O,i}D + b_O D + b_D$ is added, plus $b_E D$ in the residual network. The key biases drop out, since they shift all scores of a token equally, and the value biases enter only through γ, since the weights of a token sum to one.

Step 1: the directions that keep M, B_i and C_i. For generic parameters E and D have rank d_{model}; every $W_{Q,i}$, $W_{K,i}$, $W_{V,i}$ and $W_{O,i}^\top$ has rank d_h, since $d_h \le d_{\text{model}}$; and the queries $[W_{Q,1}, \dots, W_{Q,H}] \in \mathbb{R}^{d_{\text{model}} \times H d_h}$, the keys $[W_{K,1}, \dots, W_{K,H}]$ and the output projection $W_O \in \mathbb{R}^{H d_h \times d_{\text{model}}}$ of all heads have rank d_{model}, since $H d_h \ge d_{\text{model}}$. Write $\eta_{Q,i}$, $\eta_{K,i}$, $\eta_{V,i}$ and $\eta_{O,i}$ for the entries of a direction η at the positions of $W_{Q,i}$, $W_{K,i}$, $W_{V,i}$ and $W_{O,i}$, and consider a direction with $\mathrm{D}_\theta M\,\eta = 0$ and, for every head, $\mathrm{D}_\theta B_i\,\eta = 0$ and $\mathrm{D}_\theta C_i\,\eta = 0$. Multiplying $\mathrm{D}_\theta B_i\,\eta = 0$ on the left by the projection $I - EE^+$ onto the orthogonal complement of the column space of E, where E^+ is the pseudo-inverse, leaves $(I - EE^+)\,\eta_E\,W_{Q,i}W_{K,i}^\top E^\top = 0$. As $W_{K,i}^\top E^\top$ has full row rank, $(I - EE^+)\,\eta_E\,W_{Q,i} = 0$ for every head, and as the queries of all heads have rank d_{model}, $\eta_E = EX$ with $X = E^+\eta_E$. In the same way, multiplying $\mathrm{D}_\theta C_i\,\eta = 0$ on the right by the projection $I - D^+D$ gives $\eta_D = -YD$ with $Y = -\eta_D\,D^+$, because $EW_{V,i}$ has full column rank and the output projection has rank d_{model}. Substituting, and cancelling E on the left and $E^\top$ or D on the right,

$$(\eta_{Q,i} + XW_{Q,i})W_{K,i}^\top + W_{Q,i}(\eta_{K,i} + XW_{K,i})^\top = 0,$$
$$(\eta_{V,i} + XW_{V,i})W_{O,i} + W_{V,i}(\eta_{O,i} - W_{O,i}Y) = 0.$$

If P_1 and P_2 have full column rank and $Z_1P_2^\top + P_1Z_2^\top = 0$, then multiplying on the right by a right inverse of $P_2^\top$ gives $Z_1 = P_1\Gamma$ for a square matrix Γ, and then $Z_2 = -P_2\Gamma^\top$. Hence $\eta_{Q,i} = -XW_{Q,i} + W_{Q,i}\Gamma_i$, $\eta_{K,i} = -XW_{K,i} - W_{K,i}\Gamma_i^\top$, $\eta_{V,i} = -XW_{V,i} + W_{V,i}\Gamma_i'$ and $\eta_{O,i} = W_{O,i}Y - \Gamma_i'W_{O,i}$: the directions of the changes of basis $I + \tau X$ at h_0 and $I + \tau Y$ at h_1, and of $I + \tau\Gamma_i$ between the queries and keys and $I + \tau\Gamma_i'$ between the values and outputs of head i. In the residual network $\mathrm{D}_\theta M\,\eta = E(X - Y)D = 0$ gives $X = Y$. These directions are independent: $EX = 0$ forces $X = 0$, $YD = 0$ forces $Y = 0$, and then $W_{Q,i}\Gamma_i = 0$ and $W_{V,i}\Gamma_i' = 0$ force $\Gamma_i = \Gamma_i' = 0$. With biases, $\mathrm{D}_\theta\beta_i\,\eta = 0$ gives $\eta_{b_{Q,i}} = b_{Q,i}\Gamma_i - gW_{Q,i}$ with $g = \eta_{b_E} - b_EX$, since $W_{K,i}^\top E^\top$ has full row rank, and $\mathrm{D}_\theta\gamma\,\eta = 0$ determines η_{b_D}. This leaves g, η_{b_O} and the entries of η at the key and value biases free, $2d_{\text{model}} + 2Hd_h$ dimensions, which the translations at h_0 and h_1, the shifts of the key biases and the value biases moved into the output bias span.

Step 2: two tokens. For generic parameters and generic v the vectors $B_iv^\top$ are nonzero and pairwise non-collinear, since $d_{\text{model}} \ge 2$, and every vC_i is nonzero. For such a v, (58) is, as a function of u, a residual network with one hidden layer of $\tanh$ units, one per head, with input weights $\frac{1}{2}B_iv^\top$, biases $\frac{1}{2}(\beta_iv^\top - \log k)$, output weights $\frac{1}{2}vC_i$, linear part K and constant term $\gamma - \frac{1}{2}vS$. Along a flat direction η these have the derivatives $\frac{1}{2}(\mathrm{D}_\theta B_i\,\eta)v^\top$, $\frac{1}{2}(\mathrm{D}_\theta\beta_i\,\eta)v^\top$, $\frac{1}{2}v(\mathrm{D}_\theta C_i\,\eta)$, $\mathrm{D}_\theta K\,\eta$ and $\mathrm{D}_\theta\gamma\,\eta - \frac{1}{2}v(\mathrm{D}_\theta S\,\eta)$, and Lemma B.3 for $\tanh$ shows that all of these vanish. This holds for all v in an open set, so $\mathrm{D}_\theta B_i\,\eta = 0$, $\mathrm{D}_\theta\beta_i\,\eta = 0$ and $\mathrm{D}_\theta C_i\,\eta = 0$; hence $\mathrm{D}_\theta S\,\eta = 0$, $\mathrm{D}_\theta M\,\eta = \mathrm{D}_\theta K\,\eta - \mathrm{D}_\theta S\,\eta = 0$ and $\mathrm{D}_\theta\gamma\,\eta = 0$.

Step 3: counting. Every direction of Step 1 is flat, since it comes from a symmetry, and by Step 2 every flat direction, already on the inputs of (58), is one of them. So $\dim\mathcal{K}(\theta)$ is $2d_{\text{model}}^2 + 2Hd_h^2$ without the skip and $d_{\text{model}}^2 + 2Hd_h^2$ with it, and $2d_{\text{model}} + 2Hd_h$ more with biases.

Part (ii)

Write $h_1 = xE + g(x)$, with $g(x) = f(N_\varepsilon(xE)W_{\text{read}})W_{\text{write}}$ the output of the block, and $y = N_\varepsilon(h_1)D$, with N_ε applied to every token; for attention, x has T tokens, and for an MLP block one suffices. Let η be flat, and write $\mathrm{D}_\theta h_1\,\eta$ and $\mathrm{D}_\theta y\,\eta$ for the derivatives of h_1 and y along η at a given input, and $\mathrm{D}N_\varepsilon(h)$ for the Fréchet derivative of N_ε at h.

Step 1: the final normalization. Split $\eta_D = -YD + Z$, where $Y \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ and the rows of Z are orthogonal to the row space of D, which has dimension d_{model}. At every token, by the chain rule,

$$0 \;=\; \mathrm{D}_\theta y\,\eta \;=\; \big(\mathrm{D}N_\varepsilon(h_1)\,(\mathrm{D}_\theta h_1\,\eta) - N_\varepsilon(h_1)Y\big)D + N_\varepsilon(h_1)Z,$$

and the two terms lie in complementary subspaces, so both vanish. As x varies, the rows $N_\varepsilon(h_1)$ span $\mathbb{R}^{d_{\text{model}}}$, since at large inputs they approach $\sqrt{d_{\text{model}}}\,xE/\|xE\|$ and xE takes every direction, so $Z = 0$; and since D has full row rank, $\mathrm{D}N_\varepsilon(h_1)\,(\mathrm{D}_\theta h_1\,\eta) = N_\varepsilon(h_1)Y$. With $\rho^2 = \|h\|^2/d_{\text{model}} + \varepsilon$,

the derivative of N_ε at h is $\mathrm{D}N_\varepsilon(h)\,k = k/\rho - h\,\langle h,k\rangle/(d_{\mathrm{model}}\rho^3)$, which vanishes only for $k=0$ when $\varepsilon>0$, and the unique $\mathrm{D}_\theta h_1\,\eta$ with $\mathrm{D}N_\varepsilon(h_1)\,(\mathrm{D}_\theta h_1\,\eta) = N_\varepsilon(h_1)Y$ is

$$\mathrm{D}_\theta h_1\,\eta = h_1 Y + \frac{h_1 Y_s h_1^\top}{d_{\mathrm{model}}\varepsilon}\,h_1, \qquad Y_s = \tfrac{1}{2}(Y+Y^\top),$$

as one checks by substituting $k = h_1 Y + \mu h_1$ and solving for the number μ.

Step 2: large inputs. Fix w with $wE \neq 0$, for attention one such row for every token, and let $x = tw$ with $t\to\infty$. The block reads $N_\varepsilon(twE)$, which stays in the ball of radius $\sqrt{d_{\mathrm{model}}}$, so $g(tw)$ stays bounded, and for attention the scores are bounded too. The derivative of $g(tw)$ along η is bounded as well: by the formula of Step 1, $\|\mathrm{D}N_\varepsilon(h)\,k\| \le 2\sqrt{d_{\mathrm{model}}}\,\|k\|/\|h\|$, which is $O(1)$ for $h = twE$ and $k = tw\eta_E$. Hence $\mathrm{D}_\theta h_1\,\eta = t\,w\eta_E + O(1)$ at the input tw, while $h_1(tw) = t\,wE + O(1)$ and the right-hand side of Step 1 is $t^3\,(wEY_sE^\top w^\top)\,wE/(d_{\mathrm{model}}\varepsilon) + O(t^2)$. So $wEY_sE^\top w^\top = 0$ for all w in an open set, hence $EY_sE^\top = 0$, and $Y_s = 0$ because E has rank d_{model}. The matrix Y is antisymmetric, and $\mathrm{D}_\theta h_1\,\eta = h_1 Y$.

Step 3: the skip. An antisymmetric Y is the derivative at $\tau=0$ of the orthogonal changes of basis $I+\tau Y+O(\tau^2)$ that the residual network shares between h_0 and h_1 (Lemma A.2). They define a direction η^Y with the entries EY, $-YW_{\mathrm{read}}$, $W_{\mathrm{write}}Y$ and $-YD$ at E, W_{read}, W_{write} and D, along which the stream h_1 has the derivative h_1Y. The direction $\eta' = \eta - \eta^Y$ is flat, with $\eta'_D = 0$ and $\mathrm{D}_\theta h_1\,\eta' = 0$ at every input. At the inputs tw, $\mathrm{D}_\theta h_1\,\eta' = t\,w\eta'_E + O(1)$, so $w\eta'_E = 0$ for all w in an open set, and $\eta'_E = 0$.

Step 4: the block. With $\eta'_E = 0$, the condition $\mathrm{D}_\theta h_1\,\eta' = 0$ says that the derivative of $n \mapsto f(nW_{\mathrm{read}})W_{\mathrm{write}}$ along the entries of η' at the weights of the block vanishes for every n in the image of $x \mapsto N_\varepsilon(xE)$, which is the open ball of radius $\sqrt{d_{\mathrm{model}}}$, since E has rank d_{model} and N_ε maps $\mathbb{R}^{d_{\mathrm{model}}}$ onto that ball for $\varepsilon>0$. For $\tanh$, GELU, SwiGLU, GeGLU and attention this derivative is, along every line through the ball, a function of one variable given by a convergent power series around every point; it vanishes on an interval, hence on the whole line, so it vanishes for all inputs. For the ReLU-type activations without biases the kinks lie on hyperplanes through the center of the ball, and the arguments of Lemma B.3 are local. So Lemma B.3, with E and D replaced by the identity and $M=0$, makes the entries of η' at each hidden unit a rescaling. For attention, each token $N_\varepsilon(x_tE)$ ranges over the ball, and Step 2 for attention blocks, with E and D replaced by the identity and $M=0$, keeps every $W_{Q,i}W_{K,i}^\top$ and $W_{V,i}W_{O,i}$ fixed to first order; by Step 1 there, η' is a direction of the changes of basis inside the heads.

Step 5: counting. The flat directions are thus those of the orthogonal change of basis and of the symmetries inside the block. An antisymmetric Y with $EY=0$ vanishes, since E has rank d_{model}, and the symmetries inside the block move generic parameters in independent directions, so $\dim\mathcal{K}(\theta) = d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2 + s$. If the block has width at least d_{model}, the skipless network with the same shapes has functional dimension at most $P - d_{\mathrm{model}}(d_{\mathrm{model}}-1) - s$ by Theorem A.14(ii), Lemma A.15, so the skip adds at least $d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2$.

Part (iii)

Write $B(n) = f(nW_{\mathrm{read}})W_{\mathrm{write}}$ for the block as a function of its input n and of its weights, applied to every token for attention, so that $h_1 = xEA + g(x)$ with $g(x) = B(N_\varepsilon(xE))$, and $y = N_\varepsilon(h_1)D$. For a direction ψ of the weights of the block, $\partial_\psi B(n)$ is the derivative of $B(n)$ along ψ at fixed n, and $\mathrm{D}B(n)\,k$ is the derivative of B at n along k at fixed weights. Let η be flat. For an upper triangular A, view the network as the network with a full A restricted to triangular skips: its flat directions are those of the full network whose entry η_A is upper triangular.

Steps 1 and 2 of part (ii) run with EA in place of E, which has rank d_{model} since A is invertible for generic parameters: $\eta_D = -YD$ with Y antisymmetric, and $\mathrm{D}_\theta h_1\,\eta = h_1 Y$ at every input.

Step 3: the skip. At the inputs tw of Step 2, $\mathrm{D}_\theta h_1\,\eta = t\,w(\eta_E A + E\eta_A) + O(1)$ and $h_1 Y = t\,wEAY + O(1)$, so $\eta_E A + E\eta_A = EAY$. Multiplying on the left by $I - EE^+$ gives $(I-EE^+)\,\eta_E = 0$, as A is invertible, so $\eta_E = EX$ with $X = E^+\eta_E$, and then $\eta_A = AY - XA$. At every input

$\mathrm{D}_\theta h_1\,\eta = xEAY + \mathrm{D}_\theta g\,\eta$ and $h_1 Y = xEAY + g(x)Y$, so $\mathrm{D}_\theta g\,\eta = g(x)Y$. By the formula for $\mathrm{D}N_\varepsilon$ in Step 1 of part (ii), which gives the same for $\varepsilon = 0$, the input $n = N_\varepsilon(xE)$ of the block has the derivative

$$\mathrm{D}N_\varepsilon(xE)\,(xEX) = nX - q(n)\,n, \qquad q(n) = nX_s n^\top / d_{\mathrm{model}}, \quad X_s = \tfrac{1}{2}(X + X^\top).$$

Step 4: the symmetric part of X. Write $X = X_a + X_s$ with X_a antisymmetric. The rotations of h_0 and h_1 with the derivatives X_a and Y are symmetries. They define a flat direction η^{rot} with the entries EX_a at E, $AY - X_a A$ at A, $-X_a W_{\mathrm{read}}$ at W_{read}, $W_{\mathrm{write}} Y$ at W_{write} and $-YD$ at D. The flat direction $\eta' = \eta - \eta^{\mathrm{rot}}$ has the entries EX_s, $-X_s A$ and 0 at E, A and D, so Step 3 applies to it with X_s in place of X and 0 in place of Y, and $\mathrm{D}_\theta g\,\eta' = 0$. The change nX_s of the input moves the block as the entries $X_s W_{\mathrm{read}}$ at W_{read} do; for attention, X_s moves the queries, keys and values in the same way. So, with ψ the entries of η' at the weights of the block plus these,

$$\partial_\psi B(n) = q(n)\,\mathrm{D}B(n)\,n \tag{59}$$

for every n in the image of N_ε, the open ball of radius $\sqrt{d_{\mathrm{model}}}$. For attention, $q(n)\,n$ stands for the matrix with the rows $q(n_t)\,n_t$, one per token.

We show that $q = 0$, hence $X_s = 0$. For ReLU, ReLU^2 and ReGLU, $B(\lambda n) = \lambda^k B(n)$ for $\lambda > 0$, with $k = 1$ or 2, so the right side of (59) is $k\,q(n)B(n)$. Replacing n by λn with $0 < \lambda \le 1$ multiplies the left side by λ^k and the right side by λ^{k+2}, so both vanish, and $q = 0$ wherever $B \neq 0$. For the other blocks both sides are analytic in n, so (59) holds for all n, and we take $n = t\hat{n}$ with a fixed row $\hat{n}$ and $t \to \infty$, as in the proof of Lemma B.3. For GELU the right side is $t^3 q(\hat{n}) \max(\hat{n}W_{\mathrm{read}}, 0) W_{\mathrm{write}}$ up to terms that decay faster than any power of t, and the left side is a polynomial of degree at most one in t up to such terms, so $q(\hat{n}) \max(\hat{n}W_{\mathrm{read}}, 0) W_{\mathrm{write}} = 0$. SwiGLU and GeGLU units $\sigma(nc_i)\,(ne_i)\,a_i$ give $2t^4 q(\hat{n}) \sum_{i:\,\hat{n}c_i > 0} (\hat{n}c_i)(\hat{n}e_i)\,a_i$ against a polynomial of degree two. For $\tanh$ the right side is $4t^3 q(\hat{n}) \sum_i (\hat{n}c_i)\,e^{-2|\hat{n}c_i|t} a_i$ up to terms $O(t^3 e^{-4|\hat{n}c_i|t})$; for the unit j with the smallest $|\hat{n}c_j|$, the left side contributes a polynomial of degree one times $e^{-2|\hat{n}c_j|t}$ (proof of Lemma B.3), so $q(\hat{n})(\hat{n}c_j)\,a_j = 0$. For attention, on the two-token inputs of Step 2 for attention blocks with $u = t\hat{n}$ and v fixed, the right side is $t^3 q(\hat{n})\,\hat{n}S + O(t^2)$ and the left side is $O(t)$. In every case q vanishes on an open set for generic parameters, so $X_s = 0$.

Step 5: counting. With $X_s = 0$, the direction η' moves only the weights of the block, and by (59) $B(n)$ has derivative zero along it on the whole ball. So η' lies in the space $\mathcal{K}_B$ of such directions, which is the space found in Step 4 of part (ii), and every direction of $\mathcal{K}_B$ is flat. Hence $\mathcal{K}(\theta)$ is spanned by the directions η^{rot} and $\mathcal{K}_B$, as for the residual network with its single rotation, and for generic parameters of the block $\mathcal{K}_B$ has the same dimension in both networks. The directions η^{rot} are independent of $\mathcal{K}_B$ and of each other, since $\eta_E = EX_a$ and $\eta_D = -YD$ determine X_a and Y. For a full A they have dimension $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)$. For an upper triangular A, $AY - X_a A$ must be upper triangular, which determines Y from X_a: for generic A the map from antisymmetric Y to the entries of AY below the diagonal is invertible, as it is for $A = I$. They then have dimension $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2$. So $\mathcal{K}(\theta)$ is larger than for the residual network by $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2$, respectively 0, while the network has d_{model}^2, respectively $d_{\mathrm{model}}(d_{\mathrm{model}} + 1)/2$, more parameters. In both cases the functional dimension is larger by $d_{\mathrm{model}}(d_{\mathrm{model}} + 1)/2$. For the blocks of the theorem, $\mathcal{K}_B$ consists of the symmetries inside the block (Step 4 of part (ii)), so the equality condition holds.

The case $\varepsilon = 0$. Now N_ε maps onto the sphere of radius $\sqrt{d_{\mathrm{model}}}$, and the scalings of h_0 and h_1 are symmetries too, since $N_\varepsilon(ch) = N_\varepsilon(h)$ for $c > 0$. In Step 1, $\mathrm{D}N_\varepsilon(h)$ vanishes along h, so $\mathrm{D}_\theta h_1\,\eta = h_1 Y + \mu\,h_1$ with a number μ at each input. Y is again antisymmetric, since $N_\varepsilon(h_1)Y$ must be orthogonal to h_1, which is $t\,wEA + O(1)$ at the inputs tw. At these inputs g and its derivative along η do not depend on t, since $N_\varepsilon(twE) = N_\varepsilon(wE)$. The terms of order t give $w\Lambda = \mu_w\,wEA$ with $\Lambda = \eta_E A + E\eta_A - EAY$ and a number μ_w, for all w in an open set. Adding to w a vector w_0 with $w_0 E = 0$ leaves wEA unchanged, so $w_0\Lambda$ is a multiple of wEA for every such w, hence $w_0\Lambda = 0$, and $\Lambda = E\Lambda'$ with $\Lambda' = E^+\Lambda$. Every vector $v = wE$ of an open set is then a left eigenvector of $\Lambda' A^{-1}$, which is therefore cI for a number c, as in the proof of Lemma A.2: $\eta_E A + E\eta_A = EA(Y + cI)$. The constant terms then give $\mu = c$ wherever wEA and $g(w)$ are

linearly independent. Steps 3 and 4 run with $Y + cI$ in place of Y and with $X_s = c_0 I + X_0$, where X_0 has trace zero; the scalings, with the derivatives c_0 at h_0 and c at h_1, join the rotations in η^{rot}; and (59) holds with X_0 in place of X_s on the sphere. For the residual network the same steps give $\eta_E = E(Y + cI)$, so $X_s = cI$ and $X_0 = 0$. Its flat directions are the rotation and the scaling of all hidden states, and the space $\mathcal{K}_B$ of directions of the block along which B has derivative zero on the sphere. For the learned skip, η^{rot} and $\mathcal{K}_B$ are flat, which gives at least $d_{\mathrm{model}}(d_{\mathrm{model}} - 1) + 2$, respectively $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 2$, dimensions besides $\mathcal{K}_B$, against $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1$ for the residual network. So the difference is at most $d_{\mathrm{model}}(d_{\mathrm{model}} + 1)/2 - 1$. For ReLU, ReLU^2 and ReGLU, both sides of (59) are positively homogeneous, of degrees k and $k + 2$, so $\|n\|^2\, \partial_\psi B(n) = k d_{\mathrm{model}}\, q(n) B(n)$ for all $n \neq 0$. On each region where no nc_k changes sign, both sides are polynomials in n. Across the hyperplane $nc_i = 0$, at a point of no other such hyperplane, only the terms of unit i change, so they satisfy the identity on their own: $\|n\|^2$ times a polynomial equals $k d_{\mathrm{model}}\, q(n)$ times $(nc_i)\, a_i$, $(nc_i)^2 a_i$ or $(nc_i)(ne_i)\, a_i$. Since $\|n\|^2$ has no real linear factor, it divides $q(n)$, so X_0 is a multiple of I, and $X_0 = 0$ because its trace vanishes. Then Step 5 gives equality.

Part (iv)

With $\varepsilon = 0$, $N_\varepsilon(h) = \sqrt{d_{\mathrm{model}}}\, h/\|h\|$ depends only on the direction of h, and $B(\lambda n) = \lambda^k B(n)$ for $\lambda > 0$, so $N_\varepsilon(B(N_\varepsilon(h))) = N_\varepsilon(B(h))$: the skipless network computes $x \mapsto N_\varepsilon(B(xE))D$. Its normalization before the block drops out, which leaves the changes of basis $\mathrm{GL}(d_{\mathrm{model}})$ at h_0 and $\mathrm{CO}(d_{\mathrm{model}})$ at h_1, of dimension $d_{\mathrm{model}}^2 + d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1$, and the residual network keeps one $\mathrm{CO}(d_{\mathrm{model}})$ for both hidden states (Theorem A.5(iii)).

The residual network. The case $\varepsilon = 0$ of part (iii), with $A = I$, shows that a flat direction is the sum of a direction of the rotation and scaling of all hidden states and a direction ψ of the weights of the block along which B has derivative zero on the sphere of radius $\sqrt{d_{\mathrm{model}}}$. This derivative is positively homogeneous of degree k in n, so it vanishes for all $n \neq 0$, and Lemma B.3, with E and D replaced by the identity and $M = 0$, makes ψ a rescaling of the units. So $\dim \mathcal{K}(\theta) = d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1 + s$.

The skipless network: the bound. The changes of basis in $\mathrm{GL}(d_{\mathrm{model}})$ at h_0 and in $\mathrm{CO}(d_{\mathrm{model}})$ at h_1 and the symmetries inside the block preserve the function. They move the parameters along the directions with the entries EX at E and $-XW_{\mathrm{read}}$ at W_{read}, with the entries $W_{\mathrm{write}}(Y + cI)$ at W_{write} and $-YD$ at D for an antisymmetric Y and a number c, and along the rescalings of the units. For generic parameters these directions are independent, as in Step 3 of part (i). So $\dim \mathcal{K}(\theta) \geq d_{\mathrm{model}}^2 + d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1 + s$, which is the bound.

The skipless network: equality for ReLU and ReLU^2. Let $\sigma(z) = \max(z, 0)^k$ with $k = 1$ or 2, and write $h_1 = B(xE) = \sum_i \sigma(xc_i)\, a_i$, with c_i the columns of $C = EW_{\mathrm{read}}$ and a_i the rows of W_{write}. The network is defined on the set Ω of inputs at which some unit is active: for generic parameters $h_1 \neq 0$ there, and $h_1 = 0$ elsewhere. Indeed, let S be the set of units active at x, and let w_i be the columns of W_{read}. If $|S| < d_{\mathrm{model}}$, the rows a_i, $i \in S$, are linearly independent. If $|S| \geq d_{\mathrm{model}}$ and $k = 1$, $h_1 = xE \sum_{i \in S} w_i a_i$, and this $d_{\mathrm{model}} \times d_{\mathrm{model}}$ matrix is invertible. If $|S| \geq d_{\mathrm{model}}$ and $k = 2$, the entries of h_1 are d_{model} quadratic forms in xE. A common zero other than 0, real or complex, makes their resultant vanish. The resultant is a polynomial in the weights, and it is nonzero at $w_i = e_i$, $a_i = e_i^\top$ for d_{model} units of S and $a_i = 0$ for the others, where the forms are the squares of the coordinates. Let η be flat, and let ψ_{c_i} and ψ_{a_i} be the derivatives of c_i and a_i along η. As in the case $\varepsilon = 0$ of part (iii), $\eta_D = -YD$ with Y antisymmetric, since h_1 takes all values in an open set, and $\mathrm{D}_\theta h_1\, \eta = h_1 Y + \mu\, h_1$ with a number $\mu(x)$ at each input of Ω. So

$$J(x) := \sum_i \Big[\sigma(xc_i)\, (\psi_{a_i} - a_i Y) + \sigma'(xc_i)\, (x\psi_{c_i})\, a_i \Big] = \mu(x)\, h_1(x) \qquad \text{on } \Omega. \tag{60}$$

The kinks stay. Fix a unit i and a point x_0 with $x_0 c_i = 0$, on no other hyperplane $xc_j = 0$, at which another unit is active. For generic parameters and x_0 in an open dense set of such points, $h_1(x_0)$ is not parallel to a_i. Near x_0, on each side of the hyperplane, $\mu = \langle J, h_1 \rangle / \|h_1\|^2$ is a smooth function with limits at x_0. For $k = 1$, h_1 is continuous at x_0, so across the hyperplane the right side

of (60) jumps by a multiple of $h_1(x_0)$, and the left side by $(x_0\psi_{c_i})\,a_i$; both jumps vanish. For $k=2$, both sides, μ and the derivative of h_1 in x are continuous at x_0. The derivative in x of the left side jumps by the matrix $2\,(x_0\psi_{c_i})\,c_i a_i$, and that of the right side by a column times the row $h_1(x_0)$; two such matrices are equal only if both vanish. In both cases $x_0\psi_{c_i}=0$ on an open subset of the hyperplane, so $\psi_{c_i}=\lambda_i c_i$ for a number λ_i.

One scale. As $\sigma'(z)\,z = k\,\sigma(z)$, (60) now reads

$$\sum_i \sigma(xc_i)\,\big(b_i-\mu(x)\,a_i\big)=0 \quad \text{on } \Omega, \qquad b_i=\psi_{a_i}-a_iY+k\lambda_i a_i,$$

and $J=\sum_i \sigma(xc_i)\,b_i$ and h_1 are continuous, so μ is continuous on Ω. On a region where only unit j is active, $b_j=\mu(x)\,a_j$, so μ is constant. On a region where at least two units are active, the entries of J and of h_1 are homogeneous polynomials of degree k in x, and for generic parameters two entries h_a and h_b of h_1 have no common factor. From $J_a h_b = J_b h_a$, h_a divides J_a, so $J=c\,h_1$ for a number c, and $\mu=c$ on the region. So μ is continuous on Ω and constant on every region. The set Ω is connected: it is the union of the open half-spaces $\{x : xc_i>0\}$, and any two of them meet, as the c_i are pairwise non-collinear. Hence $\mu=c$ on Ω for one number c, and $\mathrm{D}_\theta h_1\,\eta = h_1(Y+cI)$ at every input, since both sides vanish where no unit is active.

Counting. Subtracting the change of basis $I+\tau(Y+cI)$ at h_1, with the entries $W_{\text{write}}(Y+cI)$ at W_{write} and $-YD$ at D, leaves a direction along which $f(xC)W_{\text{write}}$ has derivative zero, which moves C and W_{write} only along a rescaling of the units, by Lemma B.3 with $M=0$. After subtracting that rescaling as well, Lemma B.2 gives $\eta_E=EX$ and $\eta_{\text{read}}=-XW_{\text{read}}$ for a matrix $X\in\mathbb{R}^{d_{\text{model}}\times d_{\text{model}}}$, the change of basis at h_0. So $\dim\mathcal{K}(\theta)=d_{\text{model}}^2+d_{\text{model}}(d_{\text{model}}-1)/2+1+s$ for ReLU and ReLU^2 blocks, and the difference to the residual network is d_{model}^2.

Part (v)

With $\varepsilon=0$, $N_\varepsilon(h)=\sqrt{d_{\text{model}}}\,h/\|h\|$ ignores the length of each token, so the block reads only rows on the sphere of radius $\sqrt{d_{\text{model}}}$. Let V_θ be the space of directions that move only the weights of the block and leave its output unchanged to first order at every input whose rows lie on this sphere; it contains the symmetries inside the block. We show that the flat directions of the residual network are those of $\mathrm{CO}(d_{\text{model}})$ and those of V_θ, while the skipless network has at least two copies of $\mathrm{CO}(d_{\text{model}})$ and V_θ. So the bound on the gain holds whatever V_θ is. Lemma B.5 below shows that at generic parameters V_θ is the space of the symmetries inside the block, which gives the functional dimension of the residual network.

The residual network. Write $h_1=xE+g(x)$, with $g(x)=f(N_\varepsilon(xE)W_{\text{read}})W_{\text{write}}$, and $y=N_\varepsilon(h_1)D$, and let η be flat. Step 1 of part (ii) runs as before, except that with $\varepsilon=0$ the derivative $\mathrm{D}N_\varepsilon(h)$ projects onto the vectors orthogonal to h and scales by $\sqrt{d_{\text{model}}}/\|h\|$, so its kernel is spanned by h. Hence $N_\varepsilon(h_1)Y$ is orthogonal to h_1; as h_1 takes every value in an open set, $Y_s=0$, and $\mathrm{D}_\theta h_1\,\eta = h_1Y+\mu(x)\,h_1$ with a number $\mu(x)$, for attention one per token. Subtracting the rotation η^Y of Step 3 there leaves a flat η' with $\eta'_D=0$ and $\mathrm{D}_\theta h_1\,\eta'=\mu'(x)\,h_1$.

Take the inputs $x=tw$ with $t>0$. Since N_ε is homogeneous of degree 0, the block output $g(tw)=g(w)$ and its derivative $\gamma(w)$ along η' do not depend on t, so

$$t\,w\eta'_E+\gamma(w) \;=\; \mu'(tw)\,\big(t\,wE+g(w)\big) \qquad \text{for all } t>0.$$

For generic parameters wE and $g(w)$ are linearly independent for all w in an open set, since otherwise the block would map every point of the sphere to a multiple of itself. Comparing the parts along wE, along $g(w)$ and outside their span gives $w\eta'_E=c_w\,wE$ and $\gamma(w)=c_w\,g(w)$ with a number c_w. Then $w\eta'_E$ is parallel to wE for all w, so η'_E vanishes on the kernel of E, $\eta'_E=ET$ with vT parallel to v for every v, and $\eta'_E=cE$ with one number c. The scaling $E\mapsto(1+\tau)E$, $W_{\text{write}}\mapsto(1+\tau)W_{\text{write}}$ multiplies h_1 by $1+\tau$ and is the scale in $\mathrm{CO}(d_{\text{model}})$. Subtracting c times its direction leaves a flat η'' that moves only the weights of the block, with $\mathrm{D}_\theta g\,\eta''=0$ at the inputs tw. Their normalized rows cover an open set of the sphere, and the derivative of the block along η'' is analytic on the sphere, so $\eta''\in V_\theta$. The directions of $\mathrm{CO}(d_{\text{model}})$ move E by $E(Y+cI)$, which vanishes only for

$Y = 0$ and $c = 0$. So $\mathcal{K}(\theta)$ is the direct sum of these directions and V_θ, and the residual network has functional dimension $P - \dim \mathrm{CO}(d_{\mathrm{model}}) - \dim V_\theta$.

The skipless network. Its block also reads only rows on the sphere, so the directions of V_θ are flat, and so are the changes of basis in $\mathrm{CO}(d_{\mathrm{model}})$ at h_0 and at h_1 (Proposition A.1). These directions are independent. A combination moves E by $E(Y_0 + c_0 I)$ and D by $-Y_1 D$, so $Y_0 = 0$, $c_0 = 0$ and $Y_1 = 0$. What is left is a multiple of the scaling $W_{\mathrm{write}} \mapsto (1+\tau) W_{\mathrm{write}}$ plus a direction of V_θ, and the scaling lies in V_θ only if the block vanishes on the sphere. So the skipless network has functional dimension at most $P - 2\dim \mathrm{CO}(d_{\mathrm{model}}) - \dim V_\theta$.

Generic parameters. The dimension of V_θ depends only on the weights of the block and takes its smallest value v for generic parameters. The generic functional dimensions are therefore $P - \dim \mathrm{CO}(d_{\mathrm{model}}) - v$ with the skip and at most $P - 2\dim \mathrm{CO}(d_{\mathrm{model}}) - v$ without it, and the skip adds at least $\dim \mathrm{CO}(d_{\mathrm{model}}) = d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1$. A direction of V_θ satisfies the hypothesis of Lemma B.5 below, with $\mu_t = 0$, so at generic parameters it is a direction of the symmetries inside the block. Hence $v = s$, and the residual network has functional dimension $P - d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 - 1 - s$.

A block whose inputs lie on a sphere

With $\varepsilon = 0$ a block reads only rows of the sphere

$$\mathbb{S} \;=\; \big\{\, n \in \mathbb{R}^{1 \times d_{\mathrm{model}}} : \|n\|^2 = d_{\mathrm{model}} \,\big\}.$$

The lemma of this subsection determines the space V_θ of part (v). Its hypothesis is weaker than that of V_θ: the output of the block may move along its input. In this form Lemma A.24 uses it at any depth.

The block. Let the block g hold any number of attention heads, of width $d_h \le d_{\mathrm{model}}$, causal or not, on sequences of $T \ge 2$ tokens, and any number of hidden units, which are all $\tanh$, all GELU, all SwiGLU or all GeGLU units. Head i has the projections $W_{Q,i}$, $W_{K,i}$, $W_{V,i}$ and $W_{O,i}$, and as in part (i), with E and D replaced by the identity,

$$B_i = W_{Q,i} W_{K,i}^\top / \sqrt{d_h}, \qquad C_i = W_{V,i} W_{O,i}, \qquad S = \textstyle\sum_i C_i.$$

Hidden unit j has the column c_j of the read and the row a_j of the write, and a gated unit also the value column e_j of the read. At a row n the unit computes $\sigma(n c_j)\, a_j$, and a gated unit $\sigma(n c_j)\,(n e_j)\, a_j$, where σ is SiLU for SwiGLU and GELU for GeGLU. For a direction η of the weights of g, let ψ_{c_j}, ψ_{e_j} and ψ_{a_j} be its entries at c_j, e_j and a_j, and let

$$\psi_{B_i} = \big(\eta_{Q,i} W_{K,i}^\top + W_{Q,i} \eta_{K,i}^\top\big)/\sqrt{d_h}, \qquad \psi_{C_i} = \eta_{V,i} W_{O,i} + W_{V,i} \eta_{O,i}, \qquad \psi_S = \textstyle\sum_i \psi_{C_i}$$

be the derivatives of B_i, C_i and S along η. For a sequence $n = (n_1, \dots, n_T)$ of rows, $\mathrm{D}_\theta g(n)\,\eta$ is the derivative of the output $g(n)$ along η.

Lemma B.5 (Block on the sphere)**.** *Let $d_{\mathrm{model}} \ge 2$. For generic weights of the block g, let a direction η of its weights satisfy, for every sequence $n = (n_1, \dots, n_T)$ of rows of $\mathbb{S}$ and every token t,*

$$\big(\mathrm{D}_\theta g(n)\,\eta\big)_t \;=\; \mu_t(n)\, n_t \qquad \textit{for a number } \mu_t(n). \tag{61}$$

Then η is a direction of the symmetries inside the block:
for every hidden unit, $\psi_{c_j} = 0$;
for $\tanh$ and GELU units, $\psi_{a_j} = 0$;
for SwiGLU and GeGLU units, $\psi_{e_j} = -\alpha_j e_j$ and $\psi_{a_j} = \alpha_j a_j$ for some number α_j;
for every head, $\psi_{B_i} = 0$ and $\psi_{C_i} = 0$, that is, $\eta_{Q,i} = W_{Q,i}\Gamma_i$, $\eta_{K,i} = -W_{K,i}\Gamma_i^\top$, $\eta_{V,i} = W_{V,i}\Gamma_i'$ and $\eta_{O,i} = -\Gamma_i' W_{O,i}$ for some $d_h \times d_h$ matrices Γ_i and Γ_i'.
In particular $\mathrm{D}_\theta g(n)\,\eta = 0$ at every input.

Proof. Notation. For rows $n, v \in \mathbb{C}^{1\times d_{\text{model}}}$ write $\langle v, n\rangle = vn^\top$, without complex conjugation, and

$$\mathrm{pr}_n(v) \;=\; d_{\text{model}}\, v - \langle v, n\rangle\, n.$$

This is linear in v and polynomial in n. If $\langle n, n\rangle = d_{\text{model}}$, then $\mathrm{pr}_n(v) = 0$ exactly when v is a multiple of n, so (61) says that pr_{n_t} of its left side vanishes. For orthonormal rows $\xi, \xi' \in \mathbb{R}^{1\times d_{\text{model}}}$, the *great circle* of ξ and ξ' is

$$n(\vartheta) \;=\; \sqrt{d_{\text{model}}}\,\big(\cos\vartheta\,\xi + \sin\vartheta\,\xi'\big), \qquad \vartheta \in \mathbb{C}.$$

It has $\langle n(\vartheta), n(\vartheta)\rangle = d_{\text{model}}$ for all $\vartheta \in \mathbb{C}$, and $n(\vartheta) \in \mathbb{S}$ for real ϑ. We use two facts. (a) A meromorphic function on $\mathbb{C}$ that vanishes on $\mathbb{R}$ vanishes identically. (b) For a polynomial map Θ from $\mathbb{C}^{1\times d_{\text{model}}}$ to itself, $\vartheta \mapsto \Theta(n(\vartheta))$ is a trigonometric polynomial: for some integer l_0, $e^{\mathrm{i}l_0\vartheta}\,\Theta(n(\vartheta))$ is a polynomial in $e^{\mathrm{i}\vartheta}$. So if it vanishes at infinitely many ϑ with pairwise different $e^{\mathrm{i}\vartheta}$, it vanishes for all ϑ.

Generic weights. The columns c_j are nonzero and pairwise non-collinear, every a_j and e_j is nonzero, and $a_j e_j \neq 0$ if $d_{\text{model}} = 2$. Every B_i and C_i is nonzero, and for two heads $i \neq i'$ the rows uB_i and $uB_{i'}$ are linearly independent for some row u. Every $W_{Q,i}$, $W_{K,i}$ and $W_{V,i}$ has full column rank, and every $W_{O,i}$ has full row rank.

Step A: equal tokens. Let all tokens equal $n \in \mathbb{S}$. Head i then gives every token the output nC_i, whatever its scores are, so (61) reads $\mathrm{pr}_n(J(n)) = 0$ for all $n \in \mathbb{S}$, where $J(n)$ is the sum of $n\psi_S$ and of the terms of the hidden units. As in (57), these terms are $\sigma(nc_j)\,\psi_{a_j} + \sigma'(nc_j)\,(n\psi_{c_j})\,a_j$ for a $\tanh$ or GELU unit, and $\sigma(nc_j)\,nW_j + \sigma'(nc_j)\,(n\psi_{c_j})\,(ne_j)\,a_j$ for a gated unit, with the matrix $W_j = \psi_{e_j} a_j + e_j\psi_{a_j} \in \mathbb{R}^{d_{\text{model}}\times d_{\text{model}}}$. So

$$\mathrm{pr}_n\big(J(n)\big) \;=\; \mathrm{pr}_n(n\psi_S) + \sum_j \Big[\sigma(nc_j)\,\Theta_j(n) + \sigma'(nc_j)\,\Theta'_j(n)\Big], \tag{62}$$

with the polynomial maps

$$\begin{aligned}
&\tanh,\ \text{GELU:} && \Theta_j(n) = \mathrm{pr}_n(\psi_{a_j}), && \Theta'_j(n) = (n\psi_{c_j})\ \mathrm{pr}_n(a_j),\\
&\text{SwiGLU, GeGLU:} && \Theta_j(n) = \mathrm{pr}_n(nW_j), && \Theta'_j(n) = (n\psi_{c_j})\,(ne_j)\ \mathrm{pr}_n(a_j).
\end{aligned}$$

We show that Θ'_j and Θ_j vanish on $\mathbb{S}$, after saying what this gives.

What $\Theta'_j = 0$ and $\Theta_j = 0$ on $\mathbb{S}$ give. Let Θ'_j vanish on $\mathbb{S}$. For $n \in \mathbb{S}$ that is not a multiple of a_j, $\mathrm{pr}_n(a_j) \neq 0$, so $n\psi_{c_j} = 0$, or $(n\psi_{c_j})(ne_j) = 0$ for a gated unit. As $e_j \neq 0$, in both cases $n\psi_{c_j} = 0$ on a dense subset of $\mathbb{S}$, so $\psi_{c_j} = 0$, and $\Theta'_j = 0$ as a polynomial map. Let Θ_j vanish on $\mathbb{S}$ as well. For a $\tanh$ or GELU unit, ψ_{a_j} is then a multiple of every $n \in \mathbb{S}$, so $\psi_{a_j} = 0$. For a gated unit, every $n \in \mathbb{S}$ is a left eigenvector of W_j, so $W_j = \lambda_j I$ for a number λ_j. Let $v \neq 0$ be a row with $ve_j = 0$. Then $\lambda_j v = vW_j = (v\psi_{e_j})\,a_j$. If $\lambda_j \neq 0$, every such v is a multiple of a_j, which is impossible for $d_{\text{model}} \geq 3$ and gives $a_j e_j = 0$ for $d_{\text{model}} = 2$. So $W_j = 0$, that is, $\psi_{e_j} a_j = -e_j\psi_{a_j}$. Multiplying on the right by $a_j^\top$ gives $\psi_{e_j} = -\alpha_j e_j$ with $\alpha_j = \psi_{a_j} a_j^\top / \|a_j\|^2$, and then $e_j\,(\psi_{a_j} - \alpha_j a_j) = 0$, so $\psi_{a_j} = \alpha_j a_j$. In both cases the terms of unit j in $J(n)$ vanish at every n.

$\tanh$ and SwiGLU units: poles. Here σ is $\tanh$ or SiLU. Its poles are the points $z_m = \mathrm{i}\,y_m$, $m \in \mathbb{Z}$, with $y_m = \pi(m + \frac{1}{2})$ for $\tanh$ and $y_m = \pi(2m+1)$ for SiLU, and as $\zeta \to 0$

$$\sigma(z_m + \zeta) = \frac{\varrho_m}{\zeta} + O(1), \qquad \sigma'(z_m + \zeta) = -\frac{\varrho_m}{\zeta^2} + O(1),$$

with $\varrho_m = 1$ for $\tanh$ and $\varrho_m = z_m$ for SiLU. Fix a unit j and let $\xi = c_j^\top / \|c_j\|$. Call a unit row ξ' *admissible* if $\xi' c_j = 0$ and $\xi' c_k \neq 0$ for every unit $k \neq j$. The admissible rows are dense among the unit rows orthogonal to c_j, because no c_k with $k \neq j$ is collinear with c_j. On the great circle of ξ and an admissible ξ',

$$n(\vartheta)\,c_j \;=\; \sqrt{d_{\text{model}}}\,\|c_j\|\cos\vartheta.$$

For $m \in \mathbb{Z}$ let $\chi_m \in \mathbb{R}$ be the solution of $\sqrt{d_{\text{model}}}\,\|c_j\|\sinh\chi_m = y_m$, and let $\vartheta_m = \frac{\pi}{2} - \mathrm{i}\chi_m$. Then $\cos\vartheta_m = \mathrm{i}\sinh\chi_m$ and $\sin\vartheta_m = \cosh\chi_m$, so

$$n(\vartheta_m) \;=\; \sqrt{d_{\text{model}}}\,\big(\mathrm{i}\sinh\chi_m\,\xi + \cosh\chi_m\,\xi'\big), \qquad n(\vartheta_m)\,c_j = z_m.$$

For $k \neq j$ the number $n(\vartheta_m)\, c_k$ has the real part $\sqrt{d_{\text{model}}} \cosh \chi_m\, (\xi' c_k) \neq 0$, so it is not a pole of σ, whose poles are imaginary. The function $\zeta(\vartheta) = n(\vartheta)\, c_j - z_m$ has a simple zero at ϑ_m, as $\zeta'(\vartheta_m) = -\sqrt{d_{\text{model}}}\, \|c_j\| \cosh \chi_m \neq 0$. So by (62), near ϑ_m,

$$\operatorname{pr}_{n(\vartheta)}\big(J(n(\vartheta))\big) \;=\; -\frac{\varrho_m}{\zeta(\vartheta)^2}\, \Theta_j'\big(n(\vartheta)\big) + \frac{\varrho_m}{\zeta(\vartheta)}\, \Theta_j\big(n(\vartheta)\big) + O(1). \tag{63}$$

The left side is meromorphic in ϑ and vanishes for real ϑ, so it vanishes identically, by (a). The coefficient of $(\vartheta - \vartheta_m)^{-2}$ in (63) is $-\varrho_m\, \Theta_j'(n(\vartheta_m))/\zeta'(\vartheta_m)^2$, so $\Theta_j'(n(\vartheta_m)) = 0$ for every $m \in \mathbb{Z}$. The numbers $e^{\mathrm{i}\vartheta_m} = \mathrm{i}\, e^{\chi_m}$ are pairwise different, so by (b) $\Theta_j'(n(\vartheta)) = 0$ for all ϑ, in particular on the real great circle of ξ and ξ'. Every $n \in \mathbb{S}$ that is not a multiple of ξ lies on the great circle of ξ and of the unit row along $n - \langle n, \xi\rangle\, \xi$. So the real great circles of ξ and of the admissible ξ' cover a dense subset of $\mathbb{S}$, and $\Theta_j' = 0$ on $\mathbb{S}$. Hence $\psi_{c_j} = 0$ and $\Theta_j' = 0$ as a polynomial map. The coefficient of $(\vartheta - \vartheta_m)^{-1}$ in (63) is then $\varrho_m\, \Theta_j(n(\vartheta_m))/\zeta'(\vartheta_m)$, so $\Theta_j(n(\vartheta_m)) = 0$ for every m, and in the same way $\Theta_j = 0$ on $\mathbb{S}$.

GELU and GeGLU units: the error function. Here $\sigma(z) = \frac{z}{2}\big(1 + \operatorname{erf}\frac{z}{\sqrt{2}}\big)$ and $\sigma'(z) = \frac{1}{2}\big(1 + \operatorname{erf}\frac{z}{\sqrt{2}}\big) + \frac{z}{\sqrt{2\pi}}\, e^{-z^2/2}$. On a great circle let $z_j(\vartheta) = n(\vartheta)\, c_j$. At $n = n(\vartheta)$, (62) is

$$\begin{aligned}\operatorname{pr}_n\big(J(n)\big) \;=\; &\Big[\operatorname{pr}_n(n\psi_S) + \tfrac{1}{2}\sum_j \big(z_j\Theta_j + \Theta_j'\big)\Big] \\ &+ \sum_j \tfrac{1}{2}\big(z_j\Theta_j + \Theta_j'\big)\, \operatorname{erf}\frac{z_j}{\sqrt{2}} \;+\; \sum_j \frac{z_j}{\sqrt{2\pi}}\, \Theta_j'\, e^{-z_j^2/2},\end{aligned} \tag{64}$$

where the bracket, $z_j\Theta_j + \Theta_j'$ and $z_j\Theta_j'$ are trigonometric polynomials in ϑ. We need the following two lemmas.

Lemma B.6 (Dawson's integral)**.** *For $y > 0$ let $\operatorname{daw}(y) = e^{-y^2} \int_0^y e^{\varsigma^2}\, d\varsigma$. For every integer $m \geq 1$, as $y \to \infty$,*

$$\operatorname{daw}(y) \;=\; \sum_{l=0}^{m-1} \frac{(2l-1)!!}{2^{\,l+1}}\, \frac{1}{y^{2l+1}} \;+\; O\Big(\frac{1}{y^{2m+1}}\Big), \qquad \textit{where } (-1)!! = 1.$$

Proof. For $l \geq 0$ let $\mathcal{I}_l(y) = \int_1^y e^{\varsigma^2} \varsigma^{-2l}\, d\varsigma$. An integration by parts, with $e^{\varsigma^2} = \frac{1}{2\varsigma}\frac{d}{d\varsigma} e^{\varsigma^2}$, gives

$$\mathcal{I}_l(y) \;=\; \frac{e^{y^2}}{2\, y^{2l+1}} + \frac{2l+1}{2}\, \mathcal{I}_{l+1}(y) + O(1),$$

and applying this for $l = 0, \ldots, m-1$,

$$\mathcal{I}_0(y) \;=\; \sum_{l=0}^{m-1} \frac{(2l-1)!!}{2^{\,l+1}}\, \frac{e^{y^2}}{y^{2l+1}} + \frac{(2m-1)!!}{2^{\,m}}\, \mathcal{I}_m(y) + O(1).$$

For $y \geq 2$,

$$\mathcal{I}_m(y) \;\leq\; \int_1^{y/2} e^{\varsigma^2}\, d\varsigma + \Big(\frac{2}{y}\Big)^{2m+1} \int_{y/2}^{y} \varsigma\, e^{\varsigma^2}\, d\varsigma \;\leq\; \frac{y}{2}\, e^{y^2/4} + \frac{2^{2m}}{y^{2m+1}}\, e^{y^2} \;=\; O\Big(\frac{e^{y^2}}{y^{2m+1}}\Big).$$

As $\int_0^y e^{\varsigma^2}\, d\varsigma = \mathcal{I}_0(y) + O(1)$, multiplying by e^{-y^2} gives the claim. □

Lemma B.7 (The error function on a circle)**.** *Let $r_j > 0$ and $\delta_j \in \mathbb{R}$, for j in a finite set, be such that the points $r_j(\cos\delta_j, \sin\delta_j)$ of $\mathbb{R}^2$ are pairwise different and no two of them are opposite. Let $z_j(\vartheta) = r_j \cos(\vartheta - \delta_j)$, and let p_0, p_j and q_j be trigonometric polynomials with values in $\mathbb{C}^{1\times d_{\text{model}}}$. If*

$$p_0(\vartheta) + \sum_j \Big[p_j(\vartheta)\, \operatorname{erf}\frac{z_j(\vartheta)}{\sqrt{2}} + q_j(\vartheta)\, e^{-z_j(\vartheta)^2/2}\Big] \;=\; 0 \qquad \textit{for all real } \vartheta, \tag{65}$$

then $p_0 = 0$ and $p_j = q_j = 0$ for every j.

Proof. Both sides of (65) are entire functions of ϑ, so (65) holds for all $\vartheta \in \mathbb{C}$. We use induction on the number of indices. Let j be an index with the largest r_j, and take $\vartheta = \delta_j + \frac{\pi}{2} + \mathrm{i}t$ with $t \to +\infty$. Let l_0 bound the degrees of the trigonometric polynomials, which are then $O(e^{l_0 t})$ on this line.

The index j. Here $z_j = -\mathrm{i}\, r_j \sinh t$. With $y = r_j \sinh t/\sqrt{2} > 0$,

$$e^{-z_j^2/2} = e^{y^2}, \qquad \operatorname{erf} \frac{z_j}{\sqrt{2}} = \operatorname{erf}(-\mathrm{i}y) = -\frac{2\mathrm{i}}{\sqrt{\pi}} \int_0^y e^{\varsigma^2}\, d\varsigma = -\frac{2\mathrm{i}}{\sqrt{\pi}}\, e^{y^2}\, \operatorname{daw}(y).$$

The other indices. For $k \neq j$ let $\delta_{jk} = \delta_j - \delta_k$. Then $z_k = -r_k \sin(\delta_{jk} + \mathrm{i}t)$ and

$$\operatorname{Re}\big(z_k^2\big) = r_k^2\big(\sin^2 \delta_{jk} - \cos(2\delta_{jk}) \sinh^2 t\big), \qquad \text{so} \qquad \big|e^{-z_k^2/2}\big| \;\le\; \exp\Big(\frac{r_k^2 \cos(2\delta_{jk})}{r_j^2}\, y^2\Big).$$

As $r_k \le r_j$, and $\cos(2\delta_{jk}) < 1$ if $r_k = r_j$, there is a number $\varkappa \in [0,1)$ with $|e^{-z_k^2/2}| \le e^{\varkappa y^2}$ for all $k \neq j$. For every $z \in \mathbb{C}$, $|\operatorname{erf} z| \le \frac{2}{\sqrt{\pi}}\, |z| \max\big(1, |e^{-z^2}|\big)$, by integrating $e^{-\varsigma^2}$ along the segment from 0 to z. So $|\operatorname{erf}(z_k/\sqrt{2})| \le r_k \cosh t\; e^{\varkappa y^2}$.

Together. Dividing (65) by e^{y^2} gives, as $t \to \infty$,

$$q_j - \frac{2\mathrm{i}}{\sqrt{\pi}}\, \operatorname{daw}(y)\, p_j \;=\; O\big(e^{(l_0+1)t}\, e^{-(1-\varkappa)y^2}\big), \tag{66}$$

which tends to zero faster than every power of e^{-t}. On the line, every component of p_j and q_j is a finite sum of powers e^{lt} with $|l| \le l_0$. Suppose that a component p of p_j is not zero, and let q be the same component of q_j. Then $|p| \ge e^{-(l_0+1)t}$ for large t, and (66) gives, for every integer m,

$$\operatorname{daw}(y) \;=\; \frac{\sqrt{\pi}}{2\mathrm{i}}\, \frac{q}{p} + O\big(y^{-m}\big) \qquad (y \to \infty).$$

Now $e^t = \sqrt{2}\, y/r_j + \sqrt{2y^2/r_j^2 + 1}$ is y times a convergent power series in $1/y$. So $\frac{\sqrt{\pi}}{2\mathrm{i}}\, \frac{q}{p}$, a rational function of e^t, equals for large y a convergent series in powers of $1/y$ with finitely many positive powers of y. This series is then an asymptotic expansion of $\operatorname{daw}(y)$, and such expansions are unique, so by Lemma B.6 its nonzero coefficients are the numbers $(2l-1)!!/2^{\,l+1}$. They grow faster than every geometric sequence, so the series does not converge, a contradiction. Hence p_j vanishes on the line, and a trigonometric polynomial that vanishes on a line is zero, so $p_j = 0$. Then by (66) q_j, a finite sum of powers of e^t on the line, tends to zero faster than every power of e^{-t}, so $q_j = 0$. The identity (65) now holds for the other indices, and by induction all p_k and q_k vanish. Then $p_0 = 0$. □

Call the great circle of ξ and ξ' *admissible* if the points $(\xi c_j,\ \xi' c_j)$ of $\mathbb{R}^2$, one per hidden unit, are nonzero and no two of them are equal or opposite. On an admissible circle $z_j(\vartheta) = r_j \cos(\vartheta - \delta_j)$ with $r_j(\cos\delta_j, \sin\delta_j) \;=\; \sqrt{d_{\text{model}}}\,(\xi c_j, \xi' c_j)$, and the left side of (64) vanishes for real ϑ. By Lemma B.7, $z_j\Theta_j' = 0$ and $z_j\Theta_j + \Theta_j' = 0$ on the circle for every j, and as z_j is not identically zero, $\Theta_j'(n(\vartheta)) = 0$ and $\Theta_j(n(\vartheta)) = 0$. Every $n \in \mathbb{S}$ lies on an admissible circle: take $\xi = n/\sqrt{d_{\text{model}}}$; the circle of ξ and ξ' is admissible unless ξ and ξ' are both orthogonal to one of the finitely many nonzero columns c_j and $c_j \pm c_k$, which excludes no ξ' if $d_{\text{model}} = 2$ and a nowhere dense set of ξ' if $d_{\text{model}} \ge 3$. So Θ_j' and Θ_j vanish on $\mathbb{S}$.

Step B: the second token fixed. By Step A every hidden unit is as in the statement, and its terms vanish at every input. Take a sequence whose second token is $u \in \mathbb{S}$ and whose other tokens all equal $w \in \mathbb{S}$, and let $v = u - w$. Let $k_i = 1$ if head i is causal and $k_i = T - 1$ otherwise, and

$$\omega_i \;=\; \tfrac{1}{2}\big(uB_iv^\top - \log k_i\big).$$

By (58), with $M = 0$ and with E and D replaced by the identity, the block gives the second token the output $uS - \frac{1}{2}vS + \frac{1}{2}\sum_i \tanh(\omega_i)\, vC_i$. Its derivative along η is

$$J_2(u,w) \;=\; \tfrac{1}{2}(u+w)\,\psi_S + \tfrac{1}{2}\sum_i \Big[\tanh(\omega_i)\, v\psi_{C_i} + \tfrac{1}{2}\tanh'(\omega_i)\,\big(u\psi_{B_i}v^\top\big)\, vC_i\Big],$$

and (61) at the second token is $\mathrm{pr}_u(J_2(u,w)) = 0$ for all $u, w \in \mathbb{S}$. We fix u and vary w. With the polynomial maps of w

$$\Theta_{u,i}(w) = \mathrm{pr}_u\big(v\psi_{C_i}\big), \qquad \Theta'_{u,i}(w) = \big(u\psi_{B_i}v^\top\big)\ \mathrm{pr}_u\big(vC_i\big), \qquad v = u - w,$$

$$\mathrm{pr}_u\big(J_2(u,w)\big) \;=\; \tfrac{1}{2}\,\mathrm{pr}_u\big((u+w)\,\psi_S\big) + \tfrac{1}{2}\sum_i \Big[\tanh(\omega_i)\,\Theta_{u,i}(w) + \tfrac{1}{2}\tanh'(\omega_i)\,\Theta'_{u,i}(w)\Big]. \quad (67)$$

Let $\mathcal{U} \subset \mathbb{S}$ be the set of u such that $uB_i \neq 0$ for every head, uB_i and $uB_{i'}$ are linearly independent for all $i \neq i'$, and no C_i has all its rows in $\mathbb{R}\,u$. It is open and dense in $\mathbb{S}$.

Fix $u \in \mathcal{U}$ and a head i, and let $\xi = uB_i/\|uB_i\|$. Call a unit row ξ' *admissible* if $\xi'(uB_i)^\top = 0$ and $\xi'(uB_{i'})^\top \neq 0$ for every head $i' \neq i$; the admissible rows are dense among the unit rows orthogonal to uB_i. Let $w(\vartheta) = \sqrt{d_{\mathrm{model}}}\,(\cos\vartheta\ \xi + \sin\vartheta\ \xi')$ be the great circle of ξ and an admissible ξ'. On it

$$\omega_i(\vartheta) \;=\; \tfrac{1}{2}\big(uB_iu^\top - \log k_i - \sqrt{d_{\mathrm{model}}}\,\|uB_i\|\cos\vartheta\big).$$

For $m \in \mathbb{Z}$ choose $\vartheta_m \in \mathbb{C}$ with $\omega_i(\vartheta_m) = \mathrm{i}\pi(m + \frac{1}{2})$, a pole of $\tanh$, that is, with

$$\sqrt{d_{\mathrm{model}}}\,\|uB_i\|\cos\vartheta_m \;=\; uB_iu^\top - \log k_i - \mathrm{i}\pi(2m+1).$$

As $\cos\vartheta_m$ is not real, $\sin\vartheta_m \neq 0$, so $\zeta(\vartheta) = \omega_i(\vartheta) - \omega_i(\vartheta_m)$ has a simple zero at ϑ_m. The numbers $\cos\vartheta_m$ are pairwise different, and hence so are the $e^{\mathrm{i}\vartheta_m}$.

For large $|m|$ no other head has a pole at ϑ_m. Indeed, a pole of a head $i' \neq i$ at w requires $\mathrm{Re}\,\big(uB_{i'}w^\top\big) = uB_{i'}u^\top - \log k_{i'}$, and

$$uB_{i'}w(\vartheta_m)^\top \;=\; \sqrt{d_{\mathrm{model}}}\,\Big(\cos\vartheta_m\ uB_{i'}\xi^\top + \sin\vartheta_m\ uB_{i'}\xi'^\top\Big).$$

Here the real part of $\cos\vartheta_m$ does not depend on m, $uB_{i'}\xi'^\top \neq 0$, and $|\mathrm{Re}\sin\vartheta_m| \to \infty$ as $|m| \to \infty$, because $\sin^2\vartheta_m = 1 - \cos^2\vartheta_m$ and the imaginary part of $\cos\vartheta_m$ tends to infinity in absolute value. So there is an m_0 such that no head $i' \neq i$ has a pole at $w(\vartheta_m)$ for $|m| \geq m_0$. For these m, $\tanh(\omega_i) = 1/\zeta + O(\zeta)$ and $\tanh'(\omega_i) = -1/\zeta^2 + O(1)$ near ϑ_m, so by (67)

$$\mathrm{pr}_u\big(J_2(u,w(\vartheta))\big) \;=\; -\frac{1}{4\,\zeta(\vartheta)^2}\,\Theta'_{u,i}\big(w(\vartheta)\big) + \frac{1}{2\,\zeta(\vartheta)}\,\Theta_{u,i}\big(w(\vartheta)\big) + O(1). \quad (68)$$

The left side is meromorphic in ϑ and vanishes for real ϑ, so it vanishes identically, by (a). The coefficient of $(\vartheta - \vartheta_m)^{-2}$ gives $\Theta'_{u,i}(w(\vartheta_m)) = 0$ for $|m| \geq m_0$, so by (b) $\Theta'_{u,i}(w(\vartheta)) = 0$ for all ϑ. As in Step A, over the admissible ξ' this gives $\Theta'_{u,i} = 0$ on $\mathbb{S}$.

Now $\Theta'_{u,i}(w)$ is the product of the number $u\psi_{B_i}(u-w)^\top$ and the row $\mathrm{pr}_u\big((u-w)\,C_i\big)$, both affine in w. This row does not vanish at all $w \in \mathbb{S}$: otherwise, adding and subtracting its values at w and $-w$, $\mathrm{pr}_u(uC_i) = 0$ and $\mathrm{pr}_u(wC_i) = 0$ for all $w \in \mathbb{S}$, hence for all rows w, so every row of C_i is a multiple of u, which $u \in \mathcal{U}$ excludes. So its zeros form an affine subspace that does not contain $\mathbb{S}$, they are nowhere dense in $\mathbb{S}$, and $u\psi_{B_i}(u-w)^\top = 0$ for all $w \in \mathbb{S}$. Subtracting this at w and at $-w$ gives $u\psi_{B_i}w^\top = 0$ for all $w \in \mathbb{S}$, so

$$u\,\psi_{B_i} = 0, \qquad \text{and} \quad \Theta'_{u,i} = 0 \ \text{ as a polynomial map.}$$

The coefficient of $(\vartheta - \vartheta_m)^{-1}$ in (68) then gives $\Theta_{u,i}(w(\vartheta_m)) = 0$ for $|m| \geq m_0$, and as before $\Theta_{u,i} = 0$ on $\mathbb{S}$. Subtracting its values at w and at $-w$ gives $\mathrm{pr}_u(w\psi_{C_i}) = 0$ for all $w \in \mathbb{S}$, hence for all rows w: every row of ψ_{C_i} is a multiple of u.

Both conclusions hold for every $u \in \mathcal{U}$. The rows of $\mathcal{U}$ span $\mathbb{R}^{1\times d_{\text{model}}}$, so $\psi_{B_i} = 0$. Two non-collinear rows of $\mathcal{U}$ show that every row of ψ_{C_i} vanishes, so $\psi_{C_i} = 0$. As in Step 1 for attention blocks, with E and D replaced by the identity, $\psi_{B_i} = 0$ and $\psi_{C_i} = 0$ give $\eta_{Q,i} = W_{Q,i}\Gamma_i$, $\eta_{K,i} = -W_{K,i}\Gamma_i^\top$, $\eta_{V,i} = W_{V,i}\Gamma_i'$ and $\eta_{O,i} = -\Gamma_i' W_{O,i}$. So η is a direction of the symmetries inside the block, and the derivative of g along it vanishes at every input. □

We recall Theorem 4.8 from Section 4.3.

Theorem 4.8 (Equality for one block)**.** *Let $L = 1$, with a* tanh*, GELU, ReLU, ReLU^2, ReGLU, SwiGLU, GeGLU or attention block.*
(i) *Without normalization, the skipless and the residual network satisfy the equality condition, and the skip raises the functional dimension by exactly d_{model}^2.*
(ii) *Under RMSNorm, the residual network satisfies the equality condition, and the skip raises the functional dimension by at least $d_{\text{model}}(d_{\text{model}} - 1)/2$. A learned skip raises it by exactly $d_{\text{model}}(d_{\text{model}} + 1)/2$ more than an identity skip.*

Proof of Theorem 4.8. Part (i) is Theorem B.1(i), and part (ii) is Theorem B.1(ii) and (iii). □

Extensions of part (i)

Grouped-query attention. If the H query heads share G heads of keys and values, the same proof applies with B_i and C_i formed from the keys and values of the group of head i. Step 1 uses only the queries and the output projection of all heads, and in its last part a direction that keeps the products $W_{Q,i}W_{K,g}^\top$ over the heads i of group g has one Γ for the whole group, since $W_{K,g}$ has full column rank; likewise for the values. So $s = 2Gd_h^2$, and the biases add $2d_{\text{model}} + 2Gd_h$.

Position embeddings. Let a learned position embedding add the row e_t of E_{pos} to token t of h_0, in a network with an attention block and no biases, and write η_{e_t} for the entry of a direction η at the position of e_t. As E has rank d_{model}, choose ϵ_t with $\epsilon_t E = e_t$: the network computes the network without position embedding on the tokens $\tilde{x}_t = x_t + \epsilon_t$. Along η, $h_{0,t} = \tilde{x}_t E$ has the derivative $\tilde{x}_t\eta_E + \nu_t$ with $\nu_t = \eta_{e_t} - \epsilon_t\eta_E$. This is the derivative of the network without position embedding along the other entries of η together with a shift κ_t of each token $\tilde{x}_t$, where $\kappa_t E = \nu_t$. Fix a token t that sees at least one other token, and inputs for which the tokens $\tilde{x}_s$, $s \neq t$, that token t sees all equal one vector w. The output at token t is symmetric in these tokens, so its derivative sees their shifts only through their mean $\bar{\kappa}$, and it is (58) with $u = \tilde{x}_t$, $v = u - w$ and k the number of these tokens. The shifts move u by κ_t and v by $\kappa_t - \bar{\kappa}$, which adds $\frac{1}{2}B_i(\kappa_t - \bar{\kappa})^\top$ to the derivatives of the input weights, $\frac{1}{2}\kappa_t B_i v^\top$ to those of the biases and $\frac{1}{2}(\kappa_t - \bar{\kappa})C_i$ to those of the output weights of the tanh units, and Step 2 again makes each of these derivatives vanish. For all v this gives $\mathrm{D}_\theta B_i\,\eta = 0$, $\mathrm{D}_\theta C_i\,\eta = 0$, $\kappa_t B_i = 0$ and $B_i(\kappa_t - \bar{\kappa})^\top = 0$ for every head, and since the queries and the keys of all heads have rank d_{model}, $\nu_t = \kappa_t E = 0$ and $\bar{\kappa}E = 0$. This holds for every token $t \geq 2$, and for every token without a mask; with a mask, $\bar{\kappa} = \kappa_1$ at $t = 2$, so $\nu_1 = 0$ as well. So $\eta_{e_t} = \epsilon_t\eta_E = e_t X$ by Step 1, the change of basis at h_0 applied to E_{pos}, and the values of part (i) hold with P counting E_{pos}. A fixed encoding, which is not trained, must instead be preserved by the change of basis at h_0; if its rows span a space of dimension r, this removes rd_{model} dimensions of symmetry from both networks, and the skip still adds d_{model}^2.

C The symmetry group of a transformer with its embedding and readout

Wang and Wang (2025) characterize the symmetry group of a transformer block. For a full transformer they state that the group is the product of the groups of its blocks, since residual connections and LayerNorm prevent any coupling across layers (Wang and Wang, 2025, Theorem 7). They do not define the group of the full model. We take the model to be the map from tokens to outputs, so that the embedding and the readout are among its parameters, as in the models they name (GPT-2, BERT,

LLaMA and Qwen) and in their end-to-end checks, which compare the logits and generated tokens of GPT-2 models. In this case two transformations of the whole stream, applied at every hidden state from the embedding to the readout, are counterexamples.

Table 4: Relative change of the output under the two transformations of the whole stream, worst case over 3 random transformers ($d_{\text{model}} = 8$, $H = 2$ heads, 3 layers, GELU MLP blocks with $d_{\text{hidden}} = 16$, generic gains and biases). The notebook also checks the assumptions of Wang and Wang (2025) before and after each transformation, and that both transformations change the function of the bare stack of blocks. Section 7 of the notebook (Appendix E).

Model	Global permutation	Global rotation
pre-LayerNorm, untied readout	6×10^{-16}	2×10^{-15}
pre-RMSNorm, untied readout	5×10^{-16}	9×10^{-16}
pre-LayerNorm, tied readout	5×10^{-16}	fails, at least 0.16
post-LayerNorm, tied readout	9×10^{-16}	not applicable

1. A *global permutation* of the stream coordinates, with every read, write, normalization gain and bias permuted along. It preserves the function at every parameter value, for every placement of the normalization and with a tied or an untied readout. Verma and Elbayad (2024, Section 3.3) apply it to merge BERT models, embeddings included.
2. A *global rotation* Q, orthogonal and, for LayerNorm, fixing the all-ones vector. In a pre-norm model with an untied readout it preserves the function when every read W after a normalization with gain g becomes $\operatorname{diag}(g)^{-1}Q^{\top}\operatorname{diag}(g)\,W$ and every LayerNorm bias b becomes $b\operatorname{diag}(g)^{-1}Q\operatorname{diag}(g)$; the gains are unchanged. This is the construction behind Theorem 1 of Ashkboos et al. (2024a).

Both transformations change the token embedding: its columns, the coordinates of the stream, are permuted or rotated. No element of the product of per-block groups changes the embedding, so both lie outside that product. The LayerNorm transformers of Table 4 satisfy the assumptions of Wang and Wang (2025), including pairwise distinct gains and generic biases, and so do the transformed parameters.

Their proof transforms the output of one block and argues that the residual term prevents the transformation (Wang and Wang, 2025, Theorem 30); applied from the embedding onward, the transformation acts on the residual term as well. Their lemma on LayerNorm holds the gain and the bias fixed (Wang and Wang, 2025, Lemma 26); the permutation moves both, and in a pre-norm block the normalized vector is only read, so the rotation does not need LayerNorm to commute with it. For a stack of blocks that acts on a fixed input and is read in fixed coordinates, both transformations change the function, and the counterexample does not apply.

The two transformations are instances of the one change of basis that the skips leave (Theorem A.5(iii)), restricted to what the normalization lets through: the rotation needs a normalization that reads the whole vector, and the permutation also commutes with per-channel gains. With a tied readout the rotation fails: the readout after the final gain would have to become $\operatorname{diag}(g)^{-1}Q^{\top}\operatorname{diag}(g)\,E^{\top}$, while tying fixes it to $Q^{\top}E^{\top}$.

D Counts at model scale

Table 5 applies the exact reduction of Proposition A.9 to published models. Without skips, every hidden state keeps its own change of basis, and each turns one submatrix into a triangular matrix; with skips and an untied readout, the whole network keeps one (Section 3.3). Under RMSNorm each change of basis is orthogonal and frees $d_{\text{model}}(d_{\text{model}} - 1)/2$ parameters, and under LayerNorm the same holds with $d_{\text{model}} - 1$ in place of d_{model} (Proposition A.13). Llama-3.1 uses RMSNorm (Grattafiori et al., 2024), and its parameter counts are from the released weights; for the two ViT-B, with $d_{\text{model}} = 768$ and $L = 24$ sub-layers, we assume standard pre-LayerNorm blocks, which reduce to RMSNorm of dimension 767. The skipless ViT-B has 25 hidden states: one before each of its 24 blocks and one before the readout. One triangular 767×767 submatrix of a read per hidden state

removes $25 \cdot 767 \cdot 766/2 = 7.3$M parameters. The residual ViT-B keeps one orthogonal change of basis, and so does a pre-RMSNorm language model whose readout is not tied to its embedding. Each removes the entries below the diagonal of one $d_{\text{model}} \times d_{\text{model}}$ submatrix.

Table 5: Parameters that can be removed from trained networks without changing their function. Section 8 of the notebook (Appendix E).

Model	Skips	Parameters	Removable exactly
ViT-B of Ji et al. (2026)	no	86.6M	7.3M (8.5 %)
ViT-B	yes	86.6M	0.29M (0.3 %)
Llama-3.1-8B	yes	8.0B	8.4M (0.10 %)
Llama-3.1-70B	yes	70.6B	34M (0.05 %)
Llama-3.1-405B	yes	405.9B	134M (0.03 %)

The gap between the two ViT-B, 24 copies of $\mathrm{O}(767)$ or 7.1M dimensions of symmetry, 8.1 percent of the parameters, is what the skips remove (Theorem A.14), and under the equality condition what the residual network gains in functional dimension. In general, a transformer layer with an MLP block of $d_{\text{hidden}} = 4d_{\text{model}}$ has about $12d_{\text{model}}^2$ parameters, and under normalization the skips of its two sub-layers remove about d_{model}^2 dimensions of symmetry, a twelfth of the parameters of the blocks.

Inside the attention blocks, with or without skips, each head also has a change of basis between its queries and keys and one between its values and outputs: $12 \cdot 2Hd_h^2 = 1.2$M dimensions for ViT-B. With the one copy of $\mathrm{O}(767)$ that the residual ViT-B keeps, this gives 1.5M, the count of van Nierop (2024, Eq. 17).

E Numerical checks and reproducibility

All computations are in double precision, in a notebook that will be released in a public Git repository, coming soon; its sections are listed at the end of this appendix. Unless stated otherwise, weights have independent $\mathcal{N}(0, 1/\text{fan-in})$ entries, inputs have independent $\mathcal{N}(0, 1)$ entries, and networks compared with and without skips share their weights; the notebook gives the few exceptions.

Exactness checks. For Theorems A.5 and A.11 and Proposition A.9 the transformed weights are constructed as in the proofs, and both networks are evaluated on random inputs; the reported number is $\|f' - f\|/\|f\|$ over the batch, and values near 10^{-15} are floating-point rounding.

The sweep. At depth six, with biases, the sweep of Theorem A.5(i) turns identity units with trainable U and V into identity skips, with U and V equal to I up to 2×10^{-15} and the function unchanged up to 2×10^{-15}, and its forward version (Remark A.6) absorbs learned affine skips with skip biases up to 10^{-15}. In Theorem A.11(ii) a scaled orthogonal skip is absorbed to 10^{-9}, which is the RMSNorm constant $\varepsilon = 10^{-8}$ acting on the overall scale (Lemma A.2), and a generic skip fails at 0.43.

Homogeneous blocks under RMSNorm. For Lemma A.3, on skipless ReLU, ReLU^2 and ReGLU networks of the size of Figure 4 with $\varepsilon = 10^{-1}$, 10^{-3} and 10^{-6}, a change of basis $e^{0.3S}$ at h_0, h_1 or h_2, or the scaling $e^{0.3}$ at h_3, multiplies each output by the factor Λ of the proof up to 5×10^{-12}, and with $\varepsilon = 0$ leaves the function unchanged up to 10^{-13}; the derivatives match part (ii) up to 4×10^{-12} of the output (Section 17 of the notebook).

Triangular skips and tied readouts. The checks of Section 3 use a pre-RMSNorm network with $d_{\text{model}} = 8$, $L = 5$ $\tanh$ blocks with $d_{\text{hidden}} = 16$ and learned skips near the identity, with $\varepsilon = 10^{-2}$. Making every skip triangular changes the function by 5×10^{-16}, and making it symmetric positive definite by 4×10^{-15}. With a readout tied to the embedding through the gain of the final normalization, making the first four skips triangular and leaving the last one full changes it by 7×10^{-16} and removes 112 parameters; making all five triangular and inserting one 8×8 matrix before the gain also changes it by 7×10^{-16} and removes $140 - 64 = 76$, and using also the rotation that this matrix frees at h_0, on a triangular block of the embedding, and the gain, which it absorbs,

changes it by 10^{-15} and removes 112, no more than without the matrix. Without normalization and with a tied readout, setting the first four skips and 8 rows of the embedding to the identity changes the function by 10^{-13} and removes 320 parameters. For identity skips, a global rotation changes the tied network by 0.22, and one inserted matrix restores it to 8×10^{-16}. Setting $r = 2$ rows of each low-rank factor U to the identity changes the function by 3×10^{-15}, and without normalization the sweep of Theorem A.5 turns the low-rank skips into identity skips with a change of 6×10^{-16}.

Changes of basis that differ across a skip. In a separate residual network of $L = 2$ blocks without normalization ($d_{\text{model}} = 8$, $d_{\text{hidden}} = 16$), a change of basis G with $\|I - G\| = 0.41$ at the two hidden states after the first block, so that only the skip of that block joins two bases, changes the output by 0.28, 0.25, 0.050 and 0.0046 when the branch of the first block is scaled by 0.1, 1, 10 and 100, which makes it about 0.05, 0.5, 5 and 50 times the size of the skip, of the order of $\|h\|\|I - G\|/\|B(h)\|$ when the branch dominates (Section 3.3).

Functional dimension. The parameter-to-output map is evaluated on a fixed batch of inputs, and its Jacobian with respect to all parameters is formed exactly by automatic differentiation, or by central finite differences with step 10^{-6} for the small $\tanh$ networks of Table 6. For exact Jacobians the rank is read at the largest gap of the spectrum, the largest ratio s_i/s_{i+1} of consecutive singular values, 10^5 to 10^{12} for the rows of Table 6; for finite differences at the relative threshold 10^{-7}, which lies inside the gap between the genuine singular values, above 10^{-6} of the largest, and the noise of the differences, below 10^{-8}. The rank on a batch is a lower bound on the functional dimension, and where P minus it equals the symmetry count, it is the functional dimension (Theorem A.14(ii)). The values of Theorem B.1(i) hold exactly for all five activations at $d_{\text{model}} = 4$, $d_{\text{hidden}} = 8$, in three seeds, with and without biases, with gaps of 10^{11} to 10^{14}; with biases the inputs must cross the kink of every unit, so these checks use inputs of scale 5. For one attention block ($d_{\text{model}} = 4$, input and output dimension 6, sequences of three tokens, $(H, d_h) = (2, 2)$, $(4, 1)$ and $(1, 4)$, causal and bidirectional, with and without biases, three seeds each), all 72 networks have the values of Theorem B.1(i), with gaps of at least 4×10^{11}. Under pre-RMSNorm with $\varepsilon = 10^{-2}$ ($d_{\text{model}} = 4$, three seeds each), the residual networks with one block have the value of Theorem B.1(ii) for all five activations and for attention, and the skipless ones have $P - d_{\text{model}}(d_{\text{model}} - 1) - s$, so the skip adds exactly $d_{\text{model}}(d_{\text{model}} - 1)/2$ in these checks. A direction that the network uses only on rare inputs, such as a rarely active unit, has small singular values, and resolving it takes far more than the minimum of about P/d_{out} inputs.

The networks of Figure 4. Every network of Figure 4 is drawn 20 times, and so are the same networks with GeGLU blocks and with GLU blocks, whose gate is the logistic sigmoid; both have the counts of SwiGLU. Except for the skipless ReGLU networks under RMSNorm, the rank equals the count in every draw, at gaps of at least 6×10^4 without normalization and 10^7 under RMSNorm with $\varepsilon = 10^{-5}$. In the skipless ReGLU networks under RMSNorm, at gaps between 40 and 2×10^4, the rank read at the largest gap is 263 in 14 draws, the count 264 in 5 and 253 in one, and with hidden units added 283 in 19 draws and the count 284 in one. With $\varepsilon = 10^{-12}$ the skipless ReGLU network has rank 233 in every draw, at gaps of at least 10^4: this is $P - Ld_{\text{model}}^2 - (d_{\text{model}}(d_{\text{model}} - 1)/2 + 1) - s$, the count of the network in which every normalization but the last is removed. The residual ReGLU network then has rank $281 = P - (d_{\text{model}}(d_{\text{model}} - 1)/2 + 1) - s$ in every draw (Section 4.5). With $\varepsilon = 10^{-2}$ the rank equals the count in every draw for all six activations, ReGLU included, at gaps of at least 2.8×10^9; the notebook also plots these spectra as in Figure 4.

The size of ε. At $\varepsilon = 0$ the skipless ReGLU network has 31 more dimensions of symmetry (Lemma A.3), and a small ε breaks them only weakly, which makes its rank hard to read at $\varepsilon = 10^{-5}$. RMSNorm divides a row $h^{(t)}$ by $\sqrt{r^2 + \varepsilon}$, where $r^2 = \|h^{(t)}\|^2/d_{\text{model}}$ is its mean square, so ε acts relative to r^2. With inputs of unit variance, the median of r^2 over the inputs, and then over the draws, is 0.74 at h_0 and between 0.15 and 0.33 after the blocks of the skipless networks, and it grows along the stream of the residual networks to between 1.6 and 1.9 at h_3; at these medians ε/r^2 lies between 5×10^{-6} and 7×10^{-5} for $\varepsilon = 10^{-5}$, and is about a thousand times larger for $\varepsilon = 10^{-2}$. In a skipless network every block reads a normalized state, so the scale of the inputs enters only at h_0; the stream of a residual network carries it to every hidden state. At $\varepsilon = 10^{-5}$ the singular values with indices 234 to 264 of the skipless ReGLU network, those of these 31 directions, lie between $10^{-4.5}$

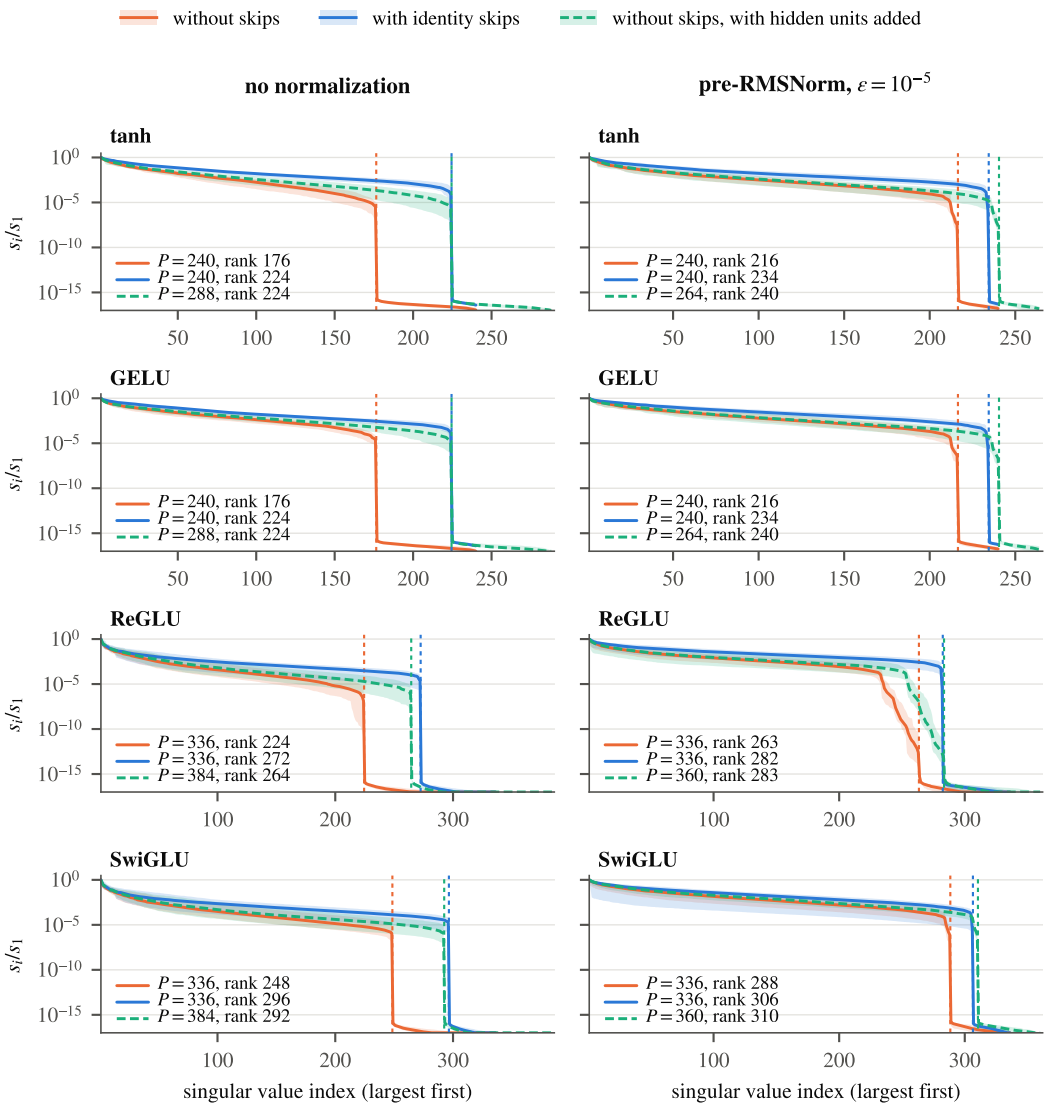


Figure 4: Singular values of the Jacobian on 4000 inputs, for the networks of Figure 1 with 4 kinds of blocks, and without skips with hidden units added; left without normalization, right with pre-RMSNorm. Dashed lines mark the rank. It equals the symmetry count in every draw, except for skipless ReGLU networks under RMSNorm.

and $10^{-12.6}$ of the largest in the median over the draws, except the last, which lies at the floor. The factors $\varepsilon/(r^2 + \varepsilon)$ that break these directions depend on the input, and they are large for the inputs whose hidden states have a small mean square: in every draw, between 11 and 592 of the 4000 inputs have a hidden state after a block with $r^2 < 10^{-3}$, where $\varepsilon/r^2 > 10^{-2}$, and in 13 of the 20 draws up to 378 inputs have one that is exactly zero, with every gate closed.

Trained networks. We also train the networks of Figure 4, with $\tanh$, GELU, ReGLU and SwiGLU blocks, on four classification datasets with at least four classes: two generated with scikit-learn, and satimage and segmentation from the Penn Machine Learning Benchmarks, with the features as inputs and one output per class. Training uses AdamW in single precision, a learning rate chosen on a validation set, early stopping and five seeds. A network counts as trained when it beats logistic regression with a tuned penalty on the test set, as 436 of 480 do. On inputs of the dataset and on Gaussian inputs alike, the rank equals the count in 398 of the 436, and the skips add 48 and 18 (Figure 5). Of the 38 exceptions, 36 are skipless ReGLU networks under RMSNorm; one residual

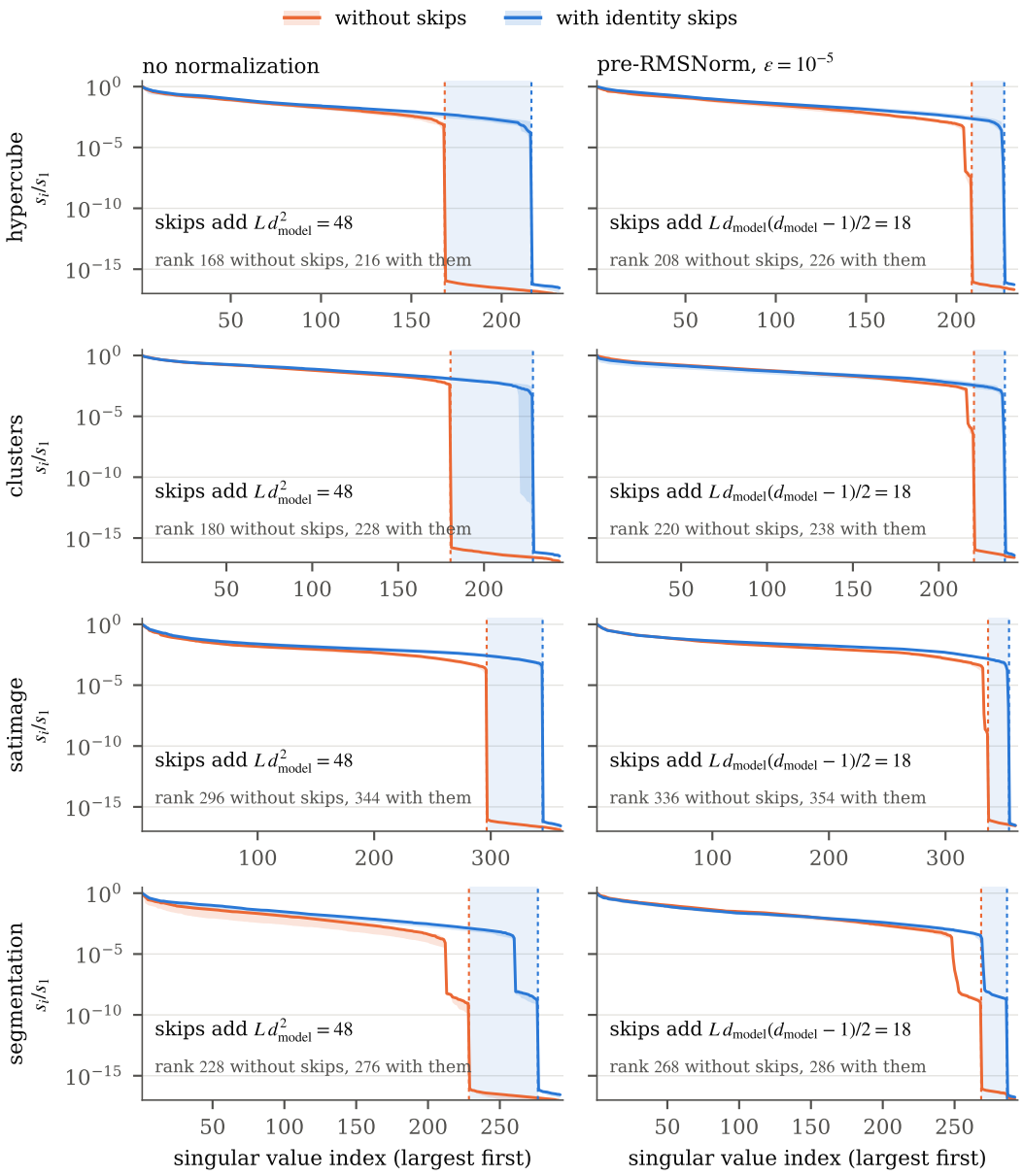


Figure 5: Singular values of the Jacobian at trained weights, on 1000 inputs of each dataset: the GELU networks of Figure 4, trained on 4 datasets (rows); median and range over 5 seeds. The skips add Ld_{model}^2, and $Ld_{\text{model}}(d_{\text{model}}-1)/2$ under RMSNorm, as at random weights.

GELU network is 8 below the count and one residual tanh network 1 below. Two further notebooks train these networks, store them and compute their Jacobians.

Learned skips. The tanh networks of Table 6 with learned skips or with trainable identity units have the functional dimension of the residual network, 224, as Theorem A.5(ii) requires. Under RMSNorm with $\varepsilon = 10^{-2}$, learned skips have $264 - 234 = 30 = 3\,d_{\text{model}}(d_{\text{model}}+1)/2$ more functional dimension than identity skips, as Theorem A.11(iii) predicts. On residual networks without biases or normalization ($d_{\text{model}} = 4$, $L = 3$, $d_{\text{hidden}} = 8$), using the last change of basis on d_{model} columns of the first read leaves exactly the redundancy s of the symmetries inside the blocks, the last row of Table 3.

Low-rank skips. Networks with skips $I + UV^\top$ ($L = 3$ tanh blocks with $d_{\text{hidden}} = 2d_{\text{model}}$, $d_{\text{model}} = 4$ with $r = 1, 2$ and $d_{\text{model}} = 6$ with $r = 2, 3$, exact Jacobians) have redundancy

Table 6: Functional-dimension checks. *Rank* is the rank of the Jacobian on the batch; where the redundancy, P minus it, equals the prediction, the rank meets the upper bound of Theorem A.14(ii) and is the functional dimension. Every network has an embedding and a separate readout, and no biases unless marked; normalization is pre-norm. A *mixed* skip is a pair of trainable identity units U, V per block. The MLP networks have $d_{\text{model}} = 4$, $d_{\text{hidden}} = 8$, $L = 3$ unless marked, and input and output dimension 6, except the rows marked *large*, which have $d_{\text{model}} = 16$, $L = 6$, $d_{\text{hidden}} = 16$ and input and output dimension 20. The transformers have $d_{\text{model}} = 8$, $H = 2$ heads of width $d_h = 4$, MLP blocks with $d_{\text{hidden}} = 16$, a vocabulary of 16 tokens and 2 layers; each attention block has $2Hd_h^2$ symmetries inside. †The exact rank modulo a prime certifies the value at two parameter points; without skips at $L = 3$, the rank in floating point sits at the double-precision floor. ‡The symmetry count is 268; an embedding table read through RMSNorm adds $d_{\text{model}}(d_{\text{model}} + 1)/2 = 36$, because N_ε maps $\mathbb{R}^{d_{\text{model}}}$ onto an open ball, so the rows of the table can absorb any invertible map near the identity, and with continuous inputs the measured value is 268.

Network	Skip	Normalization	P	Rank	Redundancy	Prediction
tanh	none	none	240	176	64	$(L{+}1)d_{\text{model}}^2 = 64$
tanh	identity	none	240	224	16	$d_{\text{model}}^2 = 16$
tanh	learned	none	288	224	64	$(L{+}1)d_{\text{model}}^2 = 64$
tanh	mixed	none	336	224	112	$(2L{+}1)d_{\text{model}}^2 = 112$
tanh, biases	none	none	286	206	80	$(L{+}1)(d_{\text{model}}^2{+}d_{\text{model}}) = 80$
tanh, biases	identity	none	286	254	32	$d_{\text{model}}^2 + (L{+}1)d_{\text{model}} = 32$
tanh	none	RMSNorm, $\varepsilon = 10^{-9}$	240	212	28	$(L{+}1)\dim \mathrm{CO}(4) = 28$
tanh	identity	RMSNorm, $\varepsilon = 10^{-9}$	240	233	7	$\dim \mathrm{CO}(4) = 7$
tanh	learned	RMSNorm, $\varepsilon = 10^{-9}$	288	260	28	$(L{+}1)\dim \mathrm{CO}(4) = 28$
tanh	none	RMSNorm, $\varepsilon = 10^{-2}$	240	216	24	$(L{+}1)\dim \mathrm{O}(4) = 24$
tanh	identity	RMSNorm, $\varepsilon = 10^{-2}$	240	234	6	$\dim \mathrm{O}(4) = 6$
tanh	learned	RMSNorm, $\varepsilon = 10^{-2}$	288	264	24	$(L{+}1)\dim \mathrm{O}(4) = 24$
ReLU^2	none	none	240	152	88†	$(L{+}1)d_{\text{model}}^2 + Lm = 88$
ReLU^2	identity	none	240	200	40†	$d_{\text{model}}^2 + Lm = 40$
ReGLU	none	none	336	224	112†	$(L{+}1)d_{\text{model}}^2 + 2Lm = 112$
ReGLU	identity	none	336	272	64†	$d_{\text{model}}^2 + 2Lm = 64$
ReLU^2, $L = 8$	none	none	560	352	208†	$(L{+}1)d_{\text{model}}^2 + Lm = 208$
ReLU^2, $L = 8$	identity	none	560	480	80†	$d_{\text{model}}^2 + Lm = 80$
ReGLU, $L = 8$	none	none	816	544	272†	$(L{+}1)d_{\text{model}}^2 + 2Lm = 272$
ReGLU, $L = 8$	identity	none	816	672	144†	$d_{\text{model}}^2 + 2Lm = 144$
tanh, large	none	none	3712	1920	1792	$(L{+}1)d_{\text{model}}^2 = 1792$
tanh, large	identity	none	3712	3456	256	$d_{\text{model}}^2 = 256$
tanh, large	none	RMSNorm, $\varepsilon = 10^{-2}$	3712	2872	840	$(L{+}1)\dim \mathrm{O}(16) = 840$
tanh, large	identity	RMSNorm, $\varepsilon = 10^{-2}$	3712	3592	120	$\dim \mathrm{O}(16) = 120$
transformer	identity	RMSNorm, $\varepsilon = 10^{-2}$	1280	1124	156	$\dim \mathrm{O}(8) + 4Hd_h^2 = 156$
transformer	none	RMSNorm, $\varepsilon = 10^{-2}$	1280	976	304‡	$5\dim \mathrm{O}(8) + 4Hd_h^2 = 268$

$d_{\text{model}}^2 + 2Ld_{\text{model}}r$ without normalization, all of U and V besides the global change of basis, and $d_{\text{model}}(d_{\text{model}} - 1)/2 + Lr(2r - 1)$ under RMSNorm with $\varepsilon = 10^{-2}$. Besides the global rotation and the r^2 dimensions of the $\mathrm{GL}(r)$ inside each product, each skip adds $r(r - 1)$. To first order, rotations $I + \Omega_{\ell-1}$ and $I + \Omega_\ell$ of the two hidden states that it joins change it by $(\Omega_\ell - \Omega_{\ell-1}) - \Omega_{\ell-1}UV^\top + UV^\top\Omega_\ell$, which keeps it within rank r of the identity when $\Omega_\ell - \Omega_{\ell-1}$ lies in the tangent space $\{XV^\top + UY^\top\}$ of the matrices of rank r at $UV^\top$; the antisymmetric matrices in this space have dimension $r(r - 1)$ when $2r \le d_{\text{model}}$ and U, V are generic, which the notebook confirms in every case.

Where the redundancy exceeds the count. For ReLU networks the exact Jacobian is taken with the activation pattern held fixed. On skipless ReLU networks without biases ($d_{\text{model}} = 4$, $L = 3$, $d_{\text{hidden}} = 8$), 7 of 8 random draws attain the bound of Theorem A.14(ii), redundancy 88 including the rescalings of hidden units, and all 8 residual draws attain their bound, 40. In the eighth skipless draw, five hidden units of the last block are active on exactly the same inputs, those on which one hidden unit of the block before is active, so there they act as one linear map of $\mathbb{R}^4$: their 40 parameters carry $40 - 16 = 24$ dimensions of redundancy instead of the 5 rescalings of the count, and the redundancy is $88 + 19 = 107$. Such coincidences need a block that reads the output of another block, so they do not occur for one block, where Theorem B.1 applies; this is consistent with Grigsby et al. (2023).

On the 3000 inputs of Section 1 of the notebook, the redundancy of a second draw also exceeds the count, by 6, because one of its hidden units is active on only 0.08 percent of the inputs; with $30{,}000$ inputs or more it attains the bound, and the eighth draw stays at 107 (Section 15 of the notebook). Skipless networks whose first hidden state is an embedding table read through RMSNorm are also more redundant than their symmetries (Table 6).

Certificates at depth. The rows of Table 6 at $L = 8$ are certified as those at $L = 3$, with the activation pattern from ball arithmetic and the Jacobian from reverse mode (Section 16 of the notebook). Without skips these networks start from odd pairs of hidden units, plus noise: ReLU^2 hidden units (u, w) and $(-u, -w)$, and ReGLU hidden units with gates u and $-u$ that share their value and write. With identity skips the weights are Gaussian, with the writes scaled by $1/L$. From Gaussian weights, the skipless ReGLU networks at $L = 8$ attain the count on $12{,}000$ inputs, and the ReLU^2 ones do not. In each ReLU^2 draw, 1 to 5 hidden units of the last three blocks stay off on 2×10^6 fresh inputs, and a unit that is never on leaves its 8 parameters redundant, 7 more than its rescaling. With inputs added on which the rarely active units fire, the rank stays below the count by 35, 7 and 29: 7 for each unit that stays off, and one more in the third draw.

DyT and nanoGPT. DyT replaces a normalization by $\tanh(\alpha h)$ with a learnable scale α (Zhu et al., 2025). DyT networks ($d_{\text{model}} = 4$, $L = 3$ blocks with $d_{\text{hidden}} = 8$, exact Jacobians, 3 draws) lose $L = 3$ dimensions to the skips, the scale of each hidden state, which the learnable α of the next DyT absorbs. The nanoGPT models of Table 7 follow the `model.py` of nanoGPT (Karpathy, 2023), with token and position embeddings, pre-LayerNorm blocks with $\varepsilon = 10^{-5}$, causal attention with $H = 4$ heads of width $d_h = 2$, GELU MLP blocks with $d_{\text{hidden}} = 4d_{\text{model}}$, a final LayerNorm and a readout, and no biases; without normalization every LayerNorm is the identity. The Jacobians are exact, over all logits of 64 random token sequences per layer. With skips the redundancy is, per layer, 144 with LayerNorm: 32 from the changes of basis inside the heads, 16 from the LayerNorm gains against the next matrix, 56 from the zero mean of each LayerNorm output, seen by every column of the next matrix, and 40 from the mean of each write, which every later LayerNorm erases. Once per model come 60 more: the global rotation, 21; the means of the 16 embedding rows, 16; the final LayerNorm against the readout, 16; and a vector moved from the position to the token embedding, 7. Without normalization it is 32 per layer and $72 = 64 + 8$ per model. For one and two layers we construct all of these symmetries, and in every draw they span the null space of the Jacobian.

Table 7: Redundancy of nanoGPT ($d_{\text{model}} = 8$, $H = 4$ heads of width $d_h = 2$, a vocabulary and a context of 8, no biases, untied readout): P minus the rank of the exact Jacobian, the same in all 4 draws of every network and equal to the prediction. N layers make $L = 2N$ blocks. With skips, each layer is redundant by 144 with LayerNorm and by 32 without normalization, and the model once more by 60 and 72 (Appendix E).

	$N = 1$	2	3	4	Prediction
LayerNorm					
P	984	1768	2552	3336	
without skips	246	432	618	804	$144N + 60 + 21L$
with skips	204	348	492	636	$144N + 60$
No normalization					
P	960	1728	2496	3264	
without skips	232	392	552	712	$32N + 72 + 64L$
with skips	104	136	168	200	$32N + 72$

Notebook. The notebook runs on a CPU in an environment pinned with uv, with Python 3.11. Its sections follow the paper and are listed below, with their times on a CPU with two cores. The two long experiments save what Figure 4 and Table 7 need, from which the notebook draws the figure and the table in seconds unless it is asked to rerun them; the times below are those of the full computation.

Section of the notebook	What it reproduces	Time
1. The gauge fix, the counts and RMSNorm	exactness checks of Theorems A.5 and A.11 and Proposition A.9; finite-difference rows of Table 6; exact ReLU and degree-two Jacobians	1 min
2. The last row of Table 3	the last row of Table 3	23 s
3. A dominant block, and learned, tied and low-rank skips	the checks of Sections 3, 3.3 and 3.4	under 1 s
4. The redundancy of low-rank skips	the low-rank skips of Section 3.4	36 s
5. DyT reads	the DyT networks	9 s
6. Exact reductions after training	the reductions of Section 3.1 and Proposition A.13	under 1 s
7. The symmetry group of a transformer is not a product	Table 4	under 1 s
8. Counts at model scale	Table 5	under 1 s
9. One block	Theorem B.1, for MLP and attention blocks	44 s
10. Figure 4	Figures 1 and 4 and the checks of their networks	28 min
11. Exact Jacobians	the rows at $d_{\mathrm{model}} = 16$, the transformers and the embedding table of Table 6	4 min
12. Certificates modulo a prime	the certificates at $L = 3$ in Table 6	19 s
13. nanoGPT	Table 7 and the constructed symmetries	38 min
14. Skipless networks as limits of residual networks	the limits of the companion paper	under 1 s
15. ReLU draws above the count	the ReLU draws above the count in Appendix E	2 min
16. Certificates at depth	the certificates at $L = 8$ in Table 6	5 min
17. Homogeneous blocks under RMSNorm	the checks of Lemma A.3	2 s